Ответы и официальные разборы

Том 10. Официальные оригиналы

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52nd International Mathematical Olympiad

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Problem Shortlist with Solutions

IMO2011 Amsterdam

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Problem shortlist with solutions

IMPORTANT

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Mexico, Mongolia, Montenegro, Pakistan, Poland, Romania,
Russian Federation, Saudi Arabia, Serbia, Slovakia, Slovenia,
Sweden, Taiwan, Thailand, Turkey, Ukraine, United Kingdom,
United States of America

Algebra

A1

A1

For any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers with sum $s_A = a_1 + a_2 + a_3 + a_4$, let p_A denote the number of pairs (i, j) with $1 \leq i < j \leq 4$ for which $a_i + a_j$ divides s_A. Among all sets of four distinct positive integers, determine those sets A for which p_A is maximal.

A2

A2

Determine all sequences $(x_1, x_2, \ldots, x_{2011})$ of positive integers such that for every positive integer n there is an integer a with

$$x_1^n + 2x_2^n + \cdots + 2011x_{2011}^n = a^{n+1} + 1.$$

A3

A3

Determine all pairs (f, g) of functions from the set of real numbers to itself that satisfy

$$g(f(x + y)) = f(x) + (2x + y)g(y)$$

for all real numbers x and y.

A4

A4

Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy

$$f^{g(n)+1}(n) + g^{f(n)}(n) = f(n + 1) - g(n + 1) + 1$$

for every positive integer n. Here, $f^k(n)$ means $\underbrace{f(f(\ldots f}_{k}(n)\ldots))$.

A5

A5

Prove that for every positive integer n, the set $\{2, 3, 4, \ldots, 3n + 1\}$ can be partitioned into n triples in such a way that the numbers from each triple are the lengths of the sides of some obtuse triangle.

A6

A6

Let f be a function from the set of real numbers to itself that satisfies

$$f(x+y) \leq yf(x) + f(f(x))$$

for all real numbers x and y. Prove that $f(x) = 0$ for all $x \leq 0$.

A7

A7

Let a, b, and c be positive real numbers satisfying $\min(a+b, b+c, c+a) > \sqrt{2}$ and $a^2+b^2+c^2 = 3$. Prove that

$$\frac{a}{(b+c-a)^2} + \frac{b}{(c+a-b)^2} + \frac{c}{(a+b-c)^2} \geq \frac{3}{(abc)^2}.$$

Combinatorics

C1

C1

Let $n > 0$ be an integer. We are given a balance and n weights of weight $2^0, 2^1, \ldots, 2^{n-1}$. In a sequence of n moves we place all weights on the balance. In the first move we choose a weight and put it on the left pan. In each of the following moves we choose one of the remaining weights and we add it either to the left or to the right pan. Compute the number of ways in which we can perform these n moves in such a way that the right pan is never heavier than the left pan.

C2

C2

Suppose that 1000 students are standing in a circle. Prove that there exists an integer k with $100 \leq k \leq 300$ such that in this circle there exists a contiguous group of $2k$ students, for which the first half contains the same number of girls as the second half.

C3

C3

Let $\mathcal{S}$ be a finite set of at least two points in the plane. Assume that no three points of $\mathcal{S}$ are collinear. By a *windmill* we mean a process as follows. Start with a line ℓ going through a point $P \in \mathcal{S}$. Rotate ℓ clockwise around the *pivot* P until the line contains another point Q of $\mathcal{S}$. The point Q now takes over as the new pivot. This process continues indefinitely, with the pivot always being a point from $\mathcal{S}$.

Show that for a suitable $P \in \mathcal{S}$ and a suitable starting line ℓ containing P, the resulting windmill will visit each point of $\mathcal{S}$ as a pivot infinitely often.

C4

C4

Determine the greatest positive integer k that satisfies the following property: The set of positive integers can be partitioned into k subsets $A_1, A_2, \ldots, A_k$ such that for all integers $n \geq 15$ and all $i \in \{1, 2, \ldots, k\}$ there exist two distinct elements of A_i whose sum is n.

C5

C5

Let m be a positive integer and consider a checkerboard consisting of m by m unit squares. At the midpoints of some of these unit squares there is an ant. At time 0, each ant starts moving with speed 1 parallel to some edge of the checkerboard. When two ants moving in opposite directions meet, they both turn 90° clockwise and continue moving with speed 1. When more than two ants meet, or when two ants moving in perpendicular directions meet, the ants continue moving in the same direction as before they met. When an ant reaches one of the edges of the checkerboard, it falls off and will not re-appear.

Considering all possible starting positions, determine the latest possible moment at which the last ant falls off the checkerboard or prove that such a moment does not necessarily exist.

C6

C6

Let n be a positive integer and let $W = \ldots x_{-1}x_0x_1x_2\ldots$ be an infinite periodic word consisting of the letters a and b. Suppose that the minimal period N of W is greater than 2^n.

A finite nonempty word U is said to *appear* in W if there exist indices $k \leq \ell$ such that $U = x_kx_{k+1}\ldots x_\ell$. A finite word U is called *ubiquitous* if the four words Ua, Ub, aU, and bU all appear in W. Prove that there are at least n ubiquitous finite nonempty words.

C7

C7

On a square table of 2011 by 2011 cells we place a finite number of napkins that each cover a square of 52 by 52 cells. In each cell we write the number of napkins covering it, and we record the maximal number k of cells that all contain the same nonzero number. Considering all possible napkin configurations, what is the largest value of k?

Geometry

G1

G1

Let ABC be an acute triangle. Let ω be a circle whose center L lies on the side BC. Suppose that ω is tangent to AB at B' and to AC at C'. Suppose also that the circumcenter O of the triangle ABC lies on the shorter arc $B'C'$ of ω. Prove that the circumcircle of ABC and ω meet at two points.

G2

G2

Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcenter and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

G3

G3

Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC, and CD. Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA, and AB. Prove that the midpoint of the segment EF lies on the line through the two intersection points of ω_E and ω_F.

G4

G4

Let ABC be an acute triangle with circumcircle Ω. Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB. Let D be the foot of the altitude from A, and let G be the centroid of the triangle ABC. Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G, and X are collinear.

G5

G5

Let ABC be a triangle with incenter I and circumcircle ω. Let D and E be the second intersection points of ω with the lines AI and BI, respectively. The chord DE meets AC at a point F, and BC at a point G. Let P be the intersection point of the line through F parallel to AD and the line through G parallel to BE. Suppose that the tangents to ω at A and at B meet at a point K. Prove that the three lines AE, BD, and KP are either parallel or concurrent.

G6

G6

Let ABC be a triangle with $AB = AC$, and let D be the midpoint of AC. The angle bisector of $\angle BAC$ intersects the circle through D, B, and C in a point E inside the triangle ABC. The line BD intersects the circle through A, E, and B in two points B and F. The lines AF and BE meet at a point I, and the lines CI and BD meet at a point K. Show that I is the incenter of triangle KAB.

G7

G7

Let $ABCDEF$ be a convex hexagon all of whose sides are tangent to a circle ω with center O. Suppose that the circumcircle of triangle ACE is concentric with ω. Let J be the foot of the perpendicular from B to CD. Suppose that the perpendicular from B to DF intersects the line EO at a point K. Let L be the foot of the perpendicular from K to DE. Prove that $DJ = DL$.

G8

G8

Let ABC be an acute triangle with circumcircle ω. Let t be a tangent line to ω. Let t_a, t_b, and t_c be the lines obtained by reflecting t in the lines BC, CA, and AB, respectively. Show that the circumcircle of the triangle determined by the lines t_a, t_b, and t_c is tangent to the circle ω.

Number Theory

N1

N1

For any integer $d > 0$, let $f(d)$ be the smallest positive integer that has exactly d positive divisors (so for example we have $f(1) = 1$, $f(5) = 16$, and $f(6) = 12$). Prove that for every integer $k \geq 0$ the number $f(2^k)$ divides $f(2^{k+1})$.

N2

N2

Consider a polynomial $P(x) = (x + d_1)(x + d_2) \cdot \ldots \cdot (x + d_9)$, where d_1, $d_2, \ldots$, d_9 are nine distinct integers. Prove that there exists an integer N such that for all integers $x \geq N$ the number $P(x)$ is divisible by a prime number greater than 20.

N3

N3

Let $n \geq 1$ be an odd integer. Determine all functions f from the set of integers to itself such that for all integers x and y the difference $f(x) - f(y)$ divides $x^n - y^n$.

N4

N4

For each positive integer k, let $t(k)$ be the largest odd divisor of k. Determine all positive integers a for which there exists a positive integer n such that all the differences

$$t(n + a) - t(n), \quad t(n + a + 1) - t(n + 1), \quad \ldots, \quad t(n + 2a - 1) - t(n + a - 1)$$

are divisible by 4.

N5

N5

Let f be a function from the set of integers to the set of positive integers. Suppose that for any two integers m and n, the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that for all integers m, n with $f(m) \leq f(n)$ the number $f(n)$ is divisible by $f(m)$.

N6

N6

Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients such that no nonconstant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)} - 1$ divides $3^{P(n)} - 1$. Prove that $Q(x)$ is a constant polynomial.

N7

N7

Let p be an odd prime number. For every integer a, define the number

$$S_a = \frac{a}{1} + \frac{a^2}{2} + \cdots + \frac{a^{p-1}}{p-1}.$$

Let m and n be integers such that

$$S_3 + S_4 - 3S_2 = \frac{m}{n}.$$

Prove that p divides m.

N8

N8

Let k be a positive integer and set $n = 2^k + 1$. Prove that n is a prime number if and only if the following holds: there is a permutation $a_1, \ldots, a_{n-1}$ of the numbers $1, 2, \ldots, n-1$ and a sequence of integers $g_1, g_2, \ldots, g_{n-1}$ such that n divides $g_i^{a_i} - a_{i+1}$ for every $i \in \{1, 2, \ldots, n-1\}$, where we set $a_n = a_1$.

A1

For any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers with sum $s_A = a_1 + a_2 + a_3 + a_4$, let p_A denote the number of pairs (i, j) with $1 \le i < j \le 4$ for which $a_i + a_j$ divides s_A. Among all sets of four distinct positive integers, determine those sets A for which p_A is maximal.

Answer. The sets A for which p_A is maximal are the sets the form $\{d, 5d, 7d, 11d\}$ and $\{d, 11d, 19d, 29d\}$, where d is any positive integer. For all these sets p_A is 4.

Solution. Firstly, we will prove that the maximum value of p_A is at most 4. Without loss of generality, we may assume that $a_1 < a_2 < a_3 < a_4$. We observe that for each pair of indices (i, j) with $1 \le i < j \le 4$, the sum $a_i + a_j$ divides s_A if and only if $a_i + a_j$ divides $s_A - (a_i + a_j) = a_k + a_l$, where k and l are the other two indices. Since there are 6 distinct pairs, we have to prove that at least two of them do not satisfy the previous condition. We claim that two such pairs are (a_2, a_4) and (a_3, a_4). Indeed, note that $a_2 + a_4 > a_1 + a_3$ and $a_3 + a_4 > a_1 + a_2$. Hence $a_2 + a_4$ and $a_3 + a_4$ do not divide s_A. This proves $p_A \le 4$.

Now suppose $p_A = 4$. By the previous argument we have

$$\begin{aligned} a_1 + a_4 \mid a_2 + a_3 \quad &\text{and} \quad a_2 + a_3 \mid a_1 + a_4, \\ a_1 + a_2 \mid a_3 + a_4 \quad &\text{and} \quad a_3 + a_4 \nmid a_1 + a_2, \\ a_1 + a_3 \mid a_2 + a_4 \quad &\text{and} \quad a_2 + a_4 \nmid a_1 + a_3. \end{aligned}$$

Hence, there exist positive integers m and n with $m > n \ge 2$ such that

$$\begin{cases} a_1 + a_4 = a_2 + a_3 \\ m(a_1 + a_2) = a_3 + a_4 \\ n(a_1 + a_3) = a_2 + a_4. \end{cases}$$

Adding up the first equation and the third one, we get $n(a_1 + a_3) = 2a_2 + a_3 - a_1$. If $n \ge 3$, then $n(a_1 + a_3) > 3a_3 > 2a_2 + a_3 > 2a_2 + a_3 - a_1$. This is a contradiction. Therefore $n = 2$. If we multiply by 2 the sum of the first equation and the third one, we obtain

$$6a_1 + 2a_3 = 4a_2,$$

while the sum of the first one and the second one is

$$(m + 1)a_1 + (m - 1)a_2 = 2a_3.$$

Adding up the last two equations we get

$$(m + 7)a_1 = (5 - m)a_2.$$

It follows that $5 - m \geq 1$, because the left-hand side of the last equation and a_2 are positive. Since we have $m > n = 2$, the integer m can be equal only to either 3 or 4. Substituting $(3, 2)$ and $(4, 2)$ for (m, n) and solving the previous system of equations, we find the families of solutions $\{d, 5d, 7d, 11d\}$ and $\{d, 11d, 19d, 29d\}$, where d is any positive integer.

A2

Determine all sequences $(x_1, x_2, \ldots, x_{2011})$ of positive integers such that for every positive integer n there is an integer a with

$$x_1^n + 2x_2^n + \cdots + 2011x_{2011}^n = a^{n+1} + 1.$$

Answer. The only sequence that satisfies the condition is

$$(x_1, \ldots, x_{2011}) = (1, k, \ldots, k) \quad \text{with } k = 2 + 3 + \cdots + 2011 = 2023065.$$

Solution. Throughout this solution, the set of positive integers will be denoted by $\mathbb{Z}_+$.

Put $k = 2 + 3 + \cdots + 2011 = 2023065$. We have

$$1^n + 2k^n + \cdots 2011k^n = 1 + k \cdot k^n = k^{n+1} + 1$$

for all n, so $(1, k, \ldots, k)$ is a valid sequence. We shall prove that it is the only one.

Let a valid sequence $(x_1, \ldots, x_{2011})$ be given. For each $n \in \mathbb{Z}_+$ we have some $y_n \in \mathbb{Z}_+$ with

$$x_1^n + 2x_2^n + \cdots + 2011x_{2011}^n = y_n^{n+1} + 1.$$

Note that $x_1^n + 2x_2^n + \cdots + 2011x_{2011}^n < (x_1 + 2x_2 + \cdots + 2011x_{2011})^{n+1}$, which implies that the sequence (y_n) is bounded. In particular, there is some $y \in \mathbb{Z}_+$ with $y_n = y$ for infinitely many n.

Let m be the maximum of all the x_i. Grouping terms with equal x_i together, the sum $x_1^n + 2x_2^n + \cdots + 2011x_{2011}^n$ can be written as

$$x_1^n + 2x_2^n + \cdots + x_{2011}^n = a_m m^n + a_{m-1}(m-1)^n + \cdots + a_1$$

with $a_i \geq 0$ for all i and $a_1 + \cdots + a_m = 1 + 2 + \cdots + 2011$. So there exist arbitrarily large values of n, for which

$$a_m m^n + \cdots + a_1 - 1 - y \cdot y^n = 0. \tag{1}$$

The following lemma will help us to determine the a_i and y:

Lemma. Let integers $b_1, \ldots, b_N$ be given and assume that there are arbitrarily large positive integers n with $b_1 + b_2 2^n + \cdots + b_N N^n = 0$. Then $b_i = 0$ for all i.

Proof. Suppose that not all b_i are zero. We may assume without loss of generality that $b_N \neq 0$.

Dividing through by N^n gives

$$|b_N| = \left| b_{N-1}\left(\frac{N-1}{N}\right)^n + \cdots + b_1\left(\frac{1}{N}\right)^n \right| \leq (|b_{N-1}| + \cdots + |b_1|)\left(\frac{N-1}{N}\right)^n .$$

The expression $\left(\frac{N-1}{N}\right)^n$ can be made arbitrarily small for n large enough, contradicting the assumption that b_N be non-zero. □

We obviously have $y > 1$. Applying the lemma to (1) we see that $a_m = y = m$, $a_1 = 1$, and all the other a_i are zero. This implies $(x_1, \ldots, x_{2011}) = (1, m, \ldots, m)$. But we also have $1 + m = a_1 + \cdots + a_m = 1 + \cdots + 2011 = 1 + k$ so $m = k$, which is what we wanted to show.

A3

Determine all pairs (f, g) of functions from the set of real numbers to itself that satisfy

$$g(f(x+y)) = f(x) + (2x+y)g(y)$$

for all real numbers x and y.

Answer. Either both f and g vanish identically, or there exists a real number C such that $f(x) = x^2 + C$ and $g(x) = x$ for all real numbers x.

Solution. Clearly all these pairs of functions satisfy the functional equation in question, so it suffices to verify that there cannot be any further ones. Substituting $-2x$ for y in the given functional equation we obtain

$$g(f(-x)) = f(x). \tag{1}$$

Using this equation for $-x-y$ in place of x we obtain

$$f(-x-y) = g(f(x+y)) = f(x) + (2x+y)g(y). \tag{2}$$

Now for any two real numbers a and b, setting $x = -b$ and $y = a+b$ we get

$$f(-a) = f(-b) + (a-b)g(a+b).$$

If c denotes another arbitrary real number we have similarly

$$f(-b) = f(-c) + (b-c)g(b+c)$$

as well as

$$f(-c) = f(-a) + (c-a)g(c+a).$$

Adding all these equations up, we obtain

$$\big((a+c)-(b+c)\big)g(a+b) + \big((a+b)-(a+c)\big)g(b+c) + \big((b+c)-(a+b)\big)g(a+c) = 0.$$

Now given any three real numbers x, y, and z one may determine three reals a, b, and c such that $x = b+c$, $y = c+a$, and $z = a+b$, so that we get

$$(y-x)g(z) + (z-y)g(x) + (x-z)g(y) = 0.$$

This implies that the three points $(x, g(x))$, $(y, g(y))$, and $(z, g(z))$ from the graph of g are collinear. Hence that graph is a line, i.e., g is either a constant or a linear function.

Let us write $g(x) = Ax + B$, where A and B are two real numbers. Substituting $(0, -y)$ for (x, y) in (2) and denoting $C = f(0)$, we have $f(y) = Ay^2 - By + C$. Now, comparing the coefficients of x^2 in (1) we see that $A^2 = A$, so $A = 0$ or $A = 1$.

If $A = 0$, then (1) becomes $B = -Bx + C$ and thus $B = C = 0$, which provides the first of the two solutions mentioned above.

Now suppose $A = 1$. Then (1) becomes $x^2 - Bx + C + B = x^2 - Bx + C$, so $B = 0$. Thus, $g(x) = x$ and $f(x) = x^2 + C$, which is the second solution from above.

Comment. Another way to show that $g(x)$ is either a constant or a linear function is the following. If we interchange x and y in the given functional equation and subtract this new equation from the given one, we obtain

$$f(x) - f(y) = (2y + x)g(x) - (2x + y)g(y).$$

Substituting $(x, 0)$, $(1, x)$, and $(0, 1)$ for (x, y), we get

$$\begin{aligned} f(x) - f(0) &= xg(x) - 2xg(0), \\ f(1) - f(x) &= (2x + 1)g(1) - (x + 2)g(x), \\ f(0) - f(1) &= 2g(0) - g(1). \end{aligned}$$

Taking the sum of these three equations and dividing by 2, we obtain

$$g(x) = x\big(g(1) - g(0)\big) + g(0).$$

This proves that $g(x)$ is either a constant of a linear function.

A4

Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy

$$f^{g(n)+1}(n) + g^{f(n)}(n) = f(n+1) - g(n+1) + 1$$

for every positive integer n. Here, $f^k(n)$ means $\underbrace{f(f(\ldots f}_{k}(n)\ldots))$.

Answer. The only pair (f, g) of functions that satisfies the equation is given by $f(n) = n$ and $g(n) = 1$ for all n.

Solution. The given relation implies

$$f\left(f^{g(n)}(n)\right) < f(n+1) \quad \text{for all } n, \tag{1}$$

which will turn out to be sufficient to determine f.

Let $y_1 < y_2 < \ldots$ be all the values attained by f (this sequence might be either finite or infinite). We will prove that for every positive n the function f attains at least n values, and we have $(\mathrm{i})_n$: $f(x) = y_n$ if and only if $x = n$, and $(\mathrm{ii})_n$: $y_n = n$. The proof will follow the scheme

$$(\mathrm{i})_1,\ (\mathrm{ii})_1,\ (\mathrm{i})_2,\ (\mathrm{ii})_2,\ \ldots,\ (\mathrm{i})_n,\ (\mathrm{ii})_n,\ \ldots \tag{2}$$

To start, consider any x such that $f(x) = y_1$. If $x > 1$, then (1) reads $f\left(f^{g(x-1)}(x-1)\right) < y_1$, contradicting the minimality of y_1. So we have that $f(x) = y_1$ is equivalent to $x = 1$, establishing $(\mathrm{i})_1$.

Next, assume that for some n statement $(\mathrm{i})_n$ is established, as well as all the previous statements in (2). Note that these statements imply that for all $k \geq 1$ and $a < n$ we have $f^k(x) = a$ if and only if $x = a$.

Now, each value y_i with $1 \leq i \leq n$ is attained at the unique integer i, so y_{n+1} exists. Choose an arbitrary x such that $f(x) = y_{n+1}$; we necessarily have $x > n$. Substituting $x - 1$ into (1) we have $f\left(f^{g(x-1)}(x-1)\right) < y_{n+1}$, which implies

$$f^{g(x-1)}(x-1) \in \{1, \ldots, n\} \tag{3}$$

Set $b = f^{g(x-1)}(x-1)$. If $b < n$ then we would have $x - 1 = b$ which contradicts $x > n$. So $b = n$, and hence $y_n = n$, which proves $(\mathrm{ii})_n$. Next, from $(\mathrm{i})_n$ we now get $f(k) = n \iff k = n$, so removing all the iterations of f in (3) we obtain $x - 1 = b = n$, which proves $(\mathrm{i})_{n+1}$.

So, all the statements in (2) are valid and hence $f(n) = n$ for all n. The given relation between f and g now reads $n + g^n(n) = n + 1 - g(n+1) + 1$ or $g^n(n) + g(n+1) = 2$, from which it

immediately follows that we have $g(n) = 1$ for all n.

Comment. Several variations of the above solution are possible. For instance, one may first prove by induction that the smallest n values of f are exactly $f(1) < \cdots < f(n)$ and proceed as follows. We certainly have $f(n) \geq n$ for all n. If there is an n with $f(n) > n$, then $f(x) > x$ for all $x \geq n$. From this we conclude $f^{g(n)+1}(n) > f^{g(n)}(n) > \cdots > f(n)$. But we also have $f^{g(n)+1} < f(n+1)$. Having squeezed in a function value between $f(n)$ and $f(n+1)$, we arrive at a contradiction.

In any case, the inequality (1) plays an essential rôle.

A5

Prove that for every positive integer n, the set $\{2, 3, 4, \ldots, 3n + 1\}$ can be partitioned into n triples in such a way that the numbers from each triple are the lengths of the sides of some obtuse triangle.

Solution. Throughout the solution, we denote by $[a, b]$ the set $\{a, a + 1, \ldots, b\}$. We say that $\{a, b, c\}$ is an *obtuse triple* if a, b, c are the sides of some obtuse triangle.

We prove by induction on n that there exists a partition of $[2, 3n + 1]$ into n obtuse triples A_i $(2 \le i \le n + 1)$ having the form $A_i = \{i, a_i, b_i\}$. For the base case $n = 1$, one can simply set $A_2 = \{2, 3, 4\}$. For the induction step, we need the following simple lemma.

Lemma. Suppose that the numbers $a < b < c$ form an obtuse triple, and let x be any positive number. Then the triple $\{a, b + x, c + x\}$ is also obtuse.

Proof. The numbers $a < b + x < c + x$ are the sides of a triangle because $(c + x) - (b + x) = c - b < a$. This triangle is obtuse since $(c+x)^2-(b+x)^2 = (c-b)(c+b+2x) > (c-b)(c+b) > a^2$. □

Now we turn to the induction step. Let $n > 1$ and put $t = \lfloor n/2 \rfloor < n$. By the induction hypothesis, there exists a partition of the set $[2, 3t + 1]$ into t obtuse triples $A'_i = \{i, a'_i, b'_i\}$ $(i \in [2, t + 1])$. For the same values of i, define $A_i = \{i, a'_i + (n - t), b'_i + (n - t)\}$. The constructed triples are obviously disjoint, and they are obtuse by the lemma. Moreover, we have

$$\bigcup_{i=2}^{t+1} A_i = [2, t + 1] \cup [n + 2, n + 2t + 1].$$

Next, for each $i \in [t + 2, n + 1]$, define $A_i = \{i, n + t + i, 2n + i\}$. All these sets are disjoint, and

$$\bigcup_{i=t+2}^{n+1} A_i = [t + 2, n + 1] \cup [n + 2t + 2, 2n + t + 1] \cup [2n + t + 2, 3n + 1],$$

so

$$\bigcup_{i=2}^{n+1} A_i = [2, 3n + 1].$$

Thus, we are left to prove that the triple A_i is obtuse for each $i \in [t + 2, n + 1]$.

Since $(2n + i) - (n + t + i) = n - t < t + 2 \le i$, the elements of A_i are the sides of a triangle. Next, we have

$$(2n+i)^2 - (n+t+i)^2 = (n-t)(3n+t+2i) \ge \frac{n}{2} \cdot (3n+3(t+1)+1) > \frac{n}{2} \cdot \frac{9n}{2} \ge (n+1)^2 \ge i^2,$$

so this triangle is obtuse. The proof is completed.

A6

Let f be a function from the set of real numbers to itself that satisfies

$$f(x+y) \leq yf(x) + f(f(x)) \tag{1}$$

for all real numbers x and y. Prove that $f(x) = 0$ for all $x \leq 0$.

Solution 1. Substituting $y = t - x$, we rewrite (1) as

$$f(t) \leq tf(x) - xf(x) + f(f(x)). \tag{2}$$

Consider now some real numbers a, b and use (2) with $t = f(a)$, $x = b$ as well as with $t = f(b)$, $x = a$. We get

$$f(f(a)) - f(f(b)) \leq f(a)f(b) - bf(b),$$
$$f(f(b)) - f(f(a)) \leq f(a)f(b) - af(a).$$

Adding these two inequalities yields

$$2f(a)f(b) \geq af(a) + bf(b).$$

Now, substitute $b = 2f(a)$ to obtain $2f(a)f(b) \geq af(a) + 2f(a)f(b)$, or $af(a) \leq 0$. So, we get

$$f(a) \geq 0 \quad \text{for all } a < 0. \tag{3}$$

Now suppose $f(x) > 0$ for some real number x. From (2) we immediately get that for every $t < \dfrac{xf(x) - f(f(x))}{f(x)}$ we have $f(t) < 0$. This contradicts (3); therefore

$$f(x) \leq 0 \quad \text{for all real } x, \tag{4}$$

and by (3) again we get $f(x) = 0$ for all $x < 0$.

We are left to find $f(0)$. Setting $t = x < 0$ in (2) we get

$$0 \leq 0 - 0 + f(0),$$

so $f(0) \geq 0$. Combining this with (4) we obtain $f(0) = 0$.

Solution 2. We will also use the condition of the problem in form (2). For clarity we divide the argument into four steps.

Step 1. We begin by proving that f attains nonpositive values only. Assume that there exist some real number z with $f(z) > 0$. Substituting $x = z$ into (2) and setting $A = f(z)$, $B = -zf(z) - f(f(z))$ we get $f(t) \le At + B$ for all real t. Hence, if for any positive real number t we substitute $x = -t$, $y = t$ into (1), we get

$$\begin{aligned} f(0) &\le tf(-t) + f(f(-t)) \le t(-At + B) + Af(-t) + B \\ &\le -t(At - B) + A(-At + B) + B = -At^2 - (A^2 - B)t + (A + 1)B. \end{aligned}$$

But surely this cannot be true if we take t to be large enough. This contradiction proves that we have indeed $f(x) \le 0$ for all real numbers x. Note that for this reason (1) entails

$$f(x + y) \le yf(x) \tag{5}$$

for all real numbers x and y.

Step 2. We proceed by proving that f has at least one zero. If $f(0) = 0$, we are done. Otherwise, in view of Step 1 we get $f(0) < 0$. Observe that (5) tells us now $f(y) \le yf(0)$ for all real numbers y. Thus we can specify a positive real number a that is so large that $f(a)^2 > -f(0)$. Put $b = f(a)$ and substitute $x = b$ and $y = -b$ into (5); we learn $-b^2 < f(0) \le -bf(b)$, i.e. $b < f(b)$. Now we apply (2) to $x = b$ and $t = f(b)$, which yields

$$f(f(b)) \le \big(f(b) - b\big)f(b) + f(f(b)),$$

i.e. $f(b) \ge 0$. So in view of Step 1, b is a zero of f.

Step 3. Next we show that if $f(a) = 0$ and $b < a$, then $f(b) = 0$ as well. To see this, we just substitute $x = b$ and $y = a - b$ into (5), thus getting $f(b) \ge 0$, which suffices by Step 1.

Step 4. By Step 3, the solution of the problem is reduced to showing $f(0) = 0$. Pick any zero r of f and substitute $x = r$ and $y = -1$ into (1). Because of $f(r) = f(r - 1) = 0$ this gives $f(0) \ge 0$ and hence $f(0) = 0$ by Step 1 again.

Comment 1. Both of these solutions also show $f(x) \le 0$ for all real numbers x. As one can see from Solution 1, this task gets much easier if one already knows that f takes nonnegative values for sufficiently small arguments. Another way of arriving at this statement, suggested by the proposer, is as follows:

Put $a = f(0)$ and substitute $x = 0$ into (1). This gives $f(y) \le ay + f(a)$ for all real numbers y. Thus if for any real number x we plug $y = a - x$ into (1), we obtain

$$f(a) \le (a - x)f(x) + f(f(x)) \le (a - x)f(x) + af(x) + f(a)$$

and hence $0 \le (2a - x)f(x)$. In particular, if $x < 2a$, then $f(x) \ge 0$.

Having reached this point, one may proceed almost exactly as in the first solution to deduce $f(x) \le 0$ for all x. Afterwards the problem can be solved in a few lines as shown in steps 3 and 4 of the second

solution.

Comment 2. The original problem also contained the question whether a nonzero function satisfying the problem condition exists. Here we present a family of such functions.

Notice first that if $g : (0, \infty) \longrightarrow [0, \infty)$ denotes any function such that

$$g(x + y) \geq yg(x) \tag{6}$$

for all positive real numbers x and y, then the function f given by

$$f(x) = \begin{cases} -g(x) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \tag{7}$$

automatically satisfies (1). Indeed, we have $f(x) \leq 0$ and hence also $f(f(x)) = 0$ for all real numbers x. So (1) reduces to (5); moreover, this inequality is nontrivial only if x and y are positive. In this last case it is provided by (6).

Now it is not hard to come up with a nonzero function g obeying (6). E.g. $g(z) = Ce^z$ (where C is a positive constant) fits since the inequality $e^y > y$ holds for all (positive) real numbers y. One may also consider the function $g(z) = e^z - 1$; in this case, we even have that f is continuous.

A7

Let a, b, and c be positive real numbers satisfying $\min(a+b, b+c, c+a) > \sqrt{2}$ and $a^2+b^2+c^2 = 3$. Prove that

$$\frac{a}{(b+c-a)^2} + \frac{b}{(c+a-b)^2} + \frac{c}{(a+b-c)^2} \geq \frac{3}{(abc)^2}. \tag{1}$$

Throughout both solutions, we denote the sums of the form $f(a,b,c) + f(b,c,a) + f(c,a,b)$ by $\sum f(a,b,c)$.

Solution 1. The condition $b+c > \sqrt{2}$ implies $b^2+c^2 > 1$, so $a^2 = 3-(b^2+c^2) < 2$, i.e. $a < \sqrt{2} < b+c$. Hence we have $b+c-a > 0$, and also $c+a-b > 0$ and $a+b-c > 0$ for similar reasons.

We will use the variant of Hölder's inequality

$$\frac{x_1^{p+1}}{y_1^p} + \frac{x_1^{p+1}}{y_1^p} + \ldots + \frac{x_n^{p+1}}{y_n^p} \geq \frac{(x_1+x_2+\ldots+x_n)^{p+1}}{(y_1+y_2+\ldots+y_n)^p},$$

which holds for all positive real numbers $p, x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n$. Applying it to the left-hand side of (1) with $p=2$ and $n=3$, we get

$$\sum \frac{a}{(b+c-a)^2} = \sum \frac{(a^2)^3}{a^5(b+c-a)^2} \geq \frac{(a^2+b^2+c^2)^3}{\left(\sum a^{5/2}(b+c-a)\right)^2} = \frac{27}{\left(\sum a^{5/2}(b+c-a)\right)^2}. \tag{2}$$

To estimate the denominator of the right-hand part, we use an instance of Schur's inequality, namely

$$\sum a^{3/2}(a-b)(a-c) \geq 0,$$

which can be rewritten as

$$\sum a^{5/2}(b+c-a) \leq abc(\sqrt{a}+\sqrt{b}+\sqrt{c}).$$

Moreover, by the inequality between the arithmetic mean and the fourth power mean we also have

$$\left(\frac{\sqrt{a}+\sqrt{b}+\sqrt{c}}{3}\right)^4 \leq \frac{a^2+b^2+c^2}{3} = 1,$$

i.e., $\sqrt{a}+\sqrt{b}+\sqrt{c} \leq 3$. Hence, (2) yields

$$\sum \frac{a}{(b+c-a)^2} \geq \frac{27}{\left(abc(\sqrt{a}+\sqrt{b}+\sqrt{c})\right)^2} \geq \frac{3}{a^2b^2c^2},$$

thus solving the problem.

Comment. In this solution, one may also start from the following version of HÖLDER's inequality

$$\left(\sum_{i=1}^{n} a_i^3\right)\left(\sum_{i=1}^{n} b_i^3\right)\left(\sum_{i=1}^{n} c_i^3\right) \geq \left(\sum_{i=1}^{n} a_i b_i c_i\right)^3$$

applied as

$$\sum \frac{a}{(b+c-a)^2} \cdot \sum a^3(b+c-a) \cdot \sum a^2(b+c-a) \geq 27.$$

After doing that, one only needs the slightly better known instances

$$\sum a^3(b+c-a) \leq (a+b+c)abc \qquad \text{and} \qquad \sum a^2(b+c-a) \leq 3abc$$

of SCHUR's Inequality.

Solution 2. As in Solution 1, we mention that all the numbers $b+c-a$, $a+c-b$, $a+b-c$ are positive. We will use only this restriction and the condition

$$a^5 + b^5 + c^5 \geq 3, \tag{3}$$

which is weaker than the given one. Due to the symmetry we may assume that $a \geq b \geq c$.

In view of (3), it suffices to prove the inequality

$$\sum \frac{a^3b^2c^2}{(b+c-a)^2} \geq \sum a^5,$$

or, moving all the terms into the left-hand part,

$$\sum \frac{a^3}{(b+c-a)^2}\left((bc)^2 - (a(b+c-a))^2\right) \geq 0. \tag{4}$$

Note that the signs of the expressions $(yz)^2-(x(y+z-x))^2$ and $yz-x(y+z-x) = (x-y)(x-z)$ are the same for every positive x, y, z satisfying the triangle inequality. So the terms in (4) corresponding to a and c are nonnegative, and hence it is sufficient to prove that the sum of the terms corresponding to a and b is nonnegative. Equivalently, we need the relation

$$\frac{a^3}{(b+c-a)^2}(a-b)(a-c)(bc+a(b+c-a)) \geq \frac{b^3}{(a+c-b)^2}(a-b)(b-c)(ac+b(a+c-b)).$$

Obviously, we have

$$a^3 \geq b^3 \geq 0, \quad 0 < b+c-a \leq a+c-b, \quad \text{and} \quad a-c \geq b-c \geq 0,$$

hence it suffices to prove that

$$\frac{ab+ac+bc-a^2}{b+c-a} \geq \frac{ab+ac+bc-b^2}{c+a-b}.$$

Since all the denominators are positive, it is equivalent to

$$(c+a-b)(ab+ac+bc-a^2)-(ab+ac+bc-b^2)(b+c-a)\geq 0,$$

or

$$(a-b)(2ab-a^2-b^2+ac+bc)\geq 0.$$

Since $a\geq b$, the last inequality follows from

$$c(a+b)>(a-b)^2$$

which holds since $c>a-b\geq 0$ and $a+b>a-b\geq 0$.

C1

Let $n > 0$ be an integer. We are given a balance and n weights of weight $2^0, 2^1, \ldots, 2^{n-1}$. In a sequence of n moves we place all weights on the balance. In the first move we choose a weight and put it on the left pan. In each of the following moves we choose one of the remaining weights and we add it either to the left or to the right pan. Compute the number of ways in which we can perform these n moves in such a way that the right pan is never heavier than the left pan.

Answer. The number $f(n)$ of ways of placing the n weights is equal to the product of all odd positive integers less than or equal to $2n-1$, i.e. $f(n) = (2n-1)!! = 1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)$.

Solution 1. Assume $n \geq 2$. We claim

$$f(n) = (2n-1)f(n-1). \tag{1}$$

Firstly, note that after the first move the left pan is always at least 1 heavier than the right one. Hence, any valid way of placing the n weights on the scale gives rise, by not considering weight 1, to a valid way of placing the weights $2, 2^2, \ldots, 2^{n-1}$.

If we divide the weight of each weight by 2, the answer does not change. So these $n-1$ weights can be placed on the scale in $f(n-1)$ valid ways. Now we look at weight 1. If it is put on the scale in the first move, then it has to be placed on the left side, otherwise it can be placed either on the left or on the right side, because after the first move the difference between the weights on the left pan and the weights on the right pan is at least 2. Hence, there are exactly $2n-1$ different ways of inserting weight 1 in each of the $f(n-1)$ valid sequences for the $n-1$ weights in order to get a valid sequence for the n weights. This proves the claim.

Since $f(1) = 1$, by induction we obtain for all positive integers n

$$f(n) = (2n-1)!! = 1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1).$$

Comment 1. The word "compute" in the statement of the problem is probably too vague. An alternative but more artificial question might ask for the smallest n for which the number of valid ways is divisible by 2011. In this case the answer would be 1006.

Comment 2. It is useful to remark that the answer is the same for any set of weights where each weight is heavier than the sum of the lighter ones. Indeed, in such cases the given condition is equivalent to asking that during the process the heaviest weight on the balance is always on the left pan.

Comment 3. Instead of considering the lightest weight, one may also consider the last weight put on the balance. If this weight is 2^{n-1} then it should be put on the left pan. Otherwise it may be put on

any pan; the inequality would not be violated since at this moment the heaviest weight is already put onto the left pan. In view of the previous comment, in each of these $2n-1$ cases the number of ways to place the previous weights is exactly $f(n-1)$, which yields (1).

Solution 2. We present a different way of obtaining (1). Set $f(0)=1$. Firstly, we find a recurrent formula for $f(n)$.

Assume $n \geq 1$. Suppose that weight 2^{n-1} is placed on the balance in the i-th move with $1 \leq i \leq n$. This weight has to be put on the left pan. For the previous moves we have $\binom{n-1}{i-1}$ choices of the weights and from Comment 2 there are $f(i-1)$ valid ways of placing them on the balance. For later moves there is no restriction on the way in which the weights are to be put on the pans. Therefore, all $(n-i)!2^{n-i}$ ways are possible. This gives

$$f(n)=\sum_{i=1}^{n}\binom{n-1}{i-1}f(i-1)(n-i)!2^{n-i}=\sum_{i=1}^{n}\frac{(n-1)!f(i-1)2^{n-i}}{(i-1)!}. \tag{2}$$

Now we are ready to prove (1). Using $n-1$ instead of n in (2) we get

$$f(n-1)=\sum_{i=1}^{n-1}\frac{(n-2)!f(i-1)2^{n-1-i}}{(i-1)!}.$$

Hence, again from (2) we get

$$\begin{aligned} f(n) &= 2(n-1)\sum_{i=1}^{n-1}\frac{(n-2)!f(i-1)2^{n-1-i}}{(i-1)!}+f(n-1) \\ &= (2n-2)f(n-1)+f(n-1)=(2n-1)f(n-1), \end{aligned}$$

QED.

Comment. There exist different ways of obtaining the formula (2). Here we show one of them.

Suppose that in the first move we use weight 2^{n-i+1}. Then the lighter $n-i$ weights may be put on the balance at any moment and on either pan. This gives $2^{n-i}\cdot(n-1)!/(i-1)!$ choices for the moves (moments and choices of pan) with the lighter weights. The remaining $i-1$ moves give a valid sequence for the $i-1$ heavier weights and this is the only requirement for these moves, so there are $f(i-1)$ such sequences. Summing over all $i=1,2,\ldots,n$ we again come to (2).

C2

Suppose that 1000 students are standing in a circle. Prove that there exists an integer k with $100 \le k \le 300$ such that in this circle there exists a contiguous group of $2k$ students, for which the first half contains the same number of girls as the second half.

Solution. Number the students consecutively from 1 to 1000. Let $a_i = 1$ if the ith student is a girl, and $a_i = 0$ otherwise. We expand this notion for all integers i by setting $a_{i+1000} = a_{i-1000} = a_i$. Next, let

$$S_k(i) = a_i + a_{i+1} + \cdots + a_{i+k-1}.$$

Now the statement of the problem can be reformulated as follows:

There exist an integer k with $100 \le k \le 300$ and an index i such that $S_k(i) = S_k(i+k)$.

Assume now that this statement is false. Choose an index i such that $S_{100}(i)$ attains the maximal possible value. In particular, we have $S_{100}(i-100) - S_{100}(i) < 0$ and $S_{100}(i) - S_{100}(i+100) > 0$, for if we had an equality, then the statement would hold. This means that the function $S(j) - S(j+100)$ changes sign somewhere on the segment $[i-100, i]$, so there exists some index $j \in [i-100, i-1]$ such that

$$S_{100}(j) \le S_{100}(j+100) - 1, \quad \text{but} \quad S_{100}(j+1) \ge S_{100}(j+101) + 1. \tag{1}$$

Subtracting the first inequality from the second one, we get $a_{j+100} - a_j \ge a_{j+200} - a_{j+100} + 2$, so

$$a_j = 0, \quad a_{j+100} = 1, \quad a_{j+200} = 0.$$

Substituting this into the inequalities of (1), we also obtain $S_{99}(j+1) \le S_{99}(j+101) \le S_{99}(j+1)$, which implies

$$S_{99}(j+1) = S_{99}(j+101). \tag{2}$$

Now let k and ℓ be the least positive integers such that $a_{j-k} = 1$ and $a_{j+200+\ell} = 1$. By symmetry, we may assume that $k \ge \ell$. If $k \ge 200$ then we have $a_j = a_{j-1} = \cdots = a_{j-199} = 0$, so $S_{100}(j-199) = S_{100}(j-99) = 0$, which contradicts the initial assumption. Hence $\ell \le k \le 199$.

Finally, we have

$$S_{100+\ell}(j-\ell+1) = (a_{j-\ell+1} + \cdots + a_j) + S_{99}(j+1) + a_{j+100} = S_{99}(j+1) + 1,$$
$$S_{100+\ell}(j+101) = S_{99}(j+101) + (a_{j+200} + \cdots + a_{j+200+\ell-1}) + a_{j+200+\ell} = S_{99}(j+101) + 1.$$

Comparing with (2) we get $S_{100+\ell}(j-\ell+1) = S_{100+\ell}(j+101)$ and $100 + \ell \le 299$, which again contradicts our assumption.

Comment. It may be seen from the solution that the number 300 from the problem statement can be

replaced by 299. Here we consider some improvements of this result. Namely, we investigate which interval can be put instead of $[100, 300]$ in order to keep the problem statement valid.

First of all, the two examples

$$\underbrace{1,1,\ldots,1}_{167},\underbrace{0,0,\ldots,0}_{167},\underbrace{1,1,\ldots,1}_{167},\underbrace{0,0,\ldots,0}_{167},\underbrace{1,1,\ldots,1}_{167},\underbrace{0,0,\ldots,0}_{165}$$

and

$$\underbrace{1,1,\ldots,1}_{249},\underbrace{0,0,\ldots,0}_{251},\underbrace{1,1,\ldots,1}_{249},\underbrace{0,0,\ldots,0}_{251}$$

show that the interval can be changed neither to $[84, 248]$ nor to $[126, 374]$.

On the other hand, we claim that this interval can be changed to $[125, 250]$. Note that this statement is invariant under replacing all 1's by 0's and vice versa. Assume, to the contrary, that there is no admissible $k \in [125, 250]$. The arguments from the solution easily yield the following lemma.

Lemma. Under our assumption, suppose that for some indices $i < j$ we have $S_{125}(i) \le S_{125}(i+125)$ but $S_{125}(j) \ge S_{125}(j+125)$. Then there exists some $t \in [i, j-1]$ such that $a_t = a_{t-1} = \cdots = a_{t-125} = 0$ and $a_{t+250} = a_{t+251} = \cdots = a_{t+375} = 0$. □

Let us call a segment $[i, j]$ of indices a *crowd*, if (a) $a_i = a_{i+1} = \cdots = a_j$, but $a_{i-1} \ne a_i \ne a_{j+1}$, and (b) $j - i \ge 125$. Now, using the lemma, one can get in the same way as in the solution that there exists some crowd. Take all the crowds in the circle, and enumerate them in cyclic order as $A_1, \ldots, A_d$. We also assume always that $A_{s+d} = A_{s-d} = A_s$.

Consider one of the crowds, say A_1. We have $A_1 = [i, i+t]$ with $125 \le t \le 248$ (if $t \ge 249$, then $a_i = a_{i+1} = \cdots = a_{i+249}$ and therefore $S_{125}(i) = S_{125}(i+125)$, which contradicts our assumption). We may assume that $a_i = 1$. Then we have $S_{125}(i+t-249) \le 125 = S_{125}(i+t-124)$ and $S_{125}(i) = 125 \ge S_{125}(i+125)$, so by the lemma there exists some index $j \in [i+t-249, i-1]$ such that the segments $[j-125, j]$ and $[j+250, j+375]$ are contained in some crowds.

Let us fix such j and denote the segment $[j+1, j+249]$ by B_1. Clearly, $A_1 \subseteq B_1$. Moreover, B_1 cannot contain any crowd other than A_1 since $|B_1| = 249 < 2 \cdot 126$. Hence it is clear that $j \in A_d$ and $j + 250 \in A_2$. In particular, this means that the genders of A_d and A_2 are different from that of A_1.

Performing this procedure for every crowd A_s, we find segments $B_s = [j_s + 1, j_s + 249]$ such that $|B_s| = 249$, $A_s \subseteq B_s$, and $j_s \in A_{s-1}$, $j_s + 250 \in A_{s+1}$. So, B_s covers the whole segment between A_{s-1} and A_{s+1}, hence the sets $B_1, \ldots, B_d$ cover some 1000 consecutive indices. This implies $249d \ge 1000$, and $d \ge 5$. Moreover, the gender of A_i is alternating, so d is even; therefore $d \ge 6$.

Consider now three segments $A_1 = [i_1, i_1']$, $B_2 = [j_2 + 1, j_2 + 249]$, $A_3 = [i_3, i_3']$. By construction, we have $[j_2 - 125, j_2] \subseteq A_1$ and $[j_2 + 250, j_2 + 375] \subseteq A_3$, whence $i_1 \le j_2 - 125$, $i_3' \ge j_2 + 375$. Therefore $i_3' - i_1 \ge 500$. Analogously, if $A_4 = [i_4, i_4']$, $A_6 = [i_6, i_6']$ then $i_6' - i_4 \ge 500$. But from $d \ge 6$ we get $i_1 < i_3' < i_4 < i_6' < i_1 + 1000$, so $1000 > (i_3' - i_1) + (i_6' - i_4) \ge 500 + 500$. This final contradiction shows that our claim holds.

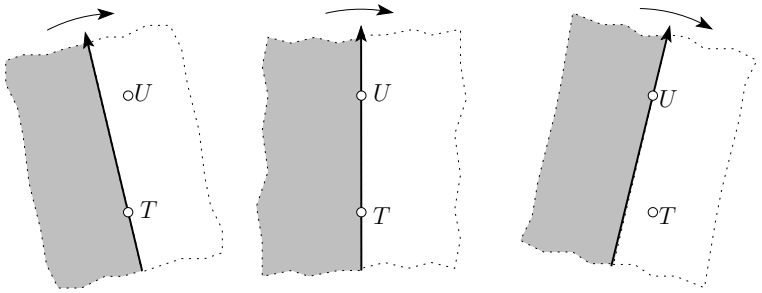
One may even show that the interval in the statement of the problem may be replaced by $[125, 249]$ (both these numbers cannot be improved due to the examples above). But a proof of this fact is a bit messy, and we do not present it here.

C3

Let $\mathcal{S}$ be a finite set of at least two points in the plane. Assume that no three points of $\mathcal{S}$ are collinear. By a *windmill* we mean a process as follows. Start with a line ℓ going through a point $P \in \mathcal{S}$. Rotate ℓ clockwise around the *pivot* P until the line contains another point Q of $\mathcal{S}$. The point Q now takes over as the new pivot. This process continues indefinitely, with the pivot always being a point from $\mathcal{S}$.

Show that for a suitable $P \in \mathcal{S}$ and a suitable starting line ℓ containing P, the resulting windmill will visit each point of $\mathcal{S}$ as a pivot infinitely often.

Solution. Give the rotating line an orientation and distinguish its sides as the *oranje side* and the *blue side*. Notice that whenever the pivot changes from some point T to another point U, after the change, T is on the same side as U was before. Therefore, the number of elements of $\mathcal{S}$ on the oranje side and the number of those on the blue side remain the same throughout the whole process (except for those moments when the line contains two points).



First consider the case that $|\mathcal{S}| = 2n + 1$ is odd. We claim that through any point $T \in \mathcal{S}$, there is a line that has n points on each side. To see this, choose an oriented line through T containing no other point of $\mathcal{S}$ and suppose that it has $n + r$ points on its oranje side. If $r = 0$ then we have established the claim, so we may assume that $r \neq 0$. As the line rotates through 180° around T, the number of points of $\mathcal{S}$ on its oranje side changes by 1 whenever the line passes through a point; after 180°, the number of points on the oranje side is $n - r$. Therefore there is an intermediate stage at which the oranje side, and thus also the blue side, contains n points.

Now select the point P arbitrarily, and choose a line through P that has n points of $\mathcal{S}$ on each side to be the initial state of the windmill. We will show that during a rotation over 180°, the line of the windmill visits each point of $\mathcal{S}$ as a pivot. To see this, select any point T of $\mathcal{S}$ and select a line ℓ through T that separates $\mathcal{S}$ into equal halves. The point T is the unique point of $\mathcal{S}$ through which a line in this direction can separate the points of $\mathcal{S}$ into equal halves (parallel translation would disturb the balance). Therefore, when the windmill line is parallel to ℓ, it must be ℓ itself, and so pass through T.

Next suppose that $|\mathcal{S}| = 2n$. Similarly to the odd case, for every $T \in \mathcal{S}$ there is an oriented

line through T with $n-1$ points on its oranje side and n points on its blue side. Select such an oriented line through an arbitrary P to be the initial state of the windmill.

We will now show that during a rotation over 360°, the line of the windmill visits each point of $\mathcal{S}$ as a pivot. To see this, select any point T of $\mathcal{S}$ and an oriented line ℓ through T that separates $\mathcal{S}$ into two subsets with $n-1$ points on its oranje and n points on its blue side. Again, parallel translation would change the numbers of points on the two sides, so when the windmill line is parallel to ℓ with the same orientation, the windmill line must pass through T.

Comment. One may shorten this solution in the following way.

Suppose that $|\mathcal{S}| = 2n+1$. Consider any line ℓ that separates $\mathcal{S}$ into equal halves; this line is unique given its direction and contains some point $T \in \mathcal{S}$. Consider the windmill starting from this line. When the line has made a rotation of 180°, it returns to the same location but the oranje side becomes blue and vice versa. So, for each point there should have been a moment when it appeared as pivot, as this is the only way for a point to pass from on side to the other.

Now suppose that $|\mathcal{S}| = 2n$. Consider a line having $n-1$ and n points on the two sides; it contains some point T. Consider the windmill starting from this line. After having made a rotation of 180°, the windmill line contains some different point R, and each point different from T and R has changed the color of its side. So, the windmill should have passed through all the points.

C4

Determine the greatest positive integer k that satisfies the following property: The set of positive integers can be partitioned into k subsets $A_1, A_2, \ldots, A_k$ such that for all integers $n \geq 15$ and all $i \in \{1, 2, \ldots, k\}$ there exist two distinct elements of A_i whose sum is n.

Answer. The greatest such number k is 3.

Solution 1. There are various examples showing that $k = 3$ does indeed have the property under consideration. E.g. one can take

$$A_1 = \{1, 2, 3\} \cup \{3m \mid m \geq 4\},$$
$$A_2 = \{4, 5, 6\} \cup \{3m - 1 \mid m \geq 4\},$$
$$A_3 = \{7, 8, 9\} \cup \{3m - 2 \mid m \geq 4\}.$$

To check that this partition fits, we notice first that the sums of two distinct elements of A_i obviously represent all numbers $n \geq 1 + 12 = 13$ for $i = 1$, all numbers $n \geq 4 + 11 = 15$ for $i = 2$, and all numbers $n \geq 7 + 10 = 17$ for $i = 3$. So, we are left to find representations of the numbers 15 and 16 as sums of two distinct elements of A_3. These are $15 = 7+8$ and $16 = 7+9$.

Let us now suppose that for some $k \geq 4$ there exist sets $A_1, A_2, \ldots, A_k$ satisfying the given property. Obviously, the sets $A_1, A_2, A_3, A_4 \cup \cdots \cup A_k$ also satisfy the same property, so one may assume $k = 4$.

Put $B_i = A_i \cap \{1, 2, \ldots, 23\}$ for $i = 1, 2, 3, 4$. Now for any index i each of the ten numbers $15, 16, \ldots, 24$ can be written as sum of two distinct elements of B_i. Therefore this set needs to contain at least five elements. As we also have $|B_1| + |B_2| + |B_3| + |B_4| = 23$, there has to be some index j for which $|B_j| = 5$. Let $B_j = \{x_1, x_2, x_3, x_4, x_5\}$. Finally, now the sums of two distinct elements of A_j representing the numbers $15, 16, \ldots, 24$ should be exactly all the pairwise sums of the elements of B_j. Calculating the sum of these numbers in two different ways, we reach

$$4(x_1 + x_2 + x_3 + x_4 + x_5) = 15 + 16 + \ldots + 24 = 195.$$

Thus the number 195 should be divisible by 4, which is false. This contradiction completes our solution.

Comment. There are several variation of the proof that k should not exceed 3. E.g., one may consider the sets $C_i = A_i \cap \{1, 2, \ldots, 19\}$ for $i = 1, 2, 3, 4$. As in the previous solution one can show that for some index j one has $|C_j| = 4$, and the six pairwise sums of the elements of C_j should represent all numbers $15, 16, \ldots, 20$. Let $C_j = \{y_1, y_2, y_3, y_4\}$ with $y_1 < y_2 < y_3 < y_4$. It is not hard to deduce

$C_j = \{7, 8, 9, 11\}$, so in particular we have $1 \notin C_j$. Hence it is impossible to represent 21 as sum of two distinct elements of A_j, which completes our argument.

Solution 2. Again we only prove that $k \leq 3$. Assume that $A_1, A_2, \ldots, A_k$ is a partition satisfying the given property. We construct a graph $\mathcal{G}$ on the set $V = \{1, 2, \ldots, 18\}$ of *vertices* as follows. For each $i \in \{1, 2, \ldots, k\}$ and each $d \in \{15, 16, 17, 19\}$ we choose one pair of distinct elements $a, b \in A_i$ with $a + b = d$, and we draw an *edge* in the ith color connecting a with b. By hypothesis, $\mathcal{G}$ has exactly 4 edges of each color.

Claim. The graph $\mathcal{G}$ contains at most one circuit.

Proof. Note that all the connected components of $\mathcal{G}$ are monochromatic and hence contain at most four edges. Thus also all circuits of $\mathcal{G}$ are monochromatic and have length at most four. Moreover, each component contains at most one circuit since otherwise it should contain at least five edges.

Suppose that there is a 4-cycle in $\mathcal{G}$, say with vertices a, b, c, and d in order. Then $\{a+b, b+c, c+d, d+a\} = \{15, 16, 17, 19\}$. Taking sums we get $2(a+b+c+d) = 15+16+17+19$ which is impossible for parity reasons. Thus all circuits of $\mathcal{G}$ are triangles.

Now if the vertices a, b, and c form such a triangle, then by a similar reasoning the set $\{a+b, b+c, c+a\}$ coincides with either $\{15, 16, 17\}$, or $\{15, 16, 19\}$, or $\{16, 17, 19\}$, or $\{15, 17, 19\}$. The last of these alternatives can be excluded for parity reasons again, whilst in the first three cases the set $\{a, b, c\}$ appears to be either $\{7, 8, 9\}$, or $\{6, 9, 10\}$, or $\{7, 9, 10\}$, respectively. Thus, a component containing a circuit should contain 9 as a vertex. Therefore there is at most one such component and hence at most one circuit. □

By now we know that $\mathcal{G}$ is a graph with $4k$ edges, at least k components and at most one circuit. Consequently, $\mathcal{G}$ must have at least $4k + k - 1$ vertices. Thus $5k - 1 \leq 18$, and $k \leq 3$.

C5

Let m be a positive integer and consider a checkerboard consisting of m by m unit squares. At the midpoints of some of these unit squares there is an ant. At time 0, each ant starts moving with speed 1 parallel to some edge of the checkerboard. When two ants moving in opposite directions meet, they both turn 90° clockwise and continue moving with speed 1. When more than two ants meet, or when two ants moving in perpendicular directions meet, the ants continue moving in the same direction as before they met. When an ant reaches one of the edges of the checkerboard, it falls off and will not re-appear.

Considering all possible starting positions, determine the latest possible moment at which the last ant falls off the checkerboard or prove that such a moment does not necessarily exist.

Antswer. The latest possible moment for the last ant to fall off is $\frac{3m}{2} - 1$.

Solution. For $m = 1$ the answer is clearly correct, so assume $m > 1$. In the sequel, the word *collision* will be used to denote meeting of exactly two ants, moving in opposite directions.

If at the beginning we place an ant on the southwest corner square facing east and an ant on the southeast corner square facing west, then they will meet in the middle of the bottom row at time $\frac{m-1}{2}$. After the collision, the ant that moves to the north will stay on the board for another $m - \frac{1}{2}$ time units and thus we have established an example in which the last ant falls off at time $\frac{m-1}{2} + m - \frac{1}{2} = \frac{3m}{2} - 1$. So, we are left to prove that this is the latest possible moment.

Consider any collision of two ants a and a'. Let us change the rule for this collision, and enforce these two ants to turn anticlockwise. Then the succeeding behavior of all the ants does not change; the only difference is that a and a' swap their positions. These arguments may be applied to any collision separately, so we may assume that at any collision, either both ants rotate clockwise or both of them rotate anticlockwise by our own choice.

For instance, we may assume that there are only two types of ants, depending on their initial direction: *NE-ants*, which move only north or east, and *SW-ants*, moving only south and west. Then we immediately obtain that all ants will have fallen off the board after $2m - 1$ time units. However, we can get a better bound by considering the last moment at which a given ant collides with another ant.

Choose a coordinate system such that the corners of the checkerboard are $(0, 0)$, $(m, 0)$, (m, m) and $(0, m)$. At time t, there will be no NE-ants in the region $\{(x, y) \colon x + y < t + 1\}$ and no SW-ants in the region $\{(x, y) \colon x + y > 2m - t - 1\}$. So if two ants collide at (x, y) at time t, we have

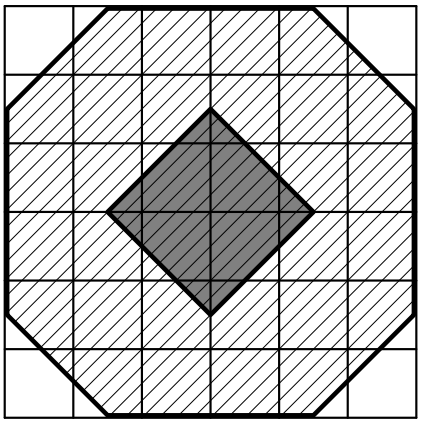
$$t + 1 \leq x + y \leq 2m - t - 1. \tag{1}$$

Analogously, we may change the rules so that each ant would move either alternatingly north and west, or alternatingly south and east. By doing so, we find that apart from (1) we also have $|x-y| \le m-t-1$ for each collision at point (x,y) and time t.

To visualize this, put

$$B(t) = \left\{(x,y) \in [0,m]^2 \colon t+1 \le x+y \le 2m-t-1 \text{ and } |x-y| \le m-t-1\right\}.$$

An ant can thus only collide with another ant at time t if it happens to be in the region $B(t)$. The following figure displays $B(t)$ for $t = \frac{1}{2}$ and $t = \frac{7}{2}$ in the case $m = 6$:



Now suppose that an NE-ant has its last collision at time t and that it does so at the point (x,y) (if the ant does not collide at all, it will fall off the board within $m-\frac{1}{2} < \frac{3m}{2}-1$ time units, so this case can be ignored). Then we have $(x,y) \in B(t)$ and thus $x+y \ge t+1$ and $x-y \ge -(m-t-1)$. So we get

$$x \ge \frac{(t+1)-(m-t-1)}{2} = t+1-\frac{m}{2}.$$

By symmetry we also have $y \ge t+1-\frac{m}{2}$, and hence $\min\{x,y\} \ge t+1-\frac{m}{2}$. After this collision, the ant will move directly to an edge, which will take at most $m-\min\{x,y\}$ units of time. In sum, the total amount of time the ant stays on the board is at most

$$t+(m-\min\{x,y\}) \le t+m-\left(t+1-\frac{m}{2}\right) = \frac{3m}{2}-1.$$

By symmetry, the same bound holds for SW-ants as well.

C6

Let n be a positive integer and let $W = \ldots x_{-1}x_0x_1x_2\ldots$ be an infinite periodic word consisting of the letters a and b. Suppose that the minimal period N of W is greater than 2^n.

A finite nonempty word U is said to *appear* in W if there exist indices $k \le \ell$ such that $U = x_kx_{k+1}\ldots x_\ell$. A finite word U is called *ubiquitous* if the four words Ua, Ub, aU, and bU all appear in W. Prove that there are at least n ubiquitous finite nonempty words.

Solution. Throughout the solution, all the words are nonempty. For any word R of length m, we call the number of indices $i \in \{1, 2, \ldots, N\}$ for which R coincides with the subword $x_{i+1}x_{i+2}\ldots x_{i+m}$ of W the *multiplicity* of R and denote it by $\mu(R)$. Thus a word R appears in W if and only if $\mu(R) > 0$. Since each occurrence of a word in W is both succeeded by either the letter a or the letter b and similarly preceded by one of those two letters, we have

$$\mu(R) = \mu(Ra) + \mu(Rb) = \mu(aR) + \mu(bR) \tag{1}$$

for all words R.

We claim that the condition that N is in fact the minimal period of W guarantees that each word of length N has multiplicity 1 or 0 depending on whether it appears or not. Indeed, if the words $x_{i+1}x_{i+2}\ldots x_{i+N}$ and $x_{j+1}\ldots x_{j+N}$ are equal for some $1 \le i < j \le N$, then we have $x_{i+a} = x_{j+a}$ for every integer a, and hence $j - i$ is also a period.

Moreover, since $N > 2^n$, at least one of the two words a and b has a multiplicity that is strictly larger than 2^{n-1}.

For each $k = 0, 1, \ldots, n-1$, let U_k be a subword of W whose multiplicity is strictly larger than 2^k and whose length is maximal subject to this property. Note that such a word exists in view of the two observations made in the two previous paragraphs.

Fix some index $k \in \{0, 1, \ldots, n-1\}$. Since the word U_kb is longer than U_k, its multiplicity can be at most 2^k, so in particular $\mu(U_kb) < \mu(U_k)$. Therefore, the word U_ka has to appear by (1). For a similar reason, the words U_kb, aU_k, and bU_k have to appear as well. Hence, the word U_k is ubiquitous. Moreover, if the multiplicity of U_k were strictly greater than 2^{k+1}, then by (1) at least one of the two words U_ka and U_kb would have multiplicity greater than 2^k and would thus violate the maximality condition imposed on U_k.

So we have $\mu(U_0) \le 2 < \mu(U_1) \le 4 < \ldots \le 2^{n-1} < \mu(U_{n-1})$, which implies in particular that the words $U_0, U_1, \ldots, U_{n-1}$ have to be distinct. As they have been proved to be ubiquitous as well, the problem is solved.

Comment 1. There is an easy construction for obtaining ubiquitous words from appearing words whose multiplicity is at least two. Starting with any such word U we may simply extend one of its occurrences in W forwards and backwards as long as its multiplicity remains fixed, thus arriving at a

word that one might call the *ubiquitous prolongation* $p(U)$ of U.

There are several variants of the argument in the second half of the solution using the concept of prolongation. For instance, one may just take all ubiquitous words $U_1, U_2, \ldots, U_\ell$ ordered by increasing multiplicity and then prove for $i \in \{1, 2, \ldots, \ell\}$ that $\mu(U_i) \leq 2^i$. Indeed, assume that i is a minimal counterexample to this statement; then by the arguments similar to those presented above, the ubiquitous prolongation of one of the words $U_i a$, $U_i b$, aU_i or bU_i violates the definition of U_i.

Now the multiplicity of one of the two letters a and b is strictly greater than 2^{n-1}, so passing to ubiquitous prolongations once more we obtain $2^{n-1} < \mu(U_\ell) \leq 2^\ell$, which entails $\ell \geq n$, as needed.

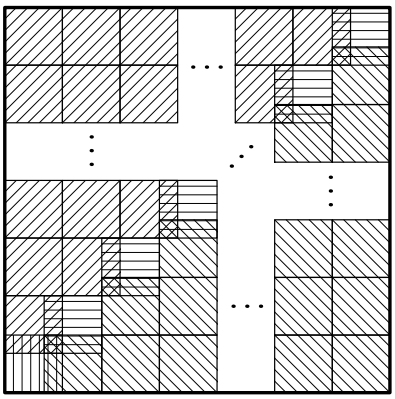
Comment 2. The bound n for the number of ubiquitous subwords in the problem statement is not optimal, but it is close to an optimal one in the following sense. There is a universal constant $C > 0$ such that for each positive integer n there exists an infinite periodic word W whose minimal period is greater than 2^n but for which there exist fewer than Cn ubiquitous words.

C7

On a square table of 2011 by 2011 cells we place a finite number of napkins that each cover a square of 52 by 52 cells. In each cell we write the number of napkins covering it, and we record the maximal number k of cells that all contain the same nonzero number. Considering all possible napkin configurations, what is the largest value of k?

Answer. $2011^2 - \left((52^2 - 35^2) \cdot 39 - 17^2\right) = 4044121 - 57392 = 3986729$.

Solution 1. Let $m = 39$, then $2011 = 52m - 17$. We begin with an example showing that there can exist 3986729 cells carrying the same positive number.



To describe it, we number the columns from the left to the right and the rows from the bottom to the top by $1, 2, \ldots, 2011$. We will denote each napkin by the coordinates of its lower-left cell. There are four kinds of napkins: first, we take all napkins $(52i + 36, 52j + 1)$ with $0 \le j \le i \le m - 2$; second, we use all napkins $(52i + 1, 52j + 36)$ with $0 \le i \le j \le m - 2$; third, we use all napkins $(52i + 36, 52i + 36)$ with $0 \le i \le m - 2$; and finally the napkin $(1, 1)$. Different groups of napkins are shown by different types of hatchings in the picture.

Now except for those squares that carry two or more different hatchings, all squares have the number 1 written into them. The number of these exceptional cells is easily computed to be $(52^2 - 35^2)m - 17^2 = 57392$.

We are left to prove that 3986729 is an upper bound for the number of cells containing the same number. Consider any configuration of napkins and any positive integer M. Suppose there are g cells with a number different from M. Then it suffices to show $g \ge 57392$. Throughout the solution, a *line* will mean either a row or a column.

Consider any line ℓ. Let $a_1, \ldots, a_{52m-17}$ be the numbers written into its consecutive cells. For $i = 1, 2, \ldots, 52$, let $s_i = \sum_{t \equiv i \pmod{52}} a_t$. Note that $s_1, \ldots, s_{35}$ have m terms each, while $s_{36}, \ldots, s_{52}$ have $m - 1$ terms each. Every napkin intersecting ℓ contributes exactly 1 to each s_i;

hence the number s of all those napkins satisfies $s_1 = \cdots = s_{52} = s$. Call the line ℓ *rich* if $s > (m-1)M$ and *poor* otherwise.

Suppose now that ℓ is rich. Then in each of the sums $s_{36}, \ldots, s_{52}$ there exists a term greater than M; consider all these terms and call the corresponding cells the *rich bad cells* for this line. So, each rich line contains at least 17 cells that are bad for this line.

If, on the other hand, ℓ is poor, then certainly $s < mM$ so in each of the sums $s_1, \ldots, s_{35}$ there exists a term less than M; consider all these terms and call the corresponding cells the *poor bad cells* for this line. So, each poor line contains at least 35 cells that are bad for this line.

Let us call all indices congruent to $1, 2, \ldots,$ or 35 modulo 52 *small*, and all other indices, i.e. those congruent to $36, 37, \ldots,$ or 52 modulo 52, *big*. Recall that we have numbered the columns from the left to the right and the rows from the bottom to the top using the numbers $1, 2, \ldots, 52m - 17$; we say that a line is *big* or *small* depending on whether its index is big or small. By definition, all rich bad cells for the rows belong to the big columns, while the poor ones belong to the small columns, and vice versa.

In each line, we put a *strawberry* on each cell that is bad for this line. In addition, for each **small rich** line we put an *extra* strawberry on each of its (rich) bad cells. A cell gets the strawberries from its row and its column independently.

Notice now that a cell with a strawberry on it contains a number different from M. If this cell gets a strawberry by the extra rule, then it contains a number greater than M. Moreover, it is either in a small row and in a big column, or vice versa. Suppose that it is in a small row, then it is not bad for its column. So it has not more than two strawberries in this case. On the other hand, if the extra rule is not applied to some cell, then it also has not more than two strawberries. So, the total number N of strawberries is at most $2g$.

We shall now estimate N in a different way. For each of the $2 \cdot 35m$ small lines, we have introduced at least 34 strawberries if it is rich and at least 35 strawberries if it is poor, so at least 34 strawberries in any case. Similarly, for each of the $2 \cdot 17(m-1)$ big lines, we put at least $\min(17, 35) = 17$ strawberries. Summing over all lines we obtain

$$2g \geq N \geq 2(35m \cdot 34 + 17(m-1) \cdot 17) = 2(1479m - 289) = 2 \cdot 57392,$$

as desired.

Comment. The same reasoning applies also if we replace 52 by R and 2011 by $Rm - H$, where m, R, and H are integers with $m, R \geq 1$ and $0 \leq H \leq \frac{1}{3}R$. More detailed information is provided after the next solution.

Solution 2. We present a different proof of the estimate which is the hard part of the problem. Let $S = 35$, $H = 17$, $m = 39$; so the table size is $2011 = Sm + H(m-1)$, and the napkin size is $52 = S + H$. Fix any positive integer M and call a cell *vicious* if it contains a number distinct

from M. We will prove that there are at least $H^2(m-1)+2SHm$ vicious cells.

Firstly, we introduce some terminology. As in the previous solution, we number rows and columns and we use the same notions of *small* and *big* indices and lines; so, an index is small if it is congruent to one of the numbers $1, 2, \ldots, S$ modulo $(S+H)$. The numbers $1, 2, \ldots, S+H$ will be known as *residues*. For two residues i and j, we say that a cell is of *type* (i, j) if the index of its row is congruent to i and the index of its column to j modulo $(S+H)$. The number of vicious cells of this type is denoted by v_{ij}.

Let s, s' be two variables ranging over small residues and let h, h' be two variables ranging over big residues. A cell is said to be of *class* A, B, C, or D if its type is of shape (s, s'), (s, h), (h, s), or (h, h'), respectively. The numbers of vicious cells belonging to these classes are denoted in this order by a, b, c, and d. Observe that each cell belongs to exactly one class.

Claim 1. We have

$$m \le \frac{a}{S^2} + \frac{b+c}{2SH}. \tag{1}$$

Proof. Consider an arbitrary small row r. Denote the numbers of vicious cells on r belonging to the classes A and B by α and β, respectively. As in the previous solution, we obtain that $\alpha \ge S$ or $\beta \ge H$. So in each case we have $\frac{\alpha}{S} + \frac{\beta}{H} \ge 1$.

Performing this argument separately for each small row and adding up all the obtained inequalities, we get $\frac{a}{S} + \frac{b}{H} \ge mS$. Interchanging rows and columns we similarly get $\frac{a}{S} + \frac{c}{H} \ge mS$. Summing these inequalities and dividing by $2S$ we get what we have claimed. □

Claim 2. Fix two small residue s, s' and two big residues h, h'. Then $2m-1 \le v_{ss'}+v_{sh'}+v_{hh'}$.

Proof. Each napkin covers exactly one cell of type (s, s'). Removing all napkins covering a vicious cell of this type, we get another collection of napkins, which covers each cell of type (s, s') either 0 or M times depending on whether the cell is vicious or not. Hence $(m^2 - v_{ss'})M$ napkins are left and throughout the proof of Claim 2 we will consider only these remaining napkins. Now, using a red pen, write in each cell the number of napkins covering it. Notice that a cell containing a red number greater than M is surely vicious.

We call two cells *neighbors* if they can be simultaneously covered by some napkin. So, each cell of type (h, h') has not more than four neighbors of type (s, s'), while each cell of type (s, h') has not more than two neighbors of each of the types (s, s') and (h, h'). Therefore, each red number at a cell of type (h, h') does not exceed $4M$, while each red number at a cell of type (s, h') does not exceed $2M$.

Let x, y, and z be the numbers of cells of type (h, h') whose red number belongs to $(M, 2M]$, $(2M, 3M]$, and $(3M, 4M]$, respectively. All these cells are vicious, hence $x + y + z \le v_{hh'}$. The red numbers appearing in cells of type (h, h') clearly sum up to $(m^2 - v_{ss'})M$. Bounding each of these numbers by a multiple of M we get

$$(m^2 - v_{ss'})M \le \big((m-1)^2 - (x+y+z)\big)\, M + 2xM + 3yM + 4zM,$$

i.e.

$$2m-1 \le v_{ss'} + x + 2y + 3z \le v_{ss'} + v_{hh'} + y + 2z.$$

So, to prove the claim it suffices to prove that $y + 2z \le v_{sh'}$.

For a cell δ of type (h, h') and a cell β of type (s, h') we say that δ *forces* β if there are more than M napkins covering both of them. Since each red number in a cell of type (s, h') does not exceed $2M$, it cannot be forced by more than one cell.

On the other hand, if a red number in a (h, h')-cell belongs to $(2M, 3M]$, then it forces at least one of its neighbors of type (s, h') (since the sum of red numbers in their cells is greater than $2M$). Analogously, an (h, h')-cell with the red number in $(3M, 4M]$ forces both its neighbors of type (s, h'), since their red numbers do not exceed $2M$. Therefore there are at least $y + 2z$ forced cells and clearly all of them are vicious, as desired. □

Claim 3. We have

$$2m - 1 \le \frac{a}{S^2} + \frac{b+c}{2SH} + \frac{d}{H^2}. \tag{2}$$

Proof. Averaging the previous result over all S^2H^2 possibilities for the quadruple (s, s', h, h'), we get $2m - 1 \le \frac{a}{S^2} + \frac{b}{SH} + \frac{d}{H^2}$. Due to the symmetry between rows and columns, the same estimate holds with b replaced by c. Averaging these two inequalities we arrive at our claim. □

Now let us multiply (2) by H^2, multiply (1) by $(2SH - H^2)$ and add them; we get

$$H^2(2m-1)+(2SH-H^2)m \le a\cdot\frac{H^2+2SH-H^2}{S^2}+(b+c)\frac{H^2+2SH-H^2}{2SH}+d = a\cdot\frac{2H}{S}+b+c+d.$$

The left-hand side is exactly $H^2(m-1) + 2SHm$, while the right-hand side does not exceed $a + b + c + d$ since $2H \le S$. Hence we come to the desired inequality.

Comment 1. Claim 2 is the key difference between the two solutions, because it allows to get rid of the notions of rich and poor cells. However, one may prove it by the "strawberry method" as well. It suffices to put a strawberry on each cell which is bad for an s-row, and a strawberry on each cell which is bad for an h'-column. Then each cell would contain not more than one strawberry.

Comment 2. Both solutions above work if the residue of the table size T modulo the napkin size R is at least $\frac{2}{3}R$, or equivalently if $T = Sm + H(m-1)$ and $R = S + H$ for some positive integers S, H, m such that $S \ge 2H$. Here we discuss all other possible combinations.

Case 1. If $2H \ge S \ge H/2$, then the sharp bound for the number of vicious cells is $mS^2 + (m-1)H^2$; it can be obtained by the same methods as in any of the solutions. To obtain an example showing that the bound is sharp, one may simply remove the napkins of the third kind from the example in Solution 1 (with an obvious change in the numbers).

Case 2. If $2S \le H$, the situation is more difficult. If $(S + H)^2 > 2H^2$, then the answer and the example are the same as in the previous case; otherwise the answer is $(2m-1)S^2 + 2SH(m-1)$, and the example is provided simply by $(m-1)^2$ nonintersecting napkins.

Now we sketch the proof of both estimates for Case 2. We introduce a more appropriate notation based on that from Solution 2. Denote by a_- and a_+ the number of cells of class A that contain the number which is strictly less than M and strictly greater than M, respectively. The numbers $b_\pm$, $c_\pm$, and $d_\pm$ are defined in a similar way. One may notice that the proofs of Claim 1 and Claims 2, 3 lead in fact to the inequalities

$$m-1 \le \frac{b_- + c_-}{2SH} + \frac{d_+}{H^2} \quad \text{and} \quad 2m-1 \le \frac{a}{S^2} + \frac{b_+ + c_+}{2SH} + \frac{d_+}{H^2}$$

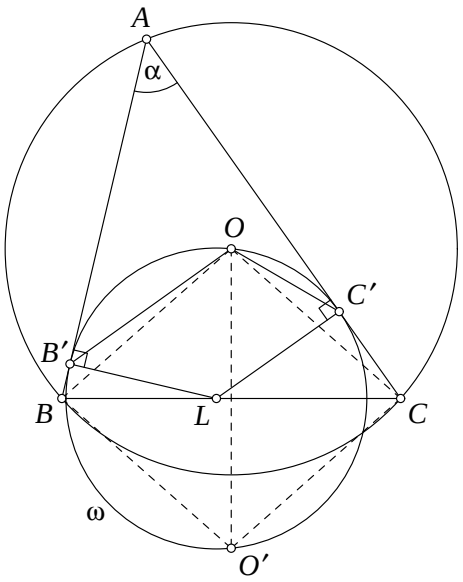
(to obtain the first one, one needs to look at the big lines instead of the small ones). Combining these inequalities, one may obtain the desired estimates.

These estimates can also be proved in some different ways, e.g. without distinguishing rich and poor cells.

G1

Let ABC be an acute triangle. Let ω be a circle whose center L lies on the side BC. Suppose that ω is tangent to AB at B' and to AC at C'. Suppose also that the circumcenter O of the triangle ABC lies on the shorter arc $B'C'$ of ω. Prove that the circumcircle of ABC and ω meet at two points.

Solution. The point B', being the perpendicular foot of L, is an interior point of side AB. Analogously, C' lies in the interior of AC. The point O is located inside the triangle $AB'C'$, hence $\angle COB < \angle C'OB'$.



Let $\alpha = \angle CAB$. The angles $\angle CAB$ and $\angle C'OB'$ are inscribed into the two circles with centers O and L, respectively, so $\angle COB = 2\angle CAB = 2\alpha$ and $2\angle C'OB' = 360^\circ - \angle C'LB'$. From the kite $AB'LC'$ we have $\angle C'LB' = 180^\circ - \angle C'AB' = 180^\circ - \alpha$. Combining these, we get

$$2\alpha = \angle COB < \angle C'OB' = \frac{360^\circ - \angle C'LB'}{2} = \frac{360^\circ - (180^\circ - \alpha)}{2} = 90^\circ + \frac{\alpha}{2},$$

so

$$\alpha < 60^\circ.$$

Let O' be the reflection of O in the line BC. In the quadrilateral $ABO'C$ we have

$$\angle CO'B + \angle CAB = \angle COB + \angle CAB = 2\alpha + \alpha < 180^\circ,$$

so the point O' is outside the circle ABC. Hence, O and O' are two points of ω such that one of them lies inside the circumcircle, while the other one is located outside. Therefore, the two circles intersect.

Comment. There are different ways of reducing the statement of the problem to the case $\alpha < 60^\circ$. E.g., since the point O lies in the interior of the isosceles triangle $AB'C'$, we have $OA < AB'$. So, if $AB' \leq 2LB'$ then $OA < 2LO$, which means that ω intersects the circumcircle of ABC. Hence the only interesting case is $AB' > 2LB'$, and this condition implies $\angle CAB = 2\angle B'AL < 2 \cdot 30^\circ = 60^\circ$.

G2

Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcenter and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

Solution 1. Let M be the point of intersection of the diagonals A_1A_3 and A_2A_4. On each diagonal choose a direction and let x, y, z, and w be the signed distances from M to the points A_1, A_2, A_3, and A_4, respectively.

Let ω_1 be the circumcircle of the triangle $A_2A_3A_4$ and let B_1 be the second intersection point of ω_1 and A_1A_3 (thus, $B_1 = A_3$ if and only if A_1A_3 is tangent to ω_1). Since the expression $O_1A_1^2 - r_1^2$ is the power of the point A_1 with respect to ω_1, we get

$$O_1A_1^2 - r_1^2 = A_1B_1 \cdot A_1A_3.$$

On the other hand, from the equality $MB_1 \cdot MA_3 = MA_2 \cdot MA_4$ we obtain $MB_1 = yw/z$. Hence, we have

$$O_1A_1^2 - r_1^2 = \left(\frac{yw}{z} - x\right)(z - x) = \frac{z - x}{z}(yw - xz).$$

Substituting the analogous expressions into the sought sum we get

$$\sum_{i=1}^{4} \frac{1}{O_iA_i^2 - r_i^2} = \frac{1}{yw - xz}\left(\frac{z}{z - x} - \frac{w}{w - y} + \frac{x}{x - z} - \frac{y}{y - w}\right) = 0,$$

as desired.

Comment. One might reformulate the problem by assuming that the quadrilateral $A_1A_2A_3A_4$ is convex. This should not really change the difficulty, but proofs that distinguish several cases may become shorter.

Solution 2. Introduce a Cartesian coordinate system in the plane. Every circle has an equation of the form $p(x, y) = x^2 + y^2 + l(x, y) = 0$, where $l(x, y)$ is a polynomial of degree at most 1. For any point $A = (x_A, y_A)$ we have $p(x_A, y_A) = d^2 - r^2$, where d is the distance from A to the center of the circle and r is the radius of the circle.

For each i in $\{1, 2, 3, 4\}$ let $p_i(x, y) = x^2 + y^2 + l_i(x, y) = 0$ be the equation of the circle with center O_i and radius r_i and let d_i be the distance from A_i to O_i. Consider the equation

$$\sum_{i=1}^{4} \frac{p_i(x, y)}{d_i^2 - r_i^2} = 1. \tag{1}$$

Since the coordinates of the points A_1, A_2, A_3, and A_4 satisfy (1) but these four points do not lie on a circle or on an line, equation (1) defines neither a circle, nor a line. Hence, the equation is an identity and the coefficient of the quadratic term x^2+y^2 also has to be zero, i.e.

$$\sum_{i=1}^{4} \frac{1}{d_i^2 - r_i^2} = 0.$$

Comment. Using the determinant form of the equation of the circle through three given points, the same solution can be formulated as follows.

For $i = 1, 2, 3, 4$ let (u_i, v_i) be the coordinates of A_i and define

$$\Delta = \begin{vmatrix} u_1^2+v_1^2 & u_1 & v_1 & 1 \\ u_2^2+v_2^2 & u_2 & v_2 & 1 \\ u_3^2+v_3^2 & u_3 & v_3 & 1 \\ u_4^2+v_4^2 & u_4 & v_4 & 1 \end{vmatrix} \qquad \text{and} \qquad \Delta_i = \begin{vmatrix} u_{i+1} & v_{i+1} & 1 \\ u_{i+2} & v_{i+2} & 1 \\ u_{i+3} & v_{i+3} & 1 \end{vmatrix},$$

where $i+1$, $i+2$, and $i+3$ have to be read modulo 4 as integers in the set $\{1, 2, 3, 4\}$.

Expanding $\begin{vmatrix} u_1 & v_1 & 1 & 1 \\ u_2 & v_2 & 1 & 1 \\ u_3 & v_3 & 1 & 1 \\ u_4 & v_4 & 1 & 1 \end{vmatrix} = 0$ along the third column, we get $\Delta_1 - \Delta_2 + \Delta_3 - \Delta_4 = 0$.

The circle through A_{i+1}, A_{i+2}, and A_{i+3} is given by the equation

$$\frac{1}{\Delta_i} \begin{vmatrix} x^2+y^2 & x & y & 1 \\ u_{i+1}^2+v_{i+1}^2 & u_{i+1} & v_{i+1} & 1 \\ u_{i+2}^2+v_{i+2}^2 & u_{i+2} & v_{i+2} & 1 \\ u_{i+3}^2+v_{i+3}^2 & u_{i+3} & v_{i+3} & 1 \end{vmatrix} = 0. \tag{2}$$

On the left-hand side, the coefficient of x^2+y^2 is equal to 1. Substituting (u_i, v_i) for (x, y) in (2) we obtain the power of point A_i with respect to the circle through A_{i+1}, A_{i+2}, and A_{i+3}:

$$d_i^2 - r_i^2 = \frac{1}{\Delta_i} \begin{vmatrix} u_i^2+v_i^2 & u_i & v_i & 1 \\ u_{i+1}^2+v_{i+1}^2 & u_{i+1} & v_{i+1} & 1 \\ u_{i+2}^2+v_{i+2}^2 & u_{i+2} & v_{i+2} & 1 \\ u_{i+3}^2+v_{i+3}^2 & u_{i+3} & v_{i+3} & 1 \end{vmatrix} = (-1)^{i+1} \frac{\Delta}{\Delta_i}.$$

Thus, we have

$$\sum_{i=1}^{4} \frac{1}{d_i^2 - r_i^2} = \frac{\Delta_1 - \Delta_2 + \Delta_3 - \Delta_4}{\Delta} = 0.$$

G3

Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC, and CD. Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA, and AB. Prove that the midpoint of the segment EF lies on the line through the two intersection points of ω_E and ω_F.

Solution. Denote by P, Q, R, and S the projections of E on the lines DA, AB, BC, and CD respectively. The points P and Q lie on the circle with diameter AE, so $\angle QPE = \angle QAE$; analogously, $\angle QRE = \angle QBE$. So $\angle QPE + \angle QRE = \angle QAE + \angle QBE = 90^\circ$. By similar reasons, we have $\angle SPE + \angle SRE = 90^\circ$, hence we get $\angle QPS + \angle QRS = 90^\circ + 90^\circ = 180^\circ$, and the quadrilateral $PQRS$ is inscribed in ω_E. Analogously, all four projections of F onto the sides of $ABCD$ lie on ω_F.

Denote by K the meeting point of the lines AD and BC. Due to the arguments above, there is no loss of generality in assuming that A lies on segment DK. Suppose that $\angle CKD \geq 90^\circ$; then the circle with diameter CD covers the whole quadrilateral $ABCD$, so the points E, F cannot lie inside this quadrilateral. Hence our assumption is wrong. Therefore, the lines EP and BC intersect at some point P', while the lines ER and AD intersect at some point R'.

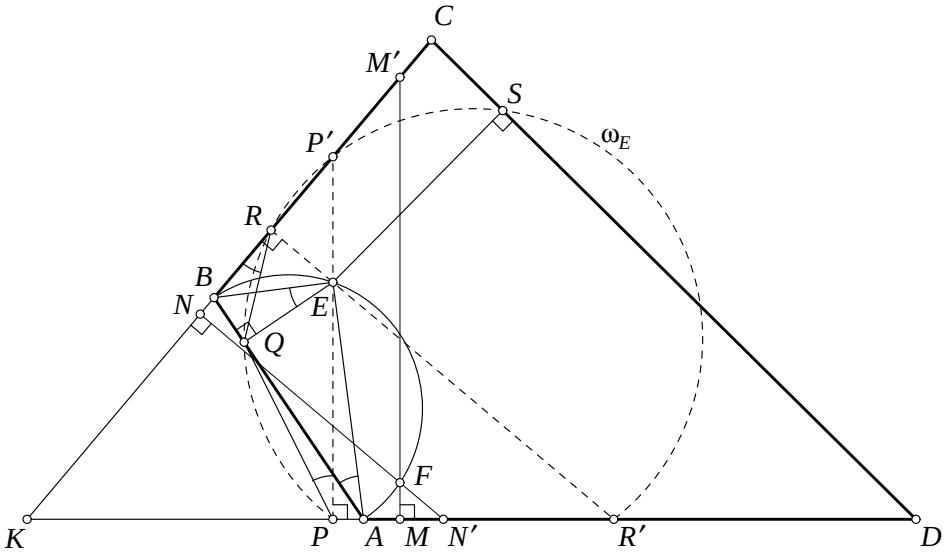


Figure 1

We claim that the points P' and R' also belong to ω_E. Since the points R, E, Q, B are concyclic, $\angle QRK = \angle QEB = 90^\circ - \angle QBE = \angle QAE = \angle QPE$. So $\angle QRK = \angle QPP'$, which means that the point P' lies on ω_E. Analogously, R' also lies on ω_E.

In the same manner, denote by M and N the projections of F on the lines AD and BC

respectively, and let $M' = FM \cap BC$, $N' = FN \cap AD$. By the same arguments, we obtain that the points M' and N' belong to ω_F.

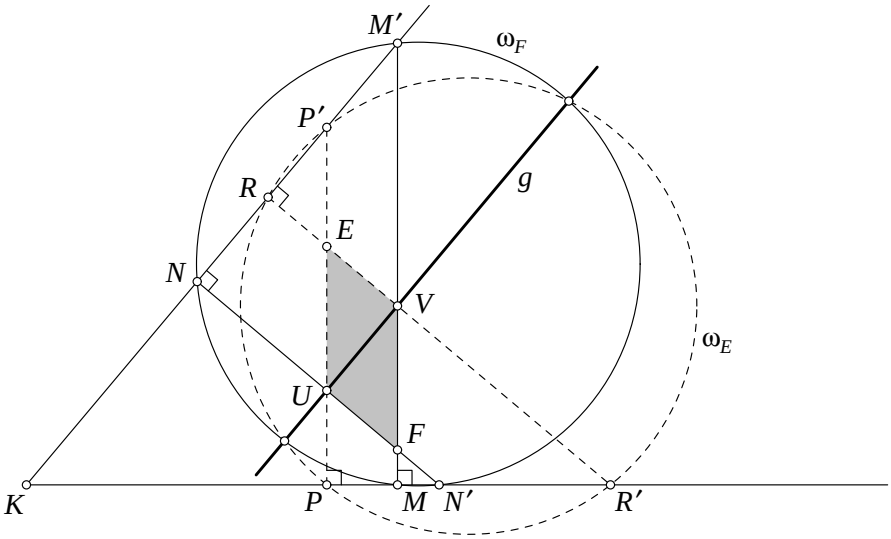


Figure 2

Now we concentrate on Figure 2, where all unnecessary details are removed. Let $U = NN' \cap PP'$, $V = MM' \cap RR'$. Due to the right angles at N and P, the points N, N', P, P' are concyclic, so $UN \cdot UN' = UP \cdot UP'$ which means that U belongs to the radical axis g of the circles ω_E and ω_F. Analogously, V also belongs to g.

Finally, since $EUFV$ is a parallelogram, the radical axis UV of ω_E and ω_F bisects EF.

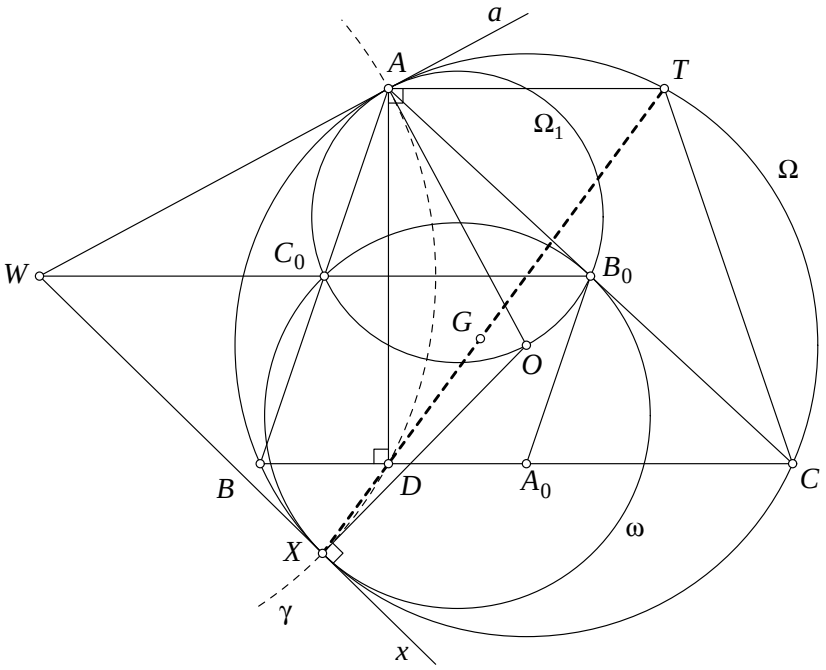
G4

Let ABC be an acute triangle with circumcircle Ω. Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB. Let D be the foot of the altitude from A, and let G be the centroid of the triangle ABC. Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G, and X are collinear.

Solution 1. If $AB = AC$, then the statement is trivial. So without loss of generality we may assume $AB < AC$. Denote the tangents to Ω at points A and X by a and x, respectively.

Let Ω_1 be the circumcircle of triangle AB_0C_0. The circles Ω and Ω_1 are homothetic with center A, so they are tangent at A, and a is their radical axis. Now, the lines a, x, and B_0C_0 are the three radical axes of the circles Ω, Ω_1, and ω. Since $a \nparallel B_0C_0$, these three lines are concurrent at some point W.

The points A and D are symmetric with respect to the line B_0C_0; hence $WX = WA = WD$. This means that W is the center of the circumcircle γ of triangle ADX. Moreover, we have $\angle WAO = \angle WXO = 90^\circ$, where O denotes the center of Ω. Hence $\angle AWX + \angle AOX = 180^\circ$.



Denote by T the second intersection point of Ω and the line DX. Note that O belongs to Ω_1. Using the circles γ and Ω, we find $\angle DAT = \angle ADX - \angle ATD = \frac{1}{2}(360^\circ - \angle AWX) - \frac{1}{2}\angle AOX = 180^\circ - \frac{1}{2}(\angle AWX + \angle AOX) = 90^\circ$. So, $AD \perp AT$, and hence $AT \parallel BC$. Thus, $ATCB$ is an isosceles trapezoid inscribed in Ω.

Denote by A_0 the midpoint of BC, and consider the image of $ATCB$ under the homothety h with center G and factor $-\frac{1}{2}$. We have $h(A) = A_0$, $h(B) = B_0$, and $h(C) = C_0$. From the

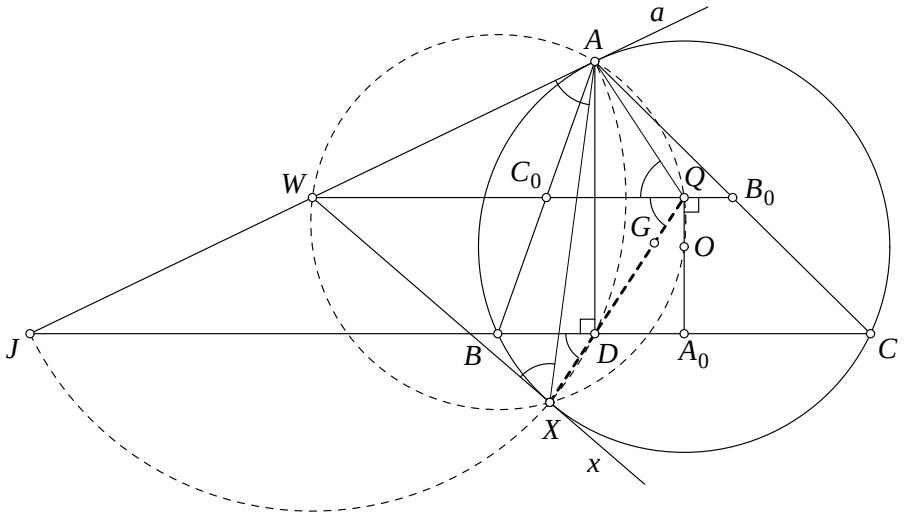
symmetry about B_0C_0, we have $\angle TCB = \angle CBA = \angle B_0C_0A = \angle DC_0B_0$. Using $AT \parallel DA_0$, we conclude $h(T) = D$. Hence the points D, G, and T are collinear, and X lies on the same line.

Solution 2. We define the points A_0, O, and W as in the previous solution and we concentrate on the case $AB < AC$. Let Q be the perpendicular projection of A_0 on B_0C_0.

Since $\angle WAO = \angle WQO = \angle OXW = 90°$, the five points A, W, X, O, and Q lie on a common circle. Furthermore, the reflections with respect to B_0C_0 and OW map A to D and X, respectively. For these reasons, we have

$$\angle WQD = \angle AQW = \angle AXW = \angle WAX = \angle WQX.$$

Thus the three points Q, D, and X lie on a common line, say ℓ.



To complete the argument, we note that the homothety centered at G sending the triangle ABC to the triangle $A_0B_0C_0$ maps the altitude AD to the altitude A_0Q. Therefore it maps D to Q, so the points D, G, and Q are collinear. Hence G lies on ℓ as well.

Comment. There are various other ways to prove the collinearity of Q, D, and X obtained in the middle part of Solution 2. Introduce for instance the point J where the lines AW and BC intersect. Then the four points A, D, X, and J lie at the same distance from W, so the quadrilateral $ADXJ$ is cyclic. In combination with the fact that $AWXQ$ is cyclic as well, this implies

$$\angle JDX = \angle JAX = \angle WAX = \angle WQX.$$

Since $BC \parallel WQ$, it follows that Q, D, and X are indeed collinear.

G5

Let ABC be a triangle with incenter I and circumcircle ω. Let D and E be the second intersection points of ω with the lines AI and BI, respectively. The chord DE meets AC at a point F, and BC at a point G. Let P be the intersection point of the line through F parallel to AD and the line through G parallel to BE. Suppose that the tangents to ω at A and at B meet at a point K. Prove that the three lines AE, BD, and KP are either parallel or concurrent.

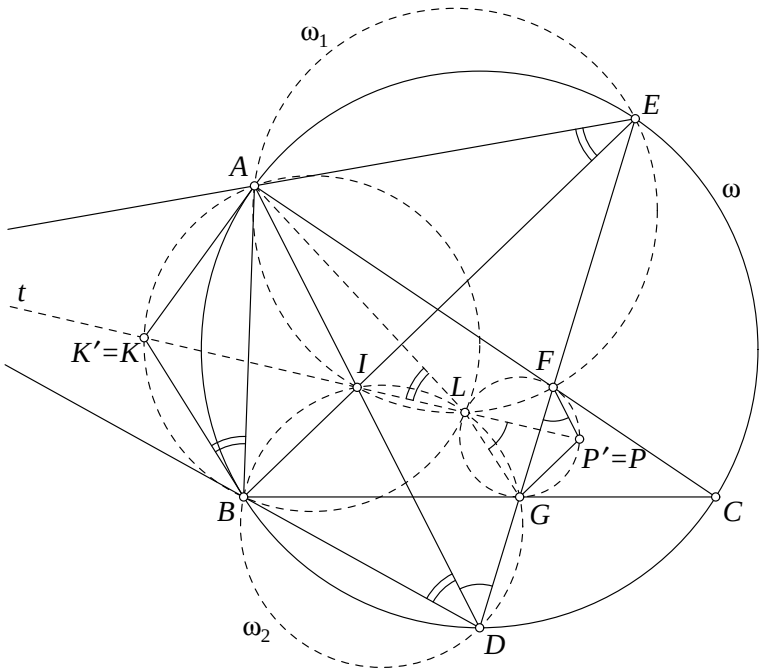
Solution 1. Since

$$\angle IAF = \angle DAC = \angle BAD = \angle BED = \angle IEF$$

the quadrilateral $AIFE$ is cyclic. Denote its circumcircle by ω_1. Similarly, the quadrilateral $BDGI$ is cyclic; denote its circumcircle by ω_2.

The line AE is the radical axis of ω and ω_1, and the line BD is the radical axis of ω and ω_2. Let t be the radical axis of ω_1 and ω_2. These three lines meet at the radical center of the three circles, or they are parallel to each other. We will show that t is in fact the line PK.

Let L be the second intersection point of ω_1 and ω_2, so $t = IL$. (If the two circles are tangent to each other then $L = I$ and t is the common tangent.)



Let the line t meet the circumcircles of the triangles ABL and FGL at $K' \neq L$ and $P' \neq L$, respectively. Using oriented angles we have

$$\angle(AB, BK') = \angle(AL, LK') = \angle(AL, LI) = \angle(AE, EI) = \angle(AE, EB) = \angle(AB, BK),$$

so $BK' \parallel BK$. Similarly we have $AK' \parallel AK$, and therefore $K' = K$. Next, we have

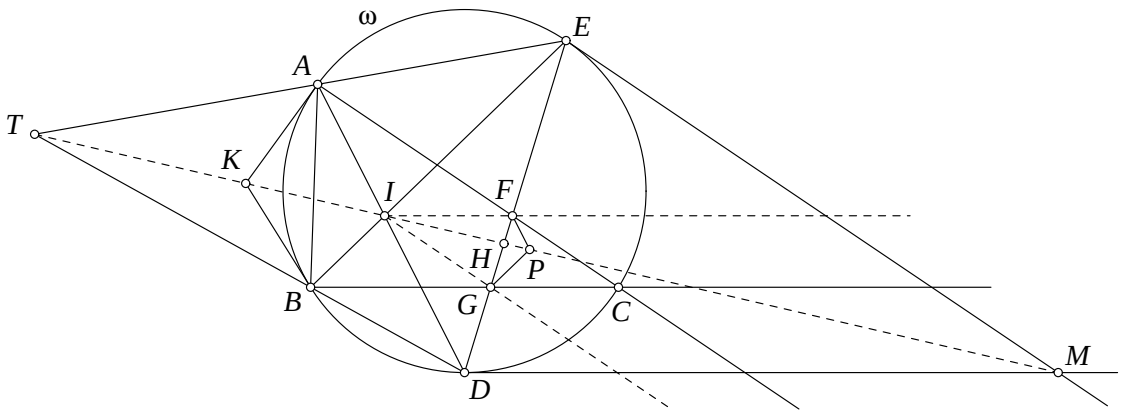
$$\angle(P'F, FG) = \angle(P'L, LG) = \angle(IL, LG) = \angle(ID, DG) = \angle(AD, DE) = \angle(PF, FG),$$

hence $P'F \parallel PF$ and similarly $P'G \parallel PG$. Therefore $P' = P$. This means that t passes through K and P, which finishes the proof.

Solution 2. Let M be the intersection point of the tangents to ω at D and E, and let the lines AE and BD meet at T; if AE and BD are parallel, then let T be their common ideal point. It is well-known that the points K and M lie on the line TI (as a consequence of Pascal's theorem, applied to the inscribed degenerate hexagons $AADBBE$ and $ADDBEE$).

The lines AD and BE are the angle bisectors of the angles $\angle CAB$ and $\angle ABC$, respectively, so D and E are the midpoints of the arcs BC and CA of the circle ω, respectively. Hence, DM is parallel to BC and EM is parallel to AC.

Apply Pascal's theorem to the degenerate hexagon $CADDEB$. By the theorem, the points $CA \cap DE = F$, $AD \cap EB = I$ and the common ideal point of lines DM and BC are collinear, therefore FI is parallel to BC and DM. Analogously, the line GI is parallel to AC and EM.



Now consider the homothety with scale factor $-\frac{FG}{ED}$ which takes E to G and D to F. Since the triangles EDM and GFI have parallel sides, the homothety takes M to I. Similarly, since the triangles DEI and FGP have parallel sides, the homothety takes I to P. Hence, the points M, I, P and the homothety center H must lie on the same line. Therefore, the point P also lies on the line $TKIM$.

Comment. One may prove that $IF \parallel BC$ and $IG \parallel AC$ in a more elementary way. Since $\angle ADE = \angle EDC$ and $\angle DEB = \angle CED$, the points I and C are symmetric about DE. Moreover, since the arcs AE and EC are equal and the arcs CD and DB are equal, we have $\angle CFG = \angle FGC$, so the triangle CFG is isosceles. Hence, the quadrilateral $IFCG$ is a rhombus.

G6

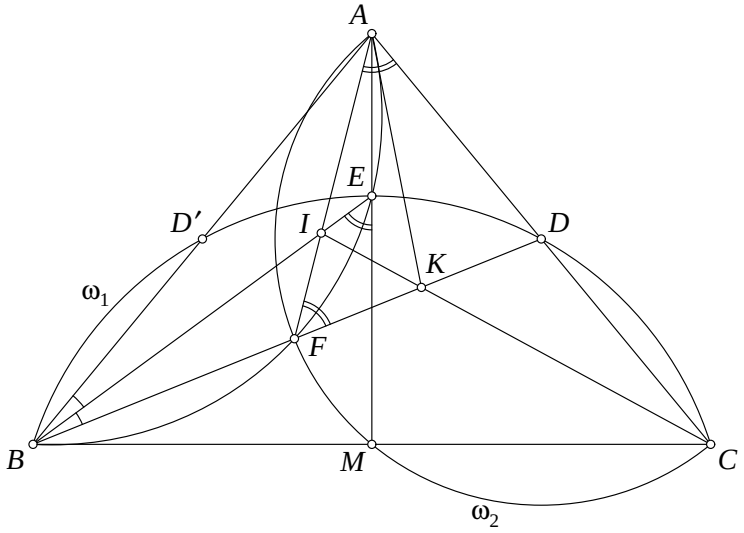
Let ABC be a triangle with $AB = AC$, and let D be the midpoint of AC. The angle bisector of $\angle BAC$ intersects the circle through D, B, and C in a point E inside the triangle ABC. The line BD intersects the circle through A, E, and B in two points B and F. The lines AF and BE meet at a point I, and the lines CI and BD meet at a point K. Show that I is the incenter of triangle KAB.

Solution 1. Let D' be the midpoint of the segment AB, and let M be the midpoint of BC. By symmetry at line AM, the point D' has to lie on the circle BCD. Since the arcs $D'E$ and ED of that circle are equal, we have $\angle ABI = \angle D'BE = \angle EBD = IBK$, so I lies on the angle bisector of $\angle ABK$. For this reason it suffices to prove in the sequel that the ray AI bisects the angle $\angle BAK$.

From

$$\angle DFA = 180^\circ - \angle BFA = 180^\circ - \angle BEA = \angle MEB = \frac{1}{2}\angle CEB = \frac{1}{2}\angle CDB$$

we derive $\angle DFA = \angle DAF$ so the triangle AFD is isosceles with $AD = DF$.



Applying Menelaus's theorem to the triangle ADF with respect to the line CIK, and applying the angle bisector theorem to the triangle ABF, we infer

$$1 = \frac{AC}{CD} \cdot \frac{DK}{KF} \cdot \frac{FI}{IA} = 2 \cdot \frac{DK}{KF} \cdot \frac{BF}{AB} = 2 \cdot \frac{DK}{KF} \cdot \frac{BF}{2 \cdot AD} = \frac{DK}{KF} \cdot \frac{BF}{AD}$$

and therefore

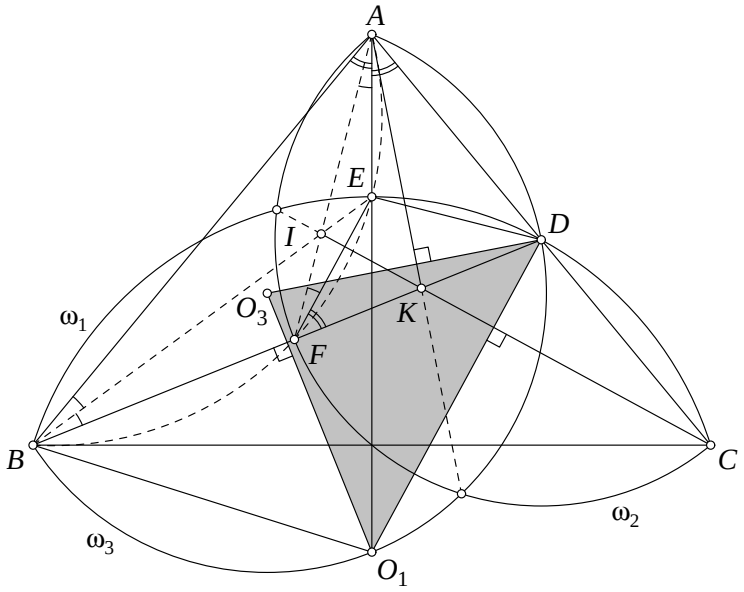
$$\frac{BD}{AD} = \frac{BF + FD}{AD} = \frac{BF}{AD} + 1 = \frac{KF}{DK} + 1 = \frac{DF}{DK} = \frac{AD}{DK}.$$

It follows that the triangles ADK and BDA are similar, hence $\angle DAK = \angle ABD$. Then

$$\angle IAB = \angle AFD - \angle ABD = \angle DAF - \angle DAK = \angle KAI$$

shows that the point K is indeed lying on the angle bisector of $\angle BAK$.

Solution 2. It can be shown in the same way as in the first solution that I lies on the angle bisector of $\angle ABK$. Here we restrict ourselves to proving that KI bisects $\angle AKB$.



Denote the circumcircle of triangle BCD and its center by ω_1 and by O_1, respectively. Since the quadrilateral $ABFE$ is cyclic, we have $\angle DFE = \angle BAE = \angle DAE$. By the same reason, we have $\angle EAF = \angle EBF = \angle ABE = \angle AFE$. Therefore $\angle DAF = \angle DFA$, and hence $DF = DA = DC$. So triangle AFC is inscribed in a circle ω_2 with center D.

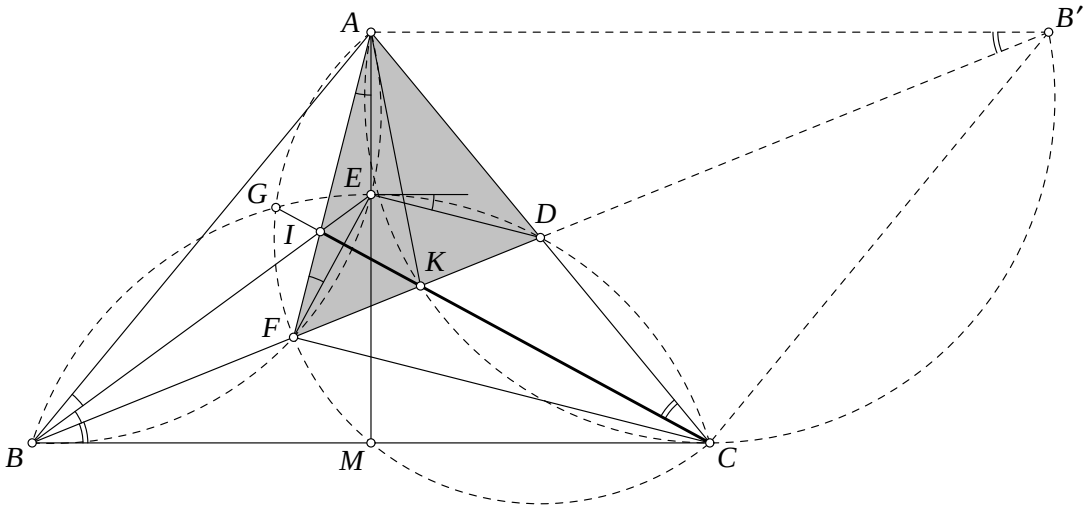
Denote the circumcircle of triangle ABD by ω_3, and let its center be O_3. Since the arcs BE and EC of circle ω_1 are equal, and the triangles ADE and FDE are congruent, we have $\angle AO_1B = 2\angle BDE = \angle BDA$, so O_1 lies on ω_3. Hence $\angle O_3O_1D = \angle O_3DO_1$.

The line BD is the radical axis of ω_1 and ω_3. Point C belongs to the radical axis of ω_1 and ω_2, and I also belongs to it since $AI \cdot IF = BI \cdot IE$. Hence $K = BD \cap CI$ is the radical center of ω_1, ω_2, and ω_3, and AK is the radical axis of ω_2 and ω_3. Now, the radical axes AK, BK and IK are perpendicular to the central lines O_3D, O_3O_1 and O_1D, respectively. By $\angle O_3O_1D = \angle O_3DO_1$, we get that KI is the angle bisector of $\angle AKB$.

Solution 3. Again, let M be the midpoint of BC. As in the previous solutions, we can deduce $\angle ABI = \angle IBK$. We show that the point I lies on the angle bisector of $\angle KAB$.

Let G be the intersection point of the circles AFC and BCD, different from C. The lines

CG, AF, and BE are the radical axes of the three circles $AGFC$, CDB, and $ABFE$, so $I = AF \cap BE$ is the radical center of the three circles and CG also passes through I.



The angle between line DE and the tangent to the circle BCD at E is equal to $\angle EBD = \angle EAF = \angle ABE = \angle AFE$. As the tangent at E is perpendicular to AM, the line DE is perpendicular to AF. The triangle AFE is isosceles, so DE is the perpendicular bisector of AF and thus $AD = DF$. Hence, the point D is the center of the circle AFC, and this circle passes through M as well since $\angle AMC = 90°$.

Let B' be the reflection of B in the point D, so $ABCB'$ is a parallelogram. Since $DC = DG$ we have $\angle GCD = \angle DBC = \angle KB'A$. Hence, the quadrilateral $AKCB'$ is cyclic and thus $\angle CAK = \angle CB'K = \angle ABD = 2\angle MAI$. Then

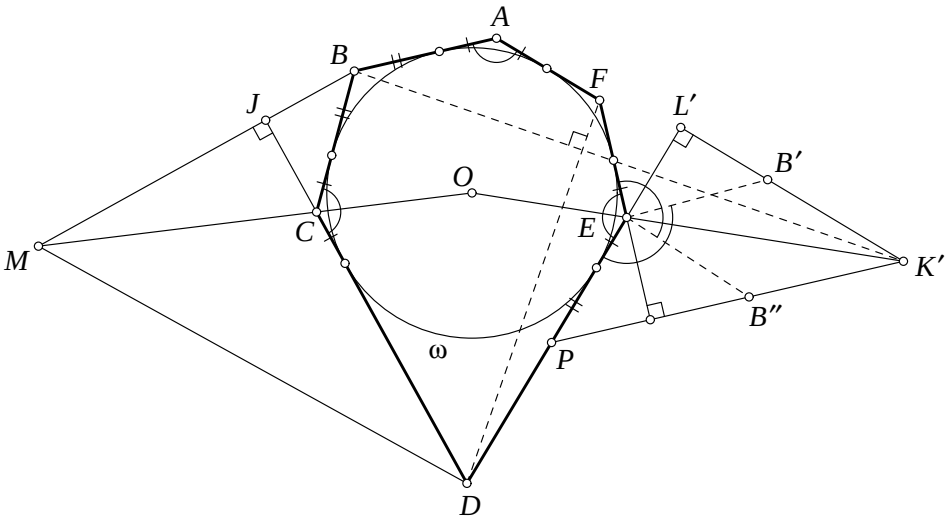
$$\angle IAB = \angle MAB - \angle MAI = \frac{1}{2}\angle CAB - \frac{1}{2}\angle CAK = \frac{1}{2}\angle KAB$$

and therefore AI is the angle bisector of $\angle KAB$.

G7

Let $ABCDEF$ be a convex hexagon all of whose sides are tangent to a circle ω with center O. Suppose that the circumcircle of triangle ACE is concentric with ω. Let J be the foot of the perpendicular from B to CD. Suppose that the perpendicular from B to DF intersects the line EO at a point K. Let L be the foot of the perpendicular from K to DE. Prove that $DJ = DL$.

Solution 1. Since ω and the circumcircle of triangle ACE are concentric, the tangents from A, C, and E to ω have equal lengths; that means that $AB = BC$, $CD = DE$, and $EF = FA$. Moreover, we have $\angle BCD = \angle DEF = \angle FAB$.



Consider the rotation around point D mapping C to E; let B' and L' be the images of the points B and J, respectively, under this rotation. Then one has $DJ = DL'$ and $B'L' \perp DE$; moreover, the triangles $B'ED$ and BCD are congruent. Since $\angle DEO < 90°$, the lines EO and $B'L'$ intersect at some point K'. We intend to prove that $K'B \perp DF$; this would imply $K = K'$, therefore $L = L'$, which proves the problem statement.

Analogously, consider the rotation around F mapping A to E; let B'' be the image of B under this rotation. Then the triangles FAB and FEB'' are congruent. We have $EB'' = AB = BC = EB'$ and $\angle FEB'' = \angle FAB = \angle BCD = \angle DEB'$, so the points B' and B'' are symmetrical with respect to the angle bisector EO of $\angle DEF$. So, from $K'B' \perp DE$ we get $K'B'' \perp EF$.

From these two relations we obtain

$$K'D^2 - K'E^2 = B'D^2 - B'E^2 \quad \text{and} \quad K'E^2 - K'F^2 = B''E^2 - B''F^2.$$

Adding these equalities and taking into account that $B'E = B''E$ we obtain

$$K'D^2 - K'F^2 = B'D^2 - B''F^2 = BD^2 - BF^2,$$

which means exactly that $K'B \perp DF$.

Comment. There are several variations of this solution. For instance, let us consider the intersection point M of the lines BJ and OC. Define the point K' as the reflection of M in the line DO. Then one has

$$DK'^2 - DB^2 = DM^2 - DB^2 = CM^2 - CB^2.$$

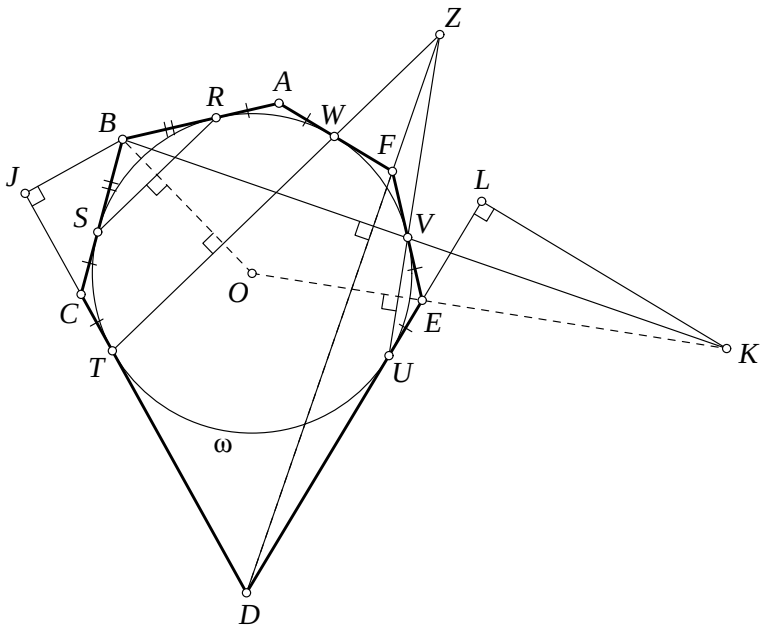
Next, consider the rotation around O which maps CM to EK'. Let P be the image of B under this rotation; so P lies on ED. Then $EF \perp K'P$, so

$$CM^2 - CB^2 = EK'^2 - EP^2 = FK'^2 - FP^2 = FK'^2 - FB^2,$$

since the triangles FEP and FAB are congruent.

Solution 2. Let us denote the points of tangency of AB, BC, CD, DE, EF, and FA to ω by R, S, T, U, V, and W, respectively. As in the previous solution, we mention that $AR = AW = CS = CT = EU = EV$.

The reflection in the line BO maps R to S, therefore A to C and thus also W to T. Hence, both lines RS and WT are perpendicular to OB, therefore they are parallel. On the other hand, the lines UV and WT are not parallel, since otherwise the hexagon $ABCDEF$ is symmetric with respect to the line BO and the lines defining the point K coincide, which contradicts the conditions of the problem. Therefore we can consider the intersection point Z of UV and WT.



Next, we recall a well-known fact that the points D, F, Z are collinear. Actually, D is the pole of the line UT, F is the pole of VW, and $Z = TW \cap UV$; so all these points belong to the polar line of $TU \cap VW$.

Now, we put O into the origin, and identify each point (say X) with the vector $\overrightarrow{OX}$. So, from now on all the products of points refer to the scalar products of the corresponding vectors.

Since $OK \perp UZ$ and $OB \perp TZ$, we have $K \cdot (Z - U) = 0 = B \cdot (Z - T)$. Next, the condition $BK \perp DZ$ can be written as $K \cdot (D - Z) = B \cdot (D - Z)$. Adding these two equalities we get

$$K \cdot (D - U) = B \cdot (D - T).$$

By symmetry, we have $D \cdot (D - U) = D \cdot (D - T)$. Subtracting this from the previous equation, we obtain $(K - D) \cdot (D - U) = (B - D) \cdot (D - T)$ and rewrite it in vector form as

$$\overrightarrow{DK} \cdot \overrightarrow{UD} = \overrightarrow{DB} \cdot \overrightarrow{TD}.$$

Finally, projecting the vectors $\overrightarrow{DK}$ and $\overrightarrow{DB}$ onto the lines UD and TD respectively, we can rewrite this equality in terms of segment lengths as $DL \cdot UD = DJ \cdot TD$, thus $DL = DJ$.

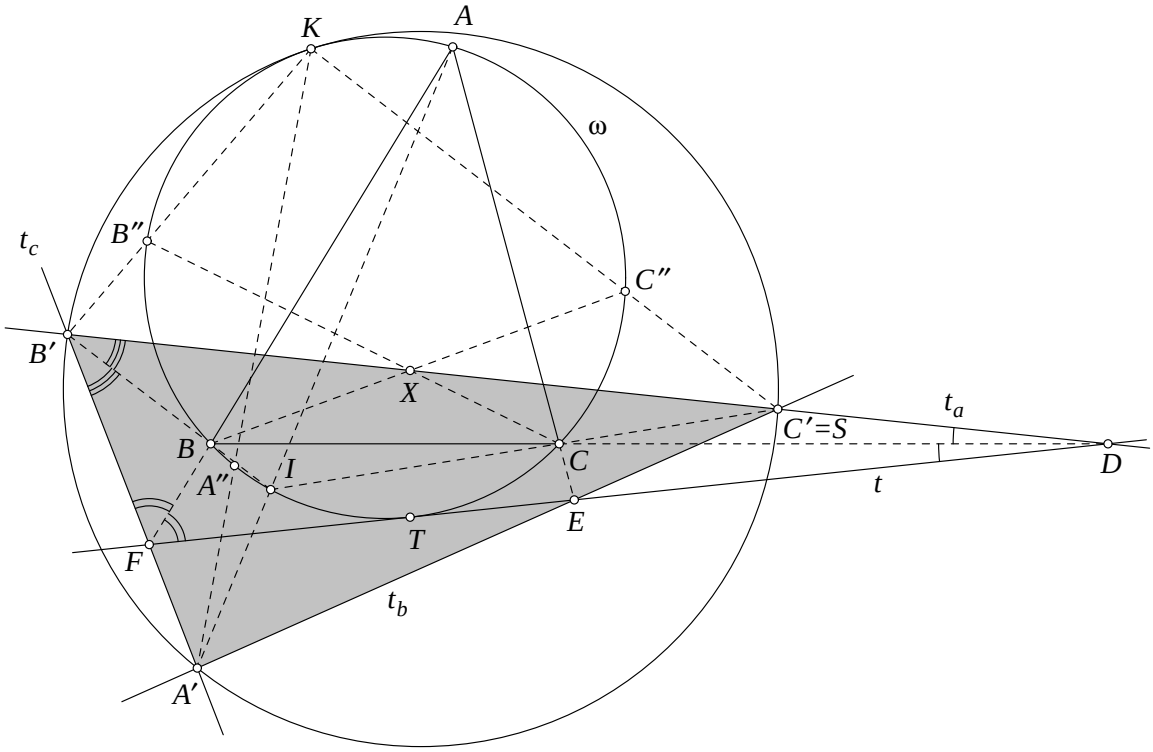
Comment. The collinearity of Z, F, and D may be shown in various more elementary ways. For instance, applying the sine theorem to the triangles DTZ and DUZ, one gets $\dfrac{\sin \angle DZT}{\sin \angle DZU} = \dfrac{\sin \angle DTZ}{\sin \angle DUZ}$; analogously, $\dfrac{\sin \angle FZW}{\sin \angle FZV} = \dfrac{\sin \angle FWZ}{\sin \angle FVZ}$. The right-hand sides are equal, hence so are the left-hand sides, which implies the collinearity of the points D, F, and Z.

There also exist purely synthetic proofs of this fact. E.g., let Q be the point of intersection of the circumcircles of the triangles ZTV and ZWU different from Z. Then QZ is the bisector of $\angle VQW$ since $\angle VQZ = \angle VTZ = \angle VUW = \angle ZQW$. Moreover, all these angles are equal to $\frac{1}{2}\angle VOW$, so $\angle VQW = \angle VOW$, hence the quadrilateral $VWOQ$ is cyclic. On the other hand, the points O, V, W lie on the circle with diameter OF due to the right angles; so Q also belongs to this circle. Since $FV = FW$, QF is also the bisector of $\angle VQW$, so F lies on QZ. Analogously, D lies on the same line.

G8

Let ABC be an acute triangle with circumcircle ω. Let t be a tangent line to ω. Let t_a, t_b, and t_c be the lines obtained by reflecting t in the lines BC, CA, and AB, respectively. Show that the circumcircle of the triangle determined by the lines t_a, t_b, and t_c is tangent to the circle ω.

To avoid a large case distinction, we will use the notion of *oriented angles*. Namely, for two lines ℓ and m, we denote by $\angle(\ell, m)$ the angle by which one may rotate ℓ anticlockwise to obtain a line parallel to m. Thus, all oriented angles are considered modulo 180°.



Solution 1. Denote by T the point of tangency of t and ω. Let $A' = t_b \cap t_c$, $B' = t_a \cap t_c$, $C' = t_a \cap t_b$. Introduce the point A'' on ω such that $TA = AA''$ ($A'' \neq T$ unless TA is a diameter). Define the points B'' and C'' in a similar way.

Since the points C and B are the midpoints of arcs TC'' and TB'', respectively, we have

$$\begin{aligned}\angle(t, B''C'') &= \angle(t, TC'') + \angle(TC'', B''C'') = 2\angle(t, TC) + 2\angle(TC'', BC'') \\ &= 2\big(\angle(t, TC) + \angle(TC, BC)\big) = 2\angle(t, BC) = \angle(t, t_a).\end{aligned}$$

It follows that t_a and $B''C''$ are parallel. Similarly, $t_b \parallel A''C''$ and $t_c \parallel A''B''$. Thus, either the triangles $A'B'C'$ and $A''B''C''$ are homothetic, or they are translates of each other. Now we will prove that they are in fact homothetic, and that the center K of the homothety belongs

to ω. It would then follow that their circumcircles are also homothetic with respect to K and are therefore tangent at this point, as desired.

We need the two following claims.

Claim 1. The point of intersection X of the lines $B''C$ and BC'' lies on t_a.

Proof. Actually, the points X and T are symmetric about the line BC, since the lines CT and CB'' are symmetric about this line, as are the lines BT and BC''. □

Claim 2. The point of intersection I of the lines BB' and CC' lies on the circle ω.

Proof. We consider the case that t is not parallel to the sides of ABC; the other cases may be regarded as limit cases. Let $D = t \cap BC$, $E = t \cap AC$, and $F = t \cap AB$.

Due to symmetry, the line DB is one of the angle bisectors of the lines $B'D$ and FD; analogously, the line FB is one of the angle bisectors of the lines $B'F$ and DF. So B is either the incenter or one of the excenters of the triangle $B'DF$. In any case we have $\angle(BD, DF) + \angle(DF, FB) + \angle(B'B, B'D) = 90°$, so

$$\angle(B'B, B'C') = \angle(B'B, B'D) = 90° - \angle(BC, DF) - \angle(DF, BA) = 90° - \angle(BC, AB).$$

Analogously, we get $\angle(C'C, B'C') = 90° - \angle(BC, AC)$. Hence,

$$\angle(BI, CI) = \angle(B'B, B'C') + \angle(B'C', C'C) = \angle(BC, AC) - \angle(BC, AB) = \angle(AB, AC),$$

which means exactly that the points A, B, I, C are concyclic. □

Now we can complete the proof. Let K be the second intersection point of $B'B''$ and ω. Applying Pascal's theorem to hexagon $KB''CIBC''$ we get that the points $B' = KB'' \cap IB$ and $X = B''C \cap BC''$ are collinear with the intersection point S of CI and $C''K$. So $S = CI \cap B'X = C'$, and the points C', C'', K are collinear. Thus K is the intersection point of $B'B''$ and $C'C''$ which implies that K is the center of the homothety mapping $A'B'C'$ to $A''B''C''$, and it belongs to ω.

Solution 2. Define the points T, A', B', and C' in the same way as in the previous solution. Let X, Y, and Z be the symmetric images of T about the lines BC, CA, and AB, respectively. Note that the projections of T on these lines form a Simson line of T with respect to ABC, therefore the points X, Y, Z are also collinear. Moreover, we have $X \in B'C'$, $Y \in C'A'$, $Z \in A'B'$.

Denote $\alpha = \angle(t, TC) = \angle(BT, BC)$. Using the symmetry in the lines AC and BC, we get

$$\angle(BC, BX) = \angle(BT, BC) = \alpha \quad \text{and} \quad \angle(XC, XC') = \angle(t, TC) = \angle(YC, YC') = \alpha.$$

Since $\angle(XC, XC') = \angle(YC, YC')$, the points X, Y, C, C' lie on some circle ω_c. Define the circles ω_a and ω_b analogously. Let ω' be the circumcircle of triangle $A'B'C'$.

Now, applying MIQUEL's theorem to the four lines $A'B'$, $A'C'$, $B'C'$, and XY, we obtain that the circles ω', ω_a, ω_b, ω_c intersect at some point K. We will show that K lies on ω, and that the tangent lines to ω and ω' at this point coincide; this implies the problem statement.

Due to symmetry, we have $XB = TB = ZB$, so the point B is the midpoint of one of the arcs XZ of circle ω_b. Therefore $\angle(KB, KX) = \angle(XZ, XB)$. Analogously, $\angle(KX, KC) = \angle(XC, XY)$. Adding these equalities and using the symmetry in the line BC we get

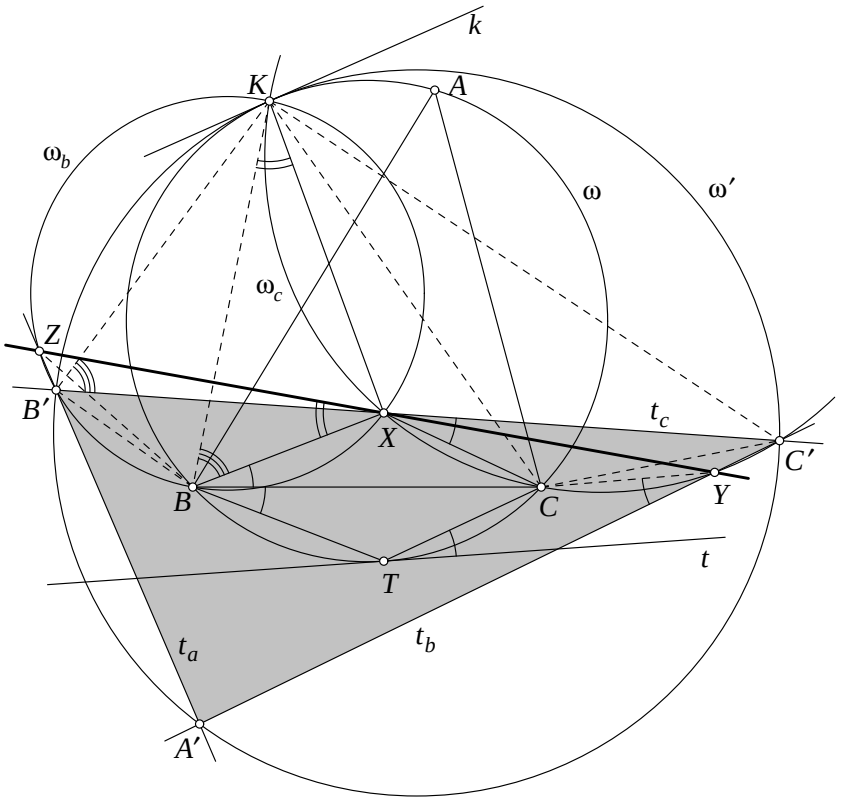
$$\angle(KB, KC) = \angle(XZ, XB) + \angle(XC, XZ) = \angle(XC, XB) = \angle(TB, TC).$$

Therefore, K lies on ω.

Next, let k be the tangent line to ω at K. We have

$$\begin{aligned}\angle(k, KC') &= \angle(k, KC) + \angle(KC, KC') = \angle(KB, BC) + \angle(XC, XC') \\ &= \big(\angle(KB, BX) - \angle(BC, BX)\big) + \alpha = \angle(KB', B'X) - \alpha + \alpha = \angle(KB', B'C'),\end{aligned}$$

which means exactly that k is tangent to ω'.



Comment. There exist various solutions combining the ideas from the two solutions presented above. For instance, one may define the point X as the reflection of T with respect to the line BC, and then introduce the point K as the second intersection point of the circumcircles of $BB'X$ and $CC'X$. Using the fact that BB' and CC' are the bisectors of $\angle(A'B', B'C')$ and $\angle(A'C', B'C')$ one can show successively that $K \in \omega$, $K \in \omega'$, and that the tangents to ω and ω' at K coincide.

N1

For any integer $d > 0$, let $f(d)$ be the smallest positive integer that has exactly d positive divisors (so for example we have $f(1) = 1$, $f(5) = 16$, and $f(6) = 12$). Prove that for every integer $k \geq 0$ the number $f(2^k)$ divides $f(2^{k+1})$.

Solution 1. For any positive integer n, let $d(n)$ be the number of positive divisors of n. Let $n = \prod_p p^{a(p)}$ be the prime factorization of n where p ranges over the prime numbers, the integers $a(p)$ are nonnegative and all but finitely many $a(p)$ are zero. Then we have $d(n) = \prod_p (a(p)+1)$. Thus, $d(n)$ is a power of 2 if and only if for every prime p there is a nonnegative integer $b(p)$ with $a(p) = 2^{b(p)} - 1 = 1 + 2 + 2^2 + \cdots + 2^{b(p)-1}$. We then have

$$n = \prod_p \prod_{i=0}^{b(p)-1} p^{2^i}, \qquad \text{and} \quad d(n) = 2^k \quad \text{with} \quad k = \sum_p b(p).$$

Let $\mathcal{S}$ be the set of all numbers of the form p^{2^r} with p prime and r a nonnegative integer. Then we deduce that $d(n)$ is a power of 2 if and only if n is the product of the elements of some finite subset $\mathcal{T}$ of $\mathcal{S}$ that satisfies the following condition: for all $t \in \mathcal{T}$ and $s \in \mathcal{S}$ with $s \mid t$ we have $s \in \mathcal{T}$. Moreover, if $d(n) = 2^k$ then the corresponding set $\mathcal{T}$ has k elements.

Note that the set $\mathcal{T}_k$ consisting of the smallest k elements from $\mathcal{S}$ obviously satisfies the condition above. Thus, given k, the smallest n with $d(n) = 2^k$ is the product of the elements of $\mathcal{T}_k$. This n is $f(2^k)$. Since obviously $\mathcal{T}_k \subset \mathcal{T}_{k+1}$, it follows that $f(2^k) \mid f(2^{k+1})$.

Solution 2. This is an alternative to the second part of the Solution 1. Suppose k is a nonnegative integer. From the first part of Solution 1 we see that $f(2^k) = \prod_p p^{a(p)}$ with $a(p) = 2^{b(p)} - 1$ and $\sum_p b(p) = k$. We now claim that for any two distinct primes p, q with $b(q) > 0$ we have

$$m = p^{2^{b(p)}} > q^{2^{b(q)-1}} = \ell. \tag{1}$$

To see this, note first that ℓ divides $f(2^k)$. With the first part of Solution 1 one can see that the integer $n = f(2^k)m/\ell$ also satisfies $d(n) = 2^k$. By the definition of $f(2^k)$ this implies that $n \geq f(2^k)$ so $m \geq \ell$. Since $p \neq q$ the inequality (1) follows.

Let the prime factorization of $f(2^{k+1})$ be given by $f(2^{k+1}) = \prod_p p^{r(p)}$ with $r(p) = 2^{s(p)} - 1$. Since we have $\sum_p s(p) = k + 1 > k = \sum_p b(p)$ there is a prime p with $s(p) > b(p)$. For any prime $q \neq p$ with $b(q) > 0$ we apply inequality (1) twice and get

$$q^{2^{s(q)}} > p^{2^{s(p)-1}} \geq p^{2^{b(p)}} > q^{2^{b(q)-1}},$$

which implies $s(q) \geq b(q)$. It follows that $s(q) \geq b(q)$ for all primes q, so $f(2^k) \mid f(2^{k+1})$.

N2

Consider a polynomial $P(x) = (x + d_1)(x + d_2) \cdot \ldots \cdot (x + d_9)$, where d_1, $d_2, \ldots$, d_9 are nine distinct integers. Prove that there exists an integer N such that for all integers $x \geq N$ the number $P(x)$ is divisible by a prime number greater than 20.

Solution 1. Note that the statement of the problem is invariant under translations of x; hence without loss of generality we may suppose that the numbers d_1, $d_2, \ldots$, d_9 are positive.

The key observation is that there are only eight primes below 20, while $P(x)$ involves more than eight factors.

We shall prove that $N = d^8$ satisfies the desired property, where $d = \max\{d_1, d_2, \ldots, d_9\}$. Suppose for the sake of contradiction that there is some integer $x \geq N$ such that $P(x)$ is composed of primes below 20 only. Then for every index $i \in \{1, 2, \ldots, 9\}$ the number $x + d_i$ can be expressed as product of powers of the first 8 primes.

Since $x + d_i > x \geq d^8$ there is some prime power $f_i > d$ that divides $x + d_i$. Invoking the pigeonhole principle we see that there are two distinct indices i and j such that f_i and f_j are powers of the same prime number. For reasons of symmetry, we may suppose that $f_i \leq f_j$. Now both of the numbers $x + d_i$ and $x + d_j$ are divisible by f_i and hence so is their difference $d_i - d_j$. But as

$$0 < |d_i - d_j| \leq \max(d_i, d_j) \leq d < f_i,$$

this is impossible. Thereby the problem is solved.

Solution 2. Observe that for each index $i \in \{1, 2, \ldots, 9\}$ the product

$$D_i = \prod_{1 \leq j \leq 9, j \neq i} |d_i - d_j|$$

is positive. We claim that $N = \max\{D_1 - d_1, D_2 - d_2, \ldots, D_9 - d_9\} + 1$ satisfies the statement of the problem. Suppose there exists an integer $x \geq N$ such that all primes dividing $P(x)$ are smaller than 20. For each index i we reduce the fraction $(x + d_i)/D_i$ to lowest terms. Since $x + d_i > D_i$ the numerator of the fraction we thereby get cannot be 1, and hence it has to be divisible by some prime number $p_i < 20$.

By the pigeonhole principle, there are a prime number p and two distinct indices i and j such that $p_i = p_j = p$. Let p^{α_i} and p^{α_j} be the greatest powers of p dividing $x + d_i$ and $x + d_j$, respectively. Due to symmetry we may suppose $\alpha_i \leq \alpha_j$. But now p^{α_i} divides $d_i - d_j$ and hence also D_i, which means that all occurrences of p in the numerator of the fraction $(x + d_i)/D_i$ cancel out, contrary to the choice of $p = p_i$. This contradiction proves our claim.

Solution 3. Given a nonzero integer N as well as a prime number p we write $v_p(N)$ for the exponent with which p occurs in the prime factorization of $|N|$.

Evidently, if the statement of the problem were not true, then there would exist an infinite sequence (x_n) of positive integers tending to infinity such that for each $n \in \mathbb{Z}_+$ the integer $P(x_n)$ is not divisible by any prime number > 20. Observe that the numbers $-d_1, -d_2, \ldots, -d_9$ do not appear in this sequence.

Now clearly there exists a prime $p_1 < 20$ for which the sequence $v_{p_1}(x_n + d_1)$ is not bounded; thinning out the sequence (x_n) if necessary we may even suppose that

$$v_{p_1}(x_n + d_1) \longrightarrow \infty.$$

Repeating this argument eight more times we may similarly choose primes $p_2, \ldots, p_9 < 20$ and suppose that our sequence (x_n) has been thinned out to such an extent that $v_{p_i}(x_n + d_i) \longrightarrow \infty$ holds for $i = 2, \ldots, 9$ as well. In view of the pigeonhole principle, there are distinct indices i and j as well as a prime $p < 20$ such that $p_i = p_j = p$. Setting $k = v_p(d_i - d_j)$ there now has to be some n for which both $v_p(x_n + d_i)$ and $v_p(x_n + d_j)$ are greater than k. But now the numbers $x_n + d_i$ and $x_n + d_j$ are divisible by p^{k+1} whilst their difference $d_i - d_j$ is not – a contradiction.

Comment. This problem is supposed to be a relatively easy one, so one might consider adding the hypothesis that the numbers $d_1, d_2, \ldots, d_9$ be positive. Then certain merely technical issues are not going to arise while the main ideas required to solve the problems remain the same.

N3

Let $n \geq 1$ be an odd integer. Determine all functions f from the set of integers to itself such that for all integers x and y the difference $f(x) - f(y)$ divides $x^n - y^n$.

Answer. All functions f of the form $f(x) = \varepsilon x^d + c$, where ε is in $\{1, -1\}$, the integer d is a positive divisor of n, and c is an integer.

Solution. Obviously, all functions in the answer satisfy the condition of the problem. We will show that there are no other functions satisfying that condition.

Let f be a function satisfying the given condition. For each integer n, the function g defined by $g(x) = f(x) + n$ also satisfies the same condition. Therefore, by subtracting $f(0)$ from $f(x)$ we may assume that $f(0) = 0$.

For any prime p, the condition on f with $(x, y) = (p, 0)$ states that $f(p)$ divides p^n. Since the set of primes is infinite, there exist integers d and ε with $0 \leq d \leq n$ and $\varepsilon \in \{1, -1\}$ such that for infinitely many primes p we have $f(p) = \varepsilon p^d$. Denote the set of these primes by P. Since a function g satisfies the given condition if and only if $-g$ satisfies the same condition, we may suppose $\varepsilon = 1$.

The case $d = 0$ is easily ruled out, because 0 does not divide any nonzero integer. Suppose $d \geq 1$ and write n as $md + r$, where m and r are integers such that $m \geq 1$ and $0 \leq r \leq d - 1$. Let x be an arbitrary integer. For each prime p in P, the difference $f(p) - f(x)$ divides $p^n - x^n$. Using the equality $f(p) = p^d$, we get

$$p^n - x^n = p^r (p^d)^m - x^n \equiv p^r f(x)^m - x^n \equiv 0 \pmod{p^d - f(x)}$$

Since we have $r < d$, for large enough primes $p \in P$ we obtain

$$|p^r f(x)^m - x^n| < p^d - f(x).$$

Hence $p^r f(x)^m - x^n$ has to be zero. This implies $r = 0$ and $x^n = (x^d)^m = f(x)^m$. Since m is odd, we obtain $f(x) = x^d$.

Comment. If n is an even positive integer, then the functions f of the form

$$f(x) = \begin{cases} x^d + c \text{ for some integers,} \\ -x^d + c \text{ for the rest of integers,} \end{cases}$$

where d is a positive divisor of $n/2$ and c is an integer, also satisfy the condition of the problem. Together with the functions in the answer, they are all functions that satisfy the condition when n is even.

N4

For each positive integer k, let $t(k)$ be the largest odd divisor of k. Determine all positive integers a for which there exists a positive integer n such that all the differences

$$t(n+a)-t(n), \quad t(n+a+1)-t(n+1), \quad \ldots, \quad t(n+2a-1)-t(n+a-1)$$

are divisible by 4.

Answer. $a = 1$, 3, or 5.

Solution. A pair (a, n) satisfying the condition of the problem will be called a *winning pair*. It is straightforward to check that the pairs $(1, 1)$, $(3, 1)$, and $(5, 4)$ are winning pairs.

Now suppose that a is a positive integer not equal to 1, 3, and 5. We will show that there are no winning pairs (a, n) by distinguishing three cases.

Case 1: a is even. In this case we have $a = 2^\alpha d$ for some positive integer α and some odd d. Since $a \geq 2^\alpha$, for each positive integer n there exists an $i \in \{0, 1, \ldots, a-1\}$ such that $n + i = 2^{\alpha-1}e$, where e is some odd integer. Then we have $t(n+i) = t(2^{\alpha-1}e) = e$ and

$$t(n+a+i) = t(2^\alpha d + 2^{\alpha-1}e) = 2d + e \equiv e + 2 \pmod 4.$$

So we get $t(n+i) - t(n+a+i) \equiv 2 \pmod 4$, and (a, n) is not a winning pair.

Case 2: a is odd and $a > 8$. For each positive integer n, there exists an $i \in \{0, 1, \ldots, a-5\}$ such that $n + i = 2d$ for some odd d. We get

$$t(n+i) = d \not\equiv d + 2 = t(n+i+4) \pmod 4$$

and

$$t(n+a+i) = n+a+i \equiv n+a+i+4 = t(n+a+i+4) \pmod 4.$$

Therefore, the integers $t(n+a+i) - t(n+i)$ and $t(n+a+i+4) - t(n+i+4)$ cannot be both divisible by 4, and therefore there are no winning pairs in this case.

Case 3: $a = 7$. For each positive integer n, there exists an $i \in \{0, 1, \ldots, 6\}$ such that $n + i$ is either of the form $8k + 3$ or of the form $8k + 6$, where k is a nonnegative integer. But we have

$$t(8k+3) \equiv 3 \not\equiv 1 \equiv 4k+5 = t(8k+3+7) \pmod 4$$

and

$$t(8k+6) = 4k+3 \equiv 3 \not\equiv 1 \equiv t(8k+6+7) \pmod 4.$$

Hence, there are no winning pairs of the form $(7, n)$.

N5

Let f be a function from the set of integers to the set of positive integers. Suppose that for any two integers m and n, the difference $f(m) - f(n)$ is divisible by $f(m-n)$. Prove that for all integers m, n with $f(m) \le f(n)$ the number $f(n)$ is divisible by $f(m)$.

Solution 1. Suppose that x and y are two integers with $f(x) < f(y)$. We will show that $f(x) \mid f(y)$. By taking $m = x$ and $n = y$ we see that

$$f(x-y) \mid |f(x) - f(y)| = f(y) - f(x) > 0,$$

so $f(x-y) \le f(y) - f(x) < f(y)$. Hence the number $d = f(x) - f(x-y)$ satisfies

$$-f(y) < -f(x-y) < d < f(x) < f(y).$$

Taking $m = x$ and $n = x - y$ we see that $f(y) \mid d$, so we deduce $d = 0$, or in other words $f(x) = f(x-y)$. Taking $m = x$ and $n = y$ we see that $f(x) = f(x-y) \mid f(x) - f(y)$, which implies $f(x) \mid f(y)$.

Solution 2. We split the solution into a sequence of claims; in each claim, the letters m and n denote arbitrary integers.

Claim 1. $f(n) \mid f(mn)$.

Proof. Since trivially $f(n) \mid f(1 \cdot n)$ and $f(n) \mid f((k+1)n) - f(kn)$ for all integers k, this is easily seen by using induction on m in both directions. □

Claim 2. $f(n) \mid f(0)$ and $f(n) = f(-n)$.

Proof. The first part follows by plugging $m = 0$ into Claim 1. Using Claim 1 twice with $m = -1$, we get $f(n) \mid f(-n) \mid f(n)$, from which the second part follows. □

From Claim 1, we get $f(1) \mid f(n)$ for all integers n, so $f(1)$ is the minimal value attained by f. Next, from Claim 2, the function f can attain only a finite number of values since all these values divide $f(0)$.

Now we prove the statement of the problem by induction on the number N_f of values attained by f. In the base case $N_f \le 2$, we either have $f(0) \ne f(1)$, in which case these two numbers are the only values attained by f and the statement is clear, or we have $f(0) = f(1)$, in which case we have $f(1) \mid f(n) \mid f(0)$ for all integers n, so f is constant and the statement is obvious again.

For the induction step, assume that $N_f \ge 3$, and let a be the least positive integer with $f(a) > f(1)$. Note that such a number exists due to the symmetry of f obtained in Claim 2.

Claim 3. $f(n) \neq f(1)$ if and only if $a \mid n$.

Proof. Since $f(1) = \cdots = f(a-1) < f(a)$, the claim follows from the fact that

$$f(n) = f(1) \iff f(n+a) = f(1).$$

So it suffices to prove this fact.

Assume that $f(n) = f(1)$. Then $f(n+a) \mid f(a) - f(-n) = f(a) - f(n) > 0$, so $f(n+a) \leq f(a) - f(n) < f(a)$; in particular the difference $f(n+a) - f(n)$ is stricly smaller than $f(a)$. Furthermore, this difference is divisible by $f(a)$ and nonnegative since $f(n) = f(1)$ is the least value attained by f. So we have $f(n+a) - f(n) = 0$, as desired. For the converse direction we only need to remark that $f(n+a) = f(1)$ entails $f(-n-a) = f(1)$, and hence $f(n) = f(-n) = f(1)$ by the forward implication. □

We return to the induction step. So let us take two arbitrary integers m and n with $f(m) \leq f(n)$. If $a \nmid m$, then we have $f(m) = f(1) \mid f(n)$. On the other hand, suppose that $a \mid m$; then by Claim 3 $a \mid n$ as well. Now define the function $g(x) = f(ax)$. Clearly, g satisfies the conditions of the problem, but $N_g < N_f - 1$, since g does not attain $f(1)$. Hence, by the induction hypothesis, $f(m) = g(m/a) \mid g(n/a) = f(n)$, as desired.

Comment. After the fact that f attains a finite number of values has been established, there are several ways of finishing the solution. For instance, let $f(0) = b_1 > b_2 > \cdots > b_k$ be all these values. One may show (essentially in the same way as in Claim 3) that the set $S_i = \{n : f(n) \geq b_i\}$ consists exactly of all numbers divisible by some integer $a_i \geq 0$. One obviously has $a_i \mid a_{i-1}$, which implies $f(a_i) \mid f(a_{i-1})$ by Claim 1. So, $b_k \mid b_{k-1} \mid \cdots \mid b_1$, thus proving the problem statement.

Moreover, now it is easy to describe all functions satisfying the conditions of the problem. Namely, all these functions can be constructed as follows. Consider a sequence of nonnegative integers $a_1, a_2, \ldots, a_k$ and another sequence of positive integers $b_1, b_2, \ldots, b_k$ such that $|a_k| = 1$, $a_i \neq a_j$ and $b_i \neq b_j$ for all $1 \leq i < j \leq k$, and $a_i \mid a_{i-1}$ and $b_i \mid b_{i-1}$ for all $i = 2, \ldots, k$. Then one may introduce the function

$$f(n) = b_{i(n)}, \qquad \text{where} \quad i(n) = \min\{i : a_i \mid n\}.$$

These are all the functions which satisfy the conditions of the problem.

N6

Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients such that no nonconstant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)}-1$ divides $3^{P(n)}-1$. Prove that $Q(x)$ is a constant polynomial.

Solution. First we show that there exists an integer d such that for all positive integers n we have $\gcd\big(P(n),Q(n)\big)\le d$.

Since $P(x)$ and $Q(x)$ are coprime (over the polynomials with rational coefficients), Euclid's algorithm provides some polynomials $R_0(x), S_0(x)$ with rational coefficients such that $P(x)R_0(x)-Q(x)S_0(x)=1$. Multiplying by a suitable positive integer d, we obtain polynomials $R(x)=d\cdot R_0(x)$ and $S(x)=d\cdot S_0(x)$ with integer coefficients for which $P(x)R(x)-Q(x)S(x)=d$. Then we have $\gcd\big(P(n),Q(n)\big)\le d$ for any integer n.

To prove the problem statement, suppose that $Q(x)$ is not constant. Then the sequence $Q(n)$ is not bounded and we can choose a positive integer m for which

$$M=2^{Q(m)}-1\ge 3^{\max\{P(1),P(2),\ldots,P(d)\}}. \tag{1}$$

Since $M=2^{Q(n)}-1\,\big|\,3^{P(n)}-1$, we have $2,3\nmid M$. Let a and b be the multiplicative orders of 2 and 3 modulo M, respectively. Obviously, $a=Q(m)$ since the lower powers of 2 do not reach M. Since M divides $3^{P(m)}-1$, we have $b\,\big|\,P(m)$. Then $\gcd(a,b)\le\gcd\big(P(m),Q(m)\big)\le d$. Since the expression $ax-by$ attains all integer values divisible by $\gcd(a,b)$ when x and y run over all nonnegative integer values, there exist some nonnegative integers x,y such that $1\le m+ax-by\le d$.

By $Q(m+ax)\equiv Q(m)\pmod a$ we have

$$2^{Q(m+ax)}\equiv 2^{Q(m)}\equiv 1\pmod M$$

and therefore

$$M\,\big|\,2^{Q(m+ax)}-1\,\big|\,3^{P(m+ax)}-1.$$

Then, by $P(m+ax-by)\equiv P(m+ax)\pmod b$ we have

$$3^{P(m+ax-by)}\equiv 3^{P(m+ax)}\equiv 1\pmod M.$$

Since $P(m+ax-by)>0$ this implies $M\le 3^{P(m+ax-by)}-1$. But $P(m+ax-by)$ is listed among $P(1),P(2),\ldots,P(d)$, so

$$M<3^{P(m+ax-by)}\le 3^{\max\{P(1),P(2),\ldots,P(d)\}}$$

which contradicts (1).

Comment. We present another variant of the solution above.

Denote the degree of P by k and its leading coefficient by p. Consider any positive integer n and let $a = Q(n)$. Again, denote by b the multiplicative order of 3 modulo $2^a - 1$. Since $2^a - 1 \mid 3^{P(n)} - 1$, we have $b \mid P(n)$. Moreover, since $2^{Q(n+at)} - 1 \mid 3^{P(n+at)} - 1$ and $a = Q(n) \mid Q(n + at)$ for each positive integer t, we have $2^a - 1 \mid 3^{P(n+at)} - 1$, hence $b \mid P(n + at)$ as well.

Therefore, b divides $\gcd\{P(n + at) : t \geq 0\}$; hence it also divides the number

$$\sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} P(n + ai) = p \cdot k! \cdot a^k.$$

Finally, we get $b \mid \gcd\big(P(n), k! \cdot p \cdot Q(n)^k\big)$, which is bounded by the same arguments as in the beginning of the solution. So $3^b - 1$ is bounded, and hence $2^{Q(n)} - 1$ is bounded as well.

N7

Let p be an odd prime number. For every integer a, define the number

$$S_a = \frac{a}{1} + \frac{a^2}{2} + \cdots + \frac{a^{p-1}}{p-1}.$$

Let m and n be integers such that

$$S_3 + S_4 - 3S_2 = \frac{m}{n}.$$

Prove that p divides m.

Solution 1. For rational numbers p_1/q_1 and p_2/q_2 with the denominators q_1, q_2 not divisible by p, we write $p_1/q_1 \equiv p_2/q_2 \pmod{p}$ if the numerator $p_1q_2 - p_2q_1$ of their difference is divisible by p.

We start with finding an explicit formula for the residue of S_a modulo p. Note first that for every $k = 1, \ldots, p-1$ the number $\binom{p}{k}$ is divisible by p, and

$$\frac{1}{p}\binom{p}{k} = \frac{(p-1)(p-2)\cdots(p-k+1)}{k!} \equiv \frac{(-1)\cdot(-2)\cdots(-k+1)}{k!} = \frac{(-1)^{k-1}}{k} \pmod{p}$$

Therefore, we have

$$S_a = -\sum_{k=1}^{p-1} \frac{(-a)^k(-1)^{k-1}}{k} \equiv -\sum_{k=1}^{p-1}(-a)^k \cdot \frac{1}{p}\binom{p}{k} \pmod{p}.$$

The number on the right-hand side is integer. Using the binomial formula we express it as

$$-\sum_{k=1}^{p-1}(-a)^k \cdot \frac{1}{p}\binom{p}{k} = -\frac{1}{p}\left(-1 - (-a)^p + \sum_{k=0}^{p}(-a)^k\binom{p}{k}\right) = \frac{(a-1)^p - a^p + 1}{p}$$

since p is odd. So, we have

$$S_a \equiv \frac{(a-1)^p - a^p + 1}{p} \pmod{p}.$$

Finally, using the obtained formula we get

$$\begin{aligned} S_3 + S_4 - 3S_2 &\equiv \frac{(2^p - 3^p + 1) + (3^p - 4^p + 1) - 3(1^p - 2^p + 1)}{p} \\ &= \frac{4\cdot 2^p - 4^p - 4}{p} = -\frac{(2^p-2)^2}{p} \pmod{p}. \end{aligned}$$

By Fermat's theorem, $p \mid 2^p - 2$, so $p^2 \mid (2^p-2)^2$ and hence $S_3 + S_4 - 3S_2 \equiv 0 \pmod{p}$.

Solution 2. One may solve the problem without finding an explicit formula for S_a. It is enough to find the following property.

Lemma. For every integer a, we have $S_{a+1} \equiv S_{-a} \pmod p$.

Proof. We expand S_{a+1} using the binomial formula as

$$S_{a+1} = \sum_{k=1}^{p-1} \frac{1}{k} \sum_{j=0}^{k} \binom{k}{j} a^j = \sum_{k=1}^{p-1} \left(\frac{1}{k} + \sum_{j=1}^{k} a^j \cdot \frac{1}{k} \binom{k}{j} \right) = \sum_{k=1}^{p-1} \frac{1}{k} + \sum_{j=1}^{p-1} a^j \sum_{k=j}^{p-1} \frac{1}{k} \binom{k}{j} a^k.$$

Note that $\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)} \equiv 0 \pmod p$ for all $1 \le k \le p-1$; hence the first sum vanishes modulo p. For the second sum, we use the relation $\frac{1}{k}\binom{k}{j} = \frac{1}{j}\binom{k-1}{j-1}$ to obtain

$$S_{a+1} \equiv \sum_{j=1}^{p-1} \frac{a^j}{j} \sum_{k=1}^{p-1} \binom{k-1}{j-1} \pmod p.$$

Finally, from the relation

$$\sum_{k=1}^{p-1} \binom{k-1}{j-1} = \binom{p-1}{j} = \frac{(p-1)(p-2)\dots(p-j)}{j!} \equiv (-1)^j \pmod p$$

we obtain

$$S_{a+1} \equiv \sum_{j=1}^{p-1} \frac{a^j (-1)^j}{j!} = S_{-a}. \qquad \square$$

Now we turn to the problem. Using the lemma we get

$$S_3 - 3S_2 \equiv S_{-2} - 3S_2 = \sum_{\substack{1 \le k \le p-1 \\ k \text{ is even}}} \frac{-2 \cdot 2^k}{k} + \sum_{\substack{1 \le k \le p-1 \\ k \text{ is odd}}} \frac{-4 \cdot 2^k}{k} \pmod p. \tag{1}$$

The first sum in (1) expands as

$$\sum_{\ell=1}^{(p-1)/2} \frac{-2 \cdot 2^{2\ell}}{2\ell} = -\sum_{\ell=1}^{(p-1)/2} \frac{4^\ell}{\ell}.$$

Next, using Fermat's theorem, we expand the second sum in (1) as

$$-\sum_{\ell=1}^{(p-1)/2} \frac{2^{2\ell+1}}{2\ell-1} \equiv -\sum_{\ell=1}^{(p-1)/2} \frac{2^{p+2\ell}}{p+2\ell-1} = -\sum_{m=(p+1)/2}^{p-1} \frac{2 \cdot 4^m}{2m} = -\sum_{m=(p+1)/2}^{p-1} \frac{4^m}{m} \pmod p$$

(here we set $m = \ell + \frac{p-1}{2}$). Hence,

$$S_3 - 3S_2 \equiv -\sum_{\ell=1}^{(p-1)/2} \frac{4^\ell}{\ell} - \sum_{m=(p+1)/2}^{p-1} \frac{4^m}{m} = -S_4 \pmod p.$$

N8

Let k be a positive integer and set $n = 2^k + 1$. Prove that n is a prime number if and only if the following holds: there is a permutation $a_1, \ldots, a_{n-1}$ of the numbers $1, 2, \ldots, n-1$ and a sequence of integers $g_1, g_2, \ldots, g_{n-1}$ such that n divides $g_i^{a_i} - a_{i+1}$ for every $i \in \{1, 2, \ldots, n-1\}$, where we set $a_n = a_1$.

Solution. Let $N = \{1, 2, \ldots, n-1\}$. For $a, b \in N$, we say that b *follows* a if there exists an integer g such that $b \equiv g^a \pmod{n}$ and denote this property as $a \to b$. This way we have a directed graph with N as set of vertices. If $a_1, \ldots, a_{n-1}$ is a permutation of $1, 2, \ldots, n-1$ such that $a_1 \to a_2 \to \ldots \to a_{n-1} \to a_1$ then this is a *Hamiltonian cycle* in the graph.

Step I. First consider the case when n is composite. Let $n = p_1^{\alpha_1} \ldots p_s^{\alpha_s}$ be its prime factorization. All primes p_i are odd.

Suppose that $\alpha_i > 1$ for some i. For all integers a, g with $a \geq 2$, we have $g^a \not\equiv p_i \pmod{p_i^2}$, because g^a is either divisible by p_i^2 or it is not divisible by p_i. It follows that in any Hamiltonian cycle p_i comes immediately after 1. The same argument shows that $2p_i$ also should come immediately after 1, which is impossible. Hence, there is no Hamiltonian cycle in the graph.

Now suppose that n is square-free. We have $n = p_1 p_2 \ldots p_s > 9$ and $s \geq 2$. Assume that there exists a Hamiltonian cycle. There are $\frac{n-1}{2}$ even numbers in this cycle, and each number which follows one of them should be a quadratic residue modulo n. So, there should be at least $\frac{n-1}{2}$ nonzero quadratic residues modulo n. On the other hand, for each p_i there exist exactly $\frac{p_i+1}{2}$ quadratic residues modulo p_i; by the Chinese Remainder Theorem, the number of quadratic residues modulo n is exactly $\frac{p_1+1}{2} \cdot \frac{p_2+1}{2} \cdot \ldots \cdot \frac{p_s+1}{2}$, including 0. Then we have a contradiction by

$$\frac{p_1+1}{2} \cdot \frac{p_2+1}{2} \cdot \ldots \cdot \frac{p_s+1}{2} \leq \frac{2p_1}{3} \cdot \frac{2p_2}{3} \cdot \ldots \cdot \frac{2p_s}{3} = \left(\frac{2}{3}\right)^s n \leq \frac{4n}{9} < \frac{n-1}{2}.$$

This proves the "if"-part of the problem.

Step II. Now suppose that n is prime. For any $a \in N$, denote by $\nu_2(a)$ the exponent of 2 in the prime factorization of a, and let $\mu(a) = \max\{t \in [0, k] \mid 2^t \to a\}$.

Lemma. For any $a, b \in N$, we have $a \to b$ if and only if $\nu_2(a) \leq \mu(b)$.

Proof. Let $\ell = \nu_2(a)$ and $m = \mu(b)$.

Suppose $\ell \leq m$. Since b follows 2^m, there exists some g_0 such that $b \equiv g_0^{2^m} \pmod{n}$. By $\gcd(a, n-1) = 2^\ell$ there exist some integers p and q such that $pa - q(n-1) = 2^\ell$. Choosing $g = g_0^{2^{m-\ell}p}$ we have $g^a = g_0^{2^{m-\ell}pa} = g_0^{2^m + 2^{m-\ell}q(n-1)} \equiv g_0^{2^m} \equiv b \pmod{n}$ by Fermat's theorem. Hence, $a \to b$.

To prove the reverse statement, suppose that $a \to b$, so $b \equiv g^a \pmod{n}$ with some g. Then $b \equiv (g^{a/2^\ell})^{2^\ell}$, and therefore $2^\ell \to b$. By the definition of $\mu(b)$, we have $\mu(b) \geq \ell$. The lemma is

proved. □

Now for every i with $0 \le i \le k$, let

$$A_i = \{a \in N \mid \nu_2(a) = i\},$$
$$B_i = \{a \in N \mid \mu(a) = i\},$$
$$\text{and } C_i = \{a \in N \mid \mu(a) \ge i\} = B_i \cup B_{i+1} \cup \ldots \cup B_k.$$

We claim that $|A_i| = |B_i|$ for all $0 \le i \le k$. Obviously we have $|A_i| = 2^{k-i-1}$ for all $i = 0, \ldots, k-1$, and $|A_k| = 1$. Now we determine $|C_i|$. We have $|C_0| = n-1$ and by FERMAT's theorem we also have $C_k = \{1\}$, so $|C_k| = 1$. Next, notice that $C_{i+1} = \{x^2 \mod n \mid x \in C_i\}$. For every $a \in N$, the relation $x^2 \equiv a \pmod n$ has at most two solutions in N. Therefore we have $2|C_{i+1}| \le |C_i|$, with the equality achieved only if for every $y \in C_{i+1}$, there exist distinct elements $x, x' \in C_i$ such that $x^2 \equiv x'^2 \equiv y \pmod n$ (this implies $x + x' = n$). Now, since $2^k|C_k| = |C_0|$, we obtain that this equality should be achieved in each step. Hence $|C_i| = 2^{k-i}$ for $0 \le i \le k$, and therefore $|B_i| = 2^{k-i-1}$ for $0 \le i \le k-1$ and $|B_k| = 1$.

From the previous arguments we can see that for each $z \in C_i$ $(0 \le i < k)$ the equation $x^2 \equiv z^2 \pmod n$ has two solutions in C_i, so we have $n - z \in C_i$. Hence, for each $i = 0, 1, \ldots, k-1$, exactly half of the elements of C_i are odd. The same statement is valid for $B_i = C_i \setminus C_{i+1}$ for $0 \le i \le k-2$. In particular, each such B_i contains an odd number. Note that $B_k = \{1\}$ also contains an odd number, and $B_{k-1} = \{2^k\}$ since C_{k-1} consists of the two square roots of 1 modulo n.

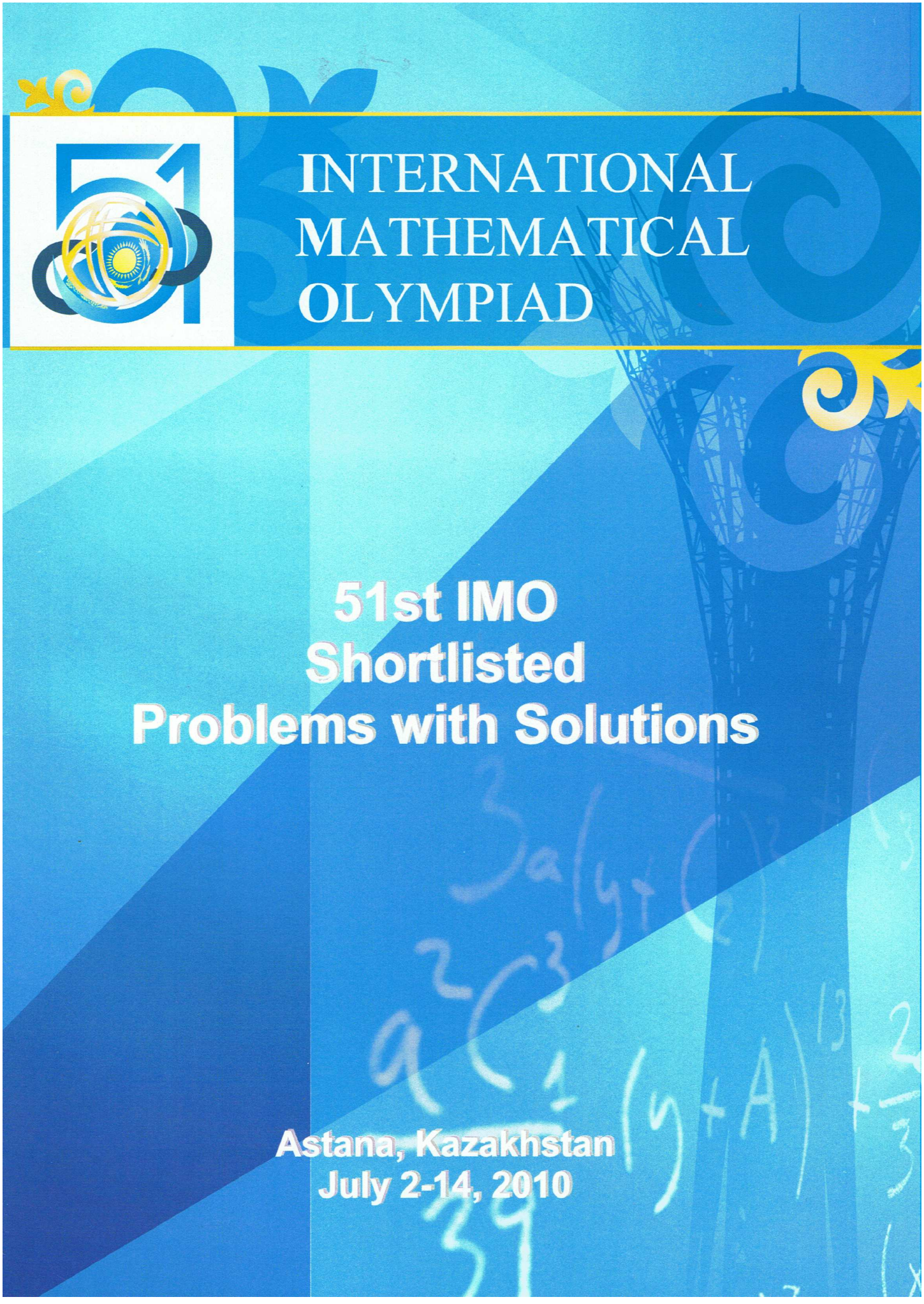
Step III. Now we construct a Hamiltonian cycle in the graph. First, for each i with $0 \le i \le k$, connect the elements of A_i to the elements of B_i by means of an arbitrary bijection. After performing this for every i, we obtain a subgraph with all vertices having in-degree 1 and out-degree 1, so the subgraph is a disjoint union of cycles. If there is a unique cycle, we are done. Otherwise, we modify the subgraph in such a way that the previous property is preserved and the number of cycles decreases; after a finite number of steps we arrive at a single cycle.

For every cycle C, let $\lambda(C) = \min_{c \in C} \nu_2(c)$. Consider a cycle C for which $\lambda(C)$ is maximal. If $\lambda(C) = 0$, then for any other cycle C' we have $\lambda(C') = 0$. Take two arbitrary vertices $a \in C$ and $a' \in C'$ such that $\nu_2(a) = \nu_2(a') = 0$; let their direct successors be b and b', respectively. Then we can unify C and C' to a single cycle by replacing the edges $a \to b$ and $a' \to b'$ by $a \to b'$ and $a' \to b$.

Now suppose that $\lambda = \lambda(C) \ge 1$; let $a \in C \cap A_\lambda$. If there exists some $a' \in A_\lambda \setminus C$, then a' lies in another cycle C' and we can merge the two cycles in exactly the same way as above. So, the only remaining case is $A_\lambda \subset C$. Since the edges from A_λ lead to B_λ, we get also $B_\lambda \subset C$. If $\lambda \ne k-1$ then B_λ contains an odd number; this contradicts the assumption $\lambda(C) > 0$. Finally, if $\lambda = k-1$, then C contains 2^{k-1} which is the only element of A_{k-1}. Since $B_{k-1} = \{2^k\} = A_k$ and $B_k = \{1\}$, the cycle C contains the path $2^{k-1} \to 2^k \to 1$ and it contains an odd number again. This completes the proof of the "only if"-part of the problem.

Comment 1. The lemma and the fact $|A_i| = |B_i|$ together show that for every edge $a \to b$ of the Hamiltonian cycle, $\nu_2(a) = \mu(b)$ must hold. After this observation, the Hamiltonian cycle can be built in many ways. For instance, it is possible to select edges from A_i to B_i for $i = k, k-1, \ldots, 1$ in such a way that they form disjoint paths; at the end all these paths will have odd endpoints. In the final step, the paths can be closed to form a unique cycle.

Comment 2. Step II is an easy consequence of some basic facts about the multiplicative group modulo the prime $n = 2^k + 1$. The Lemma follows by noting that this group has order 2^k, so the a-th powers are exactly the $2^{\nu_2(a)}$-th powers. Using the existence of a primitive root g modulo n one sees that the map from $\{1, 2, \ldots, n-1\}$ to itself that sends a to $g^a \bmod n$ is a bijection that sends A_i to B_i for each $i \in \{0, \ldots, k\}$.


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Contents

Note of Confidentiality **5**

Contributing Countries & Problem Selection Committee **5**

Algebra **7**
Problem A1 . . . 7
Problem A2 . . . 8
Problem A3 . . . 10
Problem A4 . . . 12
Problem A5 . . . 13
Problem A6 . . . 15
Problem A7 . . . 17
Problem A8 . . . 19

Combinatorics **23**
Problem C1 . . . 23
Problem C2 . . . 26
Problem C3 . . . 28
Problem C4 . . . 30
Problem C4′ . . . 30
Problem C5 . . . 32
Problem C6 . . . 35
Problem C7 . . . 38

Geometry **44**
Problem G1 . . . 44
Problem G2 . . . 46
Problem G3 . . . 50
Problem G4 . . . 52
Problem G5 . . . 54
Problem G6 . . . 56
Problem G6′ . . . 56
Problem G7 . . . 60

Number Theory **64**
Problem N1 . . . 64
Problem N1′ . . . 64
Problem N2 . . . 66
Problem N3 . . . 68
Problem N4 . . . 70
Problem N5 . . . 71
Problem N6 . . . 72

Note of Confidentiality

The Shortlisted Problems should be kept strictly confidential until IMO 2011.

Contributing Countries

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Armenia, Australia, Austria, Bulgaria, Canada, Columbia, Croatia, Cyprus, Estonia, Finland, France, Georgia, Germany, Greece, Hong Kong, Hungary, India, Indonesia, Iran, Ireland, Japan, Korea (North), Korea (South), Luxembourg, Mongolia, Netherlands, Pakistan, Panama, Poland, Romania, Russia, Saudi Arabia, Serbia, Slovakia, Slovenia, Switzerland, Thailand, Turkey, Ukraine, United Kingdom, United States of America, Uzbekistan

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Algebra

A1. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that the equality

$$f([x]y) = f(x)[f(y)]. \tag{1}$$

holds for all $x, y \in \mathbb{R}$. Here, by $[x]$ we denote the greatest integer not exceeding x.

(France)

Answer. $f(x) = \text{const} = C$, where $C = 0$ or $1 \le C < 2$.

Solution 1. First, setting $x = 0$ in (1) we get

$$f(0) = f(0)[f(y)] \tag{2}$$

for all $y \in \mathbb{R}$. Now, two cases are possible.

Case 1. Assume that $f(0) \neq 0$. Then from (2) we conclude that $[f(y)] = 1$ for all $y \in \mathbb{R}$. Therefore, equation (1) becomes $f([x]y) = f(x)$, and substituting $y = 0$ we have $f(x) = f(0) = C \neq 0$. Finally, from $[f(y)] = 1 = [C]$ we obtain that $1 \le C < 2$.

Case 2. Now we have $f(0) = 0$. Here we consider two subcases.

Subcase 2a. Suppose that there exists $0 < \alpha < 1$ such that $f(\alpha) \neq 0$. Then setting $x = \alpha$ in (1) we obtain $0 = f(0) = f(\alpha)[f(y)]$ for all $y \in \mathbb{R}$. Hence, $[f(y)] = 0$ for all $y \in \mathbb{R}$. Finally, substituting $x = 1$ in (1) provides $f(y) = 0$ for all $y \in \mathbb{R}$, thus contradicting the condition $f(\alpha) \neq 0$.

Subcase 2b. Conversely, we have $f(\alpha) = 0$ for all $0 \le \alpha < 1$. Consider any real z; there exists an integer N such that $\alpha = \dfrac{z}{N} \in [0, 1)$ (one may set $N = [z] + 1$ if $z \ge 0$ and $N = [z] - 1$ otherwise). Now, from (1) we get $f(z) = f([N]\alpha) = f(N)[f(\alpha)] = 0$ for all $z \in \mathbb{R}$.

Finally, a straightforward check shows that all the obtained functions satisfy (1).

Solution 2. Assume that $[f(y)] = 0$ for some y; then the substitution $x = 1$ provides $f(y) = f(1)[f(y)] = 0$. Hence, if $[f(y)] = 0$ for all y, then $f(y) = 0$ for all y. This function obviously satisfies the problem conditions.

So we are left to consider the case when $[f(a)] \neq 0$ for some a. Then we have

$$f([x]a) = f(x)[f(a)], \qquad \text{or} \quad f(x) = \frac{f([x]a)}{[f(a)]}. \tag{3}$$

This means that $f(x_1) = f(x_2)$ whenever $[x_1] = [x_2]$, hence $f(x) = f([x])$, and we may assume that a is an integer.

Now we have

$$f(a) = f\left(2a \cdot \tfrac{1}{2}\right) = f(2a)\left[f\left(\tfrac{1}{2}\right)\right] = f(2a)[f(0)];$$

this implies $[f(0)] \neq 0$, so we may even assume that $a = 0$. Therefore equation (3) provides

$$f(x) = \frac{f(0)}{[f(0)]} = C \neq 0$$

for each x. Now, condition (1) becomes equivalent to the equation $C = C[C]$ which holds exactly when $[C] = 1$.

8

A2. Let the real numbers a, b, c, d satisfy the relations $a+b+c+d = 6$ and $a^2+b^2+c^2+d^2 = 12$. Prove that

$$36 \le 4(a^3+b^3+c^3+d^3) - (a^4+b^4+c^4+d^4) \le 48.$$

(Ukraine)

Solution 1. Observe that

$$\begin{aligned} 4(a^3+b^3+c^3+d^3) - (a^4+b^4+c^4+d^4) &= -\left((a-1)^4+(b-1)^4+(c-1)^4+(d-1)^4\right) \\ &\quad + 6(a^2+b^2+c^2+d^2) - 4(a+b+c+d) + 4 \\ &= -\left((a-1)^4+(b-1)^4+(c-1)^4+(d-1)^4\right) + 52. \end{aligned}$$

Now, introducing $x = a-1$, $y = b-1$, $z = c-1$, $t = d-1$, we need to prove the inequalities

$$16 \ge x^4+y^4+z^4+t^4 \ge 4,$$

under the constraint

$$x^2+y^2+z^2+t^2 = (a^2+b^2+c^2+d^2) - 2(a+b+c+d) + 4 = 4 \tag{1}$$

(we will not use the value of $x+y+z+t$ though it can be found).

Now the rightmost inequality in (1) follows from the power mean inequality:

$$x^4+y^4+z^4+t^4 \ge \frac{(x^2+y^2+z^2+t^2)^2}{4} = 4.$$

For the other one, expanding the brackets we note that

$$(x^2+y^2+z^2+t^2)^2 = (x^4+y^4+z^4+t^4) + q,$$

where q is a nonnegative number, so

$$x^4+y^4+z^4+t^4 \le (x^2+y^2+z^2+t^2)^2 = 16,$$

and we are done.

Comment 1. The estimates are sharp; the lower and upper bounds are attained at $(3,1,1,1)$ and $(0,2,2,2)$, respectively.

Comment 2. After the change of variables, one can finish the solution in several different ways. The latter estimate, for instance, it can be performed by moving the variables – since we need only the second of the two shifted conditions.

Solution 2. First, we claim that $0 \le a, b, c, d \le 3$. Actually, we have

$$a+b+c = 6-d, \quad a^2+b^2+c^2 = 12-d^2,$$

hence the power mean inequality

$$a^2+b^2+c^2 \ge \frac{(a+b+c)^2}{3}$$

rewrites as

$$12-d^2 \ge \frac{(6-d)^2}{3} \quad \Longleftrightarrow \quad 2d(d-3) \le 0,$$

which implies the desired inequalities for d; since the conditions are symmetric, we also have the same estimate for the other variables.

Now, to prove the rightmost inequality, we use the obvious inequality $x^2(x-2)^2 \ge 0$ for each real x; this inequality rewrites as $4x^3 - x^4 \le 4x^2$. It follows that

$$(4a^3 - a^4) + (4b^3 - b^4) + (4c^3 - c^4) + (4d^3 - d^4) \le 4(a^2 + b^2 + c^2 + d^2) = 48,$$

as desired.

Now we prove the leftmost inequality in an analogous way. For each $x \in [0, 3]$, we have $(x+1)(x-1)^2(x-3) \le 0$ which is equivalent to $4x^3 - x^4 \ge 2x^2 + 4x - 3$. This implies that

$$(4a^3 - a^4) + (4b^3 - b^4) + (4c^3 - c^4) + (4d^3 - d^4) \ge 2(a^2 + b^2 + c^2 + d^2) + 4(a + b + c + d) - 12 = 36,$$

as desired.

Comment. It is easy to guess the extremal points $(0, 2, 2, 2)$ and $(3, 1, 1, 1)$ for this inequality. This provides a method of finding the polynomials used in Solution 2. Namely, these polynomials should have the form $x^4 - 4x^3 + ax^2 + bx + c$; moreover, the former polynomial should have roots at 2 (with an even multiplicity) and 0, while the latter should have roots at 1 (with an even multiplicity) and 3. These conditions determine the polynomials uniquely.

Solution 3. First, expanding $48 = 4(a^2 + b^2 + c^2 + d^2)$ and applying the AM–GM inequality, we have

$$\begin{aligned} a^4 + b^4 + c^4 + d^4 + 48 &= (a^4 + 4a^2) + (b^4 + 4b^2) + (c^4 + 4c^2) + (d^4 + 4d^2) \\ &\ge 2\left(\sqrt{a^4 \cdot 4a^2} + \sqrt{b^4 \cdot 4b^2} + \sqrt{c^4 \cdot 4c^2} + \sqrt{d^4 \cdot 4d^2}\right) \\ &= 4(|a^3| + |b^3| + |c^3| + |d^3|) \ge 4(a^3 + b^3 + c^3 + d^3), \end{aligned}$$

which establishes the rightmost inequality.

To prove the leftmost inequality, we first show that $a, b, c, d \in [0, 3]$ as in the previous solution. Moreover, we can assume that $0 \le a \le b \le c \le d$. Then we have $a + b \le b + c \le \frac{2}{3}(b + c + d) \le \frac{2}{3} \cdot 6 = 4$.

Next, we show that $4b - b^2 \le 4c - c^2$. Actually, this inequality rewrites as $(c-b)(b+c-4) \le 0$, which follows from the previous estimate. The inequality $4a - a^2 \le 4b - b^2$ can be proved analogously.

Further, the inequalities $a \le b \le c$ together with $4a - a^2 \le 4b - b^2 \le 4c - c^2$ allow us to apply the Chebyshev inequality obtaining

$$\begin{aligned} a^2(4a - a^2) + b^2(4b - b^2) + c^2(4c - c^2) &\ge \frac{1}{3}(a^2 + b^2 + c^2)\left(4(a + b + c) - (a^2 + b^2 + c^2)\right) \\ &= \frac{(12 - d^2)(4(6 - d) - (12 - d^2))}{3}. \end{aligned}$$

This implies that

$$\begin{aligned} (4a^3 - a^4) + (4b^3 - b^4) + (4c^3 - c^4) + (4d^3 - d^4) &\ge \frac{(12 - d^2)(d^2 - 4d + 12)}{3} + 4d^3 - d^4 \\ = \frac{144 - 48d + 16d^3 - 4d^4}{3} &= 36 + \frac{4}{3}(3 - d)(d - 1)(d^2 - 3). \end{aligned} \tag{2}$$

Finally, we have $d^2 \ge \frac{1}{4}(a^2 + b^2 + c^2 + d^2) = 3$ (which implies $d > 1$); so, the expression $\frac{4}{3}(3 - d)(d - 1)(d^2 - 3)$ in the right-hand part of (2) is nonnegative, and the desired inequality is proved.

Comment. The rightmost inequality is easier than the leftmost one. In particular, Solutions 2 and 3 show that only the condition $a^2 + b^2 + c^2 + d^2 = 12$ is needed for the former one.

10

A3. Let $x_1, \ldots, x_{100}$ be nonnegative real numbers such that $x_i + x_{i+1} + x_{i+2} \le 1$ for all $i = 1, \ldots, 100$ (we put $x_{101} = x_1$, $x_{102} = x_2$). Find the maximal possible value of the sum

$$S = \sum_{i=1}^{100} x_i x_{i+2}.$$

(Russia)

Answer. $\dfrac{25}{2}$.

Solution 1. Let $x_{2i} = 0$, $x_{2i-1} = \frac{1}{2}$ for all $i = 1, \ldots, 50$. Then we have $S = 50 \cdot \left(\frac{1}{2}\right)^2 = \frac{25}{2}$. So, we are left to show that $S \le \frac{25}{2}$ for all values of x_i's satisfying the problem conditions.

Consider any $1 \le i \le 50$. By the problem condition, we get $x_{2i-1} \le 1 - x_{2i} - x_{2i+1}$ and $x_{2i+2} \le 1 - x_{2i} - x_{2i+1}$. Hence by the AM–GM inequality we get

$$\begin{aligned} x_{2i-1}x_{2i+1} + x_{2i}x_{2i+2} &\le (1 - x_{2i} - x_{2i+1})x_{2i+1} + x_{2i}(1 - x_{2i} - x_{2i+1}) \\ &= (x_{2i} + x_{2i+1})(1 - x_{2i} - x_{2i+1}) \le \left(\frac{(x_{2i} + x_{2i+1}) + (1 - x_{2i} - x_{2i+1})}{2}\right)^2 = \frac{1}{4}. \end{aligned}$$

Summing up these inequalities for $i = 1, 2, \ldots, 50$, we get the desired inequality

$$\sum_{i=1}^{50} (x_{2i-1}x_{2i+1} + x_{2i}x_{2i+2}) \le 50 \cdot \frac{1}{4} = \frac{25}{2}.$$

Comment. This solution shows that a bit more general fact holds. Namely, consider $2n$ nonnegative numbers $x_1, \ldots, x_{2n}$ in a row (with no cyclic notation) and suppose that $x_i + x_{i+1} + x_{i+2} \le 1$ for all $i = 1, 2, \ldots, 2n-2$. Then $\sum\limits_{i=1}^{2n-2} x_i x_{i+2} \le \dfrac{n-1}{4}$.

The proof is the same as above, though if might be easier to find it (for instance, applying induction). The original estimate can be obtained from this version by considering the sequence $x_1, x_2, \ldots, x_{100}, x_1, x_2$.

Solution 2. We present another proof of the estimate. From the problem condition, we get

$$\begin{aligned} S = \sum_{i=1}^{100} x_i x_{i+2} \le \sum_{i=1}^{100} x_i(1 - x_i - x_{i+1}) &= \sum_{i=1}^{100} x_i - \sum_{i=1}^{100} x_i^2 - \sum_{i=1}^{100} x_i x_{i+1} \\ &= \sum_{i=1}^{100} x_i - \frac{1}{2}\sum_{i=1}^{100} (x_i + x_{i+1})^2. \end{aligned}$$

By the AM–QM inequality, we have $\sum (x_i + x_{i+1})^2 \ge \frac{1}{100}\big(\sum (x_i + x_{i+1})\big)^2$, so

$$\begin{aligned} S \le \sum_{i=1}^{100} x_i - \frac{1}{200}\left(\sum_{i=1}^{100} (x_i + x_{i+1})\right)^2 &= \sum_{i=1}^{100} x_i - \frac{2}{100}\left(\sum_{i=1}^{100} x_i\right)^2 \\ &= \frac{2}{100}\left(\sum_{i=1}^{100} x_i\right)\left(\frac{100}{2} - \sum_{i=1}^{100} x_i\right). \end{aligned}$$

And finally, by the AM–GM inequality

$$S \le \frac{2}{100} \cdot \left(\frac{1}{2}\left(\sum_{i=1}^{100} x_i + \frac{100}{2} - \sum_{i=1}^{100} x_i\right)\right)^2 = \frac{2}{100} \cdot \left(\frac{100}{4}\right)^2 = \frac{25}{2}.$$

Comment. These solutions are not as easy as they may seem at the first sight. There are two different optimal configurations in which the variables have different values, and not all of sums of three consecutive numbers equal 1. Although it is easy to find the value $\frac{25}{2}$, the estimates must be done with care to preserve equality in the optimal configurations.

A4. A sequence $x_1, x_2, \ldots$ is defined by $x_1 = 1$ and $x_{2k} = -x_k$, $x_{2k-1} = (-1)^{k+1}x_k$ for all $k \geq 1$. Prove that $x_1 + x_2 + \cdots + x_n \geq 0$ for all $n \geq 1$.

(Austria)

Solution 1. We start with some observations. First, from the definition of x_i it follows that for each positive integer k we have

$$x_{4k-3} = x_{2k-1} = -x_{4k-2} \quad \text{and} \quad x_{4k-1} = x_{4k} = -x_{2k} = x_k. \tag{1}$$

Hence, denoting $S_n = \sum_{i=1}^{n} x_i$, we have

$$S_{4k} = \sum_{i=1}^{k}\big((x_{4k-3} + x_{4k-2}) + (x_{4k-1} + x_{4k})\big) = \sum_{i=1}^{k}(0 + 2x_k) = 2S_k, \tag{2}$$

$$S_{4k+2} = S_{4k} + (x_{4k+1} + x_{4k+2}) = S_{4k}. \tag{3}$$

Observe also that $S_n = \sum_{i=1}^{n} x_i \equiv \sum_{i=1}^{n} 1 = n \pmod 2$.

Now we prove by induction on k that $S_i \geq 0$ for all $i \leq 4k$. The base case is valid since $x_1 = x_3 = x_4 = 1$, $x_2 = -1$. For the induction step, assume that $S_i \geq 0$ for all $i \leq 4k$. Using the relations (1)–(3), we obtain

$$S_{4k+4} = 2S_{k+1} \geq 0, \quad S_{4k+2} = S_{4k} \geq 0, \quad S_{4k+3} = S_{4k+2} + x_{4k+3} = \frac{S_{4k+2} + S_{4k+4}}{2} \geq 0.$$

So, we are left to prove that $S_{4k+1} \geq 0$. If k is odd, then $S_{4k} = 2S_k \geq 0$; since k is odd, S_k is odd as well, so we have $S_{4k} \geq 2$ and hence $S_{4k+1} = S_{4k} + x_{4k+1} \geq 1$.

Conversely, if k is even, then we have $x_{4k+1} = x_{2k+1} = x_{k+1}$, hence $S_{4k+1} = S_{4k} + x_{4k+1} = 2S_k + x_{k+1} = S_k + S_{k+1} \geq 0$. The step is proved.

Solution 2. We will use the notation of S_n and the relations (1)–(3) from the previous solution.

Assume the contrary and consider the minimal n such that $S_{n+1} < 0$; surely $n \geq 1$, and from $S_n \geq 0$ we get $S_n = 0$, $x_{n+1} = -1$. Hence, we are especially interested in the set $M = \{n : S_n = 0\}$; our aim is to prove that $x_{n+1} = 1$ whenever $n \in M$ thus coming to a contradiction.

For this purpose, we first describe the set M inductively. We claim that (i) M consists only of even numbers, (ii) $2 \in M$, and (iii) for every even $n \geq 4$ we have $n \in M \iff [n/4] \in M$. Actually, (i) holds since $S_n \equiv n \pmod 2$, (ii) is straightforward, while (iii) follows from the relations $S_{4k+2} = S_{4k} = 2S_k$.

Now, we are left to prove that $x_{n+1} = 1$ if $n \in M$. We use the induction on n. The base case is $n = 2$, that is, the minimal element of M; here we have $x_3 = 1$, as desired.

For the induction step, consider some $4 \leq n \in M$ and let $m = [n/4] \in M$; then m is even, and $x_{m+1} = 1$ by the induction hypothesis. We prove that $x_{n+1} = x_{m+1} = 1$. If $n = 4m$ then we have $x_{n+1} = x_{2m+1} = x_{m+1}$ since m is even; otherwise, $n = 4m+2$, and $x_{n+1} = -x_{2m+2} = x_{m+1}$, as desired. The proof is complete.

Comment. Using the inductive definition of set M, one can describe it explicitly. Namely, M consists exactly of all positive integers not containing digits 1 and 3 in their 4-base representation.

A5. Denote by $\mathbb{Q}^+$ the set of all positive rational numbers. Determine all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ which satisfy the following equation for all $x, y \in \mathbb{Q}^+$:

$$f\big(f(x)^2 y\big) = x^3 f(xy). \tag{1}$$

(Switzerland)

Answer. The only such function is $f(x) = \dfrac{1}{x}$.

Solution. By substituting $y = 1$, we get

$$f\big(f(x)^2\big) = x^3 f(x). \tag{2}$$

Then, whenever $f(x) = f(y)$, we have

$$x^3 = \frac{f\big(f(x)^2\big)}{f(x)} = \frac{f\big(f(y)^2\big)}{f(y)} = y^3$$

which implies $x = y$, so the function f is injective.

Now replace x by xy in (2), and apply (1) twice, second time to $\big(y, f(x)^2\big)$ instead of (x, y):

$$f\big(f(xy)^2\big) = (xy)^3 f(xy) = y^3 f\big(f(x)^2 y\big) = f\big(f(x)^2 f(y)^2\big).$$

Since f is injective, we get

$$\begin{aligned} f(xy)^2 &= f(x)^2 f(y)^2, \\ f(xy) &= f(x) f(y). \end{aligned}$$

Therefore, f is multiplicative. This also implies $f(1) = 1$ and $f(x^n) = f(x)^n$ for all integers n.

Then the function equation (1) can be re-written as

$$\begin{aligned} f\big(f(x)\big)^2 f(y) &= x^3 f(x) f(y), \\ f\big(f(x)\big) &= \sqrt{x^3 f(x)}. \end{aligned} \tag{3}$$

Let $g(x) = xf(x)$. Then, by (3), we have

$$\begin{aligned} g\big(g(x)\big) = g\big(xf(x)\big) = xf(x) \cdot f\big(xf(x)\big) = xf(x)^2 f\big(f(x)\big) = \\ = xf(x)^2 \sqrt{x^3 f(x)} = \big(xf(x)\big)^{5/2} = \big(g(x)\big)^{5/2}, \end{aligned}$$

and, by induction,

$$\underbrace{g\Big(g\big(\ldots g}_{n+1}(x)\ldots\big)\Big) = \big(g(x)\big)^{(5/2)^n} \tag{4}$$

for every positive integer n.

Consider (4) for a fixed x. The left-hand side is always rational, so $\big(g(x)\big)^{(5/2)^n}$ must be rational for every n. We show that this is possible only if $g(x) = 1$. Suppose that $g(x) \neq 1$, and let the prime factorization of $g(x)$ be $g(x) = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ where $p_1, \ldots, p_k$ are distinct primes and $\alpha_1, \ldots, \alpha_k$ are nonzero integers. Then the unique prime factorization of (4) is

$$\underbrace{g\Big(g\big(\ldots g}_{n+1}(x)\ldots\big)\Big) = \big(g(x)\big)^{(5/2)^n} = p_1^{(5/2)^n \alpha_1} \cdots p_k^{(5/2)^n \alpha_k}$$

14

where the exponents should be integers. But this is not true for large values of n, for example $(\frac{5}{2})^n\alpha_1$ cannot be a integer number when $2^n \nmid \alpha_1$. Therefore, $g(x) \neq 1$ is impossible.

Hence, $g(x) = 1$ and thus $f(x) = \dfrac{1}{x}$ for all x.

The function $f(x) = \dfrac{1}{x}$ satisfies the equation (1):

$$f(f(x)^2 y) = \frac{1}{f(x)^2 y} = \frac{1}{\left(\frac{1}{x}\right)^2 y} = \frac{x^3}{xy} = x^3 f(xy).$$

Comment. Among $\mathbb{R}^+ \to \mathbb{R}^+$ functions, $f(x) = \dfrac{1}{x}$ is not the only solution. Another solution is $f_1(x) = x^{3/2}$. Using transfinite tools, infinitely many other solutions can be constructed.

A6. Suppose that f and g are two functions defined on the set of positive integers and taking positive integer values. Suppose also that the equations $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ hold for all positive integers. Prove that $f(n) = g(n)$ for all positive integer n.

(Germany)

Solution 1. Throughout the solution, by $\mathbb{N}$ we denote the set of all positive integers. For any function $h\colon \mathbb{N} \to \mathbb{N}$ and for any positive integer k, define $h^k(x) = \underbrace{h\big(h\big(\dots h}_{k}(x)\dots\big)\big)$ (in particular, $h^0(x) = x$).

Observe that $f\big(g^k(x)\big) = f\big(g^{k-1}(x)\big) + 1 = \dots = f(x) + k$ for any positive integer k, and similarly $g\big(f^k(x)\big) = g(x) + k$. Now let a and b are the minimal values attained by f and g, respectively; say $f(n_f) = a$, $g(n_g) = b$. Then we have $f\big(g^k(n_f)\big) = a + k$, $g\big(f^k(n_g)\big) = b + k$, so the function f attains all values from the set $N_f = \{a, a+1, \dots\}$, while g attains all the values from the set $N_g = \{b, b+1, \dots\}$.

Next, note that $f(x) = f(y)$ implies $g(x) = g\big(f(x)\big) - 1 = g\big(f(y)\big) - 1 = g(y)$; surely, the converse implication also holds. Now, we say that x and y are *similar* (and write $x \sim y$) if $f(x) = f(y)$ (equivalently, $g(x) = g(y)$). For every $x \in \mathbb{N}$, we define $[x] = \{y \in \mathbb{N} : x \sim y\}$; surely, $y_1 \sim y_2$ for all $y_1, y_2 \in [x]$, so $[x] = [y]$ whenever $y \in [x]$.

Now we investigate the structure of the sets $[x]$.

Claim 1. Suppose that $f(x) \sim f(y)$; then $x \sim y$, that is, $f(x) = f(y)$. Consequently, each class $[x]$ contains at most one element from N_f, as well as at most one element from N_g.

Proof. If $f(x) \sim f(y)$, then we have $g(x) = g\big(f(x)\big) - 1 = g\big(f(y)\big) - 1 = g(y)$, so $x \sim y$. The second statement follows now from the sets of values of f and g. □

Next, we clarify which classes do not contain large elements.

Claim 2. For any $x \in \mathbb{N}$, we have $[x] \subseteq \{1, 2, \dots, b-1\}$ if and only if $f(x) = a$. Analogously, $[x] \subseteq \{1, 2, \dots, a-1\}$ if and only if $g(x) = b$.

Proof. We will prove that $[x] \not\subseteq \{1, 2, \dots, b-1\} \iff f(x) > a$; the proof of the second statement is similar.

Note that $f(x) > a$ implies that there exists some y satisfying $f(y) = f(x) - 1$, so $f\big(g(y)\big) = f(y) + 1 = f(x)$, and hence $x \sim g(y) \geq b$. Conversely, if $b \leq c \sim x$ then $c = g(y)$ for some $y \in \mathbb{N}$, which in turn follows $f(x) = f\big(g(y)\big) = f(y) + 1 \geq a + 1$, and hence $f(x) > a$. □

Claim 2 implies that there exists exactly one class contained in $\{1, \dots, a-1\}$ (that is, the class $[n_g]$), as well as exactly one class contained in $\{1, \dots, b-1\}$ (the class $[n_f]$). Assume for a moment that $a \leq b$; then $[n_g]$ is contained in $\{1, \dots, b-1\}$ as well, hence it coincides with $[n_g]$. So, we get that

$$f(x) = a \iff g(x) = b \iff x \sim n_f \sim n_g. \tag{1}$$

Claim 3. $a = b$.

Proof. By Claim 2, we have $[a] \neq [n_f]$, so $[a]$ should contain some element $a' \geq b$ by Claim 2 again. If $a \neq a'$, then $[a]$ contains two elements $\geq a$ which is impossible by Claim 1. Therefore, $a = a' \geq b$. Similarly, $b \geq a$. □

Now we are ready to prove the problem statement. First, we establish the following

Claim 4. For every integer $d \geq 0$, $f^{d+1}(n_f) = g^{d+1}(n_f) = a + d$.

Proof. Induction on d. For $d = 0$, the statement follows from (1) and Claim 3. Next, for $d > 1$ from the induction hypothesis we have $f^{d+1}(n_f) = f\big(f^d(n_f)\big) = f\big(g^d(n_f)\big) = f(n_f) + d = a + d$. The equality $g^{d+1}(n_f) = a + d$ is analogous. □

Finally, for each $x \in \mathbb{N}$, we have $f(x) = a + d$ for some $d \geq 0$, so $f(x) = f\big(g^d(n_f)\big)$ and hence $x \sim g^d(n_f)$. It follows that $g(x) = g\big(g^d(n_f)\big) = g^{d+1}(n_f) = a + d = f(x)$ by Claim 4.

Solution 2. We start with the same observations, introducing the relation $\sim$ and proving Claim 1 from the previous solution.

Note that $f(a) > a$ since otherwise we have $f(a) = a$ and hence $g(a) = g\big(f(a)\big) = g(a) + 1$, which is false.

Claim $2'$. $a = b$.

Proof. We can assume that $a \leq b$. Since $f(a) \geq a + 1$, there exists some $x \in \mathbb{N}$ such that $f(a) = f(x) + 1$, which is equivalent to $f(a) = f\big(g(x)\big)$ and $a \sim g(x)$. Since $g(x) \geq b \geq a$, by Claim 1 we have $a = g(x) \geq b$, which together with $a \leq b$ proves the Claim. □

Now, almost the same method allows to find the values $f(a)$ and $g(a)$.

Claim $3'$. $f(a) = g(a) = a + 1$.

Proof. Assume the contrary; then $f(a) \geq a + 2$, hence there exist some $x, y \in \mathbb{N}$ such that $f(x) = f(a) - 2$ and $f(y) = g(x)$ (as $g(x) \geq a = b$). Now we get $f(a) = f(x) + 2 = f\big(g^2(x)\big)$, so $a \sim g^2(x) \geq a$, and by Claim 1 we get $a = g^2(x) = g\big(f(y)\big) = 1 + g(y) \geq 1 + a$; this is impossible. The equality $g(a) = a + 1$ is similar.

Now, we are prepared for the proof of the problem statement. First, we prove it for $n \geq a$.

Claim $4'$. For each integer $x \geq a$, we have $f(x) = g(x) = x + 1$.

Proof. Induction on x. The base case $x = a$ is provided by Claim $3'$, while the induction step follows from $f(x + 1) = f\big(g(x)\big) = f(x) + 1 = (x + 1) + 1$ and the similar computation for $g(x + 1)$.

Finally, for an arbitrary $n \in \mathbb{N}$ we have $g(n) \geq a$, so by Claim $4'$ we have $f(n) + 1 = f\big(g(n)\big) = g(n) + 1$, hence $f(n) = g(n)$.

Comment. It is not hard now to describe all the functions $f\colon \mathbb{N} \to \mathbb{N}$ satisfying the property $f(f(n)) = f(n) + 1$. For each such function, there exists $n_0 \in \mathbb{N}$ such that $f(n) = n + 1$ for all $n \geq n_0$, while for each $n < n_0$, $f(n)$ is an arbitrary number greater than of equal to n_0 (these numbers may be different for different $n < n_0$).

A7. Let $a_1, \ldots, a_r$ be positive real numbers. For $n > r$, we inductively define

$$a_n = \max_{1 \le k \le n-1} (a_k + a_{n-k}). \tag{1}$$

Prove that there exist positive integers $\ell \le r$ and N such that $a_n = a_{n-\ell} + a_\ell$ for all $n \ge N$.

(Iran)

Solution 1. First, from the problem conditions we have that each a_n $(n > r)$ can be expressed as $a_n = a_{j_1} + a_{j_2}$ with $j_1, j_2 < n$, $j_1 + j_2 = n$. If, say, $j_1 > r$ then we can proceed in the same way with a_{j_1}, and so on. Finally, we represent a_n in a form

$$a_n = a_{i_1} + \cdots + a_{i_k}, \tag{2}$$

$$1 \le i_j \le r, \quad i_1 + \cdots + i_k = n. \tag{3}$$

Moreover, if a_{i_1} and a_{i_2} are the numbers in (2) obtained on the last step, then $i_1 + i_2 > r$. Hence we can adjust (3) as

$$1 \le i_j \le r, \quad i_1 + \cdots + i_k = n, \quad i_1 + i_2 > r. \tag{4}$$

On the other hand, suppose that the indices $i_1, \ldots, i_k$ satisfy the conditions (4). Then, denoting $s_j = i_1 + \cdots + i_j$, from (1) we have

$$a_n = a_{s_k} \ge a_{s_{k-1}} + a_{i_k} \ge a_{s_{k-2}} + a_{i_{k-1}} + a_{i_k} \ge \cdots \ge a_{i_1} + \cdots + a_{i_k}.$$

Summarizing these observations we get the following

Claim. For every $n > r$, we have

$$a_n = \max\{a_{i_1} + \cdots + a_{i_k} : \text{the collection } (i_1, \ldots, i_k) \text{ satisfies (4)}\}. \qquad \square$$

Now we denote

$$s = \max_{1 \le i \le r} \frac{a_i}{i}$$

and fix some index $\ell \le r$ such that $s = \dfrac{a_\ell}{\ell}$.

Consider some $n \ge r^2\ell + 2r$ and choose an expansion of a_n in the form (2), (4). Then we have $n = i_1 + \cdots + i_k \le rk$, so $k \ge n/r \ge r\ell + 2$. Suppose that none of the numbers $i_3, \ldots, i_k$ equals ℓ. Then by the pigeonhole principle there is an index $1 \le j \le r$ which appears among $i_3, \ldots, i_k$ at least ℓ times, and surely $j \ne \ell$. Let us delete these ℓ occurrences of j from $(i_1, \ldots, i_k)$, and add j occurrences of ℓ instead, obtaining a sequence $(i_1, i_2, i'_3, \ldots, i'_{k'})$ also satisfying (4). By Claim, we have

$$a_{i_1} + \cdots + a_{i_k} = a_n \ge a_{i_1} + a_{i_2} + a_{i'_3} + \cdots + a_{i'_{k'}},$$

or, after removing the coinciding terms, $\ell a_j \ge j a_\ell$, so $\dfrac{a_\ell}{\ell} \le \dfrac{a_j}{j}$. By the definition of ℓ, this means that $\ell a_j = j a_\ell$, hence

$$a_n = a_{i_1} + a_{i_2} + a_{i'_3} + \cdots + a_{i'_{k'}}.$$

Thus, for every $n \ge r^2\ell + 2r$ we have found a representation of the form (2), (4) with $i_j = \ell$ for some $j \ge 3$. Rearranging the indices we may assume that $i_k = \ell$.

Finally, observe that in this representation, the indices $(i_1, \ldots, i_{k-1})$ satisfy the conditions (4) with n replaced by $n - \ell$. Thus, from the Claim we get

$$a_{n-\ell} + a_\ell \ge (a_{i_1} + \cdots + a_{i_{k-1}}) + a_\ell = a_n,$$

which by (1) implies

$$a_n = a_{n-\ell} + a_\ell \qquad \text{for each } n \ge r^2\ell + 2r,$$

as desired.

Solution 2. As in the previous solution, we involve the expansion (2), (3), and we fix some index $1 \le \ell \le r$ such that

$$\frac{a_\ell}{\ell} = s = \max_{1 \le i \le r} \frac{a_i}{i}.$$

Now, we introduce the sequence (b_n) as $b_n = a_n - sn$; then $b_\ell = 0$.

We prove by induction on n that $b_n \le 0$, and (b_n) satisfies the same recurrence relation as (a_n). The base cases $n \le r$ follow from the definition of s. Now, for $n > r$ from the induction hypothesis we have

$$b_n = \max_{1 \le k \le n-1} (a_k + a_{n-k}) - ns = \max_{1 \le k \le n-1} (b_k + b_{n-k} + ns) - ns = \max_{1 \le k \le n-1} (b_k + b_{n-k}) \le 0,$$

as required.

Now, if $b_k = 0$ for all $1 \le k \le r$, then $b_n = 0$ for all n, hence $a_n = sn$, and the statement is trivial. Otherwise, define

$$M = \max_{1 \le i \le r} |b_i|, \quad \varepsilon = \min\{\, |b_i| : 1 \le i \le r,\ b_i < 0 \}.$$

Then for $n > r$ we obtain

$$b_n = \max_{1 \le k \le n-1} (b_k + b_{n-k}) \ge b_\ell + b_{n-\ell} = b_{n-\ell},$$

so

$$0 \ge b_n \ge b_{n-\ell} \ge b_{n-2\ell} \ge \cdots \ge -M.$$

Thus, in view of the expansion (2), (3) applied to the sequence (b_n), we get that each b_n is contained in a set

$$T = \{ b_{i_1} + b_{i_2} + \cdots + b_{i_k} :\ i_1, \ldots, i_k \le r \} \cap [-M, 0]$$

We claim that this set is finite. Actually, for any $x \in T$, let $x = b_{i_1} + \cdots + b_{i_k}$ $(i_1, \ldots, i_k \le r)$. Then among b_{i_j}'s there are at most $\dfrac{M}{\varepsilon}$ nonzero terms (otherwise $x < \dfrac{M}{\varepsilon} \cdot (-\varepsilon) < -M$). Thus x can be expressed in the same way with $k \le \dfrac{M}{\varepsilon}$, and there is only a finite number of such sums.

Finally, for every $t = 1, 2, \ldots, \ell$ we get that the sequence

$$b_{r+t}, b_{r+t+\ell}, b_{r+t+2\ell}, \ldots$$

is non-decreasing and attains the finite number of values; therefore it is constant from some index. Thus, the sequence (b_n) is periodic with period ℓ from some index N, which means that

$$b_n = b_{n-\ell} = b_{n-\ell} + b_\ell \qquad \text{for all } n > N + \ell,$$

and hence

$$a_n = b_n + ns = (b_{n-\ell} + (n-\ell)s) + (b_\ell + \ell s) = a_{n-\ell} + a_\ell \qquad \text{for all } n > N + \ell,$$

as desired.

A8. Given six positive numbers a, b, c, d, e, f such that $a < b < c < d < e < f$. Let $a+c+e = S$ and $b + d + f = T$. Prove that

$$2ST > \sqrt{3(S+T)\big(S(bd+bf+df)+T(ac+ae+ce)\big)}. \tag{1}$$

(South Korea)

Solution 1. We define also $\sigma = ac + ce + ae$, $\tau = bd + bf + df$. The idea of the solution is to interpret (1) as a natural inequality on the roots of an appropriate polynomial.

Actually, consider the polynomial

$$\begin{aligned} P(x) &= (b+d+f)(x-a)(x-c)(x-e) + (a+c+e)(x-b)(x-d)(x-f) \\ &= T(x^3 - Sx^2 + \sigma x - ace) + S(x^3 - Tx^2 + \tau x - bdf). \end{aligned} \tag{2}$$

Surely, P is cubic with leading coefficient $S + T > 0$. Moreover, we have

$$\begin{aligned} P(a) &= S(a-b)(a-d)(a-f) < 0, & P(c) &= S(c-b)(c-d)(c-f) > 0, \\ P(e) &= S(e-b)(e-d)(e-f) < 0, & P(f) &= T(f-a)(f-c)(f-e) > 0. \end{aligned}$$

Hence, each of the intervals (a, c), (c, e), (e, f) contains at least one root of $P(x)$. Since there are at most three roots at all, we obtain that there is exactly one root in each interval (denote them by $\alpha \in (a, c)$, $\beta \in (c, e)$, $\gamma \in (e, f)$). Moreover, the polynomial P can be factorized as

$$P(x) = (T+S)(x-\alpha)(x-\beta)(x-\gamma). \tag{3}$$

Equating the coefficients in the two representations (2) and (3) of $P(x)$ provides

$$\alpha + \beta + \gamma = \frac{2TS}{T+S}, \qquad \alpha\beta + \alpha\gamma + \beta\gamma = \frac{S\tau + T\sigma}{T+S}.$$

Now, since the numbers α, β, γ are distinct, we have

$$0 < (\alpha-\beta)^2 + (\alpha-\gamma)^2 + (\beta-\gamma)^2 = 2(\alpha+\beta+\gamma)^2 - 6(\alpha\beta+\alpha\gamma+\beta\gamma),$$

which implies

$$\frac{4S^2T^2}{(T+S)^2} = (\alpha+\beta+\gamma)^2 > 3(\alpha\beta+\alpha\gamma+\beta\gamma) = \frac{3(S\tau+T\sigma)}{T+S},$$

or

$$4S^2T^2 > 3(T+S)(T\sigma+S\tau),$$

which is exactly what we need.

Comment 1. In fact, one can locate the roots of $P(x)$ more narrowly: they should lie in the intervals (a, b), (c, d), (e, f).

Surely, if we change all inequality signs in the problem statement to non-strict ones, the (non-strict) inequality will also hold by continuity. One can also find when the equality is achieved. This happens in that case when $P(x)$ is a perfect cube, which immediately implies that $b = c = d = e (= \alpha = \beta = \gamma)$, together with the additional condition that $P''(b) = 0$. Algebraically,

$$\begin{aligned} 6(T+S)b - 4TS = 0 \quad &\Longleftrightarrow \quad 3b(a+4b+f) = 2(a+2b)(2b+f) \\ &\Longleftrightarrow \quad f = \frac{b(4b-a)}{2a+b} = b\left(1 + \frac{3(b-a)}{2a+b}\right) > b. \end{aligned}$$

This means that for every pair of numbers a, b such that $0 < a < b$, there exists $f > b$ such that the point (a, b, b, b, b, f) is a point of equality.

Solution 2. Let

$$U = \frac{1}{2}\big((e-a)^2 + (c-a)^2 + (e-c)^2\big) = S^2 - 3(ac + ae + ce)$$

and

$$V = \frac{1}{2}\big((f-b)^2 + (f-d)^2 + (d-b)^2\big) = T^2 - 3(bd + bf + df).$$

Then

$$\begin{aligned}(\text{L.H.S.})^2 - (\text{R.H.S.})^2 &= (2ST)^2 - (S+T)\big(S\cdot 3(bd+bf+df) + T\cdot 3(ac+ae+ce)\big) = \\ &= 4S^2T^2 - (S+T)\big(S(T^2-V) + T(S^2-U)\big) = (S+T)(SV+TU) - ST(T-S)^2,\end{aligned}$$

and the statement is equivalent with

$$(S+T)(SV+TU) > ST(T-S)^2. \tag{4}$$

By the Cauchy-Schwarz inequality,

$$(S+T)(TU+SV) \geq \big(\sqrt{S\cdot TU} + \sqrt{T\cdot SV}\big)^2 = ST\big(\sqrt{U}+\sqrt{V}\big)^2. \tag{5}$$

Estimate the quantities $\sqrt{U}$ and $\sqrt{V}$ by the QM–AM inequality with the positive terms $(e-c)^2$ and $(d-b)^2$ being omitted:

$$\begin{aligned}\sqrt{U}+\sqrt{V} &> \sqrt{\frac{(e-a)^2+(c-a)^2}{2}} + \sqrt{\frac{(f-b)^2+(f-d)^2}{2}} \\ &> \frac{(e-a)+(c-a)}{2} + \frac{(f-b)+(f-d)}{2} = \left(f - \frac{d}{2} - \frac{b}{2}\right) + \left(\frac{e}{2} + \frac{c}{2} - a\right) \\ &= (T-S) + \frac{3}{2}(e-d) + \frac{3}{2}(c-b) > T-S.\end{aligned} \tag{6}$$

The estimates (5) and (6) prove (4) and hence the statement.

Solution 3. We keep using the notations σ and τ from Solution 1. Moreover, let $s = c+e$. Note that

$$(c-b)(c-d) + (e-f)(e-d) + (e-f)(c-b) < 0,$$

since each summand is negative. This rewrites as

$$\begin{gathered}(bd+bf+df) - (ac+ce+ae) < (c+e)(b+d+f-a-c-e), \text{ or} \\ \tau - \sigma < s(T-S).\end{gathered} \tag{7}$$

Then we have

$$\begin{aligned}S\tau + T\sigma &= S(\tau-\sigma) + (S+T)\sigma < Ss(T-S) + (S+T)(ce+as) \\ &\leq Ss(T-S) + (S+T)\left(\frac{s^2}{4} + (S-s)s\right) = s\left(2ST - \frac{3}{4}(S+T)s\right).\end{aligned}$$

Using this inequality together with the AM–GM inequality we get

$$\begin{aligned}\sqrt{\frac{3}{4}(S+T)(S\tau+T\sigma)} &< \sqrt{\frac{3}{4}(S+T)s\left(2ST - \frac{3}{4}(S+T)s\right)} \\ &\leq \frac{\frac{3}{4}(S+T)s + 2ST - \frac{3}{4}(S+T)s}{2} = ST.\end{aligned}$$

Hence,

$$2ST > \sqrt{3(S+T)\big(S(bd+bf+df) + T(ac+ae+ce)\big)}.$$

Comment 2. The expression (7) can be found by considering the sum of the roots of the quadratic polynomial $q(x) = (x-b)(x-d)(x-f) - (x-a)(x-c)(x-e)$.

Solution 4. We introduce the expressions σ and τ as in the previous solutions. The idea of the solution is to change the values of variables $a, \ldots, f$ keeping the left-hand side unchanged and increasing the right-hand side; it will lead to a simpler inequality which can be proved in a direct way.

Namely, we change the variables (i) keeping the (non-strict) inequalities $a \le b \le c \le d \le e \le f$; (ii) keeping the values of sums S and T unchanged; and finally (iii) increasing the values of σ and τ. Then the left-hand side of (1) remains unchanged, while the right-hand side increases. Hence, the inequality (1) (and even a non-strict version of (1)) for the changed values would imply the same (strict) inequality for the original values.

First, we find the sufficient conditions for (ii) and (iii) to be satisfied.

Lemma. Let $x, y, z > 0$; denote $U(x,y,z) = x+y+z$, $\upsilon(x,y,z) = xy+xz+yz$. Suppose that $x'+y' = x+y$ but $|x-y| \ge |x'-y'|$; then we have $U(x',y',z) = U(x,y,z)$ and $\upsilon(x',y',z) \ge \upsilon(x,y,z)$ with equality achieved only when $|x-y| = |x'-y'|$.

Proof. The first equality is obvious. For the second, we have

$$\upsilon(x',y',z) = z(x'+y') + x'y' = z(x'+y') + \frac{(x'+y')^2 - (x'-y')^2}{4}$$
$$\ge z(x+y) + \frac{(x+y)^2 - (x-y)^2}{4} = \upsilon(x,y,z),$$

with the equality achieved only for $(x'-y')^2 = (x-y)^2 \iff |x'-y'| = |x-y|$, as desired. □

Now, we apply Lemma several times making the following changes. For each change, we denote the new values by the same letters to avoid cumbersome notations.

1. Let $k = \dfrac{d-c}{2}$. Replace (b,c,d,e) by $(b+k, c+k, d-k, e-k)$. After the change we have $a < b < c = d < e < f$, the values of S, T remain unchanged, but σ, τ strictly increase by Lemma.

2. Let $\ell = \dfrac{e-d}{2}$. Replace (c,d,e,f) by $(c+\ell, d+\ell, e-\ell, f-\ell)$. After the change we have $a < b < c = d = e < f$, the values of S, T remain unchanged, but σ, τ strictly increase by the Lemma.

3. Finally, let $m = \dfrac{c-b}{3}$. Replace (a,b,c,d,e,f) by $(a+2m, b+2m, c-m, d-m, e-m, f-m)$. After the change, we have $a < b = c = d = e < f$ and S, T are unchanged. To check (iii), we observe that our change can be considered as a composition of two changes: $(a,b,c,d) \to (a+m, b+m, c-m, d-m)$ and $(a,b,e,f) \to (a+m, b+m, e-m, f-m)$. It is easy to see that each of these two consecutive changes satisfy the conditions of the Lemma, hence the values of σ and τ increase.

Finally, we come to the situation when $a < b = c = d = e < f$, and we need to prove the inequality

$$2(a+2b)(2b+f) \ge \sqrt{3(a+4b+f)\big((a+2b)(b^2+2bf) + (2b+f)(2ab+b^2)\big)}$$
$$= \sqrt{3b(a+4b+f)\cdot\big((a+2b)(b+2f) + (2b+f)(2a+b)\big)}. \qquad (8)$$

Now, observe that

$$2\cdot 2(a+2b)(2b+f) = 3b(a+4b+f) + \big((a+2b)(b+2f) + (2a+b)(2b+f)\big).$$

Hence (4) rewrites as

$$3b(a+4b+f)+\big((a+2b)(b+2f)+(2a+b)(2b+f)\big) \\ \geq 2\sqrt{3b(a+4b+f)\cdot\big((a+2b)(b+2f)+(2b+f)(2a+b)\big)},$$

which is simply the AM–GM inequality.

Comment 3. Here, we also can find all the cases of equality. Actually, it is easy to see that if some two numbers among b, c, d, e are distinct then one can use Lemma to increase the right-hand side of (1). Further, if $b = c = d = e$, then we need equality in (4); this means that we apply AM–GM to equal numbers, that is,

$$3b(a+4b+f) = (a+2b)(b+2f)+(2a+b)(2b+f),$$

which leads to the same equality as in Comment 1.

Combinatorics

C1. In a concert, 20 singers will perform. For each singer, there is a (possibly empty) set of other singers such that he wishes to perform later than all the singers from that set. Can it happen that there are exactly 2010 orders of the singers such that all their wishes are satisfied?

(Austria)

Answer. Yes, such an example exists.

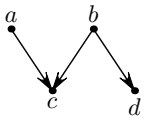
Solution. We say that an order of singers is *good* if it satisfied all their wishes. Next, we say that a number N is *realizable* by k singers (or *k-realizable*) if for some set of wishes of these singers there are exactly N good orders. Thus, we have to prove that a number 2010 is 20-realizable.

We start with the following simple

Lemma. Suppose that numbers n_1, n_2 are realizable by respectively k_1 and k_2 singers. Then the number $n_1 n_2$ is $(k_1 + k_2)$-realizable.

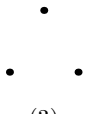
Proof. Let the singers $A_1, \ldots, A_{k_1}$ (with some wishes among them) realize n_1, and the singers B_1, $\ldots$, B_{k_2} (with some wishes among them) realize n_2. Add to each singer B_i the wish to perform later than all the singers A_j. Then, each good order of the obtained set of singers has the form $(A_{i_1}, \ldots, A_{i_{k_1}}, B_{j_1}, \ldots, B_{j_{k_2}})$, where $(A_{i_1}, \ldots, A_{i_{k_1}})$ is a good order for A_i's and $(B_{j_1}, \ldots, B_{j_{k_2}})$ is a good order for B_j's. Conversely, each order of this form is obviously good. Hence, the number of good orders is $n_1 n_2$. □

In view of Lemma, we show how to construct sets of singers containing 4, 3 and 13 singers and realizing the numbers 5, 6 and 67, respectively. Thus the number $2010 = 6 \cdot 5 \cdot 67$ will be realizable by $4 + 3 + 13 = 20$ singers. These companies of singers are shown in Figs. 1–3; the wishes are denoted by arrows, and the number of good orders for each Figure stands below in the brackets.



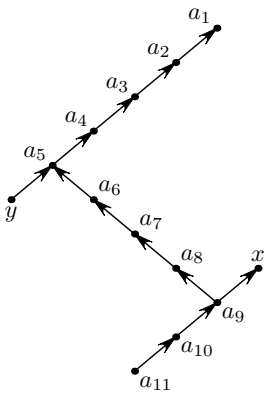
(5)

Fig. 1



(3)

Fig. 2



(67)

Fig. 3

For Fig. 1, there are exactly 5 good orders (a, b, c, d), (a, b, d, c), (b, a, c, d), (b, a, d, c), (b, d, a, c). For Fig. 2, each of 6 orders is good since there are no wishes.

Finally, for Fig. 3, the order of $a_1, \ldots, a_{11}$ is fixed; in this line, singer x can stand before each of a_i ($i \le 9$), and singer y can stand after each of a_j ($j \ge 5$), thus resulting in $9 \cdot 7 = 63$ cases. Further, the positions of x and y in this line determine the whole order uniquely unless both of them come between the same pair (a_i, a_{i+1}) (thus $5 \le i \le 8$); in the latter cases, there are two orders instead of 1 due to the order of x and y. Hence, the total number of good orders is $63 + 4 = 67$, as desired.

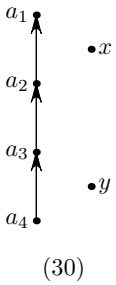
Comment. The number 20 in the problem statement is not sharp and is put there to respect the original formulation. So, if necessary, the difficulty level of this problem may be adjusted by replacing 20 by a smaller number. Here we present some improvements of the example leading to a smaller number of singers.

Surely, each example with < 20 singers can be filled with some "super-stars" who should perform at the very end in a fixed order. Hence each of these improvements provides a different solution for the problem. Moreover, the large variety of ideas standing behind these examples allows to suggest that there are many other examples.

1. Instead of building the examples realizing 5 and 6, it is more economic to make an example realizing 30; it may seem even simpler. Two possible examples consisting of 5 and 6 singers are shown in Fig. 4; hence the number 20 can be decreased to 19 or 18.

For Fig. 4a, the order of $a_1, \ldots, a_4$ is fixed, there are 5 ways to add x into this order, and there are 6 ways to add y into the resulting order of $a_1, \ldots, a_4, x$. Hence there are $5 \cdot 6 = 30$ good orders.

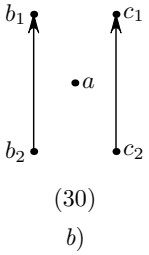
On Fig. 4b, for 5 singers a, b_1, b_2, c_1, c_2 there are $5! = 120$ orders at all. Obviously, exactly one half of them satisfies the wish $b_1 \leftarrow b_2$, and exactly one half of these orders satisfies another wish $c_1 \leftarrow c_2$; hence, there are exactly $5!/4 = 30$ good orders.



(30)

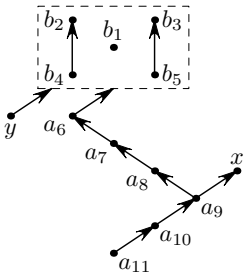
a)

Fig. 4



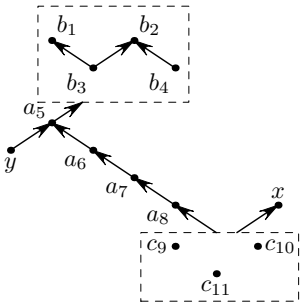
(30)

b)



(2010)

Fig. 5



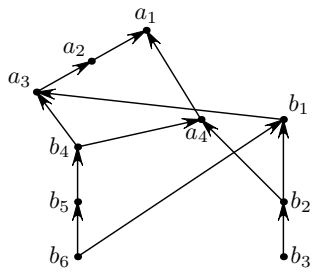
(2010)

Fig. 6

2. One can merge the examples for 30 and 67 shown in Figs. 4b and 3 in a smarter way, obtaining a set of 13 singers representing 2010. This example is shown in Fig. 5; an arrow from/to group $\{b_1, \ldots, b_5\}$ means that there exists such arrow from each member of this group.

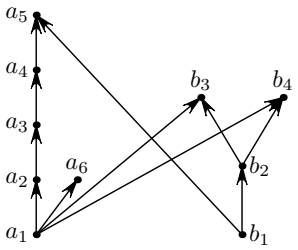
Here, as in Fig. 4b, one can see that there are exactly 30 orders of $b_1, \ldots, b_5, a_6, \ldots, a_{11}$ satisfying all their wishes among themselves. Moreover, one can prove in the same way as for Fig. 3 that each of these orders can be complemented by x and y in exactly 67 ways, hence obtaining $30 \cdot 67 = 2010$ good orders at all.

Analogously, one can merge the examples in Figs. 1–3 to represent 2010 by 13 singers as is shown in Fig. 6.



(67)

Fig. 7



(2010)

Fig. 8

3. Finally, we will present two other improvements; the proofs are left to the reader. The graph in Fig. 7 shows how 10 singers can represent 67. Moreover, even a company of 10 singers representing 2010 can be found; this company is shown in Fig. 8.

C2. On some planet, there are 2^N countries ($N \geq 4$). Each country has a flag N units wide and one unit high composed of N fields of size 1×1, each field being either yellow or blue. No two countries have the same flag.

We say that a set of N flags is *diverse* if these flags can be arranged into an $N \times N$ square so that all N fields on its main diagonal will have the same color. Determine the smallest positive integer M such that among any M distinct flags, there exist N flags forming a diverse set.

(Croatia)

Answer. $M = 2^{N-2} + 1$.

Solution. When speaking about the diagonal of a square, we will always mean the main diagonal.

Let M_N be the smallest positive integer satisfying the problem condition. First, we show that $M_N > 2^{N-2}$. Consider the collection of all 2^{N-2} flags having yellow first squares and blue second ones. Obviously, both colors appear on the diagonal of each $N \times N$ square formed by these flags.

We are left to show that $M_N \leq 2^{N-2} + 1$, thus obtaining the desired answer. We start with establishing this statement for $N = 4$.

Suppose that we have 5 flags of length 4. We decompose each flag into two parts of 2 squares each; thereby, we denote each flag as LR, where the 2×1 flags $L, R \in \mathcal{S} = \{\mathrm{BB}, \mathrm{BY}, \mathrm{YB}, \mathrm{YY}\}$ are its left and right parts, respectively. First, we make two easy observations on the flags 2×1 which can be checked manually.

(i) For each $A \in \mathcal{S}$, there exists only one 2×1 flag $C \in \mathcal{S}$ (possibly $C = A$) such that A and C cannot form a 2×2 square with monochrome diagonal (for part BB, that is YY, and for BY that is YB).

(ii) Let $A_1, A_2, A_3 \in \mathcal{S}$ be three distinct elements; then two of them can form a 2×2 square with yellow diagonal, and two of them can form a 2×2 square with blue diagonal (for all parts but BB, a pair (BY, YB) fits for both statements, while for all parts but BY, these pairs are (YB, YY) and (BB, YB)).

Now, let ℓ and r be the numbers of distinct left and right parts of our 5 flags, respectively. The total number of flags is $5 \leq r\ell$, hence one of the factors (say, r) should be at least 3. On the other hand, $\ell, r \leq 4$, so there are two flags with coinciding right part; let them be L_1R_1 and L_2R_1 ($L_1 \neq L_2$).

Next, since $r \geq 3$, there exist some flags L_3R_3 and L_4R_4 such that R_1, R_3, R_4 are distinct. Let $L'R'$ be the remaining flag. By (i), one of the pairs (L', L_1) and (L', L_2) can form a 2×2 square with monochrome diagonal; we can assume that L', L_2 form a square with a blue diagonal. Finally, the right parts of two of the flags L_1R_1, L_3R_3, L_4R_4 can also form a 2×2 square with a blue diagonal by (ii). Putting these 2×2 squares on the diagonal of a 4×4 square, we find a desired arrangement of four flags.

We are ready to prove the problem statement by induction on N; actually, above we have proved the base case $N = 4$. For the induction step, assume that $N > 4$, consider any $2^{N-2} + 1$ flags of length N, and arrange them into a large flag of size $(2^{N-2} + 1) \times N$. This flag contains a non-monochrome column since the flags are distinct; we may assume that this column is the first one. By the pigeonhole principle, this column contains at least $\left\lceil \dfrac{2^{N-2}+1}{2} \right\rceil = 2^{N-3} + 1$ squares of one color (say, blue). We call the flags with a blue first square *good*.

Consider all the good flags and remove the first square from each of them. We obtain at least $2^{N-3} + 1 \geq M_{N-1}$ flags of length $N - 1$; by the induction hypothesis, $N - 1$ of them

can form a square Q with the monochrome diagonal. Now, returning the removed squares, we obtain a rectangle $(N-1)\times N$, and our aim is to supplement it on the top by one more flag.

If Q has a yellow diagonal, then we can take each flag with a yellow first square (it exists by a choice of the first column; moreover, it is not used in Q). Conversely, if the diagonal of Q is blue then we can take any of the $\geq 2^{N-3}+1-(N-1)>0$ remaining good flags. So, in both cases we get a desired $N\times N$ square.

Solution 2. We present a different proof of the estimate $M_N\leq 2^{N-2}+1$. We do not use the induction, involving Hall's lemma on matchings instead.

Consider arbitrary $2^{N-2}+1$ distinct flags and arrange them into a large $(2^{N-2}+1)\times N$ flag. Construct two bipartite graphs $G_{\mathrm{y}}=(V\cup V',E_{\mathrm{y}})$ and $G_{\mathrm{b}}=(V\cup V',E_{\mathrm{b}})$ with the common set of vertices as follows. Let V and V' be the set of columns and the set of flags under consideration, respectively. Next, let the edge (c,f) appear in E_{y} if the intersection of column c and flag f is yellow, and $(c,f)\in E_{\mathrm{b}}$ otherwise. Then we have to prove exactly that one of the graphs G_{y} and G_{b} contains a matching with all the vertices of V involved.

Assume that these matchings do not exist. By Hall's lemma, it means that there exist two sets of columns $S_{\mathrm{y}},S_{\mathrm{b}}\subset V$ such that $|E_{\mathrm{y}}(S_{\mathrm{y}})|\leq|S_{\mathrm{y}}|-1$ and $|E_{\mathrm{b}}(S_{\mathrm{b}})|\leq|S_{\mathrm{b}}|-1$ (in the left-hand sides, $E_{\mathrm{y}}(S_{\mathrm{y}})$ and $E_{\mathrm{b}}(S_{\mathrm{b}})$ denote respectively the sets of all vertices connected to S_{y} and S_{b} in the corresponding graphs). Our aim is to prove that this is impossible. Note that $S_{\mathrm{y}},S_{\mathrm{b}}\neq V$ since $N\leq 2^{N-2}+1$.

First, suppose that $S_{\mathrm{y}}\cap S_{\mathrm{b}}\neq\varnothing$, so there exists some $c\in S_{\mathrm{y}}\cap S_{\mathrm{b}}$. Note that each flag is connected to c either in G_{y} or in G_{b}, hence $E_{\mathrm{y}}(S_{\mathrm{y}})\cup E_{\mathrm{b}}(S_{\mathrm{b}})=V'$. Hence we have $2^{N-2}+1=|V'|\leq|E_{\mathrm{y}}(S_{\mathrm{y}})|+|E_{\mathrm{b}}(S_{\mathrm{b}})|\leq|S_{\mathrm{y}}|+|S_{\mathrm{b}}|-2\leq 2N-4$; this is impossible for $N\geq 4$.

So, we have $S_{\mathrm{y}}\cap S_{\mathrm{b}}=\varnothing$. Let $y=|S_{\mathrm{y}}|$, $b=|S_{\mathrm{b}}|$. From the construction of our graph, we have that all the flags in the set $V''=V'\setminus\big(E_{\mathrm{y}}(S_{\mathrm{y}})\cup E_{\mathrm{b}}(S_{\mathrm{b}})\big)$ have blue squares in the columns of S_{y} and yellow squares in the columns of S_{b}. Hence the only undetermined positions in these flags are the remaining $N-y-b$ ones, so $2^{N-y-b}\geq|V''|\geq|V'|-|E_{\mathrm{y}}(S_{\mathrm{y}})|-|E_{\mathrm{b}}(S_{\mathrm{b}})|\geq 2^{N-2}+1-(y-1)-(b-1)$, or, denoting $c=y+b$, $2^{N-c}+c>2^{N-2}+2$. This is impossible since $N\geq c\geq 2$.

C3. 2500 chess kings have to be placed on a 100×100 chessboard so that

(i) no king can capture any other one (i.e. no two kings are placed in two squares sharing a common vertex);

(ii) each row and each column contains exactly 25 kings.

Find the number of such arrangements. (Two arrangements differing by rotation or symmetry are supposed to be different.)

(Russia)

Answer. There are two such arrangements.

Solution. Suppose that we have an arrangement satisfying the problem conditions. Divide the board into 2×2 pieces; we call these pieces *blocks*. Each block can contain not more than one king (otherwise these two kings would attack each other); hence, by the pigeonhole principle each block must contain exactly one king.

Now assign to each block a letter T or B if a king is placed in its top or bottom half, respectively. Similarly, assign to each block a letter L or R if a king stands in its left or right half. So we define *T-blocks*, *B-blocks*, *L-blocks*, and *R-blocks*. We also combine the letters; we call a block *a TL-block* if it is simultaneously T-block and L-block. Similarly we define *TR-blocks*, *BL-blocks*, and *BR-blocks*. The arrangement of blocks determines uniquely the arrangement of kings; so in the rest of the solution we consider the 50×50 system of blocks (see Fig. 1). We identify the blocks by their coordinate pairs; the pair (i, j), where $1 \le i, j \le 50$, refers to the jth block in the ith row (or the ith block in the jth column). The upper-left block is $(1, 1)$.

The system of blocks has the following properties..

(i′) If (i, j) is a B-block then $(i + 1, j)$ is a B-block: otherwise the kings in these two blocks can take each other. Similarly: if (i, j) is a T-block then $(i - 1, j)$ is a T-block; if (i, j) is an L-block then $(i, j - 1)$ is an L-block; if (i, j) is an R-block then $(i, j + 1)$ is an R-block.

(ii′) Each column contains exactly 25 L-blocks and 25 R-blocks, and each row contains exactly 25 T-blocks and 25 B-blocks. In particular, the total number of L-blocks (or R-blocks, or T-blocks, or B-blocks) is equal to $25 \cdot 50 = 1250$.

Consider any B-block of the form $(1, j)$. By (i′), all blocks in the jth column are B-blocks; so we call such a column *B-column*. By (ii′), we have 25 B-blocks in the first row, so we obtain 25 B-columns. These 25 B-columns contain 1250 B-blocks, hence all blocks in the remaining columns are T-blocks, and we obtain 25 *T-columns*. Similarly, there are exactly 25 *L-rows* and exactly 25 *R-rows*.

Now consider an arbitrary pair of a T-column and a neighboring B-column (columns with numbers j and $j + 1$).

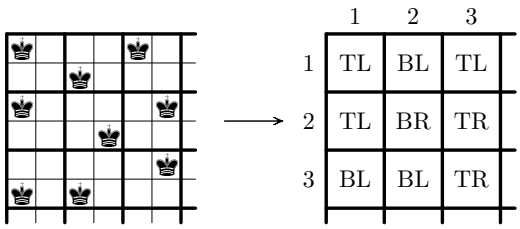


Fig. 1

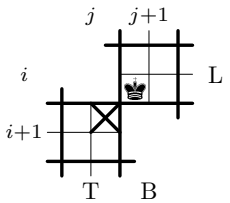


Fig. 2

Case 1. Suppose that the jth column is a T-column, and the $(j + 1)$th column is a B-column. Consider some index i such that the ith row is an L-row; then $(i, j + 1)$ is a BL-block. Therefore, $(i + 1, j)$ cannot be a TR-block (see Fig. 2), hence $(i + 1, j)$ is a TL-block, thus the

$(i+1)$th row is an L-row. Now, choosing the ith row to be the topmost L-row, we successively obtain that all rows from the ith to the 50th are L-rows. Since we have exactly 25 L-rows, it follows that the rows from the 1st to the 25th are R-rows, and the rows from the 26th to the 50th are L-rows.

Now consider the neighboring R-row and L-row (that are the rows with numbers 25 and 26). Replacing in the previous reasoning rows by columns and vice versa, the columns from the 1st to the 25th are T-columns, and the columns from the 26th to the 50th are B-columns. So we have a unique arrangement of blocks that leads to the arrangement of kings satisfying the condition of the problem (see Fig. 3).

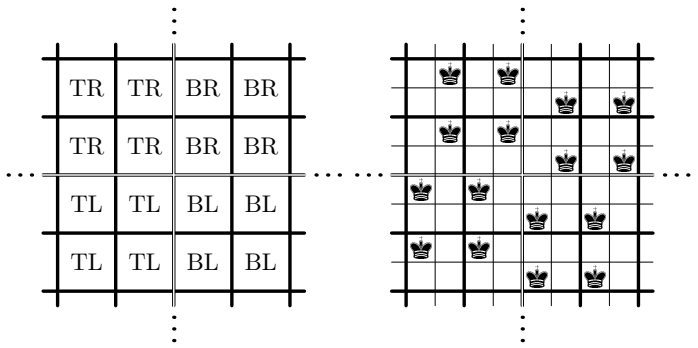


Fig. 3

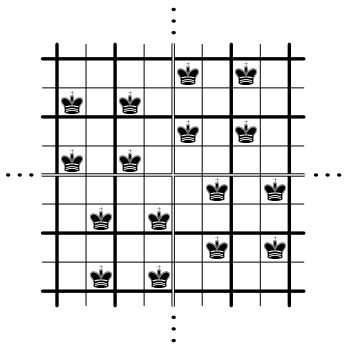


Fig. 4

Case 2. Suppose that the jth column is a B-column, and the $(j+1)$th column is a T-column. Repeating the arguments from Case 1, we obtain that the rows from the 1st to the 25th are L-rows (and all other rows are R-rows), the columns from the 1st to the 25th are B-columns (and all other columns are T-columns), so we find exactly one more arrangement of kings (see Fig. 4).

30

C4. Six stacks $S_1, \ldots, S_6$ of coins are standing in a row. In the beginning every stack contains a single coin. There are two types of allowed moves:

Move 1: If stack S_k with $1 \le k \le 5$ contains at least one coin, you may remove one coin from S_k and add two coins to S_{k+1}.

Move 2: If stack S_k with $1 \le k \le 4$ contains at least one coin, then you may remove one coin from S_k and exchange stacks S_{k+1} and S_{k+2}.

Decide whether it is possible to achieve by a sequence of such moves that the first five stacks are empty, whereas the sixth stack S_6 contains exactly $2010^{2010^{2010}}$ coins.

C4′. Same as Problem C4, but the constant $2010^{2010^{2010}}$ is replaced by 2010^{2010}.

(Netherlands)

Answer. Yes (in both variants of the problem). There exists such a sequence of moves.

Solution. Denote by $(a_1, a_2, \ldots, a_n) \to (a_1', a_2', \ldots, a_n')$ the following: if some consecutive stacks contain $a_1, \ldots, a_n$ coins, then it is possible to perform several allowed moves such that the stacks contain $a_1', \ldots, a_n'$ coins respectively, whereas the contents of the other stacks remain unchanged.

Let $A = 2010^{2010}$ or $A = 2010^{2010^{2010}}$, respectively. Our goal is to show that

$$(1, 1, 1, 1, 1, 1) \to (0, 0, 0, 0, 0, A).$$

First we prove two auxiliary observations.

Lemma 1. $(a, 0, 0) \to (0, 2^a, 0)$ for every $a \ge 1$.

Proof. We prove by induction that $(a, 0, 0) \to (a - k, 2^k, 0)$ for every $1 \le k \le a$. For $k = 1$, apply Move 1 to the first stack:

$$(a, 0, 0) \to (a - 1, 2, 0) = (a - 1, 2^1, 0).$$

Now assume that $k < a$ and the statement holds for some $k < a$. Starting from $(a-k, 2^k, 0)$, apply Move 1 to the middle stack 2^k times, until it becomes empty. Then apply Move 2 to the first stack:

$$(a - k, 2^k, 0) \to (a - k, 2^k - 1, 2) \to \cdots \to (a - k, 0, 2^{k+1}) \to (a - k - 1, 2^{k+1}, 0).$$

Hence,

$$(a, 0, 0) \to (a - k, 2^k, 0) \to (a - k - 1, 2^{k+1}, 0). \qquad \square$$

Lemma 2. For every positive integer n, let $P_n = \underbrace{2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}}_{n}$ (e.g. $P_3 = 2^{2^2} = 16$). Then $(a, 0, 0, 0) \to (0, P_a, 0, 0)$ for every $a \ge 1$.

Proof. Similarly to Lemma 1, we prove that $(a, 0, 0, 0) \to (a - k, P_k, 0, 0)$ for every $1 \le k \le a$.

For $k = 1$, apply Move 1 to the first stack:

$$(a, 0, 0, 0) \to (a - 1, 2, 0, 0) = (a - 1, P_1, 0, 0).$$

Now assume that the lemma holds for some $k < a$. Starting from $(a - k, P_k, 0, 0)$, apply Lemma 1, then apply Move 1 to the first stack:

$$(a - k, P_k, 0, 0) \to (a - k, 0, 2^{P_k}, 0) = (a - k, 0, P_{k+1}, 0) \to (a - k - 1, P_{k+1}, 0, 0).$$

Therefore,

$$(a, 0, 0, 0) \to (a - k, P_k, 0, 0) \to (a - k - 1, P_{k+1}, 0, 0). \qquad \square$$

Now we prove the statement of the problem.

First apply Move 1 to stack 5, then apply Move 2 to stacks S_4, S_3, S_2 and S_1 in this order. Then apply Lemma 2 twice:

$$(1,1,1,1,1,1) \to (1,1,1,1,0,3) \to (1,1,1,0,3,0) \to (1,1,0,3,0,0) \to (1,0,3,0,0,0) \to$$
$$\to (0,3,0,0,0,0) \to (0,0,P_3,0,0,0) = (0,0,16,0,0,0) \to (0,0,0,P_{16},0,0).$$

We already have more than A coins in stack S_4, since

$$A \le 2010^{2010^{2010}} < (2^{11})^{2010^{2010}} = 2^{11\cdot 2010^{2010}} < 2^{2010^{2011}} < 2^{(2^{11})^{2011}} = 2^{2^{11\cdot 2011}} < 2^{2^{2^{15}}} < P_{16}.$$

To decrease the number of coins in stack S_4, apply Move 2 to this stack repeatedly until its size decreases to $A/4$. (In every step, we remove a coin from S_4 and exchange the empty stacks S_5 and S_6.)

$$(0,0,0,P_{16},0,0) \to (0,0,0,P_{16}-1,0,0) \to (0,0,0,P_{16}-2,0,0) \to$$
$$\to \cdots \to (0,0,0,A/4,0,0).$$

Finally, apply Move 1 repeatedly to empty stacks S_4 and S_5:

$$(0,0,0,A/4,0,0) \to \cdots \to (0,0,0,0,A/2,0) \to \cdots \to (0,0,0,0,0,A).$$

Comment 1. Starting with only 4 stack, it is not hard to check manually that we can achieve at most 28 coins in the last position. However, around 5 and 6 stacks the maximal number of coins explodes. With 5 stacks it is possible to achieve more than $2^{2^{14}}$ coins. With 6 stacks the maximum is greater than $P_{P_{2^{14}}}$.

It is not hard to show that the numbers 2010^{2010} and $2010^{2010^{2010}}$ in the problem can be replaced by any nonnegative integer up to $P_{P_{2^{14}}}$.

Comment 2. The simpler variant C4′ of the problem can be solved without Lemma 2:

$$(1,1,1,1,1,1) \to (0,3,1,1,1,1) \to (0,1,5,1,1,1) \to (0,1,1,9,1,1) \to$$
$$\to (0,1,1,1,17,1) \to (0,1,1,1,0,35) \to (0,1,1,0,35,0) \to (0,1,0,35,0,0) \to$$
$$\to (0,0,35,0,0,0) \to (0,0,1,2^{34},0,0) \to (0,0,1,0,2^{2^{34}},0) \to (0,0,0,2^{2^{34}},0,0)$$
$$\to (0,0,0,2^{2^{34}}-1,0,0) \to \ldots \to (0,0,0,A/4,0,0) \to (0,0,0,0,A/2,0) \to (0,0,0,0,0,A).$$

For this reason, the PSC suggests to consider the problem C4 as well. Problem C4 requires more invention and technical care. On the other hand, the problem statement in C4′ hides the fact that the resulting amount of coins can be such incredibly huge and leaves this discovery to the students.

C5. $n \geq 4$ players participated in a tennis tournament. Any two players have played exactly one game, and there was no tie game. We call a company of four players *bad* if one player was defeated by the other three players, and each of these three players won a game and lost another game among themselves. Suppose that there is no bad company in this tournament. Let w_i and ℓ_i be respectively the number of wins and losses of the ith player. Prove that

$$\sum_{i=1}^{n}(w_i - \ell_i)^3 \geq 0. \tag{1}$$

(South Korea)

Solution. For any tournament T satisfying the problem condition, denote by $S(T)$ sum under consideration, namely

$$S(T) = \sum_{i=1}^{n}(w_i - \ell_i)^3.$$

First, we show that the statement holds if a tournament T has only 4 players. Actually, let $A = (a_1, a_2, a_3, a_4)$ be the number of wins of the players; we may assume that $a_1 \geq a_2 \geq a_3 \geq a_4$. We have $a_1 + a_2 + a_3 + a_4 = \binom{4}{2} = 6$, hence $a_4 \leq 1$. If $a_4 = 0$, then we cannot have $a_1 = a_2 = a_3 = 2$, otherwise the company of all players is bad. Hence we should have $A = (3, 2, 1, 0)$, and $S(T) = 3^3 + 1^3 + (-1)^3 + (-3)^3 = 0$. On the other hand, if $a_4 = 1$, then only two possibilities, $A = (3, 1, 1, 1)$ and $A = (2, 2, 1, 1)$ can take place. In the former case we have $S(T) = 3^3 + 3 \cdot (-2)^3 > 0$, while in the latter one $S(T) = 1^3 + 1^3 + (-1)^3 + (-1)^3 = 0$, as desired.

Now we turn to the general problem. Consider a tournament T with no bad companies and enumerate the players by the numbers from 1 to n. For every 4 players i_1, i_2, i_3, i_4 consider a "sub-tournament" $T_{i_1 i_2 i_3 i_4}$ consisting of only these players and the games which they performed with each other. By the abovementioned, we have $S(T_{i_1 i_2 i_3 i_4}) \geq 0$. Our aim is to prove that

$$S(T) = \sum_{i_1, i_2, i_3, i_4} S(T_{i_1 i_2 i_3 i_4}), \tag{2}$$

where the sum is taken over all 4-tuples of distinct numbers from the set $\{1, \ldots, n\}$. This way the problem statement will be established.

We interpret the number $(w_i - \ell_i)^3$ as following. For $i \neq j$, let $\varepsilon_{ij} = 1$ if the ith player wins against the jth one, and $\varepsilon_{ij} = -1$ otherwise. Then

$$(w_i - \ell_i)^3 = \left(\sum_{j \neq i} \varepsilon_{ij}\right)^3 = \sum_{j_1, j_2, j_3 \neq i} \varepsilon_{ij_1}\varepsilon_{ij_2}\varepsilon_{ij_3}.$$

Hence,

$$S(T) = \sum_{i \notin \{j_1, j_2, j_3\}} \varepsilon_{ij_1}\varepsilon_{ij_2}\varepsilon_{ij_3}.$$

To simplify this expression, consider all the terms in this sum where two indices are equal. If, for instance, $j_1 = j_2$, then the term contains $\varepsilon_{ij_1}^2 = 1$, so we can replace this term by ε_{ij_3}. Make such replacements for each such term; obviously, after this change each term of the form ε_{ij_3} will appear $P(T)$ times, hence

$$S(T) = \sum_{|\{i, j_1, j_2, j_3\}| = 4} \varepsilon_{ij_1}\varepsilon_{ij_2}\varepsilon_{ij_3} + P(T)\sum_{i \neq j} \varepsilon_{ij} = S_1(T) + P(T)S_2(T).$$

We show that $S_2(T) = 0$ and hence $S(T) = S_1(T)$ for each tournament. Actually, note that $\varepsilon_{ij} = -\varepsilon_{ji}$, and the whole sum can be split into such pairs. Since the sum in each pair is 0, so is $S_2(T)$.

Thus the desired equality (2) rewrites as

$$S_1(T) = \sum_{i_1,i_2,i_3,i_4} S_1(T_{i_1i_2i_3i_4}). \tag{3}$$

Now, if all the numbers j_1, j_2, j_3 are distinct, then the set $\{i, j_1, j_2, j_3\}$ is contained in exactly one 4-tuple, hence the term $\varepsilon_{ij_1}\varepsilon_{ij_2}\varepsilon_{ij_3}$ appears in the right-hand part of (3) exactly once, as well as in the left-hand part. Clearly, there are no other terms in both parts, so the equality is established.

Solution 2. Similarly to the first solution, we call the subsets of players as *companies*, and the k-element subsets will be called as *k-companies*.

In any company of the players, call a player *the local champion of the company* if he defeated all other members of the company. Similarly, if a player lost all his games against the others in the company then call him *the local loser of the company*. Obviously every company has at most one local champion and at most one local loser. By the condition of the problem, whenever a 4-company has a local loser, then this company has a local champion as well.

Suppose that k is some positive integer, and let us count all cases when a player is the local champion of some k-company. The ith player won against w_i other player. To be the local champion of a k-company, he must be a member of the company, and the other $k-1$ members must be chosen from those whom he defeated. Therefore, the ith player is the local champion of $\binom{w_i}{k-1}$ k-companies. Hence, the total number of local champions of all k-companies is $\sum_{i=1}^{n}\binom{w_i}{k-1}$.

Similarly, the total number of local losers of the k-companies is $\sum_{i=1}^{n}\binom{\ell_i}{k-1}$.

Now apply this for $k = 2, 3$ and 4.

Since every game has a winner and a loser, we have $\sum_{i=1}^{n} w_i = \sum_{i=1}^{n} \ell_i = \binom{n}{2}$, and hence

$$\sum_{i=1}^{n}\bigl(w_i - \ell_i\bigr) = 0. \tag{4}$$

In every 3-company, either the players defeated one another in a cycle or the company has both a local champion and a local loser. Therefore, the total number of local champions and local losers in the 3-companies is the same, $\sum_{i=1}^{n}\binom{w_i}{2} = \sum_{i=1}^{n}\binom{\ell_i}{2}$. So we have

$$\sum_{i=1}^{n}\left(\binom{w_i}{2} - \binom{\ell_i}{2}\right) = 0. \tag{5}$$

In every 4-company, by the problem's condition, the number of local losers is less than or equal to the number of local champions. Then the same holds for the total numbers of local

champions and local losers in all 4-companies, so $\sum_{i=1}^{n}\binom{w_i}{3} \geq \sum_{i=1}^{n}\binom{\ell_i}{3}$. Hence,

$$\sum_{i=1}^{n}\left(\binom{w_i}{3} - \binom{\ell_i}{3}\right) \geq 0. \tag{6}$$

Now we establish the problem statement (1) as a linear combination of (4), (5) and (6). It is easy check that

$$(x-y)^3 = 24\left(\binom{x}{3} - \binom{y}{3}\right) + 24\left(\binom{x}{2} - \binom{y}{2}\right) - \big(3(x+y)^2 - 4\big)(x-y).$$

Apply this identity to $x = w_1$ and $y = \ell_i$. Since every player played $n-1$ games, we have $w_i + \ell_i = n-1$, and thus

$$(w_i-\ell_i)^3 = 24\left(\binom{w_i}{3} - \binom{\ell_i}{3}\right) + 24\left(\binom{w_i}{2} - \binom{\ell_i}{2}\right) - \big(3(n-1)^2 - 4\big)\Big(w_i - \ell_i\Big).$$

Then

$$\sum_{i=1}^{n}(w_i-\ell_i)^3 = 24\underbrace{\sum_{i=1}^{n}\left(\binom{w_i}{3} - \binom{\ell_i}{3}\right)}_{\geq 0} + 24\underbrace{\sum_{i=1}^{n}\left(\binom{w_i}{2} - \binom{\ell_i}{2}\right)}_{0} - \big(3(n-1)^2-4\big)\underbrace{\sum_{i=1}^{n}\Big(w_i - \ell_i\Big)}_{0} \geq 0.$$

C6. Given a positive integer k and other two integers $b > w > 1$. There are two strings of pearls, a string of b black pearls and a string of w white pearls. The *length* of a string is the number of pearls on it.

One cuts these strings in some steps by the following rules. In each step:

(i) The strings are ordered by their lengths in a non-increasing order. If there are some strings of equal lengths, then the white ones precede the black ones. Then k first ones (if they consist of more than one pearl) are chosen; if there are less than k strings longer than 1, then one chooses all of them.

(ii) Next, one cuts each chosen string into two parts differing in length by at most one.

(For instance, if there are strings of 5, 4, 4, 2 black pearls, strings of 8, 4, 3 white pearls and $k = 4$, then the strings of 8 white, 5 black, 4 white and 4 black pearls are cut into the parts $(4, 4)$, $(3, 2)$, $(2, 2)$ and $(2, 2)$, respectively.)

The process stops immediately after the step when a first isolated white pearl appears. Prove that at this stage, there will still exist a string of at least two black pearls.

(Canada)

Solution 1. Denote the situation after the ith step by A_i; hence A_0 is the initial situation, and $A_{i-1} \to A_i$ is the ith step. We call a string containing m pearls an m-string; it is an *m-w-string* or a *m-b-string* if it is white or black, respectively.

We continue the process until every string consists of a single pearl. We will focus on three moments of the process: (a) the first stage A_s when the first 1-string (no matter black or white) appears; (b) the first stage A_t where the total number of strings is greater than k (if such moment does not appear then we put $t = \infty$); and (c) the first stage A_f when all black pearls are isolated. It is sufficient to prove that in A_{f-1} (or earlier), a 1-w-string appears.

We start with some easy properties of the situations under consideration. Obviously, we have $s \le f$. Moreover, all b-strings from A_{f-1} become single pearls in the fth step, hence all of them are 1- or 2-b-strings.

Next, observe that in each step $A_i \to A_{i+1}$ with $i \le t - 1$, all (>1)-strings were cut since there are not more than k strings at all; if, in addition, $i < s$, then there were no 1-string, so all the strings were cut in this step.

Now, let B_i and b_i be the lengths of the longest and the shortest b-strings in A_i, and let W_i and w_i be the same for w-strings. We show by induction on $i \le \min\{s, t\}$ that (i) the situation A_i contains exactly 2^i black and 2^i white strings, (ii) $B_i \ge W_i$, and (iii) $b_i \ge w_i$. The base case $i = 0$ is obvious. For the induction step, if $i \le \min\{s, t\}$ then in the ith step, each string is cut, thus the claim (i) follows from the induction hypothesis; next, we have $B_i = \lceil B_{i-1}/2 \rceil \ge \lceil W_{i-1}/2 \rceil = W_i$ and $b_i = \lfloor b_{i-1}/2 \rfloor \ge \lfloor w_{i-1}/2 \rfloor = w_i$, thus establishing (ii) and (iii).

For the numbers s, t, f, two cases are possible.

Case 1. Suppose that $s \le t$ or $f \le t + 1$ (and hence $s \le t + 1$); in particular, this is true when $t = \infty$. Then in A_{s-1} we have $B_{s-1} \ge W_{s-1}$, $b_{s-1} \ge w_{s-1} > 1$ as $s - 1 \le \min\{s, t\}$. Now, if $s = f$, then in A_{s-1}, there is no 1-w-string as well as no (>2)-b-string. That is, $2 = B_{s-1} \ge W_{s-1} \ge b_{s-1} \ge w_{s-1} > 1$, hence all these numbers equal 2. This means that in A_{s-1}, all strings contain 2 pearls, and there are 2^{s-1} black and 2^{s-1} white strings, which means $b = 2 \cdot 2^{s-1} = w$. This contradicts the problem conditions.

Hence we have $s \le f - 1$ and thus $s \le t$. Therefore, in the sth step each string is cut into two parts. Now, if a 1-b-string appears in this step, then from $w_{s-1} \le b_{s-1}$ we see that a

1-w-string appears as well; so, in each case in the sth step a 1-w-string appears, while not all black pearls become single, as desired.

Case 2. Now assume that $t+1 \le s$ and $t+2 \le f$. Then in A_t we have exactly 2^t white and 2^t black strings, all being larger than 1, and $2^{t+1} > k \ge 2^t$ (the latter holds since 2^t is the total number of strings in A_{t-1}). Now, in the $(t+1)$st step, exactly k strings are cut, not more than 2^t of them being black; so the number of w-strings in A_{t+1} is at least $2^t + (k - 2^t) = k$. Since the number of w-strings does not decrease in our process, in A_{f-1} we have at least k white strings as well.

Finally, in A_{f-1}, all b-strings are not larger than 2, and at least one 2-b-string is cut in the fth step. Therefore, at most $k-1$ white strings are cut in this step, hence there exists a w-string $\mathcal{W}$ which is not cut in the fth step. On the other hand, since a 2-b-string is cut, all ($\ge$2)-w-strings should also be cut in the fth step; hence $\mathcal{W}$ should be a single pearl. This is exactly what we needed.

Comment. In this solution, we used the condition $b \ne w$ only to avoid the case $b = w = 2^t$. Hence, if a number $b = w$ is not a power of 2, then the problem statement is also valid.

Solution 2. We use the same notations as introduced in the first paragraph of the previous solution. We claim that at every stage, there exist a u-b-string and a v-w-string such that either

(i) $u > v \ge 1$, or

(ii) $2 \le u \le v < 2u$, and there also exist $k-1$ of ($>v/2$)-strings other than considered above.

First, we notice that this statement implies the problem statement. Actually, in both cases (i) and (ii) we have $u > 1$, so at each stage there exists a ($\ge$2)-b-string, and for the last stage it is exactly what we need.

Now, we prove the claim by induction on the number of the stage. Obviously, for A_0 the condition (i) holds since $b > w$. Further, we suppose that the statement holds for A_i, and prove it for A_{i+1}. Two cases are possible.

Case 1. Assume that in A_i, there are a u-b-string and a v-w-string with $u > v$. We can assume that v is the length of the shortest w-string in A_i; since we are not at the final stage, we have $v \ge 2$. Now, in the $(i+1)$st step, two subcases may occur.

Subcase 1a. Suppose that either no u-b-string is cut, or both some u-b-string and some v-w-string are cut. Then in A_{i+1}, we have either a u-b-string and a ($\le v$)-w-string (and (i) is valid), or we have a $\lceil u/2 \rceil$-b-string and a $\lfloor v/2 \rfloor$-w-string. In the latter case, from $u > v$ we get $\lceil u/2 \rceil > \lfloor v/2 \rfloor$, and (i) is valid again.

Subcase 1b. Now, some u-b-string is cut, and no v-w-string is cut (and hence all the strings which are cut are longer than v). If $u' = \lceil u/2 \rceil > v$, then the condition (i) is satisfied since we have a u'-b-string and a v-w-string in A_{i+1}. Otherwise, notice that the inequality $u > v \ge 2$ implies $u' \ge 2$. Furthermore, besides a fixed u-b-string, other $k-1$ of ($\ge v+1$)-strings should be cut in the $(i+1)$st step, hence providing at least $k-1$ of ($\ge \lceil (v+1)/2 \rceil$)-strings, and $\lceil (v+1)/2 \rceil > v/2$. So, we can put $v' = v$, and we have $u' \le v < u \le 2u'$, so the condition (ii) holds for A_{i+1}.

Case 2. Conversely, assume that in A_i there exist a u-b-string, a v-w-string ($2 \le u \le v < 2u$) and a set S of $k-1$ other strings larger than $v/2$ (and hence larger than 1). In the $(i+1)$st step, three subcases may occur.

Subcase 2a. Suppose that some u-b-string is not cut, and some v-w-string is cut. The latter one results in a $\lfloor v/2 \rfloor$-w-string, we have $v' = \lfloor v/2 \rfloor < u$, and the condition (i) is valid.

Subcase 2b. Next, suppose that no v-w-string is cut (and therefore no u-b-string is cut as $u \le v$). Then all k strings which are cut have the length $> v$, so each one results in a $(>v/2)$-string. Hence in A_{i+1}, there exist $k \ge k-1$ of $(>v/2)$-strings other than the considered u- and v-strings, and the condition (ii) is satisfied.

Subcase 2c. In the remaining case, all u-b-strings are cut. This means that all $(\ge u)$-strings are cut as well, hence our v-w-string is cut. Therefore in A_{i+1} there exists a $\lceil u/2 \rceil$-b-string together with a $\lfloor v/2 \rfloor$-w-string. Now, if $u' = \lceil u/2 \rceil > \lfloor v/2 \rfloor = v'$ then the condition (i) is fulfilled. Otherwise, we have $u' \le v' < u \le 2u'$. In this case, we show that $u' \ge 2$. If, to the contrary, $u' = 1$ (and hence $u = 2$), then all black and white (≥ 2)-strings should be cut in the $(i+1)$st step, and among these strings there are at least a u-b-string, a v-w-string, and $k-1$ strings in S ($k+1$ strings altogether). This is impossible.

Hence, we get $2 \le u' \le v' < 2u'$. To reach (ii), it remains to check that in A_{i+1}, there exists a set S' of $k-1$ other strings larger than $v'/2$. These will be exactly the strings obtained from the elements of S. Namely, each $s \in S$ was either cut in the $(i+1)$st step, or not. In the former case, let us include into S' the largest of the strings obtained from s; otherwise we include s itself into S'. All $k-1$ strings in S' are greater than $v/2 \ge v'$, as desired.

C7. Let $P_1, \ldots, P_s$ be arithmetic progressions of integers, the following conditions being satisfied:

(i) each integer belongs to at least one of them;

(ii) each progression contains a number which does not belong to other progressions.

Denote by n the least common multiple of steps of these progressions; let $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k}$ be its prime factorization. Prove that

$$s \geq 1 + \sum_{i=1}^{k} \alpha_i (p_i - 1).$$

(Germany)

Solution 1. First, we prove the key lemma, and then we show how to apply it to finish the solution.

Let $n_1, \ldots, n_k$ be positive integers. By an $n_1 \times n_2 \times \cdots \times n_k$ *grid* we mean the set $N = \{(a_1, \ldots, a_k) : a_i \in \mathbb{Z},\ 0 \leq a_i \leq n_i - 1\}$; the elements of N will be referred to as *points*. In this grid, we define *a subgrid* as a subset of the form

$$L = \{(b_1, \ldots, b_k) \in N : b_{i_1} = x_{i_1}, \ldots, b_{i_t} = x_{i_t}\}, \tag{1}$$

where $I = \{i_1, \ldots, i_t\}$ is an arbitrary nonempty set of indices, and $x_{i_j} \in [0, n_{i_j} - 1]$ $(1 \leq j \leq t)$ are fixed integer numbers. Further, we say that a subgrid (1) is *orthogonal* to the ith coordinate axis if $i \in I$, and that it is *parallel* to the ith coordinate axis otherwise.

Lemma. Assume that the grid N is covered by subgrids $L_1, L_2, \ldots, L_s$ (this means $N = \bigcup_{i=1}^{s} L_i$) so that

(ii′) each subgrid contains a point which is not covered by other subgrids;

(iii) for each coordinate axis, there exists a subgrid L_i orthogonal to this axis.

Then

$$s \geq 1 + \sum_{i=1}^{k} (n_i - 1).$$

Proof. Assume to the contrary that $s \leq \sum_i (n_i - 1) = s'$. Our aim is to find a point that is not covered by $L_1, \ldots, L_s$.

The idea of the proof is the following. Imagine that we expand each subgrid to some maximal subgrid so that for the ith axis, there will be at most $n_i - 1$ maximal subgrids orthogonal to this axis. Then the desired point can be found easily: its ith coordinate should be that not covered by the maximal subgrids orthogonal to the ith axis. Surely, the conditions for existence of such expansion are provided by Hall's lemma on matchings. So, we will follow this direction, although we will apply Hall's lemma to some subgraph instead of the whole graph.

Construct a bipartite graph $G = (V \cup V', E)$ as follows. Let $V = \{L_1, \ldots, L_s\}$, and let $V' = \{v_{ij} : 1 \leq i \leq s,\ 1 \leq j \leq n_i - 1\}$ be some set of s' elements. Further, let the edge (L_m, v_{ij}) appear iff L_m is orthogonal to the ith axis.

For each subset $W \subset V$, denote

$$f(W) = \{v \in V' : (L, v) \in E \text{ for some } L \in W\}.$$

Notice that $f(V) = V'$ by (iii).

Now, consider the set $W \subset V$ containing the maximal number of elements such that $|W| > |f(W)|$; if there is no such set then we set $W = \varnothing$. Denote $W' = f(W)$, $U = V \backslash W$, $U' = V' \backslash W'$.

By our assumption and the Lemma condition, $|f(V)| = |V'| \geq |V|$, hence $W \neq V$ and $U \neq \varnothing$. Permuting the coordinates, we can assume that $U' = \{v_{ij} : 1 \leq i \leq \ell\}$, $W' = \{v_{ij} : \ell{+}1 \leq i \leq k\}$.

Consider the induced subgraph G' of G on the vertices $U \cup U'$. We claim that for every $X \subset U$, we get $|f(X) \cap U'| \geq |X|$ (so G' satisfies the conditions of Hall's lemma). Actually, we have $|W| \geq |f(W)|$, so if $|X| > |f(X) \cap U'|$ for some $X \subset U$, then we have

$$|W \cup X| = |W| + |X| > |f(W)| + |f(X) \cap U'| = |f(W) \cup (f(X) \cap U')| = |f(W \cup X)|.$$

This contradicts the maximality of $|W|$.

Thus, applying Hall's lemma, we can assign to each $L \in U$ some vertex $v_{ij} \in U'$ so that to distinct elements of U, distinct vertices of U' are assigned. In this situation, we say that $L \in U$ corresponds to the ith axis, and write $g(L) = i$. Since there are $n_i - 1$ vertices of the form v_{ij}, we get that for each $1 \leq i \leq \ell$, not more than $n_i - 1$ subgrids correspond to the ith axis.

Finally, we are ready to present the desired point. Since $W \neq V$, there exists a point $b = (b_1, b_2, \ldots, b_k) \in N \backslash (\cup_{L \in W} L)$. On the other hand, for every $1 \leq i \leq \ell$, consider any subgrid $L \in U$ with $g(L) = i$. This means exactly that L is orthogonal to the ith axis, and hence all its elements have the same ith coordinate c_L. Since there are at most $n_i - 1$ such subgrids, there exists a number $0 \leq a_i \leq n_i - 1$ which is not contained in a set $\{c_L : g(L) = i\}$. Choose such number for every $1 \leq i \leq \ell$. Now we claim that point $a = (a_1, \ldots, a_\ell, b_{\ell+1}, \ldots, b_k)$ is not covered, hence contradicting the Lemma condition.

Surely, point a cannot lie in some $L \in U$, since all the points in L have $g(L)$th coordinate $c_L \neq a_{g(L)}$. On the other hand, suppose that $a \in L$ for some $L \in W$; recall that $b \notin L$. But the points a and b differ only at first ℓ coordinates, so L should be orthogonal to at least one of the first ℓ axes, and hence our graph contains some edge (L, v_{ij}) for $i \leq \ell$. It contradicts the definition of W'. The Lemma is proved. □

Now we turn to the problem. Let d_j be the step of the progression P_j. Note that since $n = \text{l.c.m.}(d_1, \ldots, d_s)$, for each $1 \leq i \leq k$ there exists an index $j(i)$ such that $p_i^{\alpha_i} \mid d_{j(i)}$. We assume that $n > 1$; otherwise the problem statement is trivial.

For each $0 \leq m \leq n-1$ and $1 \leq i \leq k$, let m_i be the residue of m modulo $p_i^{\alpha_i}$, and let $m_i = \overline{r_{i\alpha_i} \ldots r_{i1}}$ be the base p_i representation of m_i (possibly, with some leading zeroes). Now, we put into correspondence to m the sequence $r(m) = (r_{11}, \ldots, r_{1\alpha_1}, r_{21}, \ldots, r_{k\alpha_k})$. Hence $r(m)$ lies in a $\underbrace{p_1 \times \cdots \times p_1}_{\alpha_1 \text{ times}} \times \cdots \times \underbrace{p_k \times \cdots \times p_k}_{\alpha_k \text{ times}}$ grid N.

Surely, if $r(m) = r(m')$ then $p_i^{\alpha_i} \mid m_i - m_i'$, which follows $p_i^{\alpha_i} \mid m - m'$ for all $1 \leq i \leq k$; consequently, $n \mid m - m'$. So, when m runs over the set $\{0, \ldots, n-1\}$, the sequences $r(m)$ do not repeat; since $|N| = n$, this means that r is a bijection between $\{0, \ldots, n-1\}$ and N. Now we will show that for each $1 \leq i \leq s$, the set $L_i = \{r(m) : m \in P_i\}$ is a subgrid, and that for each axis there exists a subgrid orthogonal to this axis. Obviously, these subgrids cover N, and the condition (ii′) follows directly from (ii). Hence the Lemma provides exactly the estimate we need.

Consider some $1 \leq j \leq s$ and let $d_j = p_1^{\gamma_1} \ldots p_k^{\gamma_k}$. Consider some $q \in P_j$ and let $r(q) = (r_{11}, \ldots, r_{k\alpha_k})$. Then for an arbitrary q', setting $r(q') = (r'_{11}, \ldots, r'_{k\alpha_k})$ we have

$$q' \in P_j \quad \Longleftrightarrow \quad p_i^{\gamma_i} \mid q - q' \text{ for each } 1 \leq i \leq k \quad \Longleftrightarrow \quad r_{i,t} = r'_{i,t} \text{ for all } t \leq \gamma_i.$$

Hence $L_j = \{(r'_{11}, \ldots, r'_{k\alpha_k}) \in N : r_{i,t} = r'_{i,t} \text{ for all } t \leq \gamma_i\}$ which means that L_j is a subgrid containing $r(q)$. Moreover, in $L_{j(i)}$, all the coordinates corresponding to p_i are fixed, so it is orthogonal to all of their axes, as desired.

Comment 1. The estimate in the problem is sharp for every n. One of the possible examples is the following one. For each $1 \le i \le k$, $0 \le j \le \alpha_i - 1$, $1 \le k \le p-1$, let

$$P_{i,j,k} = kp_i^j + p_i^{j+1}\mathbb{Z},$$

and add the progression $P_0 = n\mathbb{Z}$. One can easily check that this set satisfies all the problem conditions. There also exist other examples.

On the other hand, the estimate can be adjusted in the following sense. For every $1 \le i \le k$, let $0 = \alpha_{i0}, \alpha_{i1}, \ldots, \alpha_{ih_i}$ be all the numbers of the form $\mathrm{ord}_{p_i}(d_j)$ in an increasing order (we delete the repeating occurences of a number, and add a number $0 = \alpha_{i0}$ if it does not occur). Then, repeating the arguments from the solution one can obtain that

$$s \ge 1 + \sum_{i=1}^{k}\sum_{j=1}^{h_i}(p^{\alpha_j - \alpha_{j-1}} - 1).$$

Note that $p^\alpha - 1 \ge \alpha(p-1)$, and the equality is achieved only for $\alpha = 1$. Hence, for reaching the minimal number of the progressions, one should have $\alpha_{i,j} = j$ for all i, j. In other words, for each $1 \le j \le \alpha_i$, there should be an index t such that $\mathrm{ord}_{p_i}(d_t) = j$.

Solution 2. We start with introducing some notation. For positive integer r, we denote $[r] = \{1, 2, \ldots, r\}$. Next, we say that a set of progressions $\mathcal{P} = \{P_1, \ldots, P_s\}$ *cover* $\mathbb{Z}$ if each integer belongs to some of them; we say that this covering is *minimal* if no proper subset of $\mathcal{P}$ covers $\mathbb{Z}$. Obviously, each covering contains a minimal subcovering.

Next, for a minimal covering $\{P_1, \ldots, P_s\}$ and for every $1 \le i \le s$, let d_i be the step of progression P_i, and h_i be some number which is contained in P_i but in none of the other progressions. We assume that $n > 1$, otherwise the problem is trivial. This implies $d_i > 1$, otherwise the progression P_i covers all the numbers, and $n = 1$.

We will prove a more general statement, namely the following

Claim. Assume that the progressions $P_1, \ldots, P_s$ and number $n = p_1^{\alpha_1} \ldots p_k^{\alpha_k} > 1$ are chosen as in the problem statement. Moreover, choose some nonempty set of indices $I = \{i_1, \ldots, i_t\} \subseteq [k]$ and some positive integer $\beta_i \le \alpha_i$ for every $i \in I$. Consider the set of indices

$$T = \left\{ j : 1 \le j \le s, \text{and } p_i^{\alpha_i - \beta_i + 1} \;\middle|\; d_j \text{ for some } i \in I \right\}.$$

Then

$$|T| \ge 1 + \sum_{i \in I} \beta_i (p_i - 1). \tag{2}$$

Observe that the Claim for $I = [k]$ and $\beta_i = \alpha_i$ implies the problem statement, since the left-hand side in (2) is not greater than s. Hence, it suffices to prove the Claim.

1. First, we prove the Claim assuming that all d_j's are prime numbers. If for some $1 \le i \le k$ we have at least p_i progressions with the step p_i, then they do not intersect and hence cover all the integers; it means that there are no other progressions, and $n = p_i$; the Claim is trivial in this case.

Now assume that for every $1 \le i \le k$, there are not more than $p_i - 1$ progressions with step p_i; each such progression covers the numbers with a fixed residue modulo p_i, therefore there exists a residue $q_i \mod p_i$ which is not touched by these progressions. By the Chinese Remainder Theorem, there exists a number q such that $q \equiv q_i \pmod{p_i}$ for all $1 \le i \le k$; this number cannot be covered by any progression with step p_i, hence it is not covered at all. A contradiction.

2. Now, we assume that the general Claim is not valid, and hence we consider a counterexample $\{P_1, \ldots, P_s\}$ for the Claim; we can choose it to be minimal in the following sense:

• the number n is minimal possible among all the counterexamples;

• the sum $\sum_i d_i$ is minimal possible among all the counterexamples having the chosen value of n.

As was mentioned above, not all numbers d_i are primes; hence we can assume that d_1 is composite, say $p_1 \mid d_1$ and $d_1' = \frac{d_1}{p_1} > 1$. Consider a progression P_1' having the step d_1', and containing P_1. We will focus on two coverings constructed as follows.

(i) Surely, the progressions $P_1', P_2, \ldots, P_s$ cover $\mathbb{Z}$, though this covering in not necessarily minimal. So, choose some minimal subcovering $\mathcal{P}'$ in it; surely $P_1' \in \mathcal{P}'$ since h_1 is not covered by $P_2, \ldots, P_s$, so we may assume that $\mathcal{P}' = \{P_1', P_2, \ldots, P_{s'}\}$ for some $s' \le s$. Furthermore, the period of the covering $\mathcal{P}'$ can appear to be less than n; so we denote this period by

$$n' = p_1^{\alpha_1 - \sigma_1} \ldots p_k^{\alpha_k - \sigma_k} = \text{l.c.m.}\big(d_1', d_2, \ldots, d_{s'}\big).$$

Observe that for each $P_j \notin \mathcal{P}'$, we have $h_j \in P_1'$, otherwise h_j would not be covered by $\mathcal{P}$.

(ii) On the other hand, each nonempty set of the form $R_i = P_i \cap P_1'$ $(1 \le i \le s)$ is also a progression with a step $r_i = \text{l.c.m.}(d_i, d_1')$, and such sets cover P_1'. Scaling these progressions with the ratio $1/d_1'$, we obtain the progressions Q_i with steps $q_i = r_i/d_1'$ which cover $\mathbb{Z}$. Now we choose a minimal subcovering $\mathcal{Q}$ of this covering; again we should have $Q_1 \in \mathcal{Q}$ by the reasons of h_1. Now, denote the period of $\mathcal{Q}$ by

$$n'' = \text{l.c.m.}\{q_i : Q_i \in \mathcal{Q}\} = \frac{\text{l.c.m.}\{r_i : Q_i \in \mathcal{Q}\}}{d_1'} = \frac{p_1^{\gamma_1} \ldots p_k^{\gamma_k}}{d_1'}.$$

Note that if $h_j \in P_1'$, then the image of h_j under the scaling can be covered by Q_j only; so, in this case we have $Q_j \in \mathcal{Q}$.

Our aim is to find the desired number of progressions in coverings $\mathcal{P}$ and $\mathcal{Q}$. First, we have $n \ge n'$, and the sum of the steps in $\mathcal{P}'$ is less than that in $\mathcal{P}$; hence the Claim is valid for $\mathcal{P}'$. We apply it to the set of indices $I' = \{i \in I : \beta_i > \sigma_i\}$ and the exponents $\beta_i' = \beta_i - \sigma_i$; hence the set under consideration is

$$T' = \left\{ j : 1 \le j \le s', \text{and } p_i^{(\alpha_i - \sigma_i) - \beta_i' + 1} = p_i^{\alpha_i - \beta_i + 1} \mid d_j \text{ for some } i \in I' \right\} \subseteq T \cap [s'],$$

and we obtain that

$$|T \cap [s']| \ge |T'| \ge 1 + \sum_{i \in I'} (\beta_i - \sigma_i)(p_i - 1) = 1 + \sum_{i \in I} (\beta_i - \sigma_i)_+ (p_i - 1),$$

where $(x)_+ = \max\{x, 0\}$; the latter equality holds as for $i \notin I'$ we have $\beta_i \le \sigma_i$.

Observe that $x = (x - y)_+ + \min\{x, y\}$ for all x, y. So, if we find at least

$$G = \sum_{i \in I} \min\{\beta_i, \sigma_i\}(p_i - 1)$$

indices in $T \cap \{s' + 1, \ldots, s\}$, then we would have

$$|T| = |T \cap [s']| + |T \cap \{s' + 1, \ldots, s\}| \ge 1 + \sum_{i \in I} \big((\beta_i - \sigma_i)_+ + \min\{\beta_i, \sigma_i\} \big)(p_i - 1) = 1 + \sum_{i \in I} \beta_i (p_i - 1),$$

thus leading to a contradiction with the choice of $\mathcal{P}$. We will find those indices among the indices of progressions in $\mathcal{Q}$.

3. Now denote $I'' = \{i \in I : \sigma_i > 0\}$ and consider some $i \in I''$; then $p_i^{\alpha_i} \nmid n'$. On the other hand, there exists an index $j(i)$ such that $p_i^{\alpha_i} \mid d_{j(i)}$; this means that $d_{j(i)} \nmid n'$ and hence $P_{j(i)}$ cannot appear in $\mathcal{P}'$, so $j(i) > s'$. Moreover, we have observed before that in this case $h_{j(i)} \in P_1'$, hence $Q_{j(i)} \in \mathcal{Q}$. This means that $q_{j(i)} \mid n''$, therefore $\gamma_i = \alpha_i$ for each $i \in I''$ (recall here that $q_i = r_i/d_1'$ and hence $d_{j(i)} \mid r_{j(i)} \mid d_1' n''$).

Let $d_1' = p_1^{\tau_1} \dots p_k^{\tau_k}$. Then $n'' = p_1^{\gamma_1 - \tau_1} \dots p_k^{\gamma_i - \tau_i}$. Now, if $i \in I''$, then for every β the condition $p_i^{(\gamma_i - \tau_i) - \beta + 1} \mid q_j$ is equivalent to $p_i^{\alpha_i - \beta + 1} \mid r_j$.

Note that $n'' \le n/d_1' < n$, hence we can apply the Claim to the covering $\mathcal{Q}$. We perform this with the set of indices I'' and the exponents $\beta_i'' = \min\{\beta_i, \sigma_i\} > 0$. So, the set under consideration is

$$\begin{aligned} T'' &= \left\{ j : Q_j \in \mathcal{Q}, \text{and } p_i^{(\gamma_i - \tau_i) - \min\{\beta_i, \sigma_i\} + 1} \mid q_j \text{ for some } i \in I'' \right\} \\ &= \left\{ j : Q_j \in \mathcal{Q}, \text{and } p_i^{\alpha_i - \min\{\beta_i, \sigma_i\} + 1} \mid r_j \text{ for some } i \in I'' \right\}, \end{aligned}$$

and we obtain $|T''| \ge 1 + G$. Finally, we claim that $T'' \subseteq T \cap \big(\{1\} \cup \{s'+1, \dots, s\}\big)$; then we will obtain $|T \cap \{s'+1, \dots, s\}| \ge G$, which is exactly what we need.

To prove this, consider any $j \in T''$. Observe first that $\alpha_i - \min\{\beta_i, \sigma_i\} + 1 > \alpha_i - \sigma_i \ge \tau_i$, hence from $p_i^{\alpha_i - \min\{\beta_i, \sigma_i\} + 1} \mid r_j = \text{l.c.m.}(d_1', d_j)$ we have $p_i^{\alpha_i - \min\{\beta_i, \sigma_i\} + 1} \mid d_j$, which means that $j \in T$. Next, the exponent of p_i in d_j is greater than that in n', which means that $P_j \notin \mathcal{P}'$. This may appear only if $j = 1$ or $j > s'$, as desired. This completes the proof.

Comment 2. A grid analogue of the Claim is also valid. It reads as following.

Claim. Assume that the grid N is covered by subgrids $L_1, L_2, \dots, L_s$ so that

(ii′) each subgrid contains a point which is not covered by other subgrids;

(iii) for each coordinate axis, there exists a subgrid L_i orthogonal to this axis.

Choose some set of indices $I = \{i_1, \dots, i_t\} \subset [k]$, and consider the set of indices

$$T = \{ j : 1 \le j \le s, \text{and } L_j \text{ is orthogonal to the } i\text{th axis for some } i \in I \}.$$

Then

$$|T| \ge 1 + \sum_{i \in I} (n_i - 1).$$

This Claim may be proved almost in the same way as in Solution 1.

43

Geometry

G1. Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P. The lines BP and DF meet at point Q. Prove that $AP = AQ$.

(United Kingdom)

Solution 1. The line EF intersects the circumcircle at two points. Depending on the choice of P, there are two different cases to consider.

Case 1: The point P lies on the ray EF (see Fig. 1).

Let $\angle CAB = \alpha$, $\angle ABC = \beta$ and $\angle BCA = \gamma$. The quadrilaterals $BCEF$ and $CAFD$ are cyclic due to the right angles at D, E and F. So,

$$\angle BDF = 180^\circ - \angle FDC = \angle CAF = \alpha,$$
$$\angle AFE = 180^\circ - \angle EFB = \angle BCE = \gamma,$$
$$\angle DFB = 180^\circ - \angle AFD = \angle DCA = \gamma.$$

Since P lies on the arc AB of the circumcircle, $\angle PBA < \angle BCA = \gamma$. Hence, we have

$$\angle PBD + \angle BDF = \angle PBA + \angle ABD + \angle BDF < \gamma + \beta + \alpha = 180^\circ,$$

and the point Q must lie on the extensions of BP and DF beyond the points P and F, respectively.

From the cyclic quadrilateral $APBC$ we get

$$\angle QPA = 180^\circ - \angle APB = \angle BCA = \gamma = \angle DFB = \angle QFA.$$

Hence, the quadrilateral $AQPF$ is cyclic. Then $\angle AQP = 180^\circ - \angle PFA = \angle AFE = \gamma$.

We obtained that $\angle AQP = \angle QPA = \gamma$, so the triangle AQP is isosceles, $AP = AQ$.

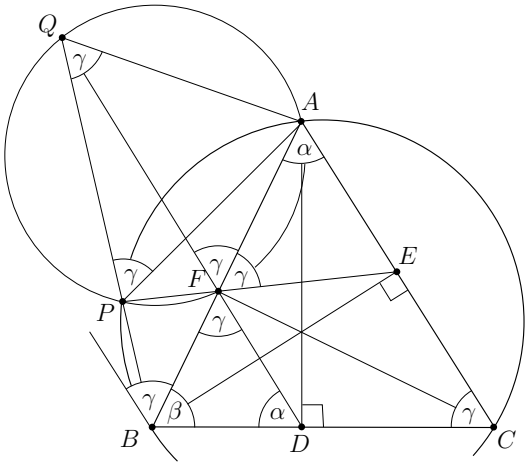


Fig. 1

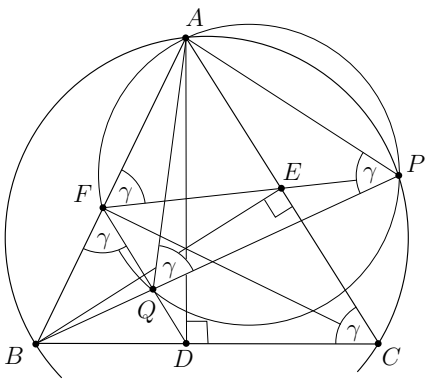


Fig. 2

Case 2: The point P lies on the ray FE (see Fig. 2). In this case the point Q lies inside the segment FD.

Similarly to the first case, we have

$$\angle QPA = \angle BCA = \gamma = \angle DFB = 180^\circ - \angle AFQ.$$

Hence, the quadrilateral $AFQP$ is cyclic.

Then $\angle AQP = \angle AFP = \angle AFE = \gamma = \angle QPA$. The triangle AQP is isosceles again, $\angle AQP = \angle QPA$ and thus $AP = AQ$.

Comment. Using signed angles, the two possible configurations can be handled simultaneously, without investigating the possible locations of P and Q.

Solution 2. For arbitrary points X, Y on the circumcircle, denote by $\widehat{XY}$ the central angle of the arc XY.

Let P and P' be the two points where the line EF meets the circumcircle; let P lie on the arc AB and let P' lie on the arc CA. Let BP and BP' meet the line DF and Q and Q', respectively (see Fig. 3). We will prove that $AP = AP' = AQ = AQ'$.

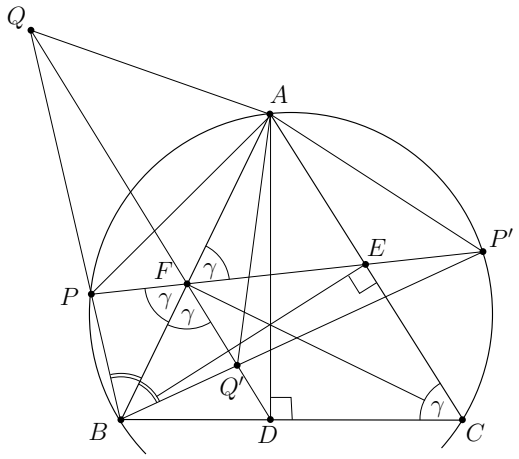


Fig. 3

Like in the first solution, we have $\angle AFE = \angle BFP = \angle DFB = \angle BCA = \gamma$ from the cyclic quadrilaterals $BCEF$ and $CAFD$.

By $\widehat{PB} + \widehat{P'A} = 2\angle AFP' = 2\gamma = 2\angle BCA = \widehat{AP} + \widehat{PB}$, we have

$$\widehat{AP} = \widehat{P'A}, \quad \angle PBA = \angle ABP' \qquad \text{and} \quad AP = AP'. \tag{1}$$

Due to $\widehat{AP} = \widehat{P'A}$, the lines BP and BQ' are symmetrical about line AB.

Similarly, by $\angle BFP = \angle Q'FB$, the lines FP and FQ' are symmetrical about AB. It follows that also the points P and P' are symmetrical to Q' and Q, respectively. Therefore,

$$AP = AQ' \quad \text{and} \quad AP' = AQ. \tag{2}$$

The relations (1) and (2) together prove $AP = AP' = AQ = AQ'$.

G2. Point P lies inside triangle ABC. Lines AP, BP, CP meet the circumcircle of ABC again at points K, L, M, respectively. The tangent to the circumcircle at C meets line AB at S. Prove that $SC = SP$ if and only if $MK = ML$.

(Poland)

Solution 1. We assume that $CA > CB$, so point S lies on the ray AB.

From the similar triangles $\triangle PKM \sim \triangle PCA$ and $\triangle PLM \sim \triangle PCB$ we get $\dfrac{PM}{KM} = \dfrac{PA}{CA}$ and $\dfrac{LM}{PM} = \dfrac{CB}{PB}$. Multiplying these two equalities, we get

$$\frac{LM}{KM} = \frac{CB}{CA} \cdot \frac{PA}{PB}.$$

Hence, the relation $MK = ML$ is equivalent to $\dfrac{CB}{CA} = \dfrac{PB}{PA}$.

Denote by E the foot of the bisector of angle B in triangle ABC. Recall that the locus of points X for which $\dfrac{XA}{XB} = \dfrac{CA}{CB}$ is the Apollonius circle Ω with the center Q on the line AB, and this circle passes through C and E. Hence, we have $MK = ML$ if and only if P lies on Ω, that is $QP = QC$.

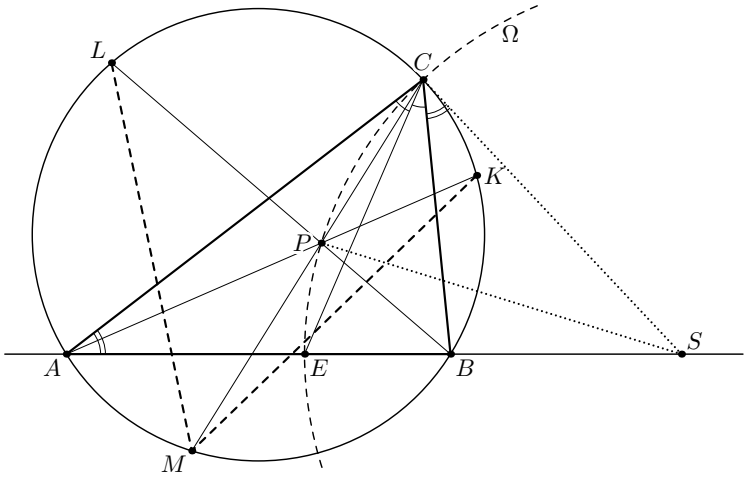


Fig. 1

Now we prove that $S = Q$, thus establishing the problem statement. We have $\angle CES = \angle CAE + \angle ACE = \angle BCS + \angle ECB = \angle ECS$, so $SC = SE$. Hence, the point S lies on AB as well as on the perpendicular bisector of CE and therefore coincides with Q.

Solution 2. As in the previous solution, we assume that S lies on the ray AB.

1. Let P be an arbitrary point inside both the circumcircle ω of the triangle ABC and the angle ASC, the points K, L, M defined as in the problem. We claim that $SP = SC$ implies $MK = ML$.

Let E and F be the points of intersection of the line SP with ω, point E lying on the segment SP (see Fig. 2).

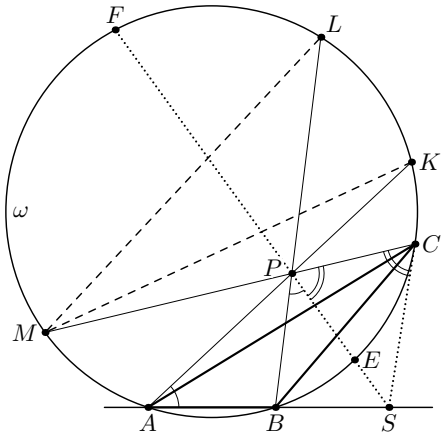


Fig. 2

We have $SP^2 = SC^2 = SA \cdot SB$, so $\dfrac{SP}{SB} = \dfrac{SA}{SP}$, and hence $\triangle PSA \sim \triangle BSP$. Then $\angle BPS = \angle SAP$. Since $2\angle BPS = \widehat{BE} + \widehat{LF}$ and $2\angle SAP = \widehat{BE} + \widehat{EK}$ we have

$$\widehat{LF} = \widehat{EK}. \tag{1}$$

On the other hand, from $\angle SPC = \angle SCP$ we have $\widehat{EC} + \widehat{MF} = \widehat{EC} + \widehat{EM}$, or

$$\widehat{MF} = \widehat{EM}. \tag{2}$$

From (1) and (2) we get $\widehat{MFL} = \widehat{MF} + \widehat{FL} = \widehat{ME} + \widehat{EK} = \widehat{MEK}$ and hence $MK = ML$. The claim is proved.

2. We are left to prove the converse. So, assume that $MK = ML$, and introduce the points E and F as above. We have $SC^2 = SE \cdot SF$; hence, there exists a point P' lying on the segment EF such that $SP' = SC$ (see Fig. 3).

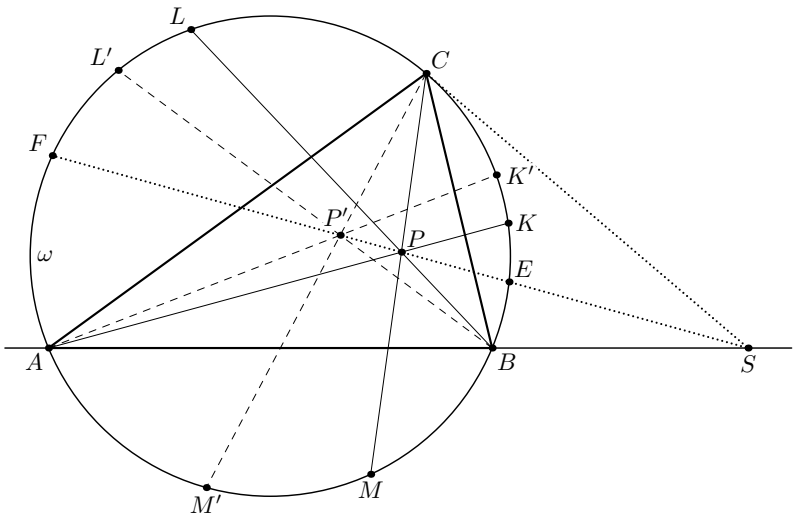


Fig. 3

Assume that $P \neq P'$. Let the lines AP', BP', CP' meet ω again at points K', L', M' respectively. Now, if P' lies on the segment PF then by the first part of the solution we have $\widehat{M'FL'} = \widehat{M'EK'}$. On the other hand, we have $\widehat{MFL} > \widehat{M'FL'} = \widehat{M'EK'} > \widehat{MEK}$, therefore $\widehat{MFL} > \widehat{MEK}$ which contradicts $MK = ML$.

Similarly, if point P' lies on the segment EP then we get $\widehat{MFL} < \widehat{MEK}$ which is impossible. Therefore, the points P and P' coincide and hence $SP = SP' = SC$.

Solution 3. We present a different proof of the converse direction, that is, $MK = ML \Rightarrow SP = SC$. As in the previous solutions we assume that $CA > CB$, and the line SP meets ω at E and F.

From $ML = MK$ we get $\widehat{MEK} = \widehat{MFL}$. Now we claim that $\widehat{ME} = \widehat{MF}$ and $\widehat{EK} = \widehat{FL}$.

To the contrary, suppose first that $\widehat{ME} > \widehat{MF}$; then $\widehat{EK} = \widehat{MEK} - \widehat{ME} < \widehat{MFL} - \widehat{MF} = \widehat{FL}$. Now, the inequality $\widehat{ME} > \widehat{MF}$ implies $2\angle SCM = \widehat{EC} + \widehat{ME} > \widehat{EC} + \widehat{MF} = 2\angle SPC$ and hence $SP > SC$. On the other hand, the inequality $\widehat{EK} < \widehat{FL}$ implies $2\angle SPK = \widehat{EK} + \widehat{AF} < \widehat{FL} + \widehat{AF} = 2\angle ABL$, hence

$$\angle SPA = 180^\circ - \angle SPK > 180^\circ - \angle ABL = \angle SBP.$$

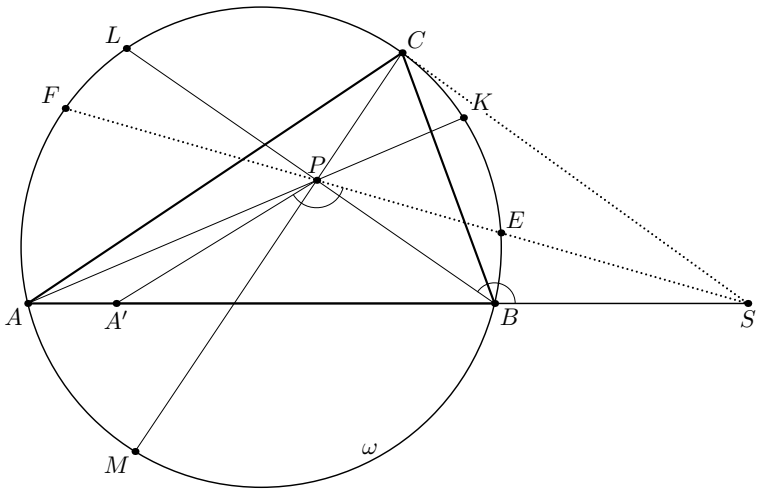


Fig. 4

Consider the point A' on the ray SA for which $\angle SPA' = \angle SBP$; in our case, this point lies on the segment SA (see Fig. 4). Then $\triangle SBP \sim \triangle SPA'$ and $SP^2 = SB \cdot SA' < SB \cdot SA = SC^2$. Therefore, $SP < SC$ which contradicts $SP > SC$.

Similarly, one can prove that the inequality $\widehat{ME} < \widehat{MF}$ is also impossible. So, we get $\widehat{ME} = \widehat{MF}$ and therefore $2\angle SCM = \widehat{EC} + \widehat{ME} = \widehat{EC} + \widehat{MF} = 2\angle SPC$, which implies $SC = SP$.

49

G3. Let $A_1A_2\ldots A_n$ be a convex polygon. Point P inside this polygon is chosen so that its projections $P_1, \ldots, P_n$ onto lines $A_1A_2, \ldots, A_nA_1$ respectively lie on the sides of the polygon. Prove that for arbitrary points $X_1, \ldots, X_n$ on sides $A_1A_2, \ldots, A_nA_1$ respectively,

$$\max\left\{\frac{X_1X_2}{P_1P_2}, \ldots, \frac{X_nX_1}{P_nP_1}\right\} \geq 1.$$

(Armenia)

Solution 1. Denote $P_{n+1} = P_1$, $X_{n+1} = X_1$, $A_{n+1} = A_1$.

Lemma. Let point Q lies inside $A_1A_2\ldots A_n$. Then it is contained in at least one of the circumcircles of triangles $X_1A_2X_2, \ldots, X_nA_1X_1$.

Proof. If Q lies in one of the triangles $X_1A_2X_2, \ldots, X_nA_1X_1$, the claim is obvious. Otherwise Q lies inside the polygon $X_1X_2\ldots X_n$ (see Fig. 1). Then we have

$$\begin{aligned}&(\angle X_1A_2X_2 + \angle X_1QX_2) + \cdots + (\angle X_nA_1X_1 + \angle X_nQX_1)\\ &\quad = (\angle X_1A_1X_2 + \cdots + \angle X_nA_1X_1) + (\angle X_1QX_2 + \cdots + \angle X_nQX_1) = (n-2)\pi + 2\pi = n\pi,\end{aligned}$$

hence there exists an index i such that $\angle X_iA_{i+1}X_{i+1} + \angle X_iQX_{i+1} \geq \frac{\pi n}{n} = \pi$. Since the quadrilateral $QX_iA_{i+1}X_{i+1}$ is convex, this means exactly that Q is contained the circumcircle of $\triangle X_iA_{i+1}X_{i+1}$, as desired. □

Now we turn to the solution. Applying lemma, we get that P lies inside the circumcircle of triangle $X_iA_{i+1}X_{i+1}$ for some i. Consider the circumcircles ω and Ω of triangles $P_iA_{i+1}P_{i+1}$ and $X_iA_{i+1}X_{i+1}$ respectively (see Fig. 2); let r and R be their radii. Then we get $2r = A_{i+1}P \leq 2R$ (since P lies inside Ω), hence

$$P_iP_{i+1} = 2r\sin\angle P_iA_{i+1}P_{i+1} \leq 2R\sin\angle X_iA_{i+1}X_{i+1} = X_iX_{i+1},$$

QED.

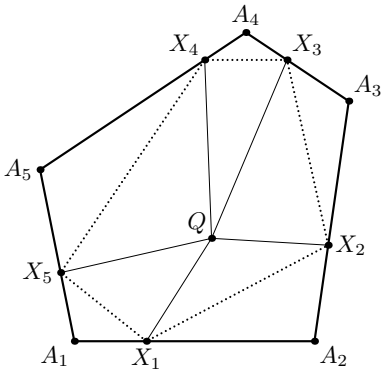


Fig. 1

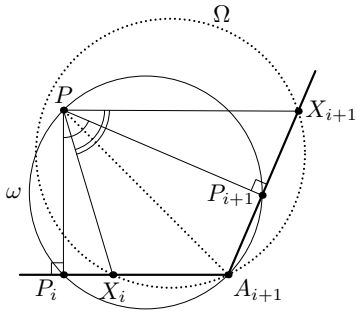


Fig. 2

Solution 2. As in Solution 1, we assume that all indices of points are considered modulo n.

We will prove a bit stronger inequality, namely

$$\max\left\{\frac{X_1X_2}{P_1P_2}\cos\alpha_1,\ldots,\frac{X_nX_1}{P_nP_1}\cos\alpha_n\right\}\geq 1,$$

where α_i $(1\leq i\leq n)$ is the angle between lines X_iX_{i+1} and P_iP_{i+1}. We denote $\beta_i=\angle A_iP_iP_{i-1}$ and $\gamma_i=\angle A_{i+1}P_iP_{i+1}$ for all $1\leq i\leq n$.

Suppose that for some $1\leq i\leq n$, point X_i lies on the segment A_iP_i, while point X_{i+1} lies on the segment $P_{i+1}A_{i+2}$. Then the projection of the segment X_iX_{i+1} onto the line P_iP_{i+1} contains segment P_iP_{i+1}, since γ_i and β_{i+1} are acute angles (see Fig. 3). Therefore, $X_iX_{i+1}\cos\alpha_i\geq P_iP_{i+1}$, and in this case the statement is proved.

So, the only case left is when point X_i lies on segment P_iA_{i+1} for all $1\leq i\leq n$ (the case when each X_i lies on segment A_iP_i is completely analogous).

Now, assume to the contrary that the inequality

$$X_iX_{i+1}\cos\alpha_i<P_iP_{i+1} \tag{1}$$

holds for every $1\leq i\leq n$. Let Y_i and Y'_{i+1} be the projections of X_i and X_{i+1} onto P_iP_{i+1}. Then inequality (1) means exactly that $Y_iY'_{i+1}<P_iP_{i+1}$, or $P_iY_i>P_{i+1}Y'_{i+1}$ (again since γ_i and β_{i+1} are acute; see Fig. 4). Hence, we have

$$X_iP_i\cos\gamma_i>X_{i+1}P_{i+1}\cos\beta_{i+1},\qquad 1\leq i\leq n.$$

Multiplying these inequalities, we get

$$\cos\gamma_1\cos\gamma_2\cdots\cos\gamma_n>\cos\beta_1\cos\beta_2\cdots\cos\beta_n. \tag{2}$$

On the other hand, the sines theorem applied to triangle PP_iP_{i+1} provides

$$\frac{PP_i}{PP_{i+1}}=\frac{\sin\left(\frac{\pi}{2}-\beta_{i+1}\right)}{\sin\left(\frac{\pi}{2}-\gamma_i\right)}=\frac{\cos\beta_{i+1}}{\cos\gamma_i}.$$

Multiplying these equalities we get

$$1=\frac{\cos\beta_2}{\cos\gamma_1}\cdot\frac{\cos\beta_3}{\cos\gamma_2}\cdots\frac{\cos\beta_1}{\cos\gamma_n}$$

which contradicts (2).

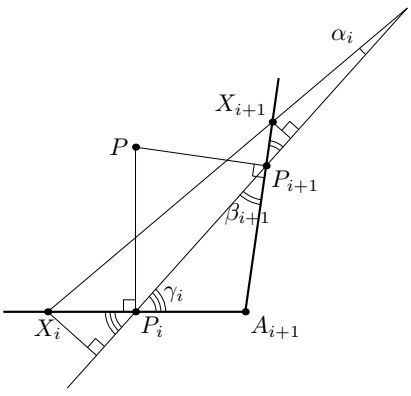


Fig. 3

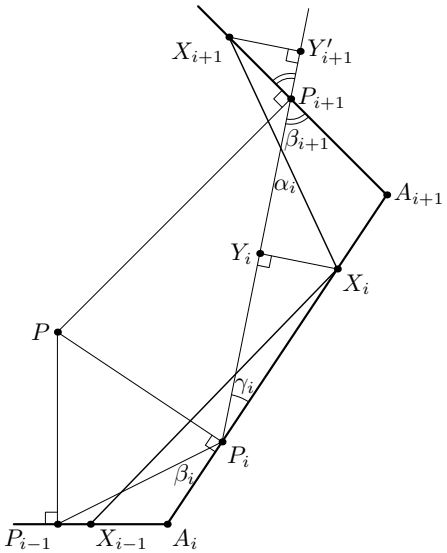


Fig. 4

G4. Let I be the incenter of a triangle ABC and Γ be its circumcircle. Let the line AI intersect Γ at a point $D \neq A$. Let F and E be points on side BC and arc BDC respectively such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. Finally, let G be the midpoint of the segment IF. Prove that the lines DG and EI intersect on Γ.

(Hong Kong)

Solution 1. Let X be the second point of intersection of line EI with Γ, and L be the foot of the bisector of angle BAC. Let G' and T be the points of intersection of segment DX with lines IF and AF, respectively. We are to prove that $G = G'$, or $IG' = G'F$. By the Menelaus theorem applied to triangle AIF and line DX, it means that we need the relation

$$1 = \frac{G'F}{IG'} = \frac{TF}{AT} \cdot \frac{AD}{ID}, \qquad \text{or} \quad \frac{TF}{AT} = \frac{ID}{AD}.$$

Let the line AF intersect Γ at point $K \neq A$ (see Fig. 1); since $\angle BAK = \angle CAE$ we have $\widehat{BK} = \widehat{CE}$, hence $KE \parallel BC$. Notice that $\angle IAT = \angle DAK = \angle EAD = \angle EXD = \angle IXT$, so the points I, A, X, T are concyclic. Hence we have $\angle ITA = \angle IXA = \angle EXA = \angle EKA$, so $IT \parallel KE \parallel BC$. Therefore we obtain $\dfrac{TF}{AT} = \dfrac{IL}{AI}$.

Since CI is the bisector of $\angle ACL$, we get $\dfrac{IL}{AI} = \dfrac{CL}{AC}$. Furthermore, $\angle DCL = \angle DCB = \angle DAB = \angle CAD = \frac{1}{2}\angle BAC$, hence the triangles DCL and DAC are similar; therefore we get $\dfrac{CL}{AC} = \dfrac{DC}{AD}$. Finally, it is known that the midpoint D of arc BC is equidistant from points I, B, C, hence $\dfrac{DC}{AD} = \dfrac{ID}{AD}$.

Summarizing all these equalities, we get

$$\frac{TF}{AT} = \frac{IL}{AI} = \frac{CL}{AC} = \frac{DC}{AD} = \frac{ID}{AD},$$

as desired.

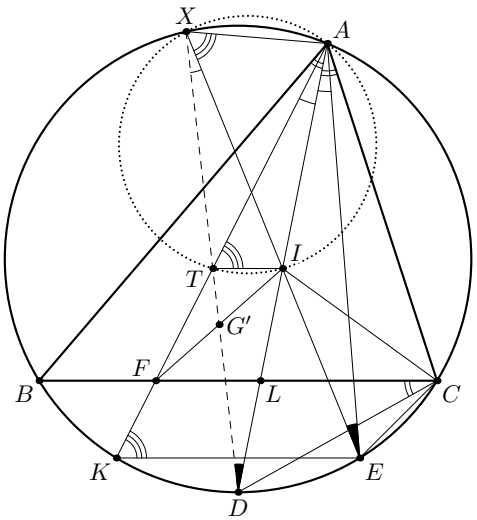


Fig. 1

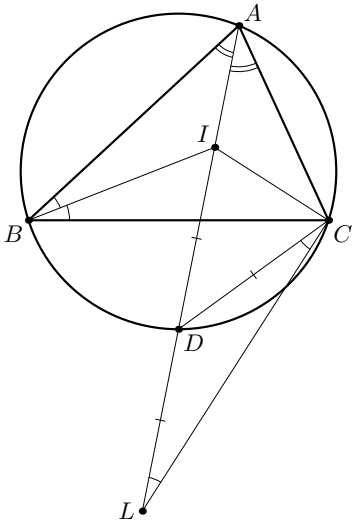


Fig. 2

Comment. The equality $\dfrac{AI}{IL} = \dfrac{AD}{DI}$ is known and can be obtained in many different ways. For instance, one can consider the inversion with center D and radius $DC = DI$. This inversion takes $\widehat{BAC}$ to the segment BC, so point A goes to L. Hence $\dfrac{IL}{DI} = \dfrac{AI}{AD}$, which is the desired equality.

Solution 2. As in the previous solution, we introduce the points X, T and K and note that it suffice to prove the equality

$$\frac{TF}{AT} = \frac{DI}{AD} \quad \Longleftrightarrow \quad \frac{TF + AT}{AT} = \frac{DI + AD}{AD} \quad \Longleftrightarrow \quad \frac{AT}{AD} = \frac{AF}{DI + AD}.$$

Since $\angle FAD = \angle EAI$ and $\angle TDA = \angle XDA = \angle XEA = \angle IEA$, we get that the triangles ATD and AIE are similar, therefore $\dfrac{AT}{AD} = \dfrac{AI}{AE}$.

Next, we also use the relation $DB = DC = DI$. Let J be the point on the extension of segment AD over point D such that $DJ = DI = DC$ (see Fig. 2). Then $\angle DJC = \angle JCD = \frac{1}{2}(\pi - \angle JDC) = \frac{1}{2}\angle ADC = \frac{1}{2}\angle ABC = \angle ABI$. Moreover, $\angle BAI = \angle JAC$, hence triangles ABI and AJC are similar, so $\dfrac{AB}{AJ} = \dfrac{AI}{AC}$, or $AB \cdot AC = AJ \cdot AI = (DI + AD) \cdot AI$.

On the other hand, we get $\angle ABF = \angle ABC = \angle AEC$ and $\angle BAF = \angle CAE$, so triangles ABF and AEC are also similar, which implies $\dfrac{AF}{AC} = \dfrac{AB}{AE}$, or $AB \cdot AC = AF \cdot AE$.

Summarizing we get

$$(DI + AD) \cdot AI = AB \cdot AC = AF \cdot AE \quad \Rightarrow \quad \frac{AI}{AE} = \frac{AF}{AD + DI} \quad \Rightarrow \quad \frac{AT}{AD} = \frac{AF}{AD + DI},$$

as desired.

Comment. In fact, point J is an excenter of triangle ABC.

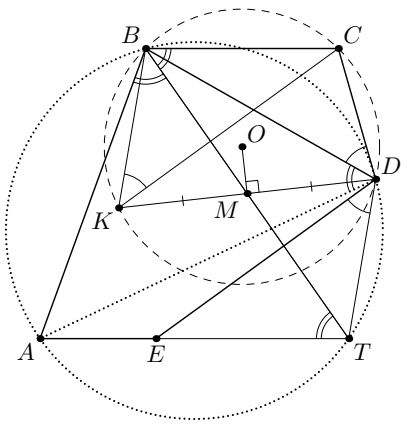
G5. Let $ABCDE$ be a convex pentagon such that $BC \parallel AE$, $AB = BC + AE$, and $\angle ABC = \angle CDE$. Let M be the midpoint of CE, and let O be the circumcenter of triangle BCD. Given that $\angle DMO = 90^\circ$, prove that $2\angle BDA = \angle CDE$.

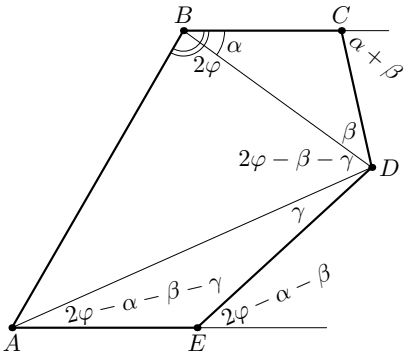
(Ukraine)

Solution 1. Choose point T on ray AE such that $AT = AB$; then from $AE \parallel BC$ we have $\angle CBT = \angle ATB = \angle ABT$, so BT is the bisector of $\angle ABC$. On the other hand, we have $ET = AT - AE = AB - AE = BC$, hence quadrilateral $BCTE$ is a parallelogram, and the midpoint M of its diagonal CE is also the midpoint of the other diagonal BT.

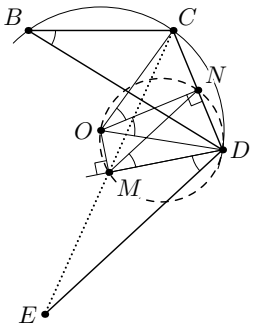
Next, let point K be symmetrical to D with respect to M. Then OM is the perpendicular bisector of segment DK, and hence $OD = OK$, which means that point K lies on the circumcircle of triangle BCD. Hence we have $\angle BDC = \angle BKC$. On the other hand, the angles BKC and TDE are symmetrical with respect to M, so $\angle TDE = \angle BKC = \angle BDC$.

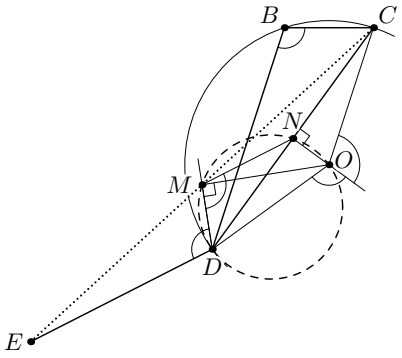
Therefore, $\angle BDT = \angle BDE + \angle EDT = \angle BDE + \angle BDC = \angle CDE = \angle ABC = 180^\circ - \angle BAT$. This means that the points A, B, D, T are concyclic, and hence $\angle ADB = \angle ATB = \frac{1}{2}\angle ABC = \frac{1}{2}\angle CDE$, as desired.





Solution 2. Let$\angle CBD = \alpha$, $\angle BDC = \beta$, $\angle ADE = \gamma$, and $\angle ABC = \angle CDE = 2\varphi$. Then we have $\angle ADB = 2\varphi - \beta - \gamma$, $\angle BCD = 180^\circ - \alpha - \beta$, $\angle AED = 360^\circ - \angle BCD - \angle CDE = 180^\circ - 2\varphi + \alpha + \beta$, and finally $\angle DAE = 180^\circ - \angle ADE - \angle AED = 2\varphi - \alpha - \beta - \gamma$.





Let N be the midpoint of CD; then $\angle DNO = 90^\circ = \angle DMO$, hence points M, N lie on the circle with diameter OD. Now, if points O and M lie on the same side of CD, we have $\angle DMN = \angle DON = \frac{1}{2}\angle DOC = \alpha$; in the other case, we have $\angle DMN = 180^\circ - \angle DON = \alpha$;

so, in both cases $\angle DMN = \alpha$ (see Figures). Next, since MN is a midline in triangle CDE, we have $\angle MDE = \angle DMN = \alpha$ and $\angle NDM = 2\varphi - \alpha$.

Now we apply the sine rule to the triangles ABD, ADE (twice), BCD and MND obtaining

$$\frac{AB}{AD} = \frac{\sin(2\varphi-\beta-\gamma)}{\sin(2\varphi-\alpha)}, \quad \frac{AE}{AD} = \frac{\sin\gamma}{\sin(2\varphi-\alpha-\beta)}, \quad \frac{DE}{AD} = \frac{\sin(2\varphi-\alpha-\beta-\gamma)}{\sin(2\varphi-\alpha-\beta)},$$
$$\frac{BC}{CD} = \frac{\sin\beta}{\sin\alpha}, \quad \frac{CD}{DE} = \frac{CD/2}{DE/2} = \frac{ND}{NM} = \frac{\sin\alpha}{\sin(2\varphi-\alpha)},$$

which implies

$$\frac{BC}{AD} = \frac{BC}{CD}\cdot\frac{CD}{DE}\cdot\frac{DE}{AD} = \frac{\sin\beta\cdot\sin(2\varphi-\alpha-\beta-\gamma)}{\sin(2\varphi-\alpha)\cdot\sin(2\varphi-\alpha-\beta)}.$$

Hence, the condition $AB = AE + BC$, or equivalently $\dfrac{AB}{AD} = \dfrac{AE+BC}{AD}$, after multiplying by the common denominator rewrites as

$$\begin{aligned}
&\sin(2\varphi-\alpha-\beta)\cdot\sin(2\varphi-\beta-\gamma) = \sin\gamma\cdot\sin(2\varphi-\alpha) + \sin\beta\cdot\sin(2\varphi-\alpha-\beta-\gamma)\\
\iff\ &\cos(\gamma-\alpha) - \cos(4\varphi-2\beta-\alpha-\gamma) = \cos(2\varphi-\alpha-2\beta-\gamma) - \cos(2\varphi+\gamma-\alpha)\\
\iff\ &\cos(\gamma-\alpha) + \cos(2\varphi+\gamma-\alpha) = \cos(2\varphi-\alpha-2\beta-\gamma) + \cos(4\varphi-2\beta-\alpha-\gamma)\\
\iff\ &\cos\varphi\cdot\cos(\varphi+\gamma-\alpha) = \cos\varphi\cdot\cos(3\varphi-2\beta-\alpha-\gamma)\\
\iff\ &\cos\varphi\cdot\big(\cos(\varphi+\gamma-\alpha) - \cos(3\varphi-2\beta-\alpha-\gamma)\big) = 0\\
\iff\ &\cos\varphi\cdot\sin(2\varphi-\beta-\alpha)\cdot\sin(\varphi-\beta-\gamma) = 0.
\end{aligned}$$

Since $2\varphi-\beta-\alpha = 180^\circ - \angle AED < 180^\circ$ and $\varphi = \frac{1}{2}\angle ABC < 90^\circ$, it follows that $\varphi = \beta+\gamma$, hence $\angle BDA = 2\varphi-\beta-\gamma = \varphi = \frac{1}{2}\angle CDE$, as desired.

G6. The vertices X, Y, Z of an equilateral triangle XYZ lie respectively on the sides BC, CA, AB of an acute-angled triangle ABC. Prove that the incenter of triangle ABC lies inside triangle XYZ.

G6′. The vertices X, Y, Z of an equilateral triangle XYZ lie respectively on the sides BC, CA, AB of a triangle ABC. Prove that if the incenter of triangle ABC lies outside triangle XYZ, then one of the angles of triangle ABC is greater than 120°.

(Bulgaria)

Solution 1 for G6. We will prove a stronger fact; namely, we will show that the incenter I of triangle ABC lies inside the incircle of triangle XYZ (and hence surely inside triangle XYZ itself). We denote by $d(U, VW)$ the distance between point U and line VW.

Denote by O the incenter of $\triangle XYZ$ and by r, r' and R' the inradii of triangles ABC, XYZ and the circumradius of XYZ, respectively. Then we have $R' = 2r'$, and the desired inequality is $OI \le r'$. We assume that $O \ne I$; otherwise the claim is trivial.

Let the incircle of $\triangle ABC$ touch its sides BC, AC, AB at points A_1, B_1, C_1 respectively. The lines IA_1, IB_1, IC_1 cut the plane into 6 acute angles, each one containing one of the points A_1, B_1, C_1 on its border. We may assume that O lies in an angle defined by lines IA_1, IC_1 and containing point C_1 (see Fig. 1). Let A' and C' be the projections of O onto lines IA_1 and IC_1, respectively.

Since $OX = R'$, we have $d(O, BC) \le R'$. Since $OA' \parallel BC$, it follows that $d(A', BC) = A'I + r \le R'$, or $A'I \le R' - r$. On the other hand, the incircle of $\triangle XYZ$ lies inside $\triangle ABC$, hence $d(O, AB) \ge r'$, and analogously we get $d(O, AB) = C'C_1 = r - IC' \ge r'$, or $IC' \le r - r'$.

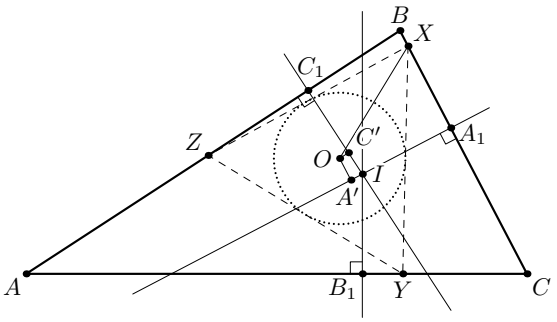


Fig. 1

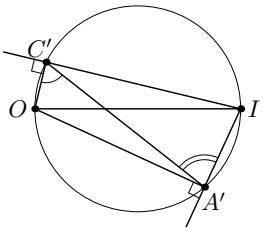


Fig. 2

Finally, the quadrilateral $IA'OC'$ is circumscribed due to the right angles at A' and C' (see Fig. 2). On its circumcircle, we have $\widehat{A'OC'} = 2\angle A'IC' < 180^\circ = \widehat{OC'I}$, hence $180^\circ \ge \widehat{IC'} > \widehat{A'O}$. This means that $IC' > A'O$. Finally, we have $OI \le IA' + A'O < IA' + IC' \le (R' - r) + (r - r') = R' - r' = r'$, as desired.

Solution 2 for G6. Assume the contrary. Then the incenter I should lie in one of triangles AYZ, BXZ, CXY — assume that it lies in $\triangle AYZ$. Let the incircle ω of $\triangle ABC$ touch sides BC, AC at point A_1, B_1 respectively. Without loss of generality, assume that point A_1 lies on segment CX. In this case we will show that $\angle C > 90^\circ$ thus leading to a contradiction.

Note that ω intersects each of the segments XY and YZ at two points; let U, U' and V, V' be the points of intersection of ω with XY and YZ, respectively ($UY > U'Y$, $VY > V'Y$; see Figs. 3 and 4). Note that $60^\circ = \angle XYZ = \frac{1}{2}(\widehat{UV} - \widehat{U'V'}) \le \frac{1}{2}\widehat{UV}$, hence $\widehat{UV} \ge 120^\circ$.

On the other hand, since I lies in $\triangle AYZ$, we get $\widehat{VUV'} < 180^\circ$, hence $\widehat{UA_1U'} \le \widehat{UA_1V'} < 180^\circ - \widehat{UV} \le 60^\circ$.

Now, two cases are possible due to the order of points Y, B_1 on segment AC.

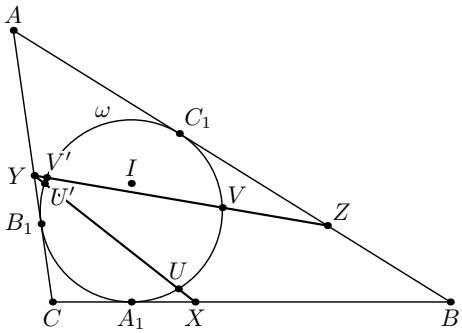


Fig. 3

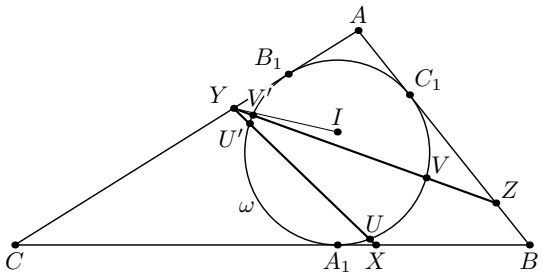


Fig. 4

Case 1. Let point Y lie on the segment AB_1 (see Fig. 3). Then we have $\angle YXC = \frac{1}{2}\left(\widehat{A_1U'} - \widehat{A_1U}\right) \le \frac{1}{2}\widehat{UA_1U'} < 30^\circ$; analogously, we get $\angle XYC \le \frac{1}{2}\widehat{UA_1U'} < 30^\circ$. Therefore, $\angle YCX = 180^\circ - \angle YXC - \angle XYC > 120^\circ$, as desired.

Case 2. Now let point Y lie on the segment CB_1 (see Fig. 4). Analogously, we obtain $\angle YXC < 30^\circ$. Next, $\angle IYX > \angle ZYX = 60^\circ$, but $\angle IYX < \angle IYB_1$, since YB_1 is a tangent and YX is a secant line to circle ω from point Y. Hence, we get $120^\circ < \angle IYB_1 + \angle IYX = \angle B_1YX = \angle YXC + \angle YCX < 30^\circ + \angle YCX$, hence $\angle YCX > 120^\circ - 30^\circ = 90^\circ$, as desired.

Comment. In the same way, one can prove a more general

Claim. Let the vertices X, Y, Z of a triangle XYZ lie respectively on the sides BC, CA, AB of a triangle ABC. Suppose that the incenter of triangle ABC lies outside triangle XYZ, and α is the least angle of $\triangle XYZ$. Then one of the angles of triangle ABC is greater than $3\alpha - 90^\circ$.

Solution for G6′. Assume the contrary. As in Solution 2, we assume that the incenter I of $\triangle ABC$ lies in $\triangle AYZ$, and the tangency point A_1 of ω and BC lies on segment CX. Surely, $\angle YZA \le 180^\circ - \angle YZX = 120^\circ$, hence points I and Y lie on one side of the perpendicular bisector to XY; therefore $IX > IY$. Moreover, ω intersects segment XY at two points, and therefore the projection M of I onto XY lies on the segment XY. In this case, we will prove that $\angle C > 120^\circ$.

Let YK, YL be two tangents from point Y to ω (points K and A_1 lie on one side of XY; if Y lies on ω, we say $K = L = Y$); one of the points K and L is in fact a tangency point B_1 of ω and AC. From symmetry, we have $\angle YIK = \angle YIL$. On the other hand, since $IX > IY$, we get $XM < XY$ which implies $\angle A_1XY < \angle KYX$.

Next, we have $\angle MIY = 90^\circ - \angle IYX < 90^\circ - \angle ZYX = 30^\circ$. Since $IA_1 \perp A_1X$, $IM \perp XY$, $IK \perp YK$ we get $\angle MIA_1 = \angle A_1XY < \angle KYX = \angle MIK$. Finally, we get

$$\begin{aligned}\angle A_1IK < \angle A_1IL &= (\angle A_1IM + \angle MIK) + (\angle KIY + \angle YIL)\\ &< 2\angle MIK + 2\angle KIY = 2\angle MIY < 60^\circ.\end{aligned}$$

Hence, $\angle A_1IB_1 < 60^\circ$, and therefore $\angle ACB = 180^\circ - \angle A_1IB_1 > 120^\circ$, as desired.

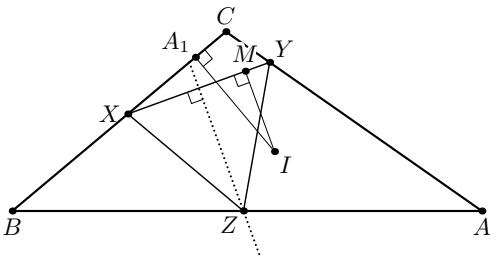


Fig. 5

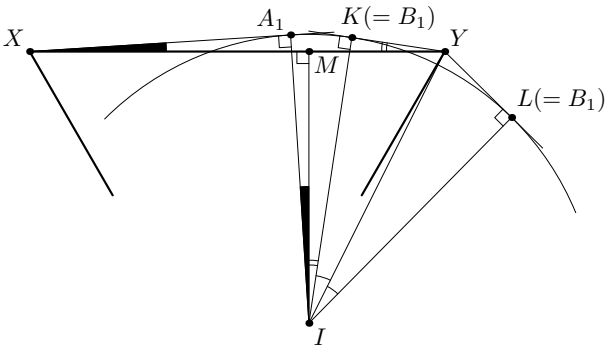


Fig. 6

Comment 1. The estimate claimed in G6′ is sharp. Actually, if $\angle BAC > 120^\circ$, one can consider an equilateral triangle XYZ with $Z = A$, $Y \in AC$, $X \in BC$ (such triangle exists since $\angle ACB < 60^\circ$). It intersects with the angle bisector of $\angle BAC$ only at point A, hence it does not contain I.

Comment 2. As in the previous solution, there is a generalization for an arbitrary triangle XYZ, but here we need some additional condition. The statement reads as follows.

Claim. Let the vertices X, Y, Z of a triangle XYZ lie respectively on the sides BC, CA, AB of a triangle ABC. Suppose that the incenter of triangle ABC lies outside triangle XYZ, α is the least angle of $\triangle XYZ$, and all sides of triangle XYZ are greater than $2r\cot\alpha$, where r is the inradius of $\triangle ABC$. Then one of the angles of triangle ABC is greater than 2α.

The additional condition is needed to verify that $XM > YM$ since it cannot be shown in the original way. Actually, we have $\angle MYI > \alpha$, $IM < r$, hence $YM < r\cot\alpha$. Now, if we have $XY = XM + YM > 2r\cot\alpha$, then surely $XM > YM$.

On the other hand, this additional condition follows easily from the conditions of the original problem. Actually, if $I \in \triangle AYZ$, then the diameter of ω parallel to YZ is contained in $\triangle AYZ$ and is thus shorter than YZ. Hence $YZ > 2r > 2r\cot 60^\circ$.

59

G7. Three circular arcs γ_1, γ_2, and γ_3 connect the points A and C. These arcs lie in the same half-plane defined by line AC in such a way that arc γ_2 lies between the arcs γ_1 and γ_3. Point B lies on the segment AC. Let h_1, h_2, and h_3 be three rays starting at B, lying in the same half-plane, h_2 being between h_1 and h_3. For $i, j = 1, 2, 3$, denote by V_{ij} the point of intersection of h_i and γ_j (see the Figure below).

Denote by $\overset{\frown}{V_{ij}V_{kj}}\overset{\frown}{V_{k\ell}V_{i\ell}}$ the curved quadrilateral, whose sides are the segments $V_{ij}V_{i\ell}$, $V_{kj}V_{k\ell}$ and arcs $V_{ij}V_{kj}$ and $V_{i\ell}V_{k\ell}$. We say that this quadrilateral is *circumscribed* if there exists a circle touching these two segments and two arcs.

Prove that if the curved quadrilaterals $\overset{\frown}{V_{11}V_{21}}\overset{\frown}{V_{22}V_{12}}$, $\overset{\frown}{V_{12}V_{22}}\overset{\frown}{V_{23}V_{13}}$, $\overset{\frown}{V_{21}V_{31}}\overset{\frown}{V_{32}V_{22}}$ are circumscribed, then the curved quadrilateral $\overset{\frown}{V_{22}V_{32}}\overset{\frown}{V_{33}V_{23}}$ is circumscribed, too.

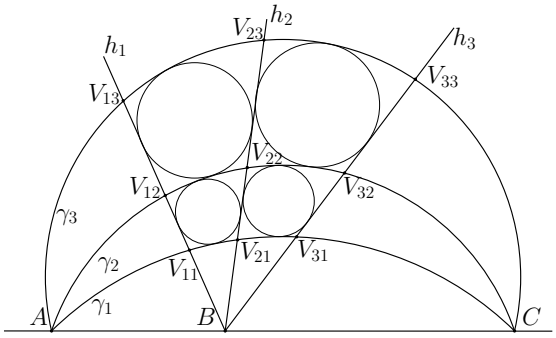


Fig. 1

(Hungary)

Solution. Denote by O_i and R_i the center and the radius of γ_i, respectively. Denote also by H the half-plane defined by AC which contains the whole configuration. For every point P in the half-plane H, denote by $d(P)$ the distance between P and line AC. Furthermore, for any $r > 0$, denote by $\Omega(P, r)$ the circle with center P and radius r.

Lemma 1. For every $1 \le i < j \le 3$, consider those circles $\Omega(P, r)$ in the half-plane H which are tangent to h_i and h_j.

(a) The locus of the centers of these circles is the angle bisector β_{ij} between h_i and h_j.

(b) There is a constant u_{ij} such that $r = u_{ij} \cdot d(P)$ for all such circles.

Proof. Part (a) is obvious. To prove part (b), notice that the circles which are tangent to h_i and h_j are homothetic with the common homothety center B (see Fig. 2). Then part (b) also becomes trivial. □

Lemma 2. For every $1 \le i < j \le 3$, consider those circles $\Omega(P, r)$ in the half-plane H which are externally tangent to γ_i and internally tangent to γ_j.

(a) The locus of the centers of these circles is an ellipse arc ε_{ij} with end-points A and C.

(b) There is a constant v_{ij} such that $r = v_{ij} \cdot d(P)$ for all such circles.

Proof. (a) Notice that the circle $\Omega(P, r)$ is externally tangent to γ_i and internally tangent to γ_j if and only if $O_iP = R_i + r$ and $O_j = R_j - r$. Therefore, for each such circle we have

$$O_iP + O_jP = O_iA + O_jA = O_iC + O_jC = R_i + R_j.$$

Such points lie on an ellipse with foci O_i and O_j; the diameter of this ellipse is $R_i + R_j$, and it passes through the points A and C. Let ε_{ij} be that arc AC of the ellipse which runs inside the half plane H (see Fig. 3.)

This ellipse arc lies between the arcs γ_i and γ_j. Therefore, if some point P lies on ε_{ij}, then $O_iP > R_i$ and $O_jP < R_j$. Now, we choose $r = O_iP - R_i = R_j - O_jP > 0$; then the

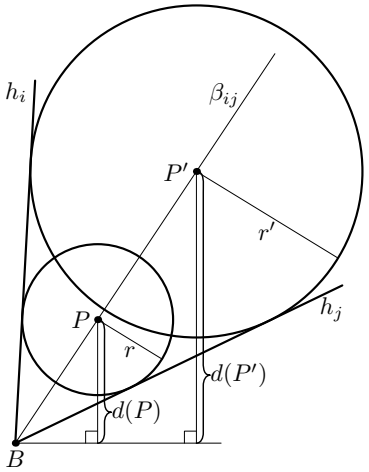


Fig. 2

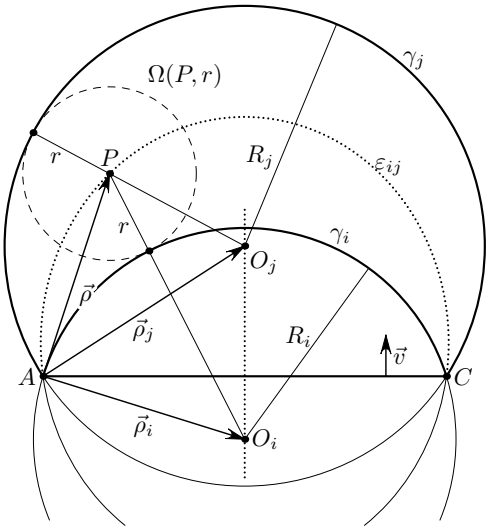


Fig. 3

circle $\Omega(P,r)$ touches γ_i externally and touches γ_j internally, so P belongs to the locus under investigation.

(b) Let $\vec{\rho} = \overrightarrow{AP}$, $\vec{\rho}_i = \overrightarrow{AO_i}$, and $\vec{\rho}_j = \overrightarrow{AO_j}$; let $d_{ij} = O_iO_j$, and let $\vec{v}$ be a unit vector orthogonal to AC and directed toward H. Then we have $|\vec{\rho}_i| = R_i$, $|\vec{\rho}_j| = R_j$, $|\overrightarrow{O_iP}| = |\vec{\rho} - \vec{\rho}_i| = R_i + r$, $|\overrightarrow{O_jP}| = |\vec{\rho} - \vec{\rho}_j| = R_j - r$, hence

$$\begin{gathered}(\vec{\rho} - \vec{\rho}_i)^2 - (\vec{\rho} - \vec{\rho}_j)^2 = (R_i + r)^2 - (R_j - r)^2,\\ (\vec{\rho}_i^2 - \vec{\rho}_j^2) + 2\vec{\rho} \cdot (\vec{\rho}_j - \vec{\rho}_i) = (R_i^2 - R_j^2) + 2r(R_i + R_j),\\ d_{ij} \cdot d(P) = d_{ij}\vec{v} \cdot \vec{\rho} = (\vec{\rho}_j - \vec{\rho}_i) \cdot \vec{\rho} = r(R_i + R_j).\end{gathered}$$

Therefore,

$$r = \frac{d_{ij}}{R_i + R_j} \cdot d(P),$$

and the value $v_{ij} = \dfrac{d_{ij}}{R_i + R_j}$ does not depend on P. □

Lemma 3. The curved quadrilateral $\mathcal{Q}_{ij} = V_{i,j}\overset{\frown}{V_{i+1,j}}V_{i+1,j}\overset{\frown}{V_{i+1,j+1}}V_{i,j+1}$ is circumscribed if and only if $u_{i,i+1} = v_{j,j+1}$.

Proof. First suppose that the curved quadrilateral $\mathcal{Q}_{ij}$ is circumscribed and $\Omega(P,r)$ is its inscribed circle. By Lemma 1 and Lemma 2 we have $r = u_{i,i+1} \cdot d(P)$ and $r = v_{j,j+1} \cdot d(P)$ as well. Hence, $u_{i,i+1} = v_{j,j+1}$.

To prove the opposite direction, suppose $u_{i,i+1} = v_{j,j+1}$. Let P be the intersection of the angle bisector $\beta_{i,i+1}$ and the ellipse arc $\varepsilon_{j,j+1}$. Choose $r = u_{i,i+1} \cdot d(P) = v_{j,j+1} \cdot d(P)$. Then the circle $\Omega(P,r)$ is tangent to the half lines h_i and h_{i+1} by Lemma 1, and it is tangent to the arcs γ_j and γ_{j+1} by Lemma 2. Hence, the curved quadrilateral $\mathcal{Q}_{ij}$ is circumscribed. □

By Lemma 3, the statement of the problem can be reformulated to an obvious fact: If the equalities $u_{12} = v_{12}$, $u_{12} = v_{23}$, and $u_{23} = v_{12}$ hold, then $u_{23} = v_{23}$ holds as well.

Comment 1. Lemma 2(b) (together with the easy Lemma 1(b)) is the key tool in this solution. If one finds this fact, then the solution can be finished in many ways. That is, one can find a circle touching three of h_2, h_3, γ_2, and γ_3, and then prove that it is tangent to the fourth one in either synthetic or analytical way. Both approaches can be successful.

Here we present some discussion about this key Lemma.

1. In the solution above we chose an analytic proof for Lemma 2(b) because we expect that most students will use coordinates or vectors to examine the locus of the centers, and these approaches are less case-sensitive.

Here we outline a synthetic proof. We consider only the case when P does not lie in the line O_iO_j. The other case can be obtained as a limit case, or computed in a direct way.

Let S be the internal homothety center between the circles of γ_i and γ_j, lying on O_iO_j; this point does not depend on P. Let U and V be the points of tangency of circle $\sigma = \Omega(P, r)$ with γ_i and γ_j, respectively (then $r = PU = PV$); in other words, points U and V are the intersection points of rays O_iP, O_jP with arcs γ_i, γ_j respectively (see Fig. 4).

Due to the theorem on three homothety centers (or just to the Menelaus theorem applied to triangle O_iO_jP), the points U, V and S are collinear. Let T be the intersection point of line AC and the common tangent to σ and γ_i at U; then T is the radical center of σ, γ_i and γ_j, hence TV is the common tangent to σ and γ_j.

Let Q be the projection of P onto the line AC. By the right angles, the points U, V and Q lie on the circle with diameter PT. From this fact and the equality $PU = PV$ we get $\angle UQP = \angle UVP = \angle VUP = \angle SUO_i$. Since $O_iS \parallel PQ$, we have $\angle SO_iU = \angle QPU$. Hence, the triangles SO_iU and UPQ are similar and thus $\dfrac{r}{d(P)} = \dfrac{PU}{PQ} = \dfrac{O_iS}{O_iU} = \dfrac{O_iS}{R_i}$; the last expression is constant since S is a constant point. □

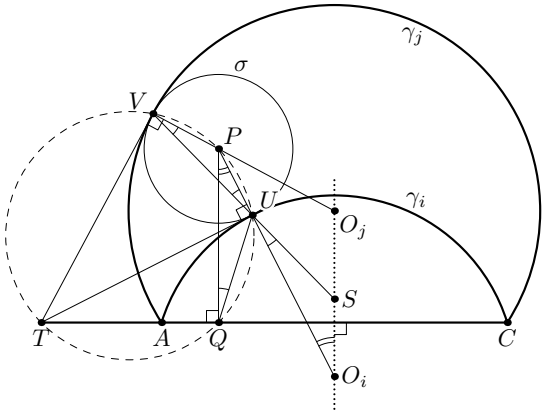


Fig. 4

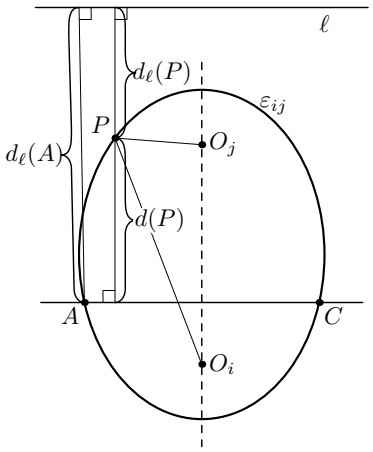


Fig. 5

2. Using some known facts about conics, the same statement can be proved in a very short way. Denote by ℓ the directrix of ellipse of ε_{ij} related to the focus O_j; since ε_{ij} is symmetrical about O_iO_j, we have $\ell \parallel AC$. Recall that for each point $P \in \varepsilon_{ij}$, we have $PO_j = \epsilon \cdot d_\ell(P)$, where $d_\ell(P)$ is the distance from P to ℓ, and ϵ is the eccentricity of ε_{ij} (see Fig. 5).

Now we have

$$r = R_j - (R_j - r) = AO_j - PO_j = \epsilon\big(d_\ell(A) - d_\ell(P)\big) = \epsilon\big(d(P) - d(A)\big) = \epsilon \cdot d(P),$$

and ϵ does not depend on P. □

Comment 2. One can find a spatial interpretations of the problem and the solution.

For every point (x, y) and radius $r > 0$, represent the circle $\Omega\big((x, y), r\big)$ by the point (x, y, r) in space. This point is the apex of the cone with base circle $\Omega\big((x, y), r\big)$ and height r. According to Lemma 1, the circles which are tangent to h_i and h_j correspond to the points of a half line β'_{ij}, starting at B.

Now we translate Lemma 2. Take some $1 \le i < j \le 3$, and consider those circles which are internally tangent to γ_j. It is easy to see that the locus of the points which represent these circles is a subset of a cone, containing γ_j. Similarly, the circles which are externally tangent to γ_i correspond to the points on the extension of another cone, which has its apex on the opposite side of the base plane Π. (See Fig. 6; for this illustration, the z-coordinates were multiplied by 2.)

The two cones are symmetric to each other (they have the same aperture, and their axes are parallel). As is well-known, it follows that the common points of the two cones are co-planar. So the intersection of the two cones is a a conic section — which is an ellipse, according to Lemma 2(a). The points which represent the circles touching γ_i and γ_j is an ellipse arc ε'_{ij} with end-points A and C.

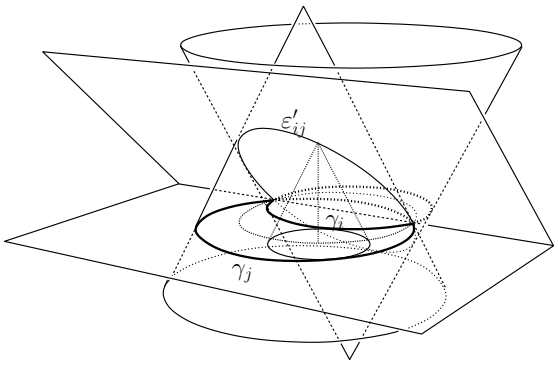


Fig. 6

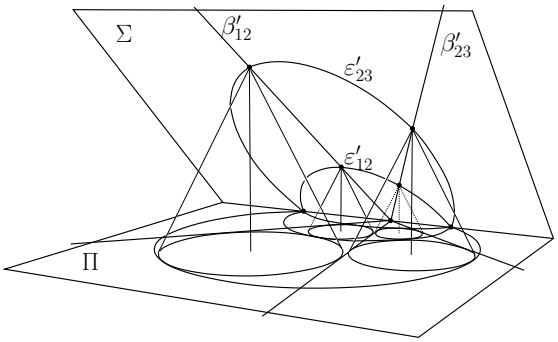
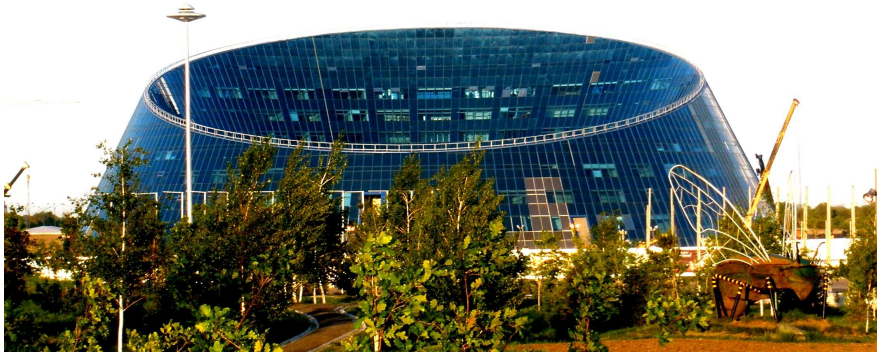


Fig. 7

Thus, the curved quadrilateral $\mathcal{Q}_{ij}$ is circumscribed if and only if $\beta'_{i,i+1}$ and $\varepsilon'_{j,j+1}$ intersect, i.e. if they are coplanar. If three of the four curved quadrilaterals are circumscribed, it means that ε'_{12}, ε'_{23}, β'_{12} and β'_{23} lie in the same plane Σ, and the fourth intersection comes to existence, too (see Fig. 7).



A connection between mathematics and real life:
the Palace of Creativity "Shabyt" ("Inspiration") in Astana

Number Theory

N1. Find the least positive integer n for which there exists a set $\{s_1, s_2, \ldots, s_n\}$ consisting of n distinct positive integers such that

$$\left(1-\frac{1}{s_1}\right)\left(1-\frac{1}{s_2}\right)\ldots\left(1-\frac{1}{s_n}\right)=\frac{51}{2010}.$$

N1′. Same as Problem N1, but the constant $\frac{51}{2010}$ is replaced by $\frac{42}{2010}$.

(Canada)

Answer for Problem N1. $n = 39$.

Solution for Problem N1. Suppose that for some n there exist the desired numbers; we may assume that $s_1 < s_2 < \cdots < s_n$. Surely $s_1 > 1$ since otherwise $1 - \frac{1}{s_1} = 0$. So we have $2 \le s_1 \le s_2 - 1 \le \cdots \le s_n - (n-1)$, hence $s_i \ge i + 1$ for each $i = 1, \ldots, n$. Therefore

$$\begin{aligned}\frac{51}{2010} &= \left(1-\frac{1}{s_1}\right)\left(1-\frac{1}{s_2}\right)\ldots\left(1-\frac{1}{s_n}\right)\\ &\ge \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\ldots\left(1-\frac{1}{n+1}\right)=\frac{1}{2}\cdot\frac{2}{3}\cdots\frac{n}{n+1}=\frac{1}{n+1},\end{aligned}$$

which implies

$$n+1 \ge \frac{2010}{51}=\frac{670}{17}>39,$$

so $n \ge 39$.

Now we are left to show that $n = 39$ fits. Consider the set $\{2, 3, \ldots, 33, 35, 36, \ldots, 40, 67\}$ which contains exactly 39 numbers. We have

$$\frac{1}{2}\cdot\frac{2}{3}\cdots\frac{32}{33}\cdot\frac{34}{35}\cdots\frac{39}{40}\cdot\frac{66}{67}=\frac{1}{33}\cdot\frac{34}{40}\cdot\frac{66}{67}=\frac{17}{670}=\frac{51}{2010}, \tag{1}$$

hence for $n = 39$ there exists a desired example.

Comment. One can show that the example (1) is unique.

Answer for Problem N1′. $n = 48$.

Solution for Problem N1′. Suppose that for some n there exist the desired numbers. In the same way we obtain that $s_i \ge i + 1$. Moreover, since the denominator of the fraction $\frac{42}{2010} = \frac{7}{335}$ is divisible by 67, some of s_i's should be divisible by 67, so $s_n \ge s_i \ge 67$. This means that

$$\frac{42}{2010}\ge\frac{1}{2}\cdot\frac{2}{3}\cdots\frac{n-1}{n}\cdot\left(1-\frac{1}{67}\right)=\frac{66}{67n},$$

which implies

$$n \geq \frac{2010 \cdot 66}{42 \cdot 67} = \frac{330}{7} > 47,$$

so $n \geq 48$.

Now we are left to show that $n = 48$ fits. Consider the set $\{2, 3, \ldots, 33, 36, 37, \ldots, 50, 67\}$ which contains exactly 48 numbers. We have

$$\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{32}{33} \cdot \frac{35}{36} \cdots \frac{49}{50} \cdot \frac{66}{67} = \frac{1}{33} \cdot \frac{35}{50} \cdot \frac{66}{67} = \frac{7}{335} = \frac{42}{2010},$$

hence for $n = 48$ there exists a desired example.

Comment 1. In this version of the problem, the estimate needs one more step, hence it is a bit harder. On the other hand, the example in this version is not unique. Another example is

$$\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{46}{47} \cdot \frac{66}{67} \cdot \frac{329}{330} = \frac{1}{67} \cdot \frac{66}{330} \cdot \frac{329}{47} = \frac{7}{67 \cdot 5} = \frac{42}{2010}.$$

Comment 2. N1′ was the Proposer's formulation of the problem. We propose N1 according to the number of current IMO.

N2. Find all pairs (m, n) of nonnegative integers for which

$$m^2 + 2 \cdot 3^n = m\left(2^{n+1} - 1\right). \tag{1}$$

(Australia)

Answer. $(6, 3)$, $(9, 3)$, $(9, 5)$, $(54, 5)$.

Solution. For fixed values of n, the equation (1) is a simple quadratic equation in m. For $n \le 5$ the solutions are listed in the following table.

case	equation	discriminant	integer roots
$n = 0$	$m^2 - m + 2 = 0$	-7	none
$n = 1$	$m^2 - 3m + 6 = 0$	-15	none
$n = 2$	$m^2 - 7m + 18 = 0$	-23	none
$n = 3$	$m^2 - 15m + 54 = 0$	9	$m = 6$ and $m = 9$
$n = 4$	$m^2 - 31m + 162 = 0$	313	none
$n = 5$	$m^2 - 63m + 486 = 0$	$2025 = 45^2$	$m = 9$ and $m = 54$

We prove that there is no solution for $n \ge 6$.

Suppose that (m, n) satisfies (1) and $n \ge 6$. Since $m \mid 2 \cdot 3^n = m\left(2^{n+1} - 1\right) - m^2$, we have $m = 3^p$ with some $0 \le p \le n$ or $m = 2 \cdot 3^q$ with some $0 \le q \le n$.

In the first case, let $q = n - p$; then

$$2^{n+1} - 1 = m + \frac{2 \cdot 3^n}{m} = 3^p + 2 \cdot 3^q.$$

In the second case let $p = n - q$. Then

$$2^{n+1} - 1 = m + \frac{2 \cdot 3^n}{m} = 2 \cdot 3^q + 3^p.$$

Hence, in both cases we need to find the nonnegative integer solutions of

$$3^p + 2 \cdot 3^q = 2^{n+1} - 1, \quad p + q = n. \tag{2}$$

Next, we prove bounds for p, q. From (2) we get

$$3^p < 2^{n+1} = 8^{\frac{n+1}{3}} < 9^{\frac{n+1}{3}} = 3^{\frac{2(n+1)}{3}}$$

and

$$2 \cdot 3^q < 2^{n+1} = 2 \cdot 8^{\frac{n}{3}} < 2 \cdot 9^{\frac{n}{3}} = 2 \cdot 3^{\frac{2n}{3}} < 2 \cdot 3^{\frac{2(n+1)}{3}},$$

so $p, q < \frac{2(n+1)}{3}$. Combining these inequalities with $p + q = n$, we obtain

$$\frac{n-2}{3} < p, q < \frac{2(n+1)}{3}. \tag{3}$$

Now let $h = \min(p, q)$. By (3) we have $h > \frac{n-2}{3}$; in particular, we have $h > 1$. On the left-hand side of (2), both terms are divisible by 3^h, therefore $9 \mid 3^h \mid 2^{n+1} - 1$. It is easy check that $\operatorname{ord}_9(2) = 6$, so $9 \mid 2^{n+1} - 1$ if and only if $6 \mid n + 1$. Therefore, $n + 1 = 6r$ for some positive integer r, and we can write

$$2^{n+1} - 1 = 4^{3r} - 1 = (4^{2r} + 4^r + 1)(2^r - 1)(2^r + 1). \tag{4}$$

Notice that the factor $4^{2r} + 4^r + 1 = (4^r - 1)^2 + 3 \cdot 4^r$ is divisible by 3, but it is never divisible by 9. The other two factors in (4), $2^r - 1$ and $2^r + 1$ are coprime: both are odd and their difference is 2. Since the whole product is divisible by 3^h, we have either $3^{h-1} \mid 2^r - 1$ or $3^{h-1} \mid 2^r + 1$. In any case, we have $3^{h-1} \leq 2^r + 1$. Then

$$3^{h-1} \leq 2^r + 1 \leq 3^r = 3^{\frac{n+1}{6}},$$
$$\frac{n-2}{3} - 1 < h - 1 \leq \frac{n+1}{6},$$
$$n < 11.$$

But this is impossible since we assumed $n \geq 6$, and we proved $6 \mid n + 1$.

N3. Find the smallest number n such that there exist polynomials f_1, f_2, ..., f_n with rational coefficients satisfying

$$x^2 + 7 = f_1(x)^2 + f_2(x)^2 + \cdots + f_n(x)^2.$$

(Poland)

Answer. The smallest n is 5.

Solution 1. The equality $x^2 + 7 = x^2 + 2^2 + 1^2 + 1^2 + 1^2$ shows that $n \le 5$. It remains to show that $x^2 + 7$ is not a sum of four (or less) squares of polynomials with rational coefficients.

Suppose by way of contradiction that $x^2 + 7 = f_1(x)^2 + f_2(x)^2 + f_3(x)^2 + f_4(x)^2$, where the coefficients of polynomials f_1, f_2, f_3 and f_4 are rational (some of these polynomials may be zero).

Clearly, the degrees of f_1, f_2, f_3 and f_4 are at most 1. Thus $f_i(x) = a_i x + b_i$ for $i = 1, 2, 3, 4$ and some rationals a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4. It follows that $x^2 + 7 = \sum_{i=1}^{4}(a_i x + b_i)^2$ and hence

$$\sum_{i=1}^{4} a_i^2 = 1, \quad \sum_{i=1}^{4} a_i b_i = 0, \quad \sum_{i=1}^{4} b_i^2 = 7. \tag{1}$$

Let $p_i = a_i + b_i$ and $q_i = a_i - b_i$ for $i = 1, 2, 3, 4$. Then

$$\sum_{i=1}^{4} p_i^2 = \sum_{i=1}^{4} a_i^2 + 2\sum_{i=1}^{4} a_i b_i + \sum_{i=1}^{4} b_i^2 = 8,$$
$$\sum_{i=1}^{4} q_i^2 = \sum_{i=1}^{4} a_i^2 - 2\sum_{i=1}^{4} a_i b_i + \sum_{i=1}^{4} b_i^2 = 8$$
$$\text{and} \quad \sum_{i=1}^{4} p_i q_i = \sum_{i=1}^{4} a_i^2 - \sum_{i=1}^{4} b_i^2 = -6,$$

which means that there exist a solution in integers x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4 and $m > 0$ of the system of equations

$$\text{(i)} \ \sum_{i=1}^{4} x_i^2 = 8m^2, \qquad \text{(ii)} \ \sum_{i=1}^{4} y_i^2 = 8m^2, \qquad \text{(iii)} \ \sum_{i=1}^{4} x_i y_i = -6m^2.$$

We will show that such a solution does not exist.

Assume the contrary and consider a solution with minimal m. Note that if an integer x is odd then $x^2 \equiv 1 \pmod 8$. Otherwise (i.e., if x is even) we have $x^2 \equiv 0 \pmod 8$ or $x^2 \equiv 4 \pmod 8$. Hence, by (i), we get that x_1, x_2, x_3 and x_4 are even. Similarly, by (ii), we get that y_1, y_2, y_3 and y_4 are even. Thus the LHS of (iii) is divisible by 4 and m is also even. It follows that $(\frac{x_1}{2}, \frac{y_1}{2}, \frac{x_2}{2}, \frac{y_2}{2}, \frac{x_3}{2}, \frac{y_3}{2}, \frac{x_4}{2}, \frac{y_4}{2}, \frac{m}{2})$ is a solution of the system of equations (i), (ii) and (iii), which contradicts the minimality of m.

Solution 2. We prove that $n \le 4$ is impossible. Define the numbers a_i, b_i for $i = 1, 2, 3, 4$ as in the previous solution.

By Euler's identity we have

$$\begin{aligned}(a_1^2 + a_2^2 + a_3^2 + a_4^2)(b_1^2 + b_2^2 + b_3^2 + b_4^2) =& (a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4)^2 + (a_1 b_2 - a_2 b_1 + a_3 b_4 - a_4 b_3)^2 \\ &+ (a_1 b_3 - a_3 b_1 + a_4 b_2 - a_2 b_4)^2 + (a_1 b_4 - a_4 b_1 + a_2 b_3 - a_3 b_2)^2.\end{aligned}$$

So, using the relations (1) from the Solution 1 we get that

$$7 = \left(\frac{m_1}{m}\right)^2 + \left(\frac{m_2}{m}\right)^2 + \left(\frac{m_3}{m}\right)^2, \tag{2}$$

where

$$\begin{aligned}
\frac{m_1}{m} &= a_1b_2 - a_2b_1 + a_3b_4 - a_4b_3,\\
\frac{m_2}{m} &= a_1b_3 - a_3b_1 + a_4b_2 - a_2b_4,\\
\frac{m_3}{m} &= a_1b_4 - a_4b_1 + a_2b_3 - a_3b_2
\end{aligned}$$

and $m_1, m_2, m_3 \in \mathbb{Z}, m \in \mathbb{N}$.

Let m be a minimum positive integer number for which (2) holds. Then

$$8m^2 = m_1^2 + m_2^2 + m_3^2 + m^2.$$

As in the previous solution, we get that m_1, m_2, m_3, m are all even numbers. Then $\left(\frac{m_1}{2}, \frac{m_2}{2}, \frac{m_3}{2}, \frac{m}{2}\right)$ is also a solution of (2) which contradicts the minimality of m. So, we have $n \geq 5$. The example with $n = 5$ is already shown in Solution 1.

N4. Let a, b be integers, and let $P(x) = ax^3 + bx$. For any positive integer n we say that the pair (a, b) is *n-good* if $n \mid P(m) - P(k)$ implies $n \mid m - k$ for all integers m, k. We say that (a, b) is *very good* if (a, b) is n-good for infinitely many positive integers n.

(a) Find a pair (a, b) which is 51-good, but not very good.

(b) Show that all 2010-good pairs are very good.

(Turkey)

Solution. (a) We show that the pair $(1, -51^2)$ is good but not very good. Let $P(x) = x^3 - 51^2 x$. Since $P(51) = P(0)$, the pair $(1, -51^2)$ is not n-good for any positive integer that does not divide 51. Therefore, $(1, -51^2)$ is not very good.

On the other hand, if $P(m) \equiv P(k) \pmod{51}$, then $m^3 \equiv k^3 \pmod{51}$. By Fermat's theorem, from this we obtain

$$m \equiv m^3 \equiv k^3 \equiv k \pmod 3 \quad \text{and} \quad m \equiv m^{33} \equiv k^{33} \equiv k \pmod{17}.$$

Hence we have $m \equiv k \pmod{51}$. Therefore $(1, -51^2)$ is 51-good.

(b) We will show that if a pair (a, b) is 2010-good then (a, b) is 67^i-good for all positive integer i.

Claim 1. If (a, b) is 2010-good then (a, b) is 67-good.

Proof. Assume that $P(m) = P(k) \pmod{67}$. Since 67 and 30 are coprime, there exist integers m' and k' such that $k' \equiv k \pmod{67}$, $k' \equiv 0 \pmod{30}$, and $m' \equiv m \pmod{67}$, $m' \equiv 0 \pmod{30}$. Then we have $P(m') \equiv P(0) \equiv P(k') \pmod{30}$ and $P(m') \equiv P(m) \equiv P(k) \equiv P(k') \pmod{67}$, hence $P(m') \equiv P(k') \pmod{2010}$. This implies $m' \equiv k' \pmod{2010}$ as (a, b) is 2010-good. It follows that $m \equiv m' \equiv k' \equiv k \pmod{67}$. Therefore, (a, b) is 67-good. □

Claim 2. If (a, b) is 67-good then $67 \mid a$.

Proof. Suppose that $67 \nmid a$. Consider the sets $\{at^2 \pmod{67} : 0 \le t \le 33\}$ and $\{-3as^2 - b \bmod 67 : 0 \le s \le 33\}$. Since $a \not\equiv 0 \pmod{67}$, each of these sets has 34 elements. Hence they have at least one element in common. If $at^2 \equiv -3as^2 - b \pmod{67}$ then for $m = t \pm s$, $k = \mp 2s$ we have

$$\begin{aligned} P(m) - P(k) = a(m^3 - k^3) + b(m - k) &= (m - k)\big(a(m^2 + mk + k^2) + b\big) \\ &= (t \pm 3s)(at^2 + 3as^2 + b) \equiv 0 \pmod{67}. \end{aligned}$$

Since (a, b) is 67-good, we must have $m \equiv k \pmod{67}$ in both cases, that is, $t \equiv 3s \pmod{67}$ and $t \equiv -3s \pmod{67}$. This means $t \equiv s \equiv 0 \pmod{67}$ and $b \equiv -3as^2 - at^2 \equiv 0 \pmod{67}$. But then $67 \mid P(7) - P(2) = 67 \cdot 5a + 5b$ and $67 \nmid 7 - 2$, contradicting that (a, b) is 67-good. □

Claim 3. If (a, b) is 2010-good then (a, b) is 67^i-good all $i \ge 1$.

Proof. By Claim 2, we have $67 \mid a$. If $67 \mid b$, then $P(x) \equiv P(0) \pmod{67}$ for all x, contradicting that (a, b) is 67-good. Hence, $67 \nmid b$.

Suppose that $67^i \mid P(m) - P(k) = (m - k)\big(a(m^2 + mk + k^2) + b\big)$. Since $67 \mid a$ and $67 \nmid b$, the second factor $a(m^2 + mk + k^2) + b$ is coprime to 67 and hence $67^i \mid m - k$. Therefore, (a, b) is 67^i-good. □

Comment 1. In the proof of Claim 2, the following reasoning can also be used. Since 3 is not a quadratic residue modulo 67, either $au^2 \equiv -b \pmod{67}$ or $3av^2 \equiv -b \pmod{67}$ has a solution. The settings $(m, k) = (u, 0)$ in the first case and $(m, k) = (v, -2v)$ in the second case lead to $b \equiv 0 \pmod{67}$.

Comment 2. The pair $(67, 30)$ is n-good if and only if $n = d \cdot 67^i$, where $d \mid 30$ and $i \ge 0$. It shows that in part (b), one should deal with the large powers of 67 to reach the solution. The key property of number 67 is that it has the form $3k + 1$, so there exists a nontrivial cubic root of unity modulo 67.

N5. Let $\mathbb{N}$ be the set of all positive integers. Find all functions $f\colon \mathbb{N} \to \mathbb{N}$ such that the number $\big(f(m)+n\big)\big(m+f(n)\big)$ is a square for all $m, n \in \mathbb{N}$.

(U.S.A.)

Answer. All functions of the form $f(n) = n + c$, where $c \in \mathbb{N} \cup \{0\}$.

Solution. First, it is clear that all functions of the form $f(n) = n + c$ with a constant nonnegative integer c satisfy the problem conditions since $\big(f(m)+n\big)\big(f(n)+m\big) = (n+m+c)^2$ is a square.

We are left to prove that there are no other functions. We start with the following

Lemma. Suppose that $p \mid f(k) - f(\ell)$ for some prime p and positive integers k, ℓ. Then $p \mid k - \ell$.

Proof. Suppose first that $p^2 \mid f(k) - f(\ell)$, so $f(\ell) = f(k) + p^2 a$ for some integer a. Take some positive integer $D > \max\{f(k), f(\ell)\}$ which is not divisible by p and set $n = pD - f(k)$. Then the positive numbers $n + f(k) = pD$ and $n + f(\ell) = pD + \big(f(\ell) - f(k)\big) = p(D + pa)$ are both divisible by p but not by p^2. Now, applying the problem conditions, we get that both the numbers $\big(f(k)+n\big)\big(f(n)+k\big)$ and $\big(f(\ell)+n\big)\big(f(n)+\ell\big)$ are squares divisible by p (and thus by p^2); this means that the multipliers $f(n) + k$ and $f(n) + \ell$ are also divisible by p, therefore $p \mid \big(f(n)+k\big) - \big(f(n)+\ell\big) = k - \ell$ as well.

On the other hand, if $f(k) - f(\ell)$ is divisible by p but not by p^2, then choose the same number D and set $n = p^3 D - f(k)$. Then the positive numbers $f(k) + n = p^3 D$ and $f(\ell) + n = p^3 D + \big(f(\ell) - f(k)\big)$ are respectively divisible by p^3 (but not by p^4) and by p (but not by p^2). Hence in analogous way we obtain that the numbers $f(n) + k$ and $f(n) + \ell$ are divisible by p, therefore $p \mid \big(f(n)+k\big) - \big(f(n)+\ell\big) = k - \ell$. □

We turn to the problem. First, suppose that $f(k) = f(\ell)$ for some $k, \ell \in \mathbb{N}$. Then by Lemma we have that $k - \ell$ is divisible by every prime number, so $k - \ell = 0$, or $k = \ell$. Therefore, the function f is injective.

Next, consider the numbers $f(k)$ and $f(k+1)$. Since the number $(k+1) - k = 1$ has no prime divisors, by Lemma the same holds for $f(k+1) - f(k)$; thus $|f(k+1) - f(k)| = 1$.

Now, let $f(2) - f(1) = q$, $|q| = 1$. Then we prove by induction that $f(n) = f(1) + q(n-1)$. The base for $n = 1, 2$ holds by the definition of q. For the step, if $n > 1$ we have $f(n+1) = f(n) \pm q = f(1) + q(n-1) \pm q$. Since $f(n) \neq f(n-2) = f(1) + q(n-2)$, we get $f(n) = f(1) + qn$, as desired.

Finally, we have $f(n) = f(1) + q(n-1)$. Then q cannot be -1 since otherwise for $n \geq f(1) + 1$ we have $f(n) \leq 0$ which is impossible. Hence $q = 1$ and $f(n) = (f(1) - 1) + n$ for each $n \in \mathbb{N}$, and $f(1) - 1 \geq 0$, as desired.

N6. The rows and columns of a $2^n \times 2^n$ table are numbered from 0 to $2^n - 1$. The cells of the table have been colored with the following property being satisfied: for each $0 \le i, j \le 2^n - 1$, the jth cell in the ith row and the $(i + j)$th cell in the jth row have the same color. (The indices of the cells in a row are considered modulo 2^n.)

Prove that the maximal possible number of colors is 2^n.

(Iran)

Solution. Throughout the solution we denote the cells of the table by coordinate pairs; (i, j) refers to the jth cell in the ith row.

Consider the directed graph, whose vertices are the cells of the board, and the edges are the arrows $(i, j) \to (j, i + j)$ for all $0 \le i, j \le 2^n - 1$. From each vertex (i, j), exactly one edge passes (to $(j, i + j \mod 2^n)$); conversely, to each cell (j, k) exactly one edge is directed (from the cell $(k - j \mod 2^n, j)$). Hence, the graph splits into cycles.

Now, in any coloring considered, the vertices of each cycle should have the same color by the problem condition. On the other hand, if each cycle has its own color, the obtained coloring obviously satisfies the problem conditions. Thus, the maximal possible number of colors is the same as the number of cycles, and we have to prove that this number is 2^n.

Next, consider any cycle $(i_1, j_1), (i_2, j_2), \dots$; we will describe it in other terms. Define a sequence $(a_0, a_1, \dots)$ by the relations $a_0 = i_1$, $a_1 = j_1$, $a_{n+1} = a_n + a_{n-1}$ for all $n \ge 1$ (we say that such a sequence is *a Fibonacci-type sequence*). Then an obvious induction shows that $i_k \equiv a_{k-1} \pmod{2^n}$, $j_k \equiv a_k \pmod{2^n}$. Hence we need to investigate the behavior of Fibonacci-type sequences modulo 2^n.

Denote by $F_0, F_1, \dots$ the Fibonacci numbers defined by $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for $n \ge 0$. We also set $F_{-1} = 1$ according to the recurrence relation.

For every positive integer m, denote by $\nu(m)$ the exponent of 2 in the prime factorization of m, i.e. for which $2^{\nu(m)} \mid m$ but $2^{\nu(m)+1} \nmid m$.

Lemma 1. For every Fibonacci-type sequence $a_0, a_1, a_2, \dots$, and every $k \ge 0$, we have $a_k = F_{k-1}a_0 + F_k a_1$.

Proof. Apply induction on k. The base cases $k = 0, 1$ are trivial. For the step, from the induction hypothesis we get

$$a_{k+1} = a_k + a_{k-1} = (F_{k-1}a_0 + F_k a_1) + (F_{k-2}a_0 + F_{k-1}a_1) = F_k a_0 + F_{k+1}a_1. \qquad \square$$

Lemma 2. For every $m \ge 3$,

(a) we have $\nu(F_{3\cdot 2^{m-2}}) = m$;

(b) $d = 3 \cdot 2^{m-2}$ is the least positive index for which $2^m \mid F_d$;

(c) $F_{3\cdot 2^{m-2}+1} \equiv 1 + 2^{m-1} \pmod{2^m}$.

Proof. Apply induction on m. In the base case $m = 3$ we have $\nu(F_{3\cdot 2^{m-2}}) = F_6 = 8$, so $\nu(F_{3\cdot 2^{m-2}}) = \nu(8) = 3$, the preceding Fibonacci-numbers are not divisible by 8, and indeed $F_{3\cdot 2^{m-2}+1} = F_7 = 13 \equiv 1 + 4 \pmod 8$.

Now suppose that $m > 3$ and let $k = 3 \cdot 2^{m-3}$. By applying Lemma 1 to the Fibonacci-type sequence $F_k, F_{k+1}, \dots$ we get

$$F_{2k} = F_{k-1}F_k + F_k F_{k+1} = (F_{k+1} - F_k)F_k + F_{k+1}F_k = 2F_{k+1}F_k - F_k^2,$$
$$F_{2k+1} = F_k \cdot F_k + F_{k+1} \cdot F_{k+1} = F_k^2 + F_{k+1}^2.$$

By the induction hypothesis, $\nu(F_k) = m - 1$, and F_{k+1} is odd. Therefore we get $\nu(F_k^2) = 2(m - 1) > (m - 1) + 1 = \nu(2F_k F_{k+1})$, which implies $\nu(F_{2k}) = m$, establishing statement (a).

Moreover, since $F_{k+1} = 1 + 2^{m-2} + a2^{m-1}$ for some integer a, we get

$$F_{2k+1} = F_k^2 + F_{k+1}^2 \equiv 0 + (1 + 2^{m-2} + a2^{m-1})^2 \equiv 1 + 2^{m-1} \pmod{2^m},$$

as desired in statement (c).

We are left to prove that $2^m \nmid F_\ell$ for $\ell < 2k$. Assume the contrary. Since $2^{m-1} \mid F_\ell$, from the induction hypothesis it follows that $\ell > k$. But then we have $F_\ell = F_{k-1}F_{\ell-k} + F_kF_{\ell-k+1}$, where the second summand is divisible by 2^{m-1} but the first one is not (since F_{k-1} is odd and $\ell - k < k$). Hence the sum is not divisible even by 2^{m-1}. A contradiction. □

Now, for every pair of integers $(a, b) \neq (0, 0)$, let $\mu(a, b) = \min\{\nu(a), \nu(b)\}$. By an obvious induction, for every Fibonacci-type sequence $A = (a_0, a_1, \dots)$ we have $\mu(a_0, a_1) = \mu(a_1, a_2) = \dots$; denote this common value by $\mu(A)$. Also denote by $p_n(A)$ the period of this sequence modulo 2^n, that is, the least $p > 0$ such that $a_{k+p} \equiv a_k \pmod{2^n}$ for all $k \geq 0$.

Lemma 3. Let $A = (a_0, a_1, \dots)$ be a Fibonacci-type sequence such that $\mu(A) = k < n$. Then $p_n(A) = 3 \cdot 2^{n-1-k}$.

Proof. First, we note that the sequence $(a_0, a_1, \dots)$ has period p modulo 2^n if and only if the sequence $(a_0/2^k, a_1/2^k, \dots)$ has period p modulo 2^{n-k}. Hence, passing to this sequence we can assume that $k = 0$.

We prove the statement by induction on n. It is easy to see that for $n = 1, 2$ the claim is true; actually, each Fibonacci-type sequence A with $\mu(A) = 0$ behaves as $0, 1, 1, 0, 1, 1, \dots$ modulo 2, and as $0, 1, 1, 2, 3, 1, 0, 1, 1, 2, 3, 1, \dots$ modulo 4 (all pairs of residues from which at least one is odd appear as a pair of consecutive terms in this sequence).

Now suppose that $n \geq 3$ and consider an arbitrary Fibonacci-type sequence $A = (a_0, a_1, \dots)$ with $\mu(A) = 0$. Obviously we should have $p_{n-1}(A) \mid p_n(A)$, or, using the induction hypothesis, $s = 3 \cdot 2^{n-2} \mid p_n(A)$. Next, we may suppose that a_0 is even; hence a_1 is odd, and $a_0 = 2b_0$, $a_1 = 2b_1 + 1$ for some integers b_0, b_1.

Consider the Fibonacci-type sequence $B = (b_0, b_1, \dots)$ starting with (b_0, b_1). Since $a_0 = 2b_0 + F_0$, $a_1 = 2b_1 + F_1$, by an easy induction we get $a_k = 2b_k + F_k$ for all $k \geq 0$. By the induction hypothesis, we have $p_{n-1}(B) \mid s$, hence the sequence $(2b_0, 2b_1, \dots)$ is s-periodic modulo 2^n. On the other hand, by Lemma 2 we have $F_{s+1} \equiv 1 + 2^{n-1} \pmod{2^n}$, $F_{2s} \equiv 0 \pmod{2^n}$, $F_{2s+1} \equiv 1 \pmod{2^n}$, hence

$$a_{s+1} = 2b_{s+1} + F_{s+1} \equiv 2b_1 + 1 + 2^{n-1} \not\equiv 2b_1 + 1 = a_1 \pmod{2^n},$$
$$a_{2s} = 2b_{2s} + F_{2s} \equiv 2b_0 + 0 = a_0 \pmod{2^n},$$
$$a_{2s+1} = 2b_{2s+1} + F_{2s+1} \equiv 2b_1 + 1 = a_1 \pmod{2^n}.$$

The first line means that A is not s-periodic, while the other two provide that $a_{2s} \equiv a_0$, $a_{2s+1} \equiv a_1$ and hence $a_{2s+t} \equiv a_t$ for all $t \geq 0$. Hence $s \mid p_n(A) \mid 2s$ and $p_n(A) \neq s$, which means that $p_n(A) = 2s$, as desired. □

Finally, Lemma 3 provides a straightforward method of counting the number of cycles. Actually, take any number $0 \leq k \leq n - 1$ and consider all the cells (i, j) with $\mu(i, j) = k$. The total number of such cells is $2^{2(n-k)} - 2^{2(n-k-1)} = 3 \cdot 2^{2n-2k-2}$. On the other hand, they are split into cycles, and by Lemma 3 the length of each cycle is $3 \cdot 2^{n-1-k}$. Hence the number of cycles consisting of these cells is exactly $\dfrac{3 \cdot 2^{2n-2k-2}}{3 \cdot 2^{n-1-k}} = 2^{n-k-1}$. Finally, there is only one cell $(0, 0)$ which is not mentioned in the previous computation, and it forms a separate cycle. So the total number of cycles is

$$1 + \sum_{k=0}^{n-1} 2^{n-1-k} = 1 + (1 + 2 + 4 + \dots + 2^{n-1}) = 2^n.$$

Comment. We outline a different proof for the essential part of Lemma 3. That is, we assume that $k = 0$ and show that in this case the period of (a_i) modulo 2^n coincides with the period of the Fibonacci numbers modulo 2^n; then the proof can be finished by the arguments from Lemma 2..

Note that p is a (not necessarily minimal) period of the sequence (a_i) modulo 2^n if and only if we have $a_0 \equiv a_p \pmod{2^n}$, $a_1 \equiv a_{p+1} \pmod{2^n}$, that is,

$$\begin{aligned} a_0 \equiv a_p \equiv F_{p-1}a_0 + F_p a_1 = F_p(a_1 - a_0) + F_{p+1}a_0 \pmod{2^n}, \\ a_1 \equiv a_{p+1} = F_p a_0 + F_{p+1}a_1 \pmod{2^n}. \end{aligned} \tag{1}$$

Now, If p is a period of (F_i) then we have $F_p \equiv F_0 = 0 \pmod{2^n}$ and $F_{p+1} \equiv F_1 = 1 \pmod{2^n}$, which by (1) implies that p is a period of (a_i) as well.

Conversely, suppose that p is a period of (a_i). Combining the relations of (1) we get

$$\begin{aligned} 0 = a_1 \cdot a_0 - a_0 \cdot a_1 &\equiv a_1\big(F_p(a_1 - a_0) + F_{p+1}a_0\big) - a_0(F_p a_0 + F_{p+1}a_1) \\ &= F_p(a_1^2 - a_1a_0 - a_0^2) \pmod{2^n}, \\ a_1^2 - a_1a_0 - a_0^2 = (a_1 - a_0)a_1 - a_0 \cdot a_0 &\equiv (a_1 - a_0)(F_p a_0 + F_{p+1}a_1) - a_0\big(F_p(a_1 - a_0) + F_{p+1}a_0\big) \\ &= F_{p+1}(a_1^2 - a_1a_0 - a_0^2) \pmod{2^n}. \end{aligned}$$

Since at least one of the numbers a_0, a_1 is odd, the number $a_1^2 - a_1a_0 - a_0^2$ is odd as well. Therefore the previous relations are equivalent with $F_p \equiv 0 \pmod{2^n}$ and $F_{p+1} \equiv 1 \pmod{2^n}$, which means exactly that p is a period of $(F_0, F_1, \dots)$ modulo 2^n.

So, the sets of periods of (a_i) and (F_i) coincide, and hence the minimal periods coincide as well.

75

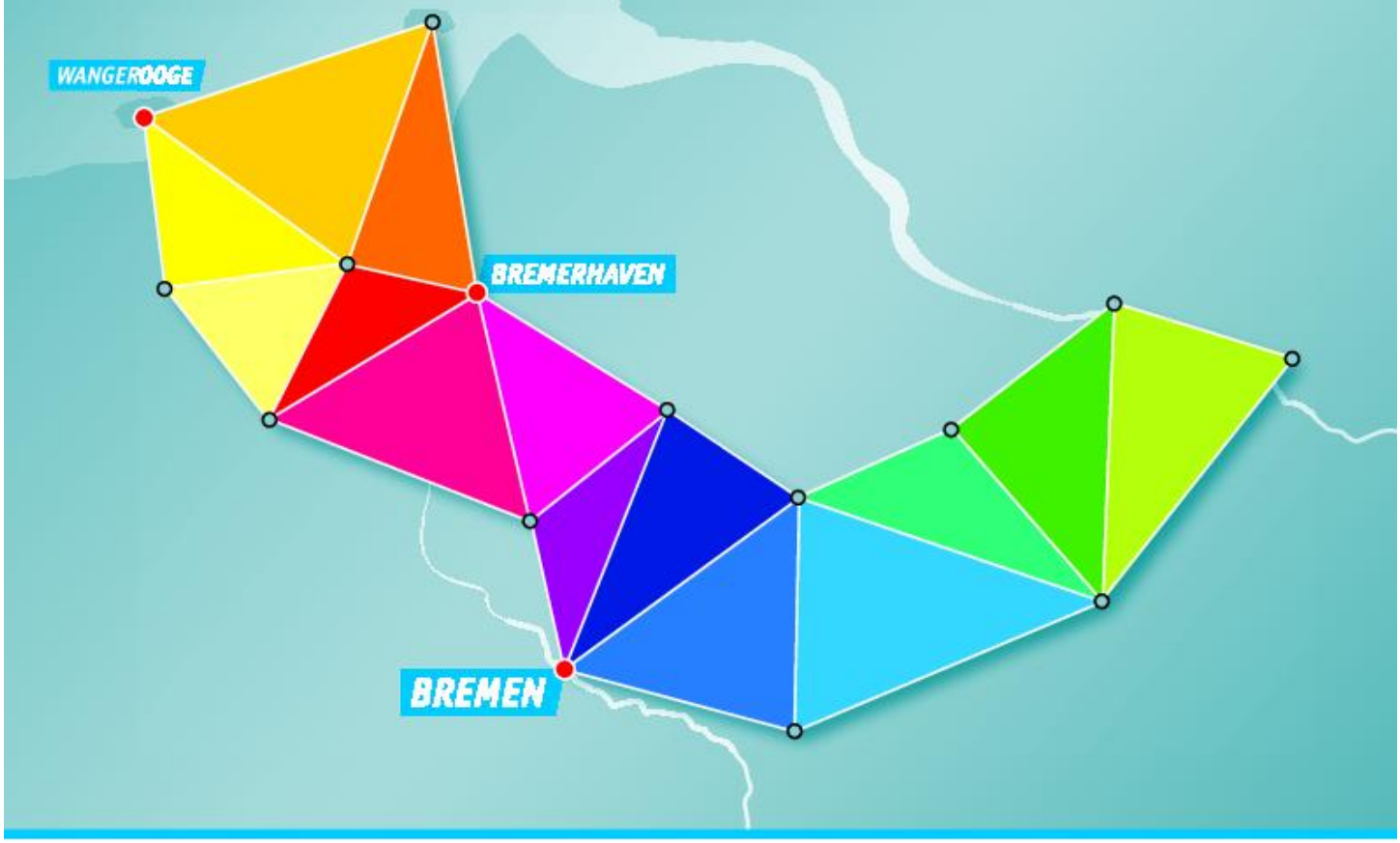
76

50th International Mathematical Olympiad

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Problem Shortlist with solutions



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Problem Shortlist with Solutions

The Problem Selection Committee

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132 problem proposals submitted by 39 countries:

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: The Problem Selection Committee

Contents

Problem Shortlist **4**

Algebra **12**

Combinatorics **26**

Geometry **47**

Number Theory **69**

Algebra

A1 CZE (Czech Republic)

Find the largest possible integer k, such that the following statement is true:

Let 2009 arbitrary non-degenerated triangles be given. In every triangle the three sides are colored, such that one is blue, one is red and one is white. Now, for every color separately, let us sort the lengths of the sides. We obtain

$$\begin{aligned} & b_1 \leq b_2 \leq \ldots \leq b_{2009} && \text{the lengths of the blue sides,} \\ & r_1 \leq r_2 \leq \ldots \leq r_{2009} && \text{the lengths of the red sides,} \\ \text{and} \quad & w_1 \leq w_2 \leq \ldots \leq w_{2009} && \text{the lengths of the white sides.} \end{aligned}$$

Then there exist k indices j such that we can form a non-degenerated triangle with side lengths b_j, r_j, w_j.

A2 EST (Estonia)

Let a, b, c be positive real numbers such that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=a+b+c$. Prove that

$$\frac{1}{(2a+b+c)^2}+\frac{1}{(2b+c+a)^2}+\frac{1}{(2c+a+b)^2}\leq\frac{3}{16}.$$

A3 FRA (France)

Determine all functions f from the set of positive integers into the set of positive integers such that for all x and y there exists a non degenerated triangle with sides of lengths

$$x, \quad f(y) \quad \text{and} \quad f(y+f(x)-1).$$

A4 BLR (Belarus)

Let a, b, c be positive real numbers such that $ab+bc+ca\leq 3abc$. Prove that

$$\sqrt{\frac{a^2+b^2}{a+b}}+\sqrt{\frac{b^2+c^2}{b+c}}+\sqrt{\frac{c^2+a^2}{c+a}}+3\leq\sqrt{2}\left(\sqrt{a+b}+\sqrt{b+c}+\sqrt{c+a}\right).$$

A5 BLR (Belarus)

Let f be any function that maps the set of real numbers into the set of real numbers. Prove that there exist real numbers x and y such that

$$f\left(x-f(y)\right)>yf(x)+x.$$

A6 USA (United States of America)

Suppose that $s_1, s_2, s_3, \ldots$ is a strictly increasing sequence of positive integers such that the subsequences

$$s_{s_1},\ s_{s_2},\ s_{s_3},\ \ldots \qquad \text{and} \qquad s_{s_1+1},\ s_{s_2+1},\ s_{s_3+1},\ \ldots$$

are both arithmetic progressions. Prove that $s_1, s_2, s_3, \ldots$ is itself an arithmetic progression.

A7 JPN (Japan)

Find all functions f from the set of real numbers into the set of real numbers which satisfy for all real x, y the identity

$$f(xf(x+y)) = f(yf(x)) + x^2.$$

Combinatorics

C1 NZL (New Zealand)

Consider 2009 cards, each having one gold side and one black side, lying in parallel on a long table. Initially all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing a block of 50 consecutive cards, the leftmost of which is showing gold, and turning them all over, so those which showed gold now show black and vice versa. The last player who can make a legal move wins.

(a) Does the game necessarily end?

(b) Does there exist a winning strategy for the starting player?

C2 ROU (Romania)

For any integer $n \geq 2$, let $N(n)$ be the maximal number of triples (a_i, b_i, c_i), $i = 1, \ldots, N(n)$, consisting of nonnegative integers a_i, b_i and c_i such that the following two conditions are satisfied:

(1) $a_i + b_i + c_i = n$ for all $i = 1, \ldots, N(n)$,

(2) If $i \neq j$, then $a_i \neq a_j, b_i \neq b_j$ and $c_i \neq c_j$.

Determine $N(n)$ for all $n \geq 2$.

Comment. The original problem was formulated for m-tuples instead for triples. The numbers $N(m, n)$ are then defined similarly to $N(n)$ in the case $m = 3$. The numbers $N(3, n)$ and $N(n, n)$ should be determined. The case $m = 3$ is the same as in the present problem. The upper bound for $N(n, n)$ can be proved by a simple generalization. The construction of a set of triples attaining the bound can be easily done by induction from n to $n + 2$.

C3 RUS (Russian Federation)

Let n be a positive integer. Given a sequence $\varepsilon_1, \ldots, \varepsilon_{n-1}$ with $\varepsilon_i = 0$ or $\varepsilon_i = 1$ for each $i = 1, \ldots, n-1$, the sequences $a_0, \ldots, a_n$ and $b_0, \ldots, b_n$ are constructed by the following rules:

$$a_0 = b_0 = 1, \quad a_1 = b_1 = 7,$$

$$a_{i+1} = \begin{cases} 2a_{i-1} + 3a_i, & \text{if } \varepsilon_i = 0, \\ 3a_{i-1} + a_i, & \text{if } \varepsilon_i = 1, \end{cases} \quad \text{for each } i = 1, \ldots, n-1,$$

$$b_{i+1} = \begin{cases} 2b_{i-1} + 3b_i, & \text{if } \varepsilon_{n-i} = 0, \\ 3b_{i-1} + b_i, & \text{if } \varepsilon_{n-i} = 1, \end{cases} \quad \text{for each } i = 1, \ldots, n-1.$$

Prove that $a_n = b_n$.

C4 NLD (Netherlands)

For an integer $m \geq 1$, we consider partitions of a $2^m \times 2^m$ chessboard into rectangles consisting of cells of the chessboard, in which each of the 2^m cells along one diagonal forms a separate rectangle of side length 1. Determine the smallest possible sum of rectangle perimeters in such a partition.

C5 NLD (Netherlands)

Five identical empty buckets of 2-liter capacity stand at the vertices of a regular pentagon. Cinderella and her wicked Stepmother go through a sequence of rounds: At the beginning of every round, the Stepmother takes one liter of water from the nearby river and distributes it arbitrarily over the five buckets. Then Cinderella chooses a pair of neighboring buckets, empties them into the river, and puts them back. Then the next round begins. The Stepmother's goal is to make one of these buckets overflow. Cinderella's goal is to prevent this. Can the wicked Stepmother enforce a bucket overflow?

C6 BGR (Bulgaria)

On a 999×999 board a *limp rook* can move in the following way: From any square it can move to any of its adjacent squares, i.e. a square having a common side with it, and every move must be a turn, i.e. the directions of any two consecutive moves must be perpendicular. A *non-intersecting route* of the limp rook consists of a sequence of pairwise different squares that the limp rook can visit in that order by an admissible sequence of moves. Such a non-intersecting route is called *cyclic*, if the limp rook can, after reaching the last square of the route, move directly to the first square of the route and start over.

How many squares does the longest possible cyclic, non-intersecting route of a limp rook visit?

C7 RUS (Russian Federation)

Variant 1. A grasshopper jumps along the real axis. He starts at point 0 and makes 2009 jumps to the right with lengths $1, 2, \ldots, 2009$ in an arbitrary order. Let M be a set of 2008 positive integers less than $1005 \cdot 2009$. Prove that the grasshopper can arrange his jumps in such a way that he never lands on a point from M.

Variant 2. Let n be a nonnegative integer. A grasshopper jumps along the real axis. He starts at point 0 and makes $n+1$ jumps to the right with pairwise different positive integral lengths $a_1, a_2, \ldots, a_{n+1}$ in an arbitrary order. Let M be a set of n positive integers in the interval $(0, s)$, where $s = a_1 + a_2 + \cdots + a_{n+1}$. Prove that the grasshopper can arrange his jumps in such a way that he never lands on a point from M.

C8 AUT (Austria)

For any integer $n \geq 2$, we compute the integer $h(n)$ by applying the following procedure to its decimal representation. Let r be the rightmost digit of n.

(1) If $r = 0$, then the decimal representation of $h(n)$ results from the decimal representation of n by removing this rightmost digit 0.

(2) If $1 \leq r \leq 9$ we split the decimal representation of n into a maximal right part R that solely consists of digits not less than r and into a left part L that either is empty or ends with a digit strictly smaller than r. Then the decimal representation of $h(n)$ consists of the decimal representation of L, followed by two copies of the decimal representation of $R-1$. For instance, for the number $n = 17{,}151{,}345{,}543$, we will have $L = 17{,}151$, $R = 345{,}543$ and $h(n) = 17{,}151{,}345{,}542{,}345{,}542$.

Prove that, starting with an arbitrary integer $n \geq 2$, iterated application of h produces the integer 1 after finitely many steps.

Geometry

G1 BEL (Belgium)

Let ABC be a triangle with $AB = AC$. The angle bisectors of A and B meet the sides BC and AC in D and E, respectively. Let K be the incenter of triangle ADC. Suppose that $\angle BEK = 45^\circ$. Find all possible values of $\angle BAC$.

G2 RUS (Russian Federation)

Let ABC be a triangle with circumcenter O. The points P and Q are interior points of the sides CA and AB, respectively. The circle k passes through the midpoints of the segments BP, CQ, and PQ. Prove that if the line PQ is tangent to circle k then $OP = OQ$.

G3 IRN (Islamic Republic of Iran)

Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y, respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms.

Prove that $GR = GS$.

G4 UNK (United Kingdom)

Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F. The midpoints of AB and CD are G and H, respectively. Show that EF is tangent at E to the circle through the points E, G, and H.

G5 POL (Poland)

Let P be a polygon that is convex and symmetric to some point O. Prove that for some parallelogram R satisfying $P \subset R$ we have

$$\frac{|R|}{|P|} \leq \sqrt{2}$$

where $|R|$ and $|P|$ denote the area of the sets R and P, respectively.

G6 UKR (Ukraine)

Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P. Points O_1 and O_2 are the circumcenters and points H_1 and H_2 are the orthocenters of triangles ABP and DCP, respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2, respectively. Prove that the perpendicular from E_1 on CD, the perpendicular from E_2 on AB and the line H_1H_2 are concurrent.

G7 IRN (Islamic Republic of Iran)

Let ABC be a triangle with incenter I and let X, Y and Z be the incenters of the triangles BIC, CIA and AIB, respectively. Let the triangle XYZ be equilateral. Prove that ABC is equilateral too.

G8 BGR (Bulgaria)

Let $ABCD$ be a circumscribed quadrilateral. Let g be a line through A which meets the segment BC in M and the line CD in N. Denote by I_1, I_2, and I_3 the incenters of $\triangle ABM$, $\triangle MNC$, and $\triangle NDA$, respectively. Show that the orthocenter of $\triangle I_1I_2I_3$ lies on g.

Number Theory

N1 AUS (Australia)

A social club has n members. They have the membership numbers $1, 2, \ldots, n$, respectively. From time to time members send presents to other members, including items they have already received as presents from other members. In order to avoid the embarrassing situation that a member might receive a present that he or she has sent to other members, the club adds the following rule to its statutes at one of its annual general meetings:

"A member with membership number a is permitted to send a present to a member with membership number b if and only if $a(b-1)$ is a multiple of n."

Prove that, if each member follows this rule, none will receive a present from another member that he or she has already sent to other members.

Alternative formulation: Let G be a directed graph with n vertices $v_1, v_2, \ldots, v_n$, such that there is an edge going from v_a to v_b if and only if a and b are distinct and $a(b-1)$ is a multiple of n. Prove that this graph does not contain a directed cycle.

N2 PER (Peru)

A positive integer N is called *balanced*, if $N = 1$ or if N can be written as a product of an even number of not necessarily distinct primes. Given positive integers a and b, consider the polynomial P defined by $P(x) = (x + a)(x + b)$.

(a) Prove that there exist distinct positive integers a and b such that all the numbers $P(1), P(2), \ldots, P(50)$ are balanced.

(b) Prove that if $P(n)$ is balanced for all positive integers n, then $a = b$.

N3 EST (Estonia)

Let f be a non-constant function from the set of positive integers into the set of positive integers, such that $a - b$ divides $f(a) - f(b)$ for all distinct positive integers a, b. Prove that there exist infinitely many primes p such that p divides $f(c)$ for some positive integer c.

N4 PRK (Democratic People's Republic of Korea)

Find all positive integers n such that there exists a sequence of positive integers $a_1, a_2, \ldots, a_n$ satisfying

$$a_{k+1} = \frac{a_k^2 + 1}{a_{k-1} + 1} - 1$$

for every k with $2 \leq k \leq n - 1$.

N5 HUN (Hungary)

Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function T from the set of integers into the set of integers such that the number of integers x with $T^n(x) = x$ is equal to $P(n)$ for every $n \geq 1$, where T^n denotes the n-fold application of T.

N6 TUR (Turkey)

Let k be a positive integer. Show that if there exists a sequence $a_0, a_1, \ldots$ of integers satisfying the condition

$$a_n = \frac{a_{n-1} + n^k}{n} \quad \text{for all } n \geq 1,$$

then $k - 2$ is divisible by 3.

N7 MNG (Mongolia)

Let a and b be distinct integers greater than 1. Prove that there exists a positive integer n such that $(a^n - 1)(b^n - 1)$ is not a perfect square.

Algebra

A1 CZE (Czech Republic)

Find the largest possible integer k, such that the following statement is true:

Let 2009 arbitrary non-degenerated triangles be given. In every triangle the three sides are colored, such that one is blue, one is red and one is white. Now, for every color separately, let us sort the lengths of the sides. We obtain

$$\begin{aligned} & b_1 \le b_2 \le \ldots \le b_{2009} && \text{the lengths of the blue sides,} \\ & r_1 \le r_2 \le \ldots \le r_{2009} && \text{the lengths of the red sides,} \\ \text{and} \quad & w_1 \le w_2 \le \ldots \le w_{2009} && \text{the lengths of the white sides.} \end{aligned}$$

Then there exist k indices j such that we can form a non-degenerated triangle with side lengths b_j, r_j, w_j.

Solution. We will prove that the largest possible number k of indices satisfying the given condition is one.

Firstly we prove that b_{2009}, r_{2009}, w_{2009} are always lengths of the sides of a triangle. Without loss of generality we may assume that $w_{2009} \ge r_{2009} \ge b_{2009}$. We show that the inequality $b_{2009} + r_{2009} > w_{2009}$ holds. Evidently, there exists a triangle with side lengths w, b, r for the white, blue and red side, respectively, such that $w_{2009} = w$. By the conditions of the problem we have $b + r > w$, $b_{2009} \ge b$ and $r_{2009} \ge r$. From these inequalities it follows

$$b_{2009} + r_{2009} \ge b + r > w = w_{2009}.$$

Secondly we will describe a sequence of triangles for which w_j, b_j, r_j with $j < 2009$ are not the lengths of the sides of a triangle. Let us define the sequence Δ_j, $j = 1, 2, \ldots, 2009$, of triangles, where Δ_j has

a blue side of length $2j$,
a red side of length j for all $j \le 2008$ and 4018 for $j = 2009$,
and a white side of length $j + 1$ for all $j \le 2007$, 4018 for $j = 2008$ and 1 for $j = 2009$.

Since

$$\begin{aligned} (j+1) + j > 2j \quad &\ge j + 1 > j, && \text{if} \quad j \le 2007, \\ 2j + j > 4018 > 2j \quad &> j, && \text{if} \quad j = 2008, \\ 4018 + 1 > 2j \quad &= 4018 > 1, && \text{if} \quad j = 2009, \end{aligned}$$

such a sequence of triangles exists. Moreover, $w_j = j$, $r_j = j$ and $b_j = 2j$ for $1 \le j \le 2008$. Then

$$w_j + r_j = j + j = 2j = b_j,$$

i.e., b_j, r_j and w_j are not the lengths of the sides of a triangle for $1 \le j \le 2008$.

A2 EST (Estonia)

Let a, b, c be positive real numbers such that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=a+b+c$. Prove that

$$\frac{1}{(2a+b+c)^2}+\frac{1}{(2b+c+a)^2}+\frac{1}{(2c+a+b)^2}\leq\frac{3}{16}.$$

Solution 1. For positive real numbers x, y, z, from the arithmetic-geometric-mean inequality,

$$2x+y+z=(x+y)+(x+z)\geq 2\sqrt{(x+y)(x+z)},$$

we obtain

$$\frac{1}{(2x+y+z)^2}\leq\frac{1}{4(x+y)(x+z)}.$$

Applying this to the left-hand side terms of the inequality to prove, we get

$$\begin{aligned}&\frac{1}{(2a+b+c)^2}+\frac{1}{(2b+c+a)^2}+\frac{1}{(2c+a+b)^2}\\&\qquad\leq\frac{1}{4(a+b)(a+c)}+\frac{1}{4(b+c)(b+a)}+\frac{1}{4(c+a)(c+b)}\\&\qquad=\frac{(b+c)+(c+a)+(a+b)}{4(a+b)(b+c)(c+a)}=\frac{a+b+c}{2(a+b)(b+c)(c+a)}.\end{aligned}\tag{1}$$

A second application of the inequality of the arithmetic-geometric mean yields

$$a^2b+a^2c+b^2a+b^2c+c^2a+c^2b\geq 6abc,$$

or, equivalently,

$$9(a+b)(b+c)(c+a)\geq 8(a+b+c)(ab+bc+ca).\tag{2}$$

The supposition $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=a+b+c$ can be written as

$$ab+bc+ca=abc(a+b+c).\tag{3}$$

Applying the arithmetic-geometric-mean inequality $x^2y^2+x^2z^2\geq 2x^2yz$ thrice, we get

$$a^2b^2+b^2c^2+c^2a^2\geq a^2bc+ab^2c+abc^2,$$

which is equivalent to

$$(ab+bc+ca)^2\geq 3abc(a+b+c).\tag{4}$$

Combining (1), (2), (3), and (4), we will finish the proof:

$$\begin{aligned}\frac{a+b+c}{2(a+b)(b+c)(c+a)} &= \frac{(a+b+c)(ab+bc+ca)}{2(a+b)(b+c)(c+a)} \cdot \frac{ab+bc+ca}{abc(a+b+c)} \cdot \frac{abc(a+b+c)}{(ab+bc+ca)^2} \\ &\leq \frac{9}{2\cdot 8} \cdot 1 \cdot \frac{1}{3} = \frac{3}{16}.\end{aligned}$$

Solution 2. Equivalently, we prove the homogenized inequality

$$\frac{(a+b+c)^2}{(2a+b+c)^2} + \frac{(a+b+c)^2}{(a+2b+c)^2} + \frac{(a+b+c)^2}{(a+b+2c)^2} \leq \frac{3}{16}(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$

for all positive real numbers a, b, c. Without loss of generality we choose $a+b+c=1$. Thus, the problem is equivalent to prove for all $a, b, c > 0$, fulfilling this condition, the inequality

$$\frac{1}{(1+a)^2} + \frac{1}{(1+b)^2} + \frac{1}{(1+c)^2} \leq \frac{3}{16}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right). \tag{5}$$

Applying Jensen's inequality to the function $f(x) = \dfrac{x}{(1+x)^2}$, which is concave for $0 \leq x \leq 2$ and increasing for $0 \leq x \leq 1$, we obtain

$$\alpha\frac{a}{(1+a)^2} + \beta\frac{b}{(1+b)^2} + \gamma\frac{c}{(1+c)^2} \leq (\alpha+\beta+\gamma)\frac{A}{(1+A)^2}, \quad \text{where} \quad A = \frac{\alpha a+\beta b+\gamma c}{\alpha+\beta+\gamma}.$$

Choosing $\alpha = \dfrac{1}{a}$, $\beta = \dfrac{1}{b}$, and $\gamma = \dfrac{1}{c}$, we can apply the harmonic-arithmetic-mean inequality

$$A = \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}} \leq \frac{a+b+c}{3} = \frac{1}{3} < 1.$$

Finally we prove (5):

$$\begin{aligned}\frac{1}{(1+a)^2} + \frac{1}{(1+b)^2} + \frac{1}{(1+c)^2} &\leq \left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\frac{A}{(1+A)^2} \\ &\leq \left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\frac{\frac{1}{3}}{\left(1+\frac{1}{3}\right)^2} = \frac{3}{16}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right).\end{aligned}$$

A3 FRA (France)

Determine all functions f from the set of positive integers into the set of positive integers such that for all x and y there exists a non degenerated triangle with sides of lengths

$$x, \quad f(y) \quad \text{and} \quad f(y + f(x) - 1).$$

Solution. The identity function $f(x) = x$ is the only solution of the problem.

If $f(x) = x$ for all positive integers x, the given three lengths are x, $y = f(y)$ and $z = f(y + f(x) - 1) = x + y - 1$. Because of $x \geq 1$, $y \geq 1$ we have $z \geq \max\{x, y\} > |x - y|$ and $z < x + y$. From this it follows that a triangle with these side lengths exists and does not degenerate. We prove in several steps that there is no other solution.

Step 1. We show $f(1) = 1$.
If we had $f(1) = 1+m > 1$ we would conclude $f(y) = f(y+m)$ for all y considering the triangle with the side lengths 1, $f(y)$ and $f(y + m)$. Thus, f would be m-periodic and, consequently, bounded. Let B be a bound, $f(x) \leq B$. If we choose $x > 2B$ we obtain the contradiction $x > 2B \geq f(y) + f(y + f(x) - 1)$.

Step 2. For all positive integers z, we have $f(f(z)) = z$.
Setting $x = z$ and $y = 1$ this follows immediately from Step 1.

Step 3. For all integers $z \geq 1$, we have $f(z) \leq z$.
Let us show, that the contrary leads to a contradiction. Assume $w + 1 = f(z) > z$ for some z. From Step 1 we know that $w \geq z \geq 2$. Let $M = \max\{f(1), f(2), \ldots, f(w)\}$ be the largest value of f for the first w integers. First we show, that no positive integer t exists with

$$f(t) > \frac{z-1}{w} \cdot t + M, \tag{1}$$

otherwise we decompose the smallest value t as $t = wr + s$ where r is an integer and $1 \leq s \leq w$. Because of the definition of M, we have $t > w$. Setting $x = z$ and $y = t - w$ we get from the triangle inequality

$$z + f(t - w) > f((t - w) + f(z) - 1) = f(t - w + w) = f(t).$$

Hence,

$$f(t - w) \geq f(t) - (z - 1) > \frac{z-1}{w}(t - w) + M,$$

a contradiction to the minimality of t.

Therefore the inequality (1) fails for all $t \geq 1$, we have proven

$$f(t) \leq \frac{z-1}{w} \cdot t + M, \tag{2}$$

instead.

Now, using (2), we finish the proof of Step 3. Because of $z \leq w$ we have $\frac{z-1}{w} < 1$ and we can choose an integer t sufficiently large to fulfill the condition

$$\left(\frac{z-1}{w}\right)^2 t + \left(\frac{z-1}{w} + 1\right) M < t.$$

Applying (2) twice we get

$$f\left(f(t)\right) \leq \frac{z-1}{w} f(t) + M \leq \frac{z-1}{w}\left(\frac{z-1}{w} t + M\right) + M < t$$

in contradiction to Step 2, which proves Step 3.

Final step. Thus, following Step 2 and Step 3, we obtain

$$z = f(f(z)) \leq f(z) \leq z$$

and $f(z) = z$ for all positive integers z is proven.

A4 BLR (Belarus)

Let a, b, c be positive real numbers such that $ab + bc + ca \leq 3abc$. Prove that

$$\sqrt{\frac{a^2+b^2}{a+b}} + \sqrt{\frac{b^2+c^2}{b+c}} + \sqrt{\frac{c^2+a^2}{c+a}} + 3 \leq \sqrt{2}\left(\sqrt{a+b} + \sqrt{b+c} + \sqrt{c+a}\right).$$

Solution. Starting with the terms of the right-hand side, the quadratic-arithmetic-mean inequality yields

$$\begin{aligned}\sqrt{2}\sqrt{a+b} &= 2\sqrt{\frac{ab}{a+b}}\sqrt{\frac{1}{2}\left(2 + \frac{a^2+b^2}{ab}\right)} \\ &\geq 2\sqrt{\frac{ab}{a+b}} \cdot \frac{1}{2}\left(\sqrt{2} + \sqrt{\frac{a^2+b^2}{ab}}\right) = \sqrt{\frac{2ab}{a+b}} + \sqrt{\frac{a^2+b^2}{a+b}}\end{aligned}$$

and, analogously,

$$\sqrt{2}\sqrt{b+c} \geq \sqrt{\frac{2bc}{b+c}} + \sqrt{\frac{b^2+c^2}{b+c}}, \qquad \sqrt{2}\sqrt{c+a} \geq \sqrt{\frac{2ca}{c+a}} + \sqrt{\frac{c^2+a^2}{c+a}}.$$

Applying the inequality between the arithmetic mean and the squared harmonic mean will finish the proof:

$$\sqrt{\frac{2ab}{a+b}} + \sqrt{\frac{2bc}{b+c}} + \sqrt{\frac{2ca}{c+a}} \geq 3 \cdot \sqrt{\frac{3}{\sqrt{\frac{a+b}{2ab}}^2 + \sqrt{\frac{b+c}{2bc}}^2 + \sqrt{\frac{c+a}{2ca}}^2}} = 3 \cdot \sqrt{\frac{3abc}{ab+bc+ca}} \geq 3.$$

A5 BLR (Belarus)

Let f be any function that maps the set of real numbers into the set of real numbers. Prove that there exist real numbers x and y such that

$$f(x - f(y)) > yf(x) + x.$$

Solution 1. Assume that

$$f(x - f(y)) \leq yf(x) + x \quad \text{for all real } x, y. \tag{1}$$

Let $a = f(0)$. Setting $y = 0$ in (1) gives $f(x - a) \leq x$ for all real x and, equivalently,

$$f(y) \leq y + a \quad \text{for all real } y. \tag{2}$$

Setting $x = f(y)$ in (1) yields in view of (2)

$$a = f(0) \leq yf(f(y)) + f(y) \leq yf(f(y)) + y + a.$$

This implies $0 \leq y(f(f(y)) + 1)$ and thus

$$f(f(y)) \geq -1 \quad \text{for all } y > 0. \tag{3}$$

From (2) and (3) we obtain $-1 \leq f(f(y)) \leq f(y) + a$ for all $y > 0$, so

$$f(y) \geq -a - 1 \quad \text{for all } y > 0. \tag{4}$$

Now we show that

$$f(x) \leq 0 \quad \text{for all real } x. \tag{5}$$

Assume the contrary, i.e. there is some x such that $f(x) > 0$. Take any y such that

$$y < x - a \quad \text{and} \quad y < \frac{-a - x - 1}{f(x)}\ .$$

Then in view of (2)

$$x - f(y) \geq x - (y + a) > 0$$

and with (1) and (4) we obtain

$$yf(x) + x \geq f(x - f(y)) \geq -a - 1,$$

whence

$$y \geq \frac{-a - x - 1}{f(x)}$$

contrary to our choice of y. Thereby, we have established (5).

Setting $x = 0$ in (5) leads to $a = f(0) \leq 0$ and (2) then yields

$$f(x) \leq x \quad \text{for all real } x. \tag{6}$$

Now choose y such that $y > 0$ and $y > -f(-1) - 1$ and set $x = f(y) - 1$. From (1), (5) and

(6) we obtain

$$f(-1) = f(x - f(y)) \le yf(x) + x = yf(f(y) - 1) + f(y) - 1 \le y(f(y) - 1) - 1 \le -y - 1,$$

i.e. $y \le -f(-1) - 1$, a contradiction to the choice of y.

Solution 2. Assume that

$$f(x - f(y)) \le yf(x) + x \quad \text{for all real } x, y. \tag{7}$$

Let $a = f(0)$. Setting $y = 0$ in (7) gives $f(x - a) \le x$ for all real x and, equivalently,

$$f(y) \le y + a \quad \text{for all real } y. \tag{8}$$

Now we show that

$$f(z) \ge 0 \quad \text{for all } z \ge 1. \tag{9}$$

Let $z \ge 1$ be fixed, set $b = f(z)$ and assume that $b < 0$. Setting $x = w + b$ and $y = z$ in (7) gives

$$f(w) - zf(w + b) \le w + b \quad \text{for all real } w. \tag{10}$$

Applying (10) to $w, w + b, \ldots, w + (n - 1)b$, where $n = 1, 2, \ldots$, leads to

$$\begin{aligned} f(w) - z^n f(w + nb) =& (f(w) - zf(w + b)) + z\,(f(w + b) - zf(w + 2b)) \\ &+ \cdots + z^{n-1}\,(f(w + (n - 1)b) - zf(w + nb)) \\ \le& (w + b) + z(w + 2b) + \cdots + z^{n-1}(w + nb). \end{aligned}$$

From (8) we obtain

$$f(w + nb) \le w + nb + a$$

and, thus, we have for all positive integers n

$$f(w) \le (1 + z + \cdots + z^{n-1} + z^n)w + (1 + 2z + \cdots + nz^{n-1} + nz^n)b + z^n a. \tag{11}$$

With $w = 0$ we get

$$a \le (1 + 2z + \cdots + nz^{n-1} + nz^n)b + az^n. \tag{12}$$

In view of the assumption $b < 0$ we find some n such that

$$a > (nb + a)z^n \tag{13}$$

because the right hand side tends to $-\infty$ as $n \to \infty$. Now (12) and (13) give the desired contradiction and (9) is established. In addition, we have for $z = 1$ the strict inequality

$$f(1) > 0. \tag{14}$$

Indeed, assume that $f(1) = 0$. Then setting $w = -1$ and $z = 1$ in (11) leads to

$$f(-1) \le -(n + 1) + a$$

which is false if n is sufficiently large.

To complete the proof we set $t = \min\{-a, -2/f(1)\}$. Setting $x = 1$ and $y = t$ in (7) gives

$$f(1 - f(t)) \le tf(1) + 1 \le -2 + 1 = -1. \tag{15}$$

On the other hand, by (8) and the choice of t we have $f(t) \leq t + a \leq 0$ and hence $1 - f(t) \geq 1$. The inequality (9) yields

$$f(1 - f(t)) \geq 0,$$

which contradicts (15).

A6 USA (United States of America)

Suppose that $s_1, s_2, s_3, \ldots$ is a strictly increasing sequence of positive integers such that the subsequences

$$s_{s_1},\ s_{s_2},\ s_{s_3},\ \ldots \qquad \text{and} \qquad s_{s_1+1},\ s_{s_2+1},\ s_{s_3+1},\ \ldots$$

are both arithmetic progressions. Prove that $s_1, s_2, s_3, \ldots$ is itself an arithmetic progression.

Solution 1. Let D be the common difference of the progression $s_{s_1}, s_{s_2}, \ldots$. Let for $n = 1, 2, \ldots$

$$d_n = s_{n+1} - s_n.$$

We have to prove that d_n is constant. First we show that the numbers d_n are bounded. Indeed, by supposition $d_n \geq 1$ for all n. Thus, we have for all n

$$d_n = s_{n+1} - s_n \leq d_{s_n} + d_{s_n+1} + \cdots + d_{s_{n+1}-1} = s_{s_{n+1}} - s_{s_n} = D.$$

The boundedness implies that there exist

$$m = \min\{d_n : n = 1, 2, \ldots\} \quad \text{and} \quad M = \max\{d_n : n = 1, 2, \ldots\}.$$

It suffices to show that $m = M$. Assume that $m < M$. Choose n such that $d_n = m$. Considering a telescoping sum of $m = d_n = s_{n+1} - s_n$ items not greater than M leads to

$$D = s_{s_{n+1}} - s_{s_n} = s_{s_n+m} - s_{s_n} = d_{s_n} + d_{s_n+1} + \cdots + d_{s_n+m-1} \leq mM \tag{1}$$

and equality holds if and only if all items of the sum are equal to M. Now choose n such that $d_n = M$. In the same way, considering a telescoping sum of M items not less than m we obtain

$$D = s_{s_{n+1}} - s_{s_n} = s_{s_n+M} - s_{s_n} = d_{s_n} + d_{s_n+1} + \cdots + d_{s_n+M-1} \geq Mm \tag{2}$$

and equality holds if and only if all items of the sum are equal to m. The inequalities (1) and (2) imply that $D = Mm$ and that

$$\begin{aligned} d_{s_n} = d_{s_n+1} = \cdots = d_{s_{n+1}-1} = M \quad &\text{if } d_n = m, \\ d_{s_n} = d_{s_n+1} = \cdots = d_{s_{n+1}-1} = m \quad &\text{if } d_n = M. \end{aligned}$$

Hence, $d_n = m$ implies $d_{s_n} = M$. Note that $s_n \geq s_1 + (n-1) \geq n$ for all n and moreover $s_n > n$ if $d_n = n$, because in the case $s_n = n$ we would have $m = d_n = d_{s_n} = M$ in contradiction to the assumption $m < M$. In the same way $d_n = M$ implies $d_{s_n} = m$ and $s_n > n$. Consequently, there is a strictly increasing sequence $n_1, n_2, \ldots$ such that

$$d_{s_{n_1}} = M, \quad d_{s_{n_2}} = m, \quad d_{s_{n_3}} = M, \quad d_{s_{n_4}} = m, \quad \ldots.$$

The sequence $d_{s_1}, d_{s_2}, \ldots$ is the sequence of pairwise differences of $s_{s_1+1}, s_{s_2+1}, \ldots$ and $s_{s_1}, s_{s_2}, \ldots$, hence also an arithmetic progression. Thus $m = M$.

Solution 2. Let the integers D and E be the common differences of the progressions $s_{s_1}, s_{s_2}, \dots$ and $s_{s_1+1}, s_{s_2+1}, \dots$, respectively. Let briefly $A = s_{s_1} - D$ and $B = s_{s_1+1} - E$. Then, for all positive integers n,

$$s_{s_n} = A + nD, \qquad s_{s_n+1} = B + nE.$$

Since the sequence $s_1, s_2, \dots$ is strictly increasing, we have for all positive integers n

$$s_{s_n} < s_{s_n+1} \le s_{s_{n+1}},$$

which implies

$$A + nD < B + nE \le A + (n+1)D,$$

and thereby

$$0 < B - A + n(E - D) \le D,$$

which implies $D - E = 0$ and thus

$$0 \le B - A \le D. \tag{3}$$

Let $m = \min\{s_{n+1} - s_n : n = 1, 2, \dots\}$. Then

$$B - A = (s_{s_1+1} - E) - (s_{s_1} - D) = s_{s_1+1} - s_{s_1} \ge m \tag{4}$$

and

$$D = A + (s_1 + 1)D - (A + s_1 D) = s_{s_{s_1+1}} - s_{s_{s_1}} = s_{B+D} - s_{A+D} \ge m(B - A). \tag{5}$$

From (3) we consider two cases.

Case 1. $B - A = D$.
Then, for each positive integer n, $s_{s_n+1} = B + nD = A + (n+1)D = s_{s_{n+1}}$, hence $s_{n+1} = s_n + 1$ and $s_1, s_2, \dots$ is an arithmetic progression with common difference 1.

Case 2. $B - A < D$. Choose some positive integer N such that $s_{N+1} - s_N = m$. Then

$$\begin{aligned} m(A - B + D - 1) &= m((A + (N+1)D) - (B + ND + 1)) \\ &\le s_{A+(N+1)D} - s_{B+ND+1} = s_{s_{s_N+1}} - s_{s_{s_N+1}+1} \\ &= (A + s_{N+1}D) - (B + (s_N + 1)D) = (s_{N+1} - s_N)D + A - B - D \\ &= mD + A - B - D, \end{aligned}$$

i.e.,

$$(B - A - m) + (D - m(B - A)) \le 0. \tag{6}$$

The inequalities (4)-(6) imply that

$$B - A = m \quad \text{and} \quad D = m(B - A).$$

Assume that there is some positive integer n such that $s_{n+1} > s_n + m$. Then

$$m(m+1) \le m(s_{n+1} - s_n) \le s_{s_{n+1}} - s_{s_n} = (A + (n+1)D) - (A + nD)) = D = m(B - A) = m^2,$$

a contradiction. Hence $s_1, s_2, \dots$ is an arithmetic progression with common difference m.

A7 JPN (Japan)

Find all functions f from the set of real numbers into the set of real numbers which satisfy for all real x, y the identity

$$f(xf(x+y)) = f(yf(x)) + x^2.$$

Solution 1. It is no hard to see that the two functions given by $f(x) = x$ and $f(x) = -x$ for all real x respectively solve the functional equation. In the sequel, we prove that there are no further solutions.

Let f be a function satisfying the given equation. It is clear that f cannot be a constant. Let us first show that $f(0) = 0$. Suppose that $f(0) \neq 0$. For any real t, substituting $(x, y) = (0, \frac{t}{f(0)})$ into the given functional equation, we obtain

$$f(0) = f(t), \tag{1}$$

contradicting the fact that f is not a constant function. Therefore, $f(0) = 0$. Next for any t, substituting $(x, y) = (t, 0)$ and $(x, y) = (t, -t)$ into the given equation, we get

$$f\left(tf(t)\right) = f(0) + t^2 = t^2,$$

and

$$f(tf(0)) = f(-tf(t)) + t^2,$$

respectively. Therefore, we conclude that

$$f(tf(t)) = t^2, \quad f(-tf(t)) = -t^2, \qquad \text{for every real } t. \tag{2}$$

Consequently, for every real v, there exists a real u, such that $f(u) = v$. We also see that if $f(t) = 0$, then $0 = f(tf(t)) = t^2$ so that $t = 0$, and thus 0 is the only real number satisfying $f(t) = 0$.

We next show that for any real number s,

$$f(-s) = -f(s). \tag{3}$$

This is clear if $f(s) = 0$. Suppose now $f(s) < 0$, then we can find a number t for which $f(s) = -t^2$. As $t \neq 0$ implies $f(t) \neq 0$, we can also find number a such that $af(t) = s$. Substituting $(x, y) = (t, a)$ into the given equation, we get

$$f(tf(t+a)) = f(af(t)) + t^2 = f(s) + t^2 = 0,$$

and therefore, $tf(t + a) = 0$, which implies $t + a = 0$, and hence $s = -tf(t)$. Consequently, $f(-s) = f(tf(t)) = t^2 = -(-t^2) = -f(s)$ holds in this case.

Finally, suppose $f(s) > 0$ holds. Then there exists a real number $t \neq 0$ for which $f(s) = t^2$. Choose a number a such that $tf(a) = s$. Substituting $(x, y) = (t, a-t)$ into the given equation, we get $f(s) = f(tf(a)) = f((a-t)f(t))+t^2 = f((a-t)f(t))+f(s)$. So we have $f((a-t)f(t)) = 0$, from which we conclude that $(a - t)f(t) = 0$. Since $f(t) \neq 0$, we get $a = t$ so that $s = tf(t)$ and thus we see $f(-s) = f(-tf(t)) = -t^2 = -f(s)$ holds in this case also. This observation finishes the proof of (3).

By substituting $(x, y) = (s, t), (x, y) = (t, -s-t)$ and $(x, y) = (-s-t, s)$ into the given equation,

we obtain

$$f(sf(s+t))) = f(tf(s)) + s^2,$$
$$f(tf(-s)) = f((-s-t)f(t)) + t^2,$$

and

$$f((-s-t)f(-t)) = f(sf(-s-t)) + (s+t)^2,$$

respectively. Using the fact that $f(-x) = -f(x)$ holds for all x to rewrite the second and the third equation, and rearranging the terms, we obtain

$$\begin{aligned} f(tf(s)) - f(sf(s+t)) &= -s^2, \\ f(tf(s)) - f((s+t)f(t)) &= -t^2, \\ f((s+t)f(t)) + f(sf(s+t)) &= (s+t)^2. \end{aligned}$$

Adding up these three equations now yields $2f(tf(s)) = 2ts$, and therefore, we conclude that $f(tf(s)) = ts$ holds for every pair of real numbers s, t. By fixing s so that $f(s) = 1$, we obtain $f(x) = sx$. In view of the given equation, we see that $s = \pm 1$. It is easy to check that both functions $f(x) = x$ and $f(x) = -x$ satisfy the given functional equation, so these are the desired solutions.

Solution 2. As in Solution 1 we obtain (1), (2) and (3).

Now we prove that f is injective. For this purpose, let us assume that $f(r) = f(s)$ for some $r \neq s$. Then, by (2)

$$r^2 = f(rf(r)) = f(rf(s)) = f((s-r)f(r)) + r^2,$$

where the last statement follows from the given functional equation with $x = r$ and $y = s - r$. Hence, $h = (s-r)f(r)$ satisfies $f(h) = 0$ which implies $h^2 = f(hf(h)) = f(0) = 0$, i.e., $h = 0$. Then, by $s \neq r$ we have $f(r) = 0$ which implies $r = 0$, and finally $f(s) = f(r) = f(0) = 0$. Analogously, it follows that $s = 0$ which gives the contradiction $r = s$.

To prove $|f(1)| = 1$ we apply (2) with $t = 1$ and also with $t = f(1)$ and obtain $f(f(1)) = 1$ and $(f(1))^2 = f(f(1) \cdot f(f(1))) = f(f(1)) = 1$.

Now we choose $\eta \in \{-1, 1\}$ with $f(1) = \eta$. Using that f is odd and the given equation with $x = 1, y = z$ (second equality) and with $x = -1, y = z + 2$ (fourth equality) we obtain

$$f(z) + 2\eta = \eta(f(z\eta) + 2) = \eta(f(f(z+1)) + 1) = \eta(-f(-f(z+1)) + 1)$$
$$= -\eta f((z+2)f(-1)) = -\eta f((z+2)(-\eta)) = \eta f((z+2)\eta) = f(z+2). \tag{4}$$

Hence,

$$f(z + 2\eta) = \eta f(\eta z + 2) = \eta(f(\eta z) + 2\eta) = f(z) + 2.$$

Using this argument twice we obtain

$$f(z + 4\eta) = f(z + 2\eta) + 2 = f(z) + 4.$$

Substituting $z = 2f(x)$ we have

$$f(2f(x)) + 4 = f(2f(x) + 4\eta) = f(2f(x+2)),$$

where the last equality follows from (4). Applying the given functional equation we proceed to

$$f(2f(x+2)) = f(xf(2)) + 4 = f(2\eta x) + 4$$

where the last equality follows again from (4) with $z = 0$, i.e., $f(2) = 2\eta$. Finally, $f(2f(x)) = f(2\eta x)$ and by injectivity of f we get $2f(x) = 2\eta x$ and hence the two solutions.

Combinatorics

C1 NZL (New Zealand)

Consider 2009 cards, each having one gold side and one black side, lying in parallel on a long table. Initially all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing a block of 50 consecutive cards, the leftmost of which is showing gold, and turning them all over, so those which showed gold now show black and vice versa. The last player who can make a legal move wins.

(a) Does the game necessarily end?

(b) Does there exist a winning strategy for the starting player?

Solution. (a) We interpret a card showing black as the digit 0 and a card showing gold as the digit 1. Thus each position of the 2009 cards, read from left to right, corresponds bijectively to a nonnegative integer written in binary notation of 2009 digits, where leading zeros are allowed. Each move decreases this integer, so the game must end.

(b) We show that there is no winning strategy for the starting player. We label the cards from right to left by $1, \ldots, 2009$ and consider the set S of cards with labels $50i$, $i = 1, 2, \ldots, 40$. Let g_n be the number of cards from S showing gold after n moves. Obviously, $g_0 = 40$. Moreover, $|g_n - g_{n+1}| = 1$ as long as the play goes on. Thus, after an odd number of moves, the non-starting player finds a card from S showing gold and hence can make a move. Consequently, this player always wins.

C2 ROU (Romania)

For any integer $n \geq 2$, let $N(n)$ be the maximal number of triples (a_i, b_i, c_i), $i = 1, \ldots, N(n)$, consisting of nonnegative integers a_i, b_i and c_i such that the following two conditions are satisfied:

(1) $a_i + b_i + c_i = n$ for all $i = 1, \ldots, N(n)$,

(2) If $i \neq j$, then $a_i \neq a_j, b_i \neq b_j$ and $c_i \neq c_j$.

Determine $N(n)$ for all $n \geq 2$.

Comment. The original problem was formulated for m-tuples instead for triples. The numbers $N(m, n)$ are then defined similarly to $N(n)$ in the case $m = 3$. The numbers $N(3, n)$ and $N(n, n)$ should be determined. The case $m = 3$ is the same as in the present problem. The upper bound for $N(n, n)$ can be proved by a simple generalization. The construction of a set of triples attaining the bound can be easily done by induction from n to $n + 2$.

Solution. Let $n \geq 2$ be an integer and let $\{T_1, \ldots, T_N\}$ be any set of triples of nonnegative integers satisfying the conditions (1) and (2). Since the a-coordinates are pairwise distinct we have

$$\sum_{i=1}^{N} a_i \geq \sum_{i=1}^{N} (i-1) = \frac{N(N-1)}{2}.$$

Analogously,

$$\sum_{i=1}^{N} b_i \geq \frac{N(N-1)}{2} \quad \text{and} \quad \sum_{i=1}^{N} c_i \geq \frac{N(N-1)}{2}.$$

Summing these three inequalities and applying (1) yields

$$3\frac{N(N-1)}{2} \leq \sum_{i=1}^{N} a_i + \sum_{i=1}^{N} b_i + \sum_{i=1}^{N} c_i = \sum_{i=1}^{N} (a_i + b_i + c_i) = nN,$$

hence $3\frac{N-1}{2} \leq n$ and, consequently,

$$N \leq \left\lfloor \frac{2n}{3} \right\rfloor + 1.$$

By constructing examples, we show that this upper bound can be attained, so $N(n) = \lfloor \frac{2n}{3} \rfloor + 1$.

We distinguish the cases $n = 3k-1$, $n = 3k$ and $n = 3k+1$ for $k \geq 1$ and present the extremal examples in form of a table.

$n = 3k-1$ $\left\lfloor \frac{2n}{3} \right\rfloor + 1 = 2k$		
a_i	b_i	c_i
0	$k+1$	$2k-2$
1	$k+2$	$2k-4$
$\vdots$	$\vdots$	$\vdots$
$k-1$	$2k$	0
k	0	$2k-1$
$k+1$	1	$2k-3$
$\vdots$	$\vdots$	$\vdots$
$2k-1$	$k-1$	1

$n = 3k$ $\left\lfloor \frac{2n}{3} \right\rfloor + 1 = 2k+1$		
a_i	b_i	c_i
0	k	$2k$
1	$k+1$	$2k-2$
$\vdots$	$\vdots$	$\vdots$
k	$2k$	0
$k+1$	0	$2k-1$
$k+2$	1	$2k-3$
$\vdots$	$\vdots$	$\vdots$
$2k$	$k-1$	1

$n = 3k+1$ $\left\lfloor \frac{2n}{3} \right\rfloor + 1 = 2k+1$		
a_i	b_i	c_i
0	k	$2k+1$
1	$k+1$	$2k-1$
$\vdots$	$\vdots$	$\vdots$
k	$2k$	1
$k+1$	0	$2k$
$k+2$	1	$2k-2$
$\vdots$	$\vdots$	$\vdots$
$2k$	$k-1$	2

It can be easily seen that the conditions (1) and (2) are satisfied and that we indeed have $\left\lfloor \frac{2n}{3} \right\rfloor + 1$ triples in each case.

Comment. A cute combinatorial model is given by an equilateral triangle, partitioned into n^2 congruent equilateral triangles by $n-1$ equidistant parallels to each of its three sides. Two chess-like bishops placed at any two vertices of the small triangles are said to *menace* one another if they lie on a same parallel. The problem is to determine the largest number of bishops that can be placed so that none menaces another. A bishop may be assigned three coordinates a, b, c, namely the numbers of sides of small triangles they are off each of the sides of the big triangle. It is readily seen that the sum of these coordinates is always n, therefore fulfilling the requirements.

C3 RUS (Russian Federation)

Let n be a positive integer. Given a sequence $\varepsilon_1, \ldots, \varepsilon_{n-1}$ with $\varepsilon_i = 0$ or $\varepsilon_i = 1$ for each $i = 1, \ldots, n-1$, the sequences $a_0, \ldots, a_n$ and $b_0, \ldots, b_n$ are constructed by the following rules:

$$a_0 = b_0 = 1, \quad a_1 = b_1 = 7,$$

$$a_{i+1} = \begin{cases} 2a_{i-1} + 3a_i, & \text{if } \varepsilon_i = 0, \\ 3a_{i-1} + a_i, & \text{if } \varepsilon_i = 1, \end{cases} \quad \text{for each } i = 1, \ldots, n-1,$$

$$b_{i+1} = \begin{cases} 2b_{i-1} + 3b_i, & \text{if } \varepsilon_{n-i} = 0, \\ 3b_{i-1} + b_i, & \text{if } \varepsilon_{n-i} = 1, \end{cases} \quad \text{for each } i = 1, \ldots, n-1.$$

Prove that $a_n = b_n$.

Solution. For a binary word $w = \sigma_1 \ldots \sigma_n$ of length n and a letter $\sigma \in \{0, 1\}$ let $w\sigma = \sigma_1 \ldots \sigma_n \sigma$ and $\sigma w = \sigma \sigma_1 \ldots \sigma_n$. Moreover let $\overline{w} = \sigma_n \ldots \sigma_1$ and let $\emptyset$ be the empty word (of length 0 and with $\overline{\emptyset} = \emptyset$). Let (u, v) be a pair of two real numbers. For binary words w we define recursively the numbers $(u, v)^w$ as follows:

$$(u, v)^{\emptyset} = v, \quad (u, v)^0 = 2u + 3v, \quad (u, v)^1 = 3u + v,$$

$$(u, v)^{w\sigma\varepsilon} = \begin{cases} 2(u, v)^w + 3(u, v)^{w\sigma}, & \text{if } \varepsilon = 0, \\ 3(u, v)^w + (u, v)^{w\sigma}, & \text{if } \varepsilon = 1. \end{cases}$$

It easily follows by induction on the length of w that for all real numbers $u_1, v_1, u_2, v_2, \lambda_1$ and λ_2

$$(\lambda_1 u_1 + \lambda_2 u_2, \lambda_1 v_1 + \lambda_2 v_2)^w = \lambda_1 (u_1, v_1)^w + \lambda_2 (u_2, v_2)^w \tag{1}$$

and that for $\varepsilon \in \{0, 1\}$

$$(u, v)^{\varepsilon w} = (v, (u, v)^{\varepsilon})^w. \tag{2}$$

Obviously, for $n \geq 1$ and $w = \varepsilon_1 \ldots \varepsilon_{n-1}$, we have $a_n = (1, 7)^w$ and $b_n = (1, 7)^{\overline{w}}$. Thus it is sufficient to prove that

$$(1, 7)^w = (1, 7)^{\overline{w}} \tag{3}$$

for each binary word w. We proceed by induction on the length of w. The assertion is obvious if w has length 0 or 1. Now let $w\sigma\varepsilon$ be a binary word of length $n \geq 2$ and suppose that the assertion is true for all binary words of length at most $n - 1$.

Note that $(2, 1)^{\sigma} = 7 = (1, 7)^{\emptyset}$ for $\sigma \in \{0, 1\}$, $(1, 7)^0 = 23$, and $(1, 7)^1 = 10$.

First let $\varepsilon = 0$. Then in view of the induction hypothesis and the equalities (1) and (2), we obtain

$$(1, 7)^{w\sigma 0} = 2(1, 7)^w + 3(1, 7)^{w\sigma} = 2(1, 7)^{\overline{w}} + 3(1, 7)^{\sigma\overline{w}} = 2(2, 1)^{\sigma\overline{w}} + 3(1, 7)^{\sigma\overline{w}}$$
$$= (7, 23)^{\sigma\overline{w}} = (1, 7)^{0\sigma\overline{w}}.$$

Now let $\varepsilon = 1$. Analogously, we obtain

$$(1,7)^{w\sigma 1} = 3(1,7)^{w} + (1,7)^{w\sigma} = 3(1,7)^{\overline{w}} + (1,7)^{\sigma\overline{w}} = 3(2,1)^{\sigma\overline{w}} + (1,7)^{\sigma\overline{w}} \\ = (7,10)^{\sigma\overline{w}} = (1,7)^{1\sigma\overline{w}}.$$

Thus the induction step is complete, (3) and hence also $a_n = b_n$ are proved.

Comment. The original solution uses the relation

$$(1,7)^{\alpha\beta w} = ((1,7)^{w}, (1,7)^{\beta w})^{\alpha}, \quad \alpha, \beta \in \{0,1\},$$

which can be proved by induction on the length of w. Then (3) also follows by induction on the length of w:

$$(1,7)^{\alpha\beta w} = ((1,7)^{w}, (1,7)^{\beta w})^{\alpha} = ((1,7)^{\overline{w}}, (1,7)^{\overline{w}\beta})^{\alpha} = (1,7)^{\overline{w}\beta\alpha}.$$

Here w may be the empty word.

C4 NLD (Netherlands)

For an integer $m \geq 1$, we consider partitions of a $2^m \times 2^m$ chessboard into rectangles consisting of cells of the chessboard, in which each of the 2^m cells along one diagonal forms a separate rectangle of side length 1. Determine the smallest possible sum of rectangle perimeters in such a partition.

Solution 1. For a $k \times k$ chessboard, we introduce in a standard way coordinates of the vertices of the cells and assume that the cell C_{ij} in row i and column j has vertices $(i-1, j-1), (i-1, j), (i, j-1), (i, j)$, where $i, j \in \{1, \ldots, k\}$. Without loss of generality assume that the cells C_{ii}, $i = 1, \ldots, k$, form a separate rectangle. Then we may consider the boards $B_k = \bigcup_{1 \leq i < j \leq k} C_{ij}$ below that diagonal and the congruent board $B'_k = \bigcup_{1 \leq j < i \leq k} C_{ij}$ above that diagonal separately because no rectangle can simultaneously cover cells from B_k and B'_k. We will show that for $k = 2^m$ the smallest total perimeter of a rectangular partition of B_k is $m2^{m+1}$. Then the overall answer to the problem is $2 \cdot m2^{m+1} + 4 \cdot 2^m = (m+1)2^{m+2}$.

First we inductively construct for $m \geq 1$ a partition of B_{2^m} with total perimeter $m2^{m+1}$. If $m = 0$, the board B_{2^m} is empty and the total perimeter is 0. For $m \geq 0$, the board $B_{2^{m+1}}$ consists of a $2^m \times 2^m$ square in the lower right corner with vertices $(2^m, 2^m), (2^m, 2^{m+1}), (2^{m+1}, 2^m)$, $(2^{m+1}, 2^{m+1})$ to which two boards congruent to B_{2^m} are glued along the left and the upper margin. The square together with the inductive partitions of these two boards yield a partition with total perimeter $4 \cdot 2^m + 2 \cdot m2^{m+1} = (m+1)2^{m+2}$ and the induction step is complete.

Let

$$D_k = 2k \log_2 k.$$

Note that $D_k = m2^{m+1}$ if $k = 2^m$. Now we show by induction on k that the total perimeter of a rectangular partition of B_k is at least D_k. The case $k = 1$ is trivial (see $m = 0$ from above). Let the assertion be true for all positive integers less than k. We investigate a fixed rectangular partition of B_k that attains the minimal total perimeter. Let R be the rectangle that covers the cell C_{1k} in the lower right corner. Let (i, j) be the upper left corner of R. First we show that $i = j$. Assume that $i < j$. Then the line from (i, j) to $(i+1, j)$ or from (i, j) to $(i, j-1)$ must belong to the boundary of some rectangle in the partition. Without loss of generality assume that this is the case for the line from (i, j) to $(i+1, j)$.

Case 1. No line from (i, l) to $(i+1, l)$ where $j < l < k$ belongs to the boundary of some rectangle of the partition.
Then there is some rectangle R' of the partition that has with R the common side from (i, j) to (i, k). If we join these two rectangles to one rectangle we get a partition with smaller total perimeter, a contradiction.

Case 2. There is some l such that $j < l < k$ and the line from (i, l) to $(i+1, l)$ belongs to the boundary of some rectangle of the partition.
Then we replace the upper side of R by the line $(i+1, j)$ to $(i+1, k)$ and for the rectangles whose lower side belongs to the line from (i, j) to (i, k) we shift the lower side upwards so that the new lower side belongs to the line from $(i+1, j)$ to $(i+1, k)$. In such a way we obtain a rectangular partition of B_k with smaller total perimeter, a contradiction.

Now the fact that the upper left corner of R has the coordinates (i, i) is established. Consequently, the partition consists of R, of rectangles of a partition of a board congruent to B_i and of rectangles of a partition of a board congruent to B_{k-i}. By the induction hypothesis, its total

perimeter is at least

$$2(k-i)+2i+D_i+D_{k-i}\geq 2k+2i\log_2 i+2(k-i)\log_2(k-i). \tag{1}$$

Since the function $f(x)=2x\log_2 x$ is convex for $x>0$, Jensen's inequality immediately shows that the minimum of the right hand sight of (1) is attained for $i=k/2$. Hence the total perimeter of the optimal partition of B_k is at least $2k+2k/2\log_2 k/2+2(k/2)\log_2(k/2)=D_k$.

Solution 2. We start as in Solution 1 and present another proof that $m2^{m+1}$ is a lower bound for the total perimeter of a partition of B_{2^m} into n rectangles. Let briefly $M=2^m$. For $1\leq i\leq M$, let r_i denote the number of rectangles in the partition that cover some cell from row i and let c_j be the number of rectangles that cover some cell from column j. Note that the total perimeter p of all rectangles in the partition is

$$p=2\left(\sum_{i=1}^{M}r_i+\sum_{i=1}^{M}c_i\right).$$

No rectangle can simultaneously cover cells from row i and from column i since otherwise it would also cover the cell C_{ii}. We classify subsets S of rectangles of the partition as follows. We say that S is of type i, $1\leq i\leq M$, if S contains all r_i rectangles that cover some cell from row i, but none of the c_i rectangles that cover some cell from column i. Altogether there are $2^{n-r_i-c_i}$ subsets of type i. Now we show that no subset S can be simultaneously of type i and of type j if $i\neq j$. Assume the contrary and let without loss of generality $i<j$. The cell C_{ij} must be covered by some rectangle R. The subset S is of type i, hence R is contained in S. S is of type j, thus R does not belong to S, a contradiction. Since there are 2^n subsets of rectangles of the partition, we infer

$$2^n\geq\sum_{i=1}^{M}2^{n-r_i-c_i}=2^n\sum_{i=1}^{M}2^{-(r_i+c_i)}. \tag{2}$$

By applying Jensen's inequality to the convex function $f(x)=2^{-x}$ we derive

$$\frac{1}{M}\sum_{i=1}^{M}2^{-(r_i+c_i)}\geq 2^{-\frac{1}{M}\sum_{i=1}^{M}(r_i+c_i)}=2^{-\frac{p}{2M}}. \tag{3}$$

From (2) and (3) we obtain

$$1\geq M2^{-\frac{p}{2M}}$$

and equivalently

$$p\geq m2^{m+1}.$$

C5 NLD (Netherlands)

Five identical empty buckets of 2-liter capacity stand at the vertices of a regular pentagon. Cinderella and her wicked Stepmother go through a sequence of rounds: At the beginning of every round, the Stepmother takes one liter of water from the nearby river and distributes it arbitrarily over the five buckets. Then Cinderella chooses a pair of neighboring buckets, empties them into the river, and puts them back. Then the next round begins. The Stepmother's goal is to make one of these buckets overflow. Cinderella's goal is to prevent this. Can the wicked Stepmother enforce a bucket overflow?

Solution 1. No, the Stepmother cannot enforce a bucket overflow and Cinderella can keep playing forever. Throughout we denote the five buckets by B_0, B_1, B_2, B_3, and B_4, where B_k is adjacent to bucket B_{k-1} and B_{k+1} $(k = 0, 1, 2, 3, 4)$ and all indices are taken modulo 5. Cinderella enforces that the following three conditions are satisfied at the beginning of every round:

(1) Two adjacent buckets (say B_1 and B_2) are empty.

(2) The two buckets standing next to these adjacent buckets (here B_0 and B_3) have total contents at most 1.

(3) The remaining bucket (here B_4) has contents at most 1.

These conditions clearly hold at the beginning of the first round, when all buckets are empty.

Assume that Cinderella manages to maintain them until the beginning of the r-th round $(r \geq 1)$. Denote by x_k $(k = 0, 1, 2, 3, 4)$ the contents of bucket B_k at the beginning of this round and by y_k the corresponding contents after the Stepmother has distributed her liter of water in this round.

By the conditions, we can assume $x_1 = x_2 = 0$, $x_0 + x_3 \leq 1$ and $x_4 \leq 1$. Then, since the Stepmother adds one liter, we conclude $y_0 + y_1 + y_2 + y_3 \leq 2$. This inequality implies $y_0 + y_2 \leq 1$ or $y_1 + y_3 \leq 1$. For reasons of symmetry, we only consider the second case.

Then Cinderella empties buckets B_0 and B_4.

At the beginning of the next round B_0 and B_4 are empty (condition (1) is fulfilled), due to $y_1 + y_3 \leq 1$ condition (2) is fulfilled and finally since $x_2 = 0$ we also must have $y_2 \leq 1$ (condition (3) is fulfilled).

Therefore, Cinderella can indeed manage to maintain the three conditions (1)–(3) also at the beginning of the $(r + 1)$-th round. By induction, she thus manages to maintain them at the beginning of every round. In particular she manages to keep the contents of every single bucket at most 1 liter. Therefore, the buckets of 2-liter capacity will never overflow.

Solution 2. We prove that Cinderella can maintain the following two conditions and hence she can prevent the buckets from overflow:

(1′) Every two non-adjacent buckets contain a total of at most 1.

(2′) The total contents of all five buckets is at most $\frac{3}{2}$.

We use the same notations as in the first solution. The two conditions again clearly hold at the beginning. Assume that Cinderella maintained these two conditions until the beginning of the r-th round. A pair of non-neighboring buckets (B_i, B_{i+2}), $i = 0, 1, 2, 3, 4$ is called *critical*

if $y_i + y_{i+2} > 1$. By condition $(2')$, after the Stepmother has distributed her water we have $y_0 + y_1 + y_2 + y_3 + y_4 \leq \frac{5}{2}$. Therefore,

$$(y_0 + y_2) + (y_1 + y_3) + (y_2 + y_4) + (y_3 + y_0) + (y_4 + y_1) = 2(y_0 + y_1 + y_2 + y_3 + y_4) \leq 5,$$

and hence there is a pair of non-neighboring buckets which is not critical, say (B_0, B_2). Now, if both of the pairs (B_3, B_0) and (B_2, B_4) are critical, we must have $y_1 < \frac{1}{2}$ and Cinderella can empty the buckets B_3 and B_4. This clearly leaves no critical pair of buckets and the total contents of all the buckets is then $y_1 + (y_0 + y_2) \leq \frac{3}{2}$. Therefore, conditions $(1')$ and $(2')$ are fulfilled.

Now suppose that without loss of generality the pair (B_3, B_0) is not critical. If in this case $y_0 \leq \frac{1}{2}$, then one of the inequalities $y_0 + y_1 + y_2 \leq \frac{3}{2}$ and $y_0 + y_3 + y_4 \leq \frac{3}{2}$ must hold. But then Cinderella can empty B_3 and B_4 or B_1 and B_2, respectively and clearly fulfill the conditions.

Finally consider the case $y_0 > \frac{1}{2}$. By $y_0 + y_1 + y_2 + y_3 + y_4 \leq \frac{5}{2}$, at least one of the pairs (B_1, B_3) and (B_2, B_4) is not critical. Without loss of generality let this be the pair (B_1, B_3). Since the pair (B_3, B_0) is not critical and $y_0 > \frac{1}{2}$, we must have $y_3 \leq \frac{1}{2}$. But then, as before, Cinderella can maintain the two conditions at the beginning of the next round by either emptying B_1 and B_2 or B_4 and B_0.

Comments on GREEDY approaches. A natural approach for Cinderella would be a GREEDY strategy as for example: *Always remove as much water as possible from the system.* It is straightforward to prove that GREEDY can avoid buckets of capacity $\frac{5}{2}$ from overflowing: If before the Stepmothers move one has $x_0 + x_1 + x_2 + x_3 + x_4 \leq \frac{3}{2}$ then after her move the inequality $Y = y_0 + y_1 + y_2 + y_3 + y_4 \leq \frac{5}{2}$ holds. If now Cinderella removes the two adjacent buckets with maximum total contents she removes at least $\frac{2Y}{5}$ and thus the remaining buckets contain at most $\frac{3}{5} \cdot Y \leq \frac{3}{2}$.

But GREEDY is in general not strong enough to settle this problem as can be seen in the following example:

- In an initial phase, the Stepmother brings all the buckets (after her move) to contents of at least $\frac{1}{2} - 2\epsilon$, where ϵ is an arbitrary small positive number. This can be done by always splitting the 1 liter she has to distribute so that all buckets have the same contents. After her r-th move the total contents of each of the buckets is then c_r with $c_1 = 1$ and $c_{r+1} = 1 + \frac{3}{5} \cdot c_r$ and hence $c_r = \frac{5}{2} - \frac{3}{2} \cdot \left(\frac{3}{5}\right)^{r-1}$. So the contents of each single bucket indeed approaches $\frac{1}{2}$ (from below). In particular, any two adjacent buckets have total contents strictly less than 1 which enables the Stepmother to always refill the buckets that Cinderella just emptied and then distribute the remaining water evenly over all buckets.
- After that phase GREEDY faces a situation like this $(\frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon)$ and leaves a situation of the form $(x_0, x_1, x_2, x_3, x_4) = (\frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon, \frac{1}{2} - 2\epsilon, 0, 0)$.
- Then the Stepmother can add the amounts $(0, \frac{1}{4} + \epsilon, \epsilon, \frac{3}{4} - 2\epsilon, 0)$ to achieve a situation like this: $(y_0, y_1, y_2, y_3, y_4) = (\frac{1}{2} - 2\epsilon, \frac{3}{4} - \epsilon, \frac{1}{2} - \epsilon, \frac{3}{4} - 2\epsilon, 0)$.
- Now B_1 and B_2 are the adjacent buckets with the maximum total contents and thus GREEDY empties them to yield $(x_0, x_1, x_2, x_3, x_4) = (\frac{1}{2} - 2\epsilon, 0, 0, \frac{3}{4} - 2\epsilon, 0)$.
- Then the Stepmother adds $(\frac{5}{8}, 0, 0, \frac{3}{8}, 0)$, which yields $(\frac{9}{8} - 2\epsilon, 0, 0, \frac{9}{8} - 2\epsilon, 0)$.
- Now GREEDY can only empty one of the two nonempty buckets and in the next step the Stepmother adds her liter to the other bucket and brings it to $\frac{17}{8} - 2\epsilon$, i.e. an overflow.

A harder variant. Five identical empty buckets of capacity b stand at the vertices of a regular pentagon. Cinderella and her wicked Stepmother go through a sequence of rounds: At the beginning of every round, the Stepmother takes one liter of water from the nearby river and distributes it arbitrarily over the five buckets. Then Cinderella chooses a pair of neighboring buckets, empties them into the river, and puts them back. Then the next round begins. The Stepmother's goal is to make one of these buckets overflow. Cinderella's goal is to prevent this. Determine all bucket capacities b for which the Stepmother can enforce a bucket to overflow.

Solution to the harder variant. The answer is $b < 2$.

The previous proof shows that for all $b \geq 2$ the Stepmother cannot enforce overflowing. Now if $b < 2$, let R be a positive integer such that $b < 2 - 2^{1-R}$. In the first R rounds the Stepmother now ensures that at least one of the (nonadjacent) buckets B_1 and B_3 have contents of at least $1 - 2^{1-r}$ at the beginning of round r ($r = 1, 2, \ldots, R$). This is trivial for $r = 1$ and if it holds at the beginning of round r, she can fill the bucket which contains at least $1 - 2^{1-r}$ liters with another 2^{-r} liters and put the rest of her water – $1 - 2^{-r}$ liters – in the other bucket. As Cinderella now can remove the water of at most one of the two buckets, the other bucket carries its contents into the next round.

At the beginning of the R-th round there are $1 - 2^{1-R}$ liters in B_1 or B_3. The Stepmother puts the entire liter into that bucket and produces an overflow since $b < 2 - 2^{1-R}$.

C6 BGR (Bulgaria)

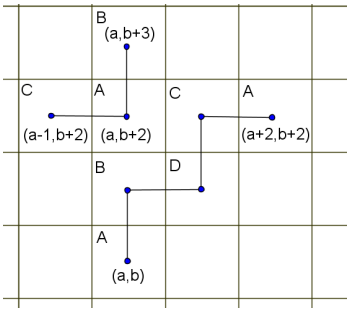
On a 999×999 board a *limp rook* can move in the following way: From any square it can move to any of its adjacent squares, i.e. a square having a common side with it, and every move must be a turn, i.e. the directions of any two consecutive moves must be perpendicular. A *non-intersecting route* of the limp rook consists of a sequence of pairwise different squares that the limp rook can visit in that order by an admissible sequence of moves. Such a non-intersecting route is called *cyclic*, if the limp rook can, after reaching the last square of the route, move directly to the first square of the route and start over.

How many squares does the longest possible cyclic, non-intersecting route of a limp rook visit?

Solution. The answer is $998^2 - 4 = 4 \cdot (499^2 - 1)$ squares.

First we show that this number is an upper bound for the number of cells a limp rook can visit. To do this we color the cells with four colors A, B, C and D in the following way: for $(i, j) \equiv (0, 0) \mod 2$ use A, for $(i, j) \equiv (0, 1) \mod 2$ use B, for $(i, j) \equiv (1, 0) \mod 2$ use C and for $(i, j) \equiv (1, 1) \mod 2$ use D. From an A-cell the rook has to move to a B-cell or a C-cell. In the first case, the order of the colors of the cells visited is given by $A, B, D, C, A, B, D, C, A, \ldots$, in the second case it is $A, C, D, B, A, C, D, B, A, \ldots$. Since the route is closed it must contain the same number of cells of each color. There are only 499^2 A-cells. In the following we will show that the rook cannot visit all the A-cells on its route and hence the maximum possible number of cells in a route is $4 \cdot (499^2 - 1)$.

Assume that the route passes through every single A-cell. Color the A-cells in black and white in a chessboard manner, i.e. color any two A-cells at distance 2 in different color. Since the number of A-cells is odd the rook cannot always alternate between visiting black and white A-cells along its route. Hence there are two A-cells of the same color which are four rook-steps apart that are visited directly one after the other. Let these two A-cells have row and column numbers (a, b) and $(a + 2, b + 2)$ respectively.



There is up to reflection only one way the rook can take from (a, b) to $(a + 2, b + 2)$. Let this way be $(a, b) \to (a, b + 1) \to (a + 1, b + 1) \to (a + 1, b + 2) \to (a + 2, b + 2)$. Also let without loss of generality the color of the cell $(a, b+1)$ be B (otherwise change the roles of columns and rows).

Now consider the A-cell $(a, b+2)$. The only way the rook can pass through it is via $(a-1, b+2) \to (a, b + 2) \to (a, b + 3)$ in this order, since according to our assumption after every A-cell the rook passes through a B-cell. Hence, to connect these two parts of the path, there must be

a path connecting the cell $(a, b+3)$ and (a, b) and also a path connecting $(a+2, b+2)$ and $(a-1, b+2)$.

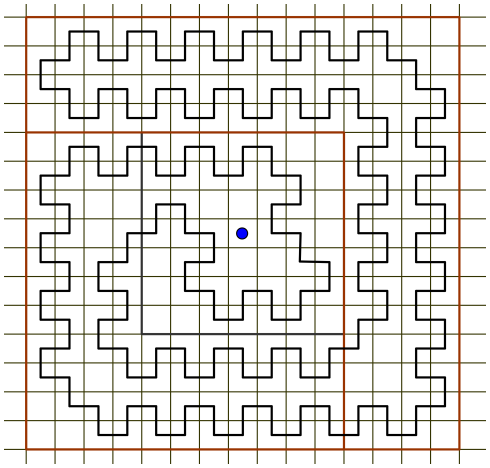
But these four cells are opposite vertices of a convex quadrilateral and the paths are outside of that quadrilateral and hence they must intersect. This is due to the following fact:

The path from (a, b) to $(a, b+3)$ together with the line segment joining these two cells form a closed loop that has one of the cells $(a-1, b+2)$ and $(a+2, b+2)$ in its inside and the other one on the outside. Thus the path between these two points must cross the previous path.

But an intersection is only possible if a cell is visited twice. This is a contradiction.

Hence the number of cells visited is at most $4 \cdot (499^2 - 1)$.

The following picture indicates a recursive construction for all $n \times n$-chessboards with $n \equiv 3 \mod 4$ which clearly yields a path that misses exactly one A-cell (marked with a dot, the center cell of the 15×15-chessboard) and hence, in the case of $n = 999$ crosses exactly $4 \cdot (499^2 - 1)$ cells.



C7 RUS (Russian Federation)

Variant 1. A grasshopper jumps along the real axis. He starts at point 0 and makes 2009 jumps to the right with lengths $1, 2, \ldots, 2009$ in an arbitrary order. Let M be a set of 2008 positive integers less than $1005 \cdot 2009$. Prove that the grasshopper can arrange his jumps in such a way that he never lands on a point from M.

Variant 2. Let n be a nonnegative integer. A grasshopper jumps along the real axis. He starts at point 0 and makes $n+1$ jumps to the right with pairwise different positive integral lengths $a_1, a_2, \ldots, a_{n+1}$ in an arbitrary order. Let M be a set of n positive integers in the interval $(0, s)$, where $s = a_1 + a_2 + \cdots + a_{n+1}$. Prove that the grasshopper can arrange his jumps in such a way that he never lands on a point from M.

Solution of Variant 1. We construct the set of landing points of the grasshopper.

Case 1. M does not contain numbers divisible by 2009.
We fix the numbers $2009k$ as landing points, $k = 1, 2, \ldots, 1005$. Consider the open intervals $I_k = (2009(k-1), 2009k)$, $k = 1, 2, \ldots, 1005$. We show that we can choose exactly one point outside of M as a landing point in 1004 of these intervals such that all lengths from 1 to 2009 are realized. Since there remains one interval without a chosen point, the length 2009 indeed will appear. Each interval has length 2009, hence a new landing point in an interval yields with a length d also the length $2009 - d$. Thus it is enough to implement only the lengths from $D = \{1, 2, \ldots, 1004\}$. We will do this in a greedy way. Let n_k, $k = 1, 2, \ldots, 1005$, be the number of elements of M that belong to the interval I_k. We order these numbers in a decreasing way, so let $p_1, p_2, \ldots, p_{1005}$ be a permutation of $\{1, 2, \ldots, 1005\}$ such that $n_{p_1} \geq n_{p_2} \geq \cdots \geq n_{p_{1005}}$. In I_{p_1} we do not choose a landing point. Assume that landing points have already been chosen in the intervals $I_{p_2}, \ldots, I_{p_m}$ and the lengths $d_2, \ldots, d_m$ from D are realized, $m = 1, \ldots, 1004$. We show that there is some $d \in D \setminus \{d_2, \ldots, d_m\}$ that can be implemented with a new landing point in $I_{p_{m+1}}$. Assume the contrary. Then the $1004 - (m-1)$ other lengths are obstructed by the $n_{p_{m+1}}$ points of M in $I_{p_{m+1}}$. Each length d can be realized by two landing points, namely $2009(p_{m+1} - 1) + d$ and $2009p_{m+1} - d$, hence

$$n_{p_{m+1}} \geq 2(1005 - m). \tag{1}$$

Moreover, since $|M| = 2008 = n_1 + \cdots + n_{1005}$,

$$2008 \geq n_{p_1} + n_{p_2} + \cdots + n_{p_{m+1}} \geq (m+1)n_{p_{m+1}}. \tag{2}$$

Consequently, by (1) and (2),

$$2008 \geq 2(m+1)(1005 - m).$$

The right hand side of the last inequality obviously attains its minimum for $m = 1004$ and this minimum value is greater than 2008, a contradiction.

Case 2. M does contain a number μ divisible by 2009.
By the pigeonhole principle there exists some $r \in \{1, \ldots, 2008\}$ such that M does not contain numbers with remainder r modulo 2009. We fix the numbers $2009(k-1) + r$ as landing points, $k = 1, 2, \ldots, 1005$. Moreover, $1005 \cdot 2009$ is a landing point. Consider the open intervals

$I_k = (2009(k-1)+r, 2009k+r)$, $k = 1, 2, \ldots, 1004$. Analogously to Case 1, it is enough to show that we can choose in 1003 of these intervals exactly one landing point outside of $M \setminus \{\mu\}$ such that each of the lengths of $D = \{1, 2, \ldots, 1004\} \setminus \{r\}$ are implemented. Note that r and $2009 - r$ are realized by the first and last jump and that choosing μ would realize these two differences again. Let n_k, $k = 1, 2, \ldots, 1004$, be the number of elements of $M \setminus \{\mu\}$ that belong to the interval I_k and $p_1, p_2, \ldots, p_{1004}$ be a permutation of $\{1, 2, \ldots, 1004\}$ such that $n_{p_1} \geq n_{p_2} \geq \cdots \geq n_{p_{1004}}$. With the same reasoning as in Case 1 we can verify that a greedy choice of the landing points in $I_{p_2}, I_{p_3}, \ldots, I_{p_{1004}}$ is possible. We only have to replace (1) by

$$n_{p_{m+1}} \geq 2(1004 - m)$$

(D has one element less) and (2) by

$$2007 \geq n_{p_1} + n_{p_2} + \cdots + n_{p_{m+1}} \geq (m+1)n_{p_{m+1}}.$$

Comment. The cardinality 2008 of M in the problem is the maximum possible value. For $M = \{1, 2, \ldots, 2009\}$, the grasshopper necessarily lands on a point from M.

Solution of Variant 2. First of all we remark that the statement in the problem implies a strengthening of itself: Instead of $|M| = n$ it is sufficient to suppose that $|M \cap (0, s - \overline{a}]| \leq n$, where $\overline{a} = \min\{a_1, a_2, \ldots, a_{n+1}\}$. This fact will be used in the proof.

We prove the statement by induction on n. The case $n = 0$ is obvious. Let $n > 0$ and let the assertion be true for all nonnegative integers less than n. Moreover let $a_1, a_2, \ldots, a_{n+1}, s$ and M be given as in the problem. Without loss of generality we may assume that $a_{n+1} < a_n < \cdots < a_2 < a_1$. Set

$$T_k = \sum_{i=1}^{k} a_i \quad \text{for } k = 0, 1, \ldots, n+1.$$

Note that $0 = T_0 < T_1 < \cdots < T_{n+1} = s$. We will make use of the induction hypothesis as follows:

Claim 1. It suffices to show that for some $m \in \{1, 2, \ldots, n+1\}$ the grasshopper is able to do at least m jumps without landing on a point of M and, in addition, after these m jumps he has jumped over at least m points of M.

Proof. Note that $m = n + 1$ is impossible by $|M| = n$. Now set $n' = n - m$. Then $0 \leq n' < n$. The remaining $n' + 1$ jumps can be carried out without landing on one of the remaining at most n' forbidden points by the induction hypothesis together with a shift of the origin. This proves the claim.

An integer $k \in \{1, 2, \ldots, n+1\}$ is called *smooth*, if the grasshopper is able to do k jumps with the lengths $a_1, a_2, \ldots, a_k$ in such a way that he never lands on a point of M except for the very last jump, when he may land on a point of M.

Obviously, 1 is smooth. Thus there is a largest number k^*, such that all the numbers $1, 2, \ldots, k^*$ are smooth. If $k^* = n + 1$, the proof is complete. In the following let $k^* \leq n$.

Claim 2. We have

$$T_{k^*} \in M \quad \text{and} \quad |M \cap (0, T_{k^*})| \geq k^*. \tag{3}$$

Proof. In the case $T_{k^*} \notin M$ any sequence of jumps that verifies the smoothness of k^* can be extended by appending a_{k^*+1}, which is a contradiction to the maximality of k^*. Therefore we have $T_{k^*} \in M$. If $|M \cap (0, T_{k^*})| < k^*$, there exists an $l \in \{1, 2, \ldots, k^*\}$ with $T_{k^*+1} - a_l \notin M$. By the induction hypothesis with $k^* - 1$ instead of n, the grasshopper is able to reach $T_{k^*+1} - a_l$

with k^* jumps of lengths from $\{a_1, a_2, \ldots, a_{k^*+1}\} \setminus \{a_l\}$ without landing on any point of M. Therefore k^*+1 is also smooth, which is a contradiction to the maximality of k^*. Thus Claim 2 is proved.

Now, by Claim 2, there exists a smallest integer $\overline{k} \in \{1, 2, \ldots, k^*\}$ with

$$T_{\overline{k}} \in M \quad \text{and} \quad |M \cap (0, T_{\overline{k}})| \geq \overline{k}.$$

Claim 3. It is sufficient to consider the case

$$|M \cap (0, T_{\overline{k}-1}]| \leq \overline{k} - 1. \tag{4}$$

Proof. If $\overline{k} = 1$, then (4) is clearly satisfied. In the following let $\overline{k} > 1$. If $T_{\overline{k}-1} \in M$, then (4) follows immediately by the minimality of $\overline{k}$. If $T_{\overline{k}-1} \notin M$, by the smoothness of $\overline{k} - 1$, we obtain a situation as in Claim 1 with $m = \overline{k} - 1$ provided that $|M \cap (0, T_{\overline{k}-1}]| \geq \overline{k} - 1$. Hence, we may even restrict ourselves to $|M \cap (0, T_{\overline{k}-1}]| \leq \overline{k} - 2$ in this case and Claim 3 is proved.

Choose an integer $v \geq 0$ with $|M \cap (0, T_{\overline{k}})| = \overline{k} + v$. Let $r_1 > r_2 > \cdots > r_l$ be exactly those indices r from $\{\overline{k}+1, \overline{k}+2, \ldots, n+1\}$ for which $T_{\overline{k}} + a_r \notin M$. Then

$$n = |M| = |M \cap (0, T_{\overline{k}})| + 1 + |M \cap (T_{\overline{k}}, s)| \geq \overline{k} + v + 1 + (n + 1 - \overline{k} - l)$$

and consequently $l \geq v + 2$. Note that

$$T_{\overline{k}} + a_{r_1} - a_1 < T_{\overline{k}} + a_{r_1} - a_2 < \cdots < T_{\overline{k}} + a_{r_1} - a_{\overline{k}} < T_{\overline{k}} + a_{r_2} - a_{\overline{k}} < \cdots < T_{\overline{k}} + a_{r_{v+2}} - a_{\overline{k}} < T_{\overline{k}}$$

and that this are $\overline{k} + v + 1$ numbers from $(0, T_{\overline{k}})$. Therefore we find some $r \in \{\overline{k}+1, \overline{k}+2, \ldots, n+1\}$ and some $s \in \{1, 2, \ldots, \overline{k}\}$ with $T_{\overline{k}} + a_r \notin M$ and $T_{\overline{k}} + a_r - a_s \notin M$. Consider the set of jump lengths $B = \{a_1, a_2, \ldots, a_{\overline{k}}, a_r\} \setminus \{a_s\}$. We have

$$\sum_{x \in B} x = T_{\overline{k}} + a_r - a_s$$

and

$$T_{\overline{k}} + a_r - a_s - \min(B) = T_{\overline{k}} - a_s \leq T_{\overline{k}-1}.$$

By (4) and the strengthening, mentioned at the very beginning with $\overline{k} - 1$ instead of n, the grasshopper is able to reach $T_{\overline{k}} + a_r - a_s$ by $\overline{k}$ jumps with lengths from B without landing on any point of M. From there he is able to jump to $T_{\overline{k}} + a_r$ and therefore we reach a situation as in Claim 1 with $m = \overline{k} + 1$, which completes the proof.

C8 AUT (Austria)

For any integer $n \geq 2$, we compute the integer $h(n)$ by applying the following procedure to its decimal representation. Let r be the rightmost digit of n.

(1) If $r = 0$, then the decimal representation of $h(n)$ results from the decimal representation of n by removing this rightmost digit 0.

(2) If $1 \leq r \leq 9$ we split the decimal representation of n into a maximal right part R that solely consists of digits not less than r and into a left part L that either is empty or ends with a digit strictly smaller than r. Then the decimal representation of $h(n)$ consists of the decimal representation of L, followed by two copies of the decimal representation of $R-1$. For instance, for the number $n = 17{,}151{,}345{,}543$, we will have $L = 17{,}151$, $R = 345{,}543$ and $h(n) = 17{,}151{,}345{,}542{,}345{,}542$.

Prove that, starting with an arbitrary integer $n \geq 2$, iterated application of h produces the integer 1 after finitely many steps.

Solution 1. We identify integers $n \geq 2$ with the digit-strings, briefly strings, of their decimal representation and extend the definition of h to all non-empty strings with digits from 0 to 9. We recursively define ten functions $f_0, \ldots, f_9$ that map some strings into integers for $k = 9, 8, \ldots, 1, 0$. The function f_9 is only defined on strings x (including the empty string ε) that entirely consist of nines. If x consists of m nines, then $f_9(x) = m+1$, $m = 0, 1, \ldots$. For $k \leq 8$, the domain of $f_k(x)$ is the set of all strings consisting only of digits that are $\geq k$. We write x in the form $x_0 k x_1 k x_2 k \ldots x_{m-1} k x_m$ where the strings x_s only consist of digits $\geq k+1$. Note that some of these strings might equal the empty string ε and that $m = 0$ is possible, i.e. the digit k does not appear in x. Then we define

$$f_k(x) = \sum_{s=0}^{m} 4^{f_{k+1}(x_s)}.$$

We will use the following obvious fact:

Fact 1. If x does not contain digits smaller than k, then $f_i(x) = 4^{f_{i+1}(x)}$ for all $i = 0, \ldots, k-1$. In particular, $f_i(\varepsilon) = 4^{9-i}$ for all $i = 0, 1, \ldots, 9$.

Moreover, by induction on $k = 9, 8, \ldots, 0$ it follows easily:

Fact 2. If the nonempty string x does not contain digits smaller than k, then $f_i(x) > f_i(\varepsilon)$ for all $i = 0, \ldots, k$.

We will show the essential fact:

Fact 3. $f_0(n) > f_0(h(n))$.

Then the empty string will necessarily be reached after a finite number of applications of h. But starting from a string without leading zeros, ε can only be reached via the strings $1 \to 00 \to 0 \to \varepsilon$. Hence also the number 1 will appear after a finite number of applications of h.

Proof of Fact 3. If the last digit r of n is 0, then we write $n = x_0 0 \ldots 0 x_{m-1} 0 \varepsilon$ where the x_i do not contain the digit 0. Then $h(n) = x_0 0 \ldots 0 x_{m-1}$ and $f_0(n) - f_0(h(n)) = f_0(\varepsilon) > 0$.

So let the last digit r of n be at least 1. Let $L = yk$ and $R = zr$ be the corresponding left and right parts where y is some string, $k \leq r-1$ and the string z consists only of digits not less

than r. Then $n = ykzr$ and $h(n) = ykz(r-1)z(r-1)$. Let $d(y)$ be the smallest digit of y. We consider two cases which do not exclude each other.

Case 1. $d(y) \geq k$.
Then

$$f_k(n) - f_k(h(n)) = f_k(zr) - f_k(z(r-1)z(r-1)).$$

In view of Fact 1 this difference is positive if and only if

$$f_{r-1}(zr) - f_{r-1}(z(r-1)z(r-1)) > 0.$$

We have, using Fact 2,

$$f_{r-1}(zr) = 4^{f_r(zr)} = 4^{f_r(z)+4^{f_{r+1}(\varepsilon)}} \geq 4 \cdot 4^{f_r(z)} > 4^{f_r(z)} + 4^{f_r(z)} + 4^{f_r(\varepsilon)} = f_{r-1}(z(r-1)z(r-1)).$$

Here we use the additional definition $f_{10}(\varepsilon) = 0$ if $r = 9$. Consequently, $f_k(n) - f_k(h(n)) > 0$ and according to Fact 1, $f_0(n) - f_0(h(n)) > 0$.

Case 2. $d(y) \leq k$.
We prove by induction on $d(y) = k, k-1, \ldots, 0$ that $f_i(n) - f_i(h(n)) > 0$ for all $i = 0, \ldots, d(y)$. By Fact 1, it suffices to do so for $i = d(y)$. The initialization $d(y) = k$ was already treated in Case 1. Let $t = d(y) < k$. Write y in the form utv where v does not contain digits $\leq t$. Then, in view of the induction hypothesis,

$$f_t(n) - f_t(h(n)) = f_t(vkzr) - f_t(vkz(r-1)z(r-1)) = 4^{f_{t+1}(vkzr)} - 4^{f_{t+1}(vkz(r-1)z(r-1))} > 0.$$

Thus the inequality $f_{d(y)}(n) - f_{d(y)}(h(n)) > 0$ is established and from Fact 1 it follows that $f_0(n) - f_0(h(n)) > 0$.

Solution 2. We identify integers $n \geq 2$ with the digit-strings, briefly strings, of their decimal representation and extend the definition of h to all non-empty strings with digits from 0 to 9. Moreover, let us define that the empty string, ε, is being mapped to the empty string. In the following all functions map the set of strings into the set of strings. For two functions f and g let $g \circ f$ be defined by $(g \circ f)(x) = g(f(x))$ for all strings x and let, for non-negative integers n, f^n denote the n-fold application of f. For any string x let $s(x)$ be the smallest digit of x, and for the empty string let $s(\varepsilon) = \infty$. We define nine functions $g_1, \ldots, g_9$ as follows: Let $k \in \{1, \ldots, 9\}$ and let x be a string. If $x = \varepsilon$ then $g_k(x) = \varepsilon$. Otherwise, write x in the form $x = yzr$ where y is either the empty string or ends with a digit smaller than k, $s(z) \geq k$ and r is the rightmost digit of x. Then $g_k(x) = zr$.

Lemma 1. We have $g_k \circ h = g_k \circ h \circ g_k$ for all $k = 1, \ldots, 9$.

Proof of Lemma 1. Let $x = yzr$ be as in the definition of g_k. If $y = \varepsilon$, then $g_k(x) = x$, whence

$$g_k(h(x)) = g_k(h(g_k(x)). \tag{1}$$

So let $y \neq \varepsilon$.

Case 1. z contains a digit smaller than r.
Let $z = uav$ where $a < r$ and $s(v) \geq r$. Then

$$h(x) = \begin{cases} yuav & \text{if } r = 0, \\ yuav(r-1)v(r-1) & \text{if } r > 0 \end{cases}$$

and

$$h(g_k(x)) = h(zr) = h(uavr) = \begin{cases} uav & \text{if } r = 0, \\ uav(r-1)v(r-1) & \text{if } r > 0. \end{cases}$$

Since y ends with a digit smaller than k, (1) is obviously true.

Case 2. z does not contain a digit smaller than r.
Let $y = uv$ where u is either the empty string or ends with a digit smaller than r and $s(v) \geq r$. We have

$$h(x) = \begin{cases} uvz & \text{if } r = 0, \\ uvz(r-1)vz(r-1) & \text{if } r > 0 \end{cases}$$

and

$$h(g_k(x)) = h(zr) = \begin{cases} z & \text{if } r = 0, \\ z(r-1)z(r-1) & \text{if } r > 0. \end{cases}$$

Recall that y and hence v ends with a digit smaller than k, but all digits of v are at least r. Now if $r > k$, then $v = \varepsilon$, whence the terminal digit of u is smaller than k, which entails

$$g_k(h(x)) = z(r-1)z(r-1) = g_k(h(g_k(x))).$$

If $r \leq k$, then

$$g_k(h(x)) = z(r-1) = g_k(h(g_k(x))) \ ,$$

so that in both cases (1) is true. Thus Lemma 1 is proved.

Lemma 2. Let $k \in \{1, \ldots, 9\}$, let x be a non-empty string and let n be a positive integer. If $h^n(x) = \varepsilon$ then $(g_k \circ h)^n(x) = \varepsilon$.

Proof of Lemma 2. We proceed by induction on n. If $n = 1$ we have

$$\varepsilon = h(x) = g_k(h(x)) = (g_k \circ h)(x).$$

Now consider the step from $n-1$ to n where $n \geq 2$. Let $h^n(x) = \varepsilon$ and let $y = h(x)$. Then $h^{n-1}(y) = \varepsilon$ and by the induction hypothesis $(g_k \circ h)^{n-1}(y) = \varepsilon$. In view of Lemma 1,

$$\begin{aligned} \varepsilon = (g_k \circ h)^{n-2}((g_k \circ h)(y)) &= (g_k \circ h)^{n-2}(g_k(h(y)) \\ &= (g_k \circ h)^{n-2}(g_k(h(g_k(y))) = (g_k \circ h)^{n-2}(g_k(h(g_k(h(x))))) = (g_k \circ h)^n(x). \end{aligned}$$

Thus the induction step is complete and Lemma 2 is proved.

We say that the non-empty string x *terminates* if $h^n(x) = \varepsilon$ for some non-negative integer n.

Lemma 3. Let $x = yzr$ where $s(y) \geq k$, $s(z) \geq k$, y ends with the digit k and z is possibly empty. If y and zr terminate then also x terminates.

Proof of Lemma 3. Suppose that y and zr terminate. We proceed by induction on k. Let $k = 0$. Obviously, $h(yw) = yh(w)$ for any non-empty string w. Let $h^n(zr) = \epsilon$. It follows easily by induction on m that $h^m(yzr) = yh^m(zr)$ for $m = 1, \ldots, n$. Consequently, $h^n(yzr) = y$. Since y terminates, also $x = yzr$ terminates.

Now let the assertion be true for all nonnegative integers less than k and let us prove it for k where $k \geq 1$. It turns out that it is sufficient to prove that $yg_k(h(zr))$ terminates. Indeed:

Case 1. $r = 0$.
Then $h(yzr) = yz = yg_k(h(zr))$.

Case 2. $0 < r \leq k$.
We have $h(zr) = z(r-1)z(r-1)$ and $g_k(h(zr)) = z(r-1)$. Then $h(yzr) = yz(r-1)yz(r-$

$1) = yg_k(h(zr))yg_k(h(zr))$ and we may apply the induction hypothesis to see that if $yg_kh(zr))$ terminates, then $h(yzr)$ terminates.

Case 3. $r > k$.
Then $h(yzr) = yh(zr) = yg_k(h(zr))$.

Note that $yg_k(h(zr))$ has the form $yz'r'$ where $s(z') \geq k$. By the same arguments it is sufficient to prove that $yg_k(h(z'r')) = y(g_k \circ h)^2(zr)$ terminates and, by induction, that $y(g_k \circ h)^m(zr)$ terminates for some positive integer m. In view of Lemma 2 there is some m such that $(g_k \circ h)^m(zr) = \epsilon$, so $x = yzr$ terminates if y terminates. Thus Lemma 3 is proved.

Now assume that there is some string x that does not terminate. We choose x minimal. If $x \geq 10$, we can write x in the form $x = yzr$ of Lemma 3 and by this lemma x terminates since y and zr are smaller than x. If $x \leq 9$, then $h(x) = (x-1)(x-1)$ and $h(x)$ terminates again by Lemma 3 and the minimal choice of x.

Solution 3. We commence by introducing some terminology. Instead of integers, we will consider the set S of all strings consisting of the digits $0, 1, \ldots, 9$, including the empty string ϵ. If $(a_1, a_2, \ldots, a_n)$ is a nonempty string, we let $\rho(a) = a_n$ denote the *terminal digit of* a and $\lambda(a)$ be the string with the last digit removed. We also define $\lambda(\epsilon) = \epsilon$ and denote the set of non-negative integers by $\mathbb{N}_0$.

Now let $k \in \{0, 1, 2, \ldots, 9\}$ denote any digit. We define a function $f_k : S \longrightarrow S$ on the set of strings: First, if the terminal digit of n belongs to $\{0, 1, \ldots, k\}$, then $f_k(n)$ is obtained from n by deleting this terminal digit, i.e $f_k(n) = \lambda(n)$. Secondly, if the terminal digit of n belongs to $\{k+1, \ldots, 9\}$, then $f_k(n)$ is obtained from n by the process described in the problem. We also define $f_k(\epsilon) = \epsilon$. Note that up to the definition for integers $n \leq 1$, the function f_0 coincides with the function h in the problem, through interpreting integers as digit strings. The argument will be roughly as follows. We begin by introducing a straightforward generalization of our claim about f_0. Then it will be easy to see that f_9 has all these stronger properties, which means that is suffices to show for $k \in \{0, 1, \ldots, 8\}$ that f_k possesses these properties provided that f_{k+1} does.

We continue to use k to denote any digit. The operation f_k is said to be *separating*, if the followings holds: Whenever a is an initial segment of b, there is some $N \in \mathbb{N}_0$ such that $f_k^N(b) = a$. The following two notions only apply to the case where f_k is indeed separating, otherwise they remain undefined. For every $a \in S$ we denote the least $N \in \mathbb{N}_0$ for which $f_k^N(a) = \epsilon$ occurs by $g_k(a)$ (because ϵ is an initial segment of a, such an N exists if f_k is separating). If for every two strings a and b such that a is a terminal segment of b one has $g_k(a) \leq g_k(b)$, we say that f_k is *coherent*. In case that f_k is separating and coherent we call the digit k *seductive*.

As $f_9(a) = \lambda(a)$ for all a, it is obvious that 9 is seductive. Hence in order to show that 0 is seductive, which clearly implies the statement of the problem, it suffices to take any $k \in \{0, 1, \ldots, 8\}$ such that $k+1$ is seductive and to prove that k has to be seductive as well. Note that in doing so, we have the function g_{k+1} at our disposal. We have to establish two things and we begin with

Step 1. f_k is separating.

Before embarking on the proof of this, we record a useful observation which is easily proved by induction on M.

Claim 1. For any strings A, B and any positive integer M such that $f_k^{M-1}(B) \neq \epsilon$, we have

$$f_k^M(AkB) = Akf_k^M(B).$$

Now we call a pair (a, b) of strings *wicked* provided that a is an initial segment of b, but there is no $N \in \mathbb{N}_0$ such that $f_k^N(b) = a$. We need to show that there are none, so assume that there were such pairs. Choose a wicked pair (a, b) for which $g_{k+1}(b)$ attains its minimal possible value. Obviously $b \neq \epsilon$ for any wicked pair (a, b). Let z denote the terminal digit of b. Observe that $a \neq b$, which means that a is also an initial segment of $\lambda(b)$. To facilitate the construction of the eventual contradiction, we prove

Claim 2. There cannot be an $N \in \mathbb{N}_0$ such that

$$f_k^N(b) = \lambda(b).$$

Proof of Claim 2. For suppose that such an N existed. Because $g_{k+1}(\lambda(b)) < g_{k+1}(b)$ in view of the coherency of f_{k+1}, the pair $(a, \lambda(b))$ is not wicked. But then there is some N' for which $f_k^{N'}(\lambda(b)) = a$ which entails $f_k^{N+N'}(b) = a$, contradiction. Hence Claim 2 is proved.

It follows that $z \leq k$ is impossible, for otherwise $N = 1$ violated Claim 2.

Also $z > k+1$ is impossible: Set $B = f_k(b)$. Then also $f_{k+1}(b) = B$, but $g_{k+1}(B) < g_{k+1}(b)$ and a is an initial segment of B. Thus the pair (a, B) is not wicked. Hence there is some $N \in \mathbb{N}_0$ with $a = f_k^N(B)$, which, however, entails $a = f_k^{N+1}(b)$.

We are left with the case $z = k + 1$. Let L denote the left part and $R = R^*(k + 1)$ the right part of b. Then we have symbolically

$$f_k(b) = LR^*kR^*k\ ,\ f_k^2(b) = LR^*kR^* \quad \text{and} \quad f_{k+1}(b) = LR^*.$$

Using that R^* is a terminal segment of LR^* and the coherency of f_{k+1}, we infer

$$g_{k+1}(R^*) \leq g_{k+1}(LR^*) < g_{k+1}(b).$$

Hence the pair (ϵ, R^*) is not wicked, so there is some minimal $M \in \mathbb{N}_0$ with $f_k^M(R^*) = \epsilon$ and by Claim 1 it follows that $f_k^{2+M}(b) = LR^*k$. Finally, we infer that $\lambda(b) = LR^* = f_k(LR^*k) = f_k^{3+M}(b)$, which yields a contradiction to Claim 2.

This final contradiction establishes that f_k is indeed separating.

Step 2. f_k is coherent.

To prepare the proof of this, we introduce some further pieces of terminology. A nonempty string $(a_1, a_2, \ldots, a_n)$ is called a *hypostasis*, if $a_n < a_i$ for all $i = 1, \ldots, n-1$. Reading an arbitrary string a backwards, we easily find a, possibly empty, sequence $(A_1, A_2, \ldots, A_m)$ of hypostases such that $\rho(A_1) \leq \rho(A_2) \leq \cdots \leq \rho(A_m)$ and, symbolically, $a = A_1A_2 \ldots A_m$. The latter sequence is referred to as the *decomposition* of a. So, for instance, $(20, 0, 9)$ is the decomposition of 2009 and the string 50 is a hypostasis. Next we explain when we say about two strings a and b that a is *injectible* into b. The definition is by induction on the length of b. Let $(B_1, B_2, \ldots, B_n)$ be the decomposition of b into hypostases. Then a is injectible into b if for the decomposition $(A_1, A_2, \ldots, A_m)$ of a there is a strictly increasing function $H : \{1, 2, \ldots, m\} \longrightarrow \{1, 2, \ldots, n\}$ satisfying

$$\rho(A_i) = \rho(B_{H(i)}) \text{ for all } i = 1, \ldots, m;$$

$\lambda(A_i)$ is injectible into $\lambda(B_{H(i)})$ for all $i = 1, \ldots, m$.

If one can choose H with $H(m) = n$, then we say that a is *strongly injectible* into b. Obviously, if a is a terminal segment of b, then a is strongly injectible into b.

Claim 3. If a and b are two nonempty strings such that a is strongly injectible into b, then $\lambda(a)$ is injectible into $\lambda(b)$.

Proof of Claim 3. Let $(B_1, B_2, \ldots, B_n)$ be the decomposition of b and let $(A_1, A_2, \ldots, A_m)$ be the decomposition of a. Take a function H exemplifying that a is strongly injectible into b. Let $(C_1, C_2, \ldots, C_r)$ be the decomposition of $\lambda(A_m)$ and let $(D_1, D_2, \ldots, D_s)$ be the decomposition of $\lambda(B_n)$. Choose a strictly increasing $H' : \{1, 2, \ldots, r\} \longrightarrow \{1, 2, \ldots s\}$ witnessing that $\lambda(A_m)$ is injectible into $\lambda(B_n)$. Clearly, $(A_1, A_2, \ldots, A_{m-1}, C_1, C_2, \ldots, C_r)$ is the decomposition of $\lambda(a)$ and $(B_1, B_2, \ldots, B_{n-1}, D_1, D_2, \ldots, D_s)$ is the decomposition of $\lambda(b)$. Then the function $H'' : \{1, 2, \ldots, m+r-1\} \longrightarrow \{1, 2, \ldots, n+s-1\}$ given by $H''(i) = H(i)$ for $i = 1, 2, \ldots, m-1$ and $H''(m-1+i) = n-1+H'(i)$ for $i = 1, 2, \ldots, r$ exemplifies that $\lambda(a)$ is injectible into $\lambda(b)$, which finishes the proof of the claim.

A pair (a, b) of strings is called *aggressive* if a is injectible into b and nevertheless $g_k(a) > g_k(b)$. Observe that if f_k was incoherent, which we shall assume from now on, then such pairs existed. Now among all aggressive pairs we choose one, say (a, b), for which $g_k(b)$ attains its least possible value. Obviously $f_k(a)$ cannot be injectible into $f_k(b)$, for otherwise the pair $(f_k(a), f_k(b))$ was aggressive and contradicted our choice of (a, b). Let $(A_1, A_2, \ldots, A_m)$ and $(B_1, B_2, \ldots, B_n)$ be the decompositions of a and b and take a function $H : \{1, 2, \ldots, m\} \longrightarrow \{1, 2, \ldots, n\}$ exemplifying that a is indeed injectible into b. If we had $H(m) < n$, then a was also injectible into the number b' whose decomposition is $(B_1, B_2, \ldots, B_{n-1})$ and by separativity of f_k we obtained $g_k(b') < g_k(b)$, whence the pair (a, b') was also aggressive, contrary to the minimality condition imposed on b. Therefore a is strongly injectible into b. In particular, a and b have a common terminal digit, say z. If we had $z \leq k$, then $f_k(a) = \lambda(a)$ and $f_k(b) = \lambda(b)$, so that by Claim 3, $f_k(a)$ was injectible into $f_k(b)$, which is a contradiction. Hence, $z \geq k+1$.

Now let r be the minimal element of $\{1, 2, \ldots, m\}$ for which $\rho(A_r) = z$. Then the maximal right part of a consisting of digits $\geq z$ is equal to R_a, the string whose decomposition is $(A_r, A_{r+1}, \ldots, A_m)$. Then $R_a - 1$ is a hypostasis and $(A_1, \ldots, A_{r-1}, R_a - 1, R_a - 1)$ is the decomposition of $f_k(a)$. Defining s and R_b in a similar fashion with respect to b, we see that $(B_1, \ldots, B_{s-1}, R_b - 1, R_b - 1)$ is the decomposition of $f_k(b)$. The definition of injectibility then easily entails that R_a is strongly injectible into R_b. It follows from Claim 3 that $\lambda(R_a) = \lambda(R_a - 1)$ is injectible into $\lambda(R_b) = \lambda(R_b - 1)$, whence the function $H' : \{1, 2, \ldots, r+1\} \longrightarrow \{1, 2, \ldots, s+1\}$, given by $H'(i) = H(i)$ for $i = 1, 2, \ldots, r-1$, $H'(r) = s$ and $H'(r+1) = s+1$ exemplifies that $f_k(a)$ is injectible into $f_k(b)$, which yields a contradiction as before.

This shows that aggressive pairs cannot exist, whence f_k is indeed coherent, which finishes the proof of the seductivity of k, whereby the problem is finally solved.

Geometry

G1 BEL (Belgium)

Let ABC be a triangle with $AB = AC$. The angle bisectors of A and B meet the sides BC and AC in D and E, respectively. Let K be the incenter of triangle ADC. Suppose that $\angle BEK = 45°$. Find all possible values of $\angle BAC$.

Solution 1. Answer: $\angle BAC = 60°$ or $\angle BAC = 90°$ are possible values and the only possible values.

Let I be the incenter of triangle ABC, then K lies on the line CI. Let F be the point, where the incircle of triangle ABC touches the side AC; then the segments IF and ID have the same length and are perpendicular to AC and BC, respectively.

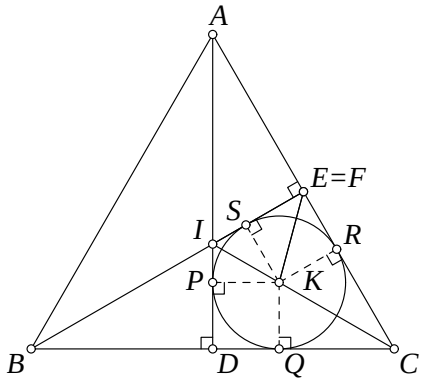


Figure 1

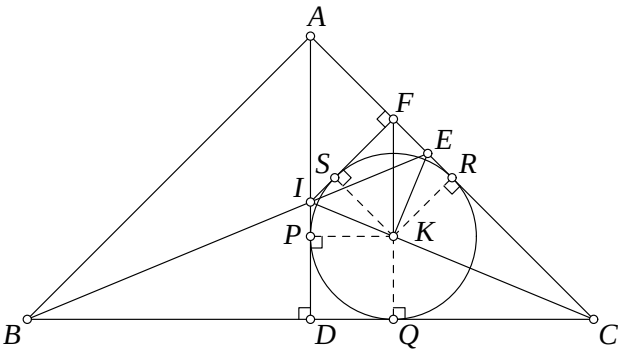


Figure 2

Let P, Q and R be the points where the incircle of triangle ADC touches the sides AD, DC and CA, respectively. Since K and I lie on the angle bisector of $\angle ACD$, the segments ID and IF are symmetric with respect to the line IC. Hence there is a point S on IF where the incircle of triangle ADC touches the segment IF. Then segments KP, KQ, KR and KS all have the same length and are perpendicular to AD, DC, CA and IF, respectively. So – regardless of the value of $\angle BEK$ – the quadrilateral $KRFS$ is a square and $\angle SFK = \angle KFC = 45°$.

Consider the case $\angle BAC = 60°$ (see Figure 1). Then triangle ABC is equilateral. Furthermore we have $F = E$, hence $\angle BEK = \angle IFK = \angle SEK = 45°$. So $60°$ is a possible value for $\angle BAC$.

Now consider the case $\angle BAC = 90°$ (see Figure 2). Then $\angle CBA = \angle ACB = 45°$. Furthermore, $\angle KIE = \frac{1}{2}\angle CBA + \frac{1}{2}\angle ACB = 45°$, $\angle AEB = 180° - 90° - 22.5° = 67.5°$ and $\angle EIA = \angle BID = 180° - 90° - 22.5° = 67.5°$. Hence triangle IEA is isosceles and a reflection of the bisector of $\angle IAE$ takes I to E and K to itself. So triangle IKE is symmetric with respect to this axis, i.e. $\angle KIE = \angle IEK = \angle BEK = 45°$. So $90°$ is a possible value for $\angle BAC$, too.

If, on the other hand, $\angle BEK = 45°$ then $\angle BEK = \angle IEK = \angle IFK = 45°$. Then

- either $F = E$, which makes the angle bisector BI be an altitude, i.e., which makes triangle ABC isosceles with base AC and hence equilateral and so $\angle BAC = 60°$,
- or E lies between F and C, which makes the points K, E, F and I concyclic, so $45° = \angle KFC = \angle KFE = \angle KIE = \angle CBI + \angle ICB = 2 \cdot \angle ICB = 90° - \frac{1}{2}\angle BAC$, and so $\angle BAC = 90°$,

- or F lies between E and C, then again, K, E, F and I are concyclic, so $45^\circ = \angle KFC = 180^\circ - \angle KFE = \angle KIE$, which yields the same result $\angle BAC = 90^\circ$. (However, for $\angle BAC = 90^\circ$ E lies, in fact, between F and C, see Figure 2. So this case does not occur.)

This proves 90° and 60° to be the only possible values for $\angle BAC$.

Solution 2. Denote angles at A, B and C as usual by α, β and γ. Since triangle ABC is isosceles, we have $\beta = \gamma = 90^\circ - \frac{\alpha}{2} < 90^\circ$, so $\angle ECK = 45^\circ - \frac{\alpha}{4} = \angle KCD$. Since K is the incenter of triangle ADC, we have $\angle CDK = \angle KDA = 45^\circ$; furthermore $\angle DIC = 45^\circ + \frac{\alpha}{4}$. Now, if $\angle BEK = 45^\circ$, easy calculations within triangles BCE and KCE yield

$$\angle KEC = 180^\circ - \tfrac{\beta}{2} - 45^\circ - \beta = 135^\circ - \tfrac{3}{2}\beta = \tfrac{3}{2}(90^\circ - \beta) = \tfrac{3}{4}\alpha,$$
$$\angle IKE = \tfrac{3}{4}\alpha + 45^\circ - \tfrac{\alpha}{4} = 45^\circ + \tfrac{\alpha}{2}.$$

So in triangles ICE, IKE, IDK and IDC we have (see Figure 3)

$$\frac{IC}{IE} = \frac{\sin\angle IEC}{\sin\angle ECI} = \frac{\sin(45^\circ + \frac{3}{4}\alpha)}{\sin(45^\circ - \frac{\alpha}{4})}, \qquad \frac{IE}{IK} = \frac{\sin\angle EKI}{\sin\angle IEK} = \frac{\sin(45^\circ + \frac{\alpha}{2})}{\sin 45^\circ},$$
$$\frac{IK}{ID} = \frac{\sin\angle KDI}{\sin\angle IKD} = \frac{\sin 45^\circ}{\sin(90^\circ - \frac{\alpha}{4})}, \qquad \frac{ID}{IC} = \frac{\sin\angle ICD}{\sin\angle CDI} = \frac{\sin(45^\circ - \frac{\alpha}{4})}{\sin 90^\circ}.$$

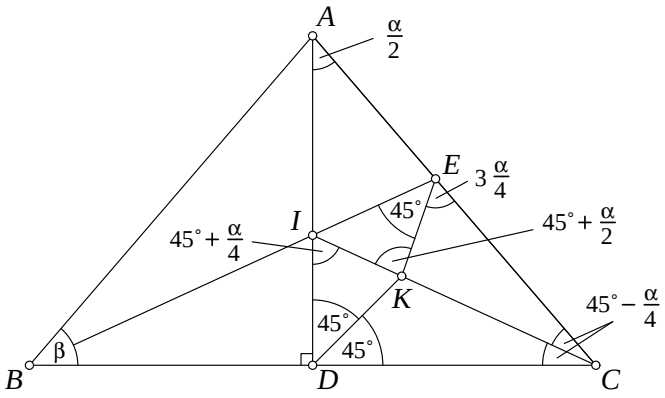


Figure 3

Multiplication of these four equations yields

$$1 = \frac{\sin(45^\circ + \frac{3}{4}\alpha)\sin(45^\circ + \frac{\alpha}{2})}{\sin(90^\circ - \frac{\alpha}{4})}.$$

But, since

$$\begin{aligned}\sin(90^\circ - \tfrac{\alpha}{4}) &= \cos\tfrac{\alpha}{4} = \cos\big((45^\circ + \tfrac{3}{4}\alpha) - (45^\circ + \tfrac{\alpha}{2})\big)\\ &= \cos\big(45^\circ + \tfrac{3}{4}\alpha\big)\cos(45^\circ + \tfrac{\alpha}{2}) + \sin(45^\circ + \tfrac{3}{4}\alpha)\sin(45^\circ + \tfrac{\alpha}{2}),\end{aligned}$$

this is equivalent to

$$\sin(45^\circ + \tfrac{3}{4}\alpha)\sin(45^\circ + \tfrac{\alpha}{2}) = \cos(45^\circ + \tfrac{3}{4}\alpha)\cos(45^\circ + \tfrac{\alpha}{2}) + \sin(45^\circ + \tfrac{3}{4}\alpha)\sin(45^\circ + \tfrac{\alpha}{2})$$

and finally

$$\cos(45^\circ + \tfrac{3}{4}\alpha)\cos(45^\circ + \tfrac{\alpha}{2}) = 0.$$

But this means $\cos(45^\circ + \frac{3}{4}\alpha) = 0$, hence $45^\circ + \frac{3}{4}\alpha = 90^\circ$, i.e. $\alpha = 60^\circ$ or $\cos(45^\circ + \frac{\alpha}{2}) = 0$, hence $45^\circ + \frac{\alpha}{2} = 90^\circ$, i.e. $\alpha = 90^\circ$. So these values are the only two possible values for α.

On the other hand, both $\alpha = 90^\circ$ and $\alpha = 60^\circ$ yield $\angle BEK = 45^\circ$, this was shown in Solution 1.

G2 RUS (Russian Federation)

Let ABC be a triangle with circumcenter O. The points P and Q are interior points of the sides CA and AB, respectively. The circle k passes through the midpoints of the segments BP, CQ, and PQ. Prove that if the line PQ is tangent to circle k then $OP = OQ$.

Solution 1. Let K, L, M, B', C' be the midpoints of BP, CQ, PQ, CA, and AB, respectively (see Figure 1). Since $CA \parallel LM$, we have $\angle LMP = \angle QPA$. Since k touches the segment PQ at M, we find $\angle LMP = \angle LKM$. Thus $\angle QPA = \angle LKM$. Similarly it follows from $AB \parallel MK$ that $\angle PQA = \angle KLM$. Therefore, triangles APQ and MKL are similar, hence

$$\frac{AP}{AQ} = \frac{MK}{ML} = \frac{\frac{QB}{2}}{\frac{PC}{2}} = \frac{QB}{PC}. \tag{1}$$

Now (1) is equivalent to $AP \cdot PC = AQ \cdot QB$ which means that the power of points P and Q with respect to the circumcircle of $\triangle ABC$ are equal, hence $OP = OQ$.

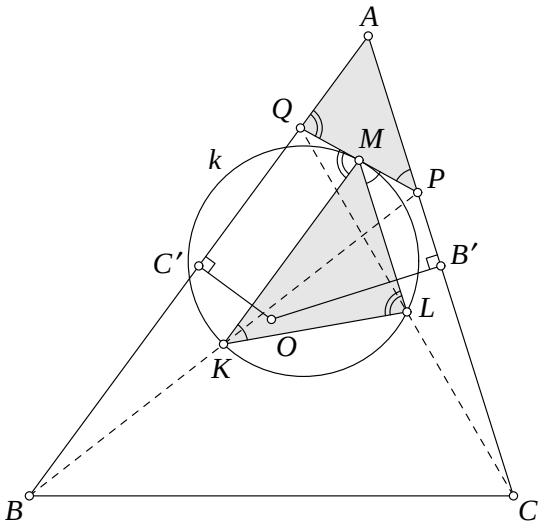


Figure 1

Comment. The last argument can also be established by the following calculation:

$$\begin{aligned} OP^2 - OQ^2 &= OB'^2 + B'P^2 - OC'^2 - C'Q^2 \\ &= (OA^2 - AB'^2) + B'P^2 - (OA^2 - AC'^2) - C'Q^2 \\ &= (AC'^2 - C'Q^2) - (AB'^2 - B'P^2) \\ &= (AC' - C'Q)(AC' + C'Q) - (AB' - B'P)(AB' + B'P) \\ &= AQ \cdot QB - AP \cdot PC. \end{aligned}$$

With (1), we conclude $OP^2 - OQ^2 = 0$, as desired.

Solution 2. Again, denote by K, L, M the midpoints of segments BP, CQ, and PQ, respectively. Let O, S, T be the circumcenters of triangles ABC, KLM, and APQ, respectively (see Figure 2). Note that MK and LM are the midlines in triangles BPQ and CPQ, respectively, so $\overrightarrow{MK} = \frac{1}{2}\overrightarrow{QB}$ and $\overrightarrow{ML} = \frac{1}{2}\overrightarrow{PC}$. Denote by $\mathrm{pr}_l(\overrightarrow{v})$ the projection of vector $\overrightarrow{v}$ onto line l. Then $\mathrm{pr}_{AB}(\overrightarrow{OT}) = \mathrm{pr}_{AB}(\overrightarrow{OA} - \overrightarrow{TA}) = \frac{1}{2}\overrightarrow{BA} - \frac{1}{2}\overrightarrow{QA} = \frac{1}{2}\overrightarrow{BQ} = \overrightarrow{KM}$ and $\mathrm{pr}_{AB}(\overrightarrow{SM}) = \mathrm{pr}_{MK}(\overrightarrow{SM}) = \frac{1}{2}\overrightarrow{KM} = \frac{1}{2}\mathrm{pr}_{AB}(\overrightarrow{OT})$. Analogously we get $\mathrm{pr}_{CA}(\overrightarrow{SM}) = \frac{1}{2}\mathrm{pr}_{CA}(\overrightarrow{OT})$. Since AB and CA are not parallel, this implies that $\overrightarrow{SM} = \frac{1}{2}\overrightarrow{OT}$.

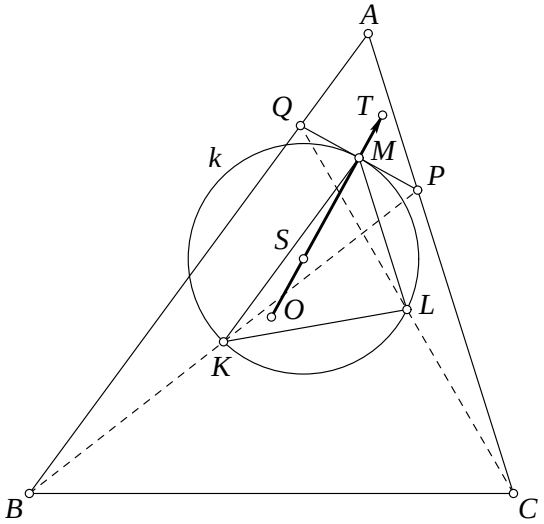


Figure 2

Now, since the circle k touches PQ at M, we get $SM \perp PQ$, hence $OT \perp PQ$. Since T is equidistant from P and Q, the line OT is a perpendicular bisector of segment PQ, and hence O is equidistant from P and Q which finishes the proof.

G3 IRN (Islamic Republic of Iran)

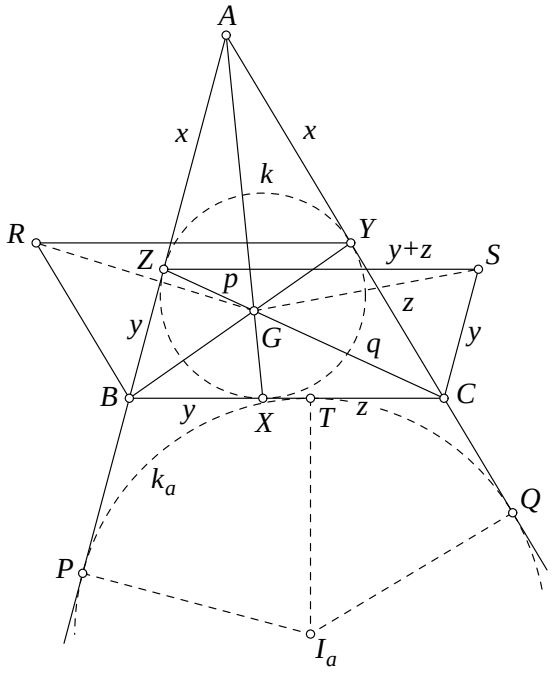
Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y, respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms.

Prove that $GR = GS$.

Solution 1. Denote by k the incircle and by k_a the excircle opposite to A of triangle ABC. Let k and k_a touch the side BC at the points X and T, respectively, let k_a touch the lines AB and AC at the points P and Q, respectively. We use several times the fact that opposing sides of a parallelogram are of equal length, that points of contact of the excircle and incircle to a side of a triangle lie symmetric with respect to the midpoint of this side and that segments on two tangents to a circle defined by the points of contact and their point of intersection have the same length. So we conclude

$$ZP = ZB + BP = XB + BT = BX + CX = ZS \quad \text{and}$$

$$CQ = CT = BX = BZ = CS.$$



So for each of the points Z, C, their distances to S equal the length of a tangent segment from this point to k_a. It is well-known, that all points with this property lie on the line ZC, which is the radical axis of S and k_a. Similar arguments yield that BY is the radical axis of R and k_a. So the point of intersection of ZC and BY, which is G by definition, is the radical center of R, S and k_a, from which the claim $GR = GS$ follows immediately.

Solution 2. Denote $x = AZ = AY$, $y = BZ = BX$, $z = CX = CY$, $p = ZG$, $q = GC$. Several lengthy calculations (Menelaos' theorem in triangle AZC, law of Cosines in triangles ABC and AZC and Stewart's theorem in triangle ZCS) give four equations for p, q, $\cos\alpha$

and GS in terms of x, y, and z that can be resolved for GS. The result is symmetric in y and z, so $GR = GS$. More in detail this means:

The line BY intersects the sides of triangle AZC, so MENELAOS' theorem yields $\frac{p}{q} \cdot \frac{z}{x} \cdot \frac{x+y}{y} = 1$, hence

$$\frac{p}{q} = \frac{xy}{yz+zx}. \tag{1}$$

Since we only want to show that the term for GS is symmetric in y and z, we abbreviate terms that are symmetric in y and z by capital letters, starting with $N = xy+yz+zx$. So (1) implies

$$\frac{p}{p+q} = \frac{xy}{xy+yz+zx} = \frac{xy}{N} \qquad \text{and} \qquad \frac{q}{p+q} = \frac{yz+zx}{xy+yz+zx} = \frac{yz+zx}{N}. \tag{2}$$

Now the law of Cosines in triangle ABC yields

$$\cos\alpha = \frac{(x+y)^2+(x+z)^2-(y+z)^2}{2(x+y)(x+z)} = \frac{2x^2+2xy+2xz-2yz}{2(x+y)(x+z)} = 1 - \frac{2yz}{(x+y)(x+z)}.$$

We use this result to apply the law of Cosines in triangle AZC:

$$\begin{aligned}(p+q)^2 &= x^2+(x+z)^2-2x(x+z)\cos\alpha \\ &= x^2+(x+z)^2-2x(x+z)\cdot\left(1-\frac{2yz}{(x+y)(x+z)}\right) \\ &= z^2+\frac{4xyz}{x+y}.\end{aligned} \tag{3}$$

Now in triangle ZCS the segment GS is a cevian, so with STEWART's theorem we have $py^2+q(y+z)^2 = (p+q)(GS^2+pq)$, hence

$$GS^2 = \frac{p}{p+q}\cdot y^2 + \frac{q}{p+q}\cdot(y+z)^2 - \frac{p}{p+q}\cdot\frac{q}{p+q}\cdot(p+q)^2.$$

Replacing the p's and q's herein by (2) and (3) yields

$$\begin{aligned}GS^2 &= \frac{xy}{N}y^2 + \frac{yz+zx}{N}(y+z)^2 - \frac{xy}{N}\cdot\frac{yz+zx}{N}\cdot\left(z^2+\frac{4xyz}{x+y}\right) \\ &= \frac{xy^3}{N} + \underbrace{\frac{yz(y+z)^2}{N}}_{M_1} + \frac{zx(y+z)^2}{N} - \frac{xyz^3(x+y)}{N^2} - \underbrace{\frac{4x^2y^2z^2}{N^2}}_{M_2} \\ &= \frac{xy^3+zx(y+z)^2}{N} - \frac{xyz^3(x+y)}{N^2} + M_1 - M_2 \\ &= \underbrace{\frac{x(y^3+y^2z+yz^2+z^3)}{N}}_{M_3} + \frac{xyz^2N}{N^2} - \frac{xyz^3(x+y)}{N^2} + M_1 - M_2 \\ &= \frac{x^2y^2z^2+xy^2z^3+x^2yz^3-x^2yz^3-xy^2z^3}{N^2} + M_1 - M_2 + M_3 \\ &= \frac{x^2y^2z^2}{N^2} + M_1 - M_2 + M_3,\end{aligned}$$

a term that is symmetric in y and z, indeed.

Comment. G is known as GERGONNE's point of $\triangle ABC$.

G4 UNK (United Kingdom)

Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F. The midpoints of AB and CD are G and H, respectively. Show that EF is tangent at E to the circle through the points E, G, and H.

Solution 1. It suffices to show that $\angle HEF = \angle HGE$ (see Figure 1), since in circle EGH the angle over the chord EH at G equals the angle between the tangent at E and EH.

First, $\angle BAD = 180^\circ - \angle DCB = \angle FCD$. Since triangles FAB and FCD have also a common interior angle at F, they are similar.

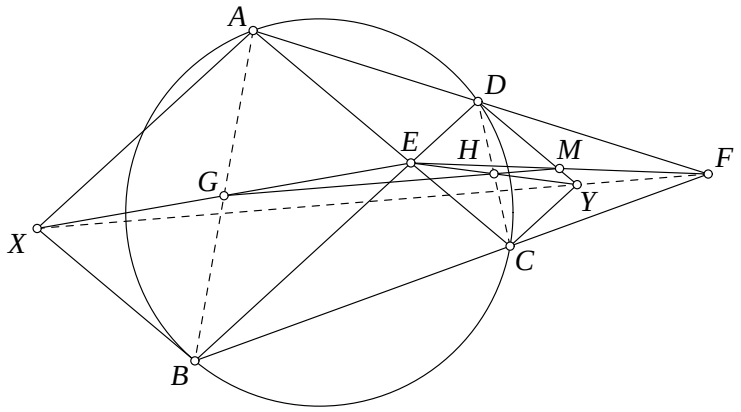


Figure 1

Denote by $\mathcal{T}$ the transformation consisting of a reflection at the bisector of $\angle DFC$ followed by a dilation with center F and factor of $\frac{FA}{FC}$. Then $\mathcal{T}$ maps F to F, C to A, D to B, and H to G. To see this, note that $\triangle FCA \sim \triangle FDB$, so $\frac{FA}{FC} = \frac{FB}{FD}$. Moreover, as $\angle ADB = \angle ACB$, the image of the line DE under $\mathcal{T}$ is parallel to AC (and passes through B) and similarly the image of CE is parallel to DB and passes through A. Hence E is mapped to the point X which is the fourth vertex of the parallelogram $BEAX$. Thus, in particular $\angle HEF = \angle FXG$.

As G is the midpoint of the diagonal AB of the parallelogram $BEAX$, it is also the midpoint of EX. In particular, E, G, X are collinear, and $EX = 2 \cdot EG$.

Denote by Y the fourth vertex of the parallelogram $DECY$. By an analogous reasoning as before, it follows that $\mathcal{T}$ maps Y to E, thus E, H, Y are collinear with $EY = 2 \cdot EH$. Therefore, by the intercept theorem, $HG \parallel XY$.

From the construction of $\mathcal{T}$ it is clear that the lines FX and FE are symmetric with respect to the bisector of $\angle DFC$, as are FY and FE. Thus, F, X, Y are collinear, which together with $HG \parallel XY$ implies $\angle FXE = \angle HGE$. This completes the proof.

Solution 2. We use the following

Lemma (Gauß). Let $ABCD$ be a quadrilateral. Let AB and CD intersect at P, and BC and DA intersect at Q. Then the midpoints K, L, M of AC, BD, and PQ, respectively, are collinear.

Proof: Let us consider the points Z that fulfill the equation

$$(ABZ) + (CDZ) = (BCZ) + (DAZ), \tag{1}$$

where (RST) denotes the oriented area of the triangle RST (see Figure 2).

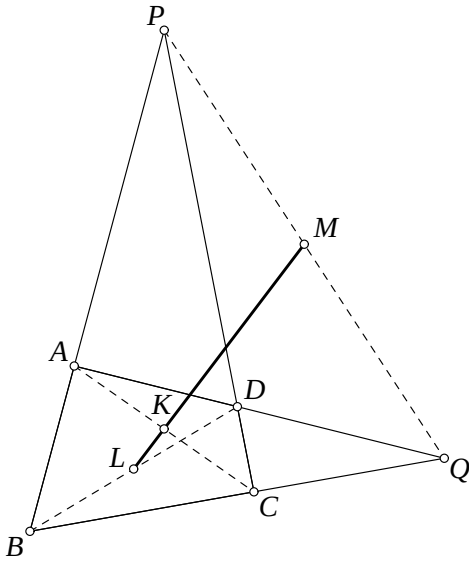


Figure 2

As (1) is linear in Z, it can either characterize a line, or be contradictory, or be trivially fulfilled for all Z in the plane. If (1) was fulfilled for all Z, then it would hold for $Z = A$, $Z = B$, which gives $(CDA) = (BCA)$, $(CDB) = (DAB)$, respectively, i.e. the diagonals of $ABCD$ would bisect each other, thus $ABCD$ would be a parallelogram. This contradicts the hypothesis that AD and BC intersect. Since E, F, G fulfill (1), it is the equation of a line which completes the proof of the lemma.

Now consider the parallelograms $EAXB$ and $ECYD$ (see Figure 1). Then G, H are the midpoints of EX, EY, respectively. Let M be the midpoint of EF. By applying the Lemma to the (re-entrant) quadrilateral $ADBC$, it is evident that G, H, and M are collinear. A dilation by a factor of 2 with center E shows that X, Y, F are collinear. Since $AX \parallel DE$ and $BX \parallel CE$, we have pairwise equal interior angles in the quadrilaterals $FDEC$ and $FBXA$. Since we have also $\angle EBA = \angle DCA = \angle CDY$, the quadrilaterals are similar. Thus, $\angle FXA = \angle CEF$.

Clearly the parallelograms $ECYD$ and $EBXA$ are similar, too, thus $\angle EXA = \angle CEY$. Consequently, $\angle FXE = \angle FXA - \angle EXA = \angle CEF - \angle CEY = \angle YEF$. By the converse of the tangent-chord angle theorem EF is tangent to the circle XEY. A dilation by a factor of $\frac{1}{2}$ completes the proof.

Solution 3. As in Solution 2, G, H, M are proven to be collinear. It suffices to show that $ME^2 = MG \cdot MH$. If $\boldsymbol{p} = \overrightarrow{OP}$ denotes the vector from circumcenter O to point P, the claim becomes

$$\left(\frac{\boldsymbol{e}-\boldsymbol{f}}{2}\right)^2 = \left(\frac{\boldsymbol{e}+\boldsymbol{f}}{2} - \frac{\boldsymbol{a}+\boldsymbol{b}}{2}\right)\left(\frac{\boldsymbol{e}+\boldsymbol{f}}{2} - \frac{\boldsymbol{c}+\boldsymbol{d}}{2}\right),$$

or equivalently

$$4\,\boldsymbol{e}\boldsymbol{f} - (\boldsymbol{e}+\boldsymbol{f})(\boldsymbol{a}+\boldsymbol{b}+\boldsymbol{c}+\boldsymbol{d}) + (\boldsymbol{a}+\boldsymbol{b})(\boldsymbol{c}+\boldsymbol{d}) = 0. \tag{2}$$

With R as the circumradius of $ABCD$, we obtain for the powers $\mathcal{P}(E)$ and $\mathcal{P}(F)$ of E and F, respectively, with respect to the circumcircle

$$\mathcal{P}(E) = (\boldsymbol{e}-\boldsymbol{a})(\boldsymbol{e}-\boldsymbol{c}) = (\boldsymbol{e}-\boldsymbol{b})(\boldsymbol{e}-\boldsymbol{d}) = \boldsymbol{e}^2 - R^2,$$
$$\mathcal{P}(F) = (\boldsymbol{f}-\boldsymbol{a})(\boldsymbol{f}-\boldsymbol{d}) = (\boldsymbol{f}-\boldsymbol{b})(\boldsymbol{f}-\boldsymbol{c}) = \boldsymbol{f}^2 - R^2,$$

hence

$$(\boldsymbol{e}-\boldsymbol{a})(\boldsymbol{e}-\boldsymbol{c}) = \boldsymbol{e}^2 - R^2, \tag{3}$$
$$(\boldsymbol{e}-\boldsymbol{b})(\boldsymbol{e}-\boldsymbol{d}) = \boldsymbol{e}^2 - R^2, \tag{4}$$
$$(\boldsymbol{f}-\boldsymbol{a})(\boldsymbol{f}-\boldsymbol{d}) = \boldsymbol{f}^2 - R^2, \tag{5}$$
$$(\boldsymbol{f}-\boldsymbol{b})(\boldsymbol{f}-\boldsymbol{c}) = \boldsymbol{f}^2 - R^2. \tag{6}$$

Since F lies on the polar to E with respect to the circumcircle, we have

$$4\,\boldsymbol{e}\boldsymbol{f} = 4R^2. \tag{7}$$

Adding up (3) to (7) yields (2), as desired.

G5 POL (Poland)

Let P be a polygon that is convex and symmetric to some point O. Prove that for some parallelogram R satisfying $P \subset R$ we have

$$\frac{|R|}{|P|} \leq \sqrt{2}$$

where $|R|$ and $|P|$ denote the area of the sets R and P, respectively.

Solution 1. We will construct two parallelograms R_1 and R_3, each of them containing P, and prove that at least one of the inequalities $|R_1| \leq \sqrt{2}\,|P|$ and $|R_3| \leq \sqrt{2}\,|P|$ holds (see Figure 1).

First we will construct a parallelogram $R_1 \supseteq P$ with the property that the midpoints of the sides of R_1 are points of the boundary of P.

Choose two points A and B of P such that the triangle OAB has maximal area. Let a be the line through A parallel to OB and b the line through B parallel to OA. Let A', B', a' and b' be the points or lines, that are symmetric to A, B, a and b, respectively, with respect to O. Now let R_1 be the parallelogram defined by a, b, a' and b'.

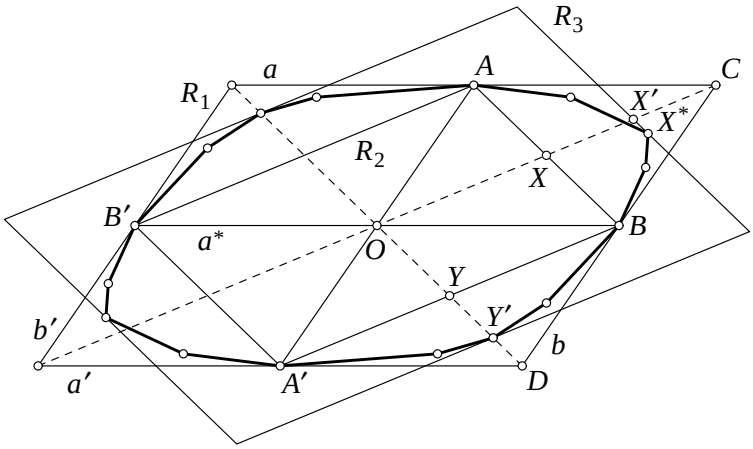


Figure 1

Obviously, A and B are located on the boundary of the polygon P, and A, B, A' and B' are midpoints of the sides of R_1. We note that $P \subseteq R_1$. Otherwise, there would be a point $Z \in P$ but $Z \notin R_1$, i.e., one of the lines a, b, a' or b' were between O and Z. If it is a, we have $|OZB| > |OAB|$, which is contradictory to the choice of A and B. If it is one of the lines b, a' or b' almost identical arguments lead to a similar contradiction.

Let R_2 be the parallelogram $ABA'B'$. Since A and B are points of P, segment $AB \subset P$ and so $R_2 \subset R_1$. Since A, B, A' and B' are midpoints of the sides of R_1, an easy argument yields

$$|R_1| = 2 \cdot |R_2|. \tag{1}$$

Let R_3 be the smallest parallelogram enclosing P defined by lines parallel to AB and BA'. Obviously $R_2 \subset R_3$ and every side of R_3 contains at least one point of the boundary of P. Denote by C the intersection point of a and b, by X the intersection point of AB and OC, and by X' the intersection point of XC and the boundary of R_3. In a similar way denote by D

the intersection point of b and a', by Y the intersection point of $A'B$ and OD, and by Y' the intersection point of YD and the boundary of R_3.

Note that $OC = 2 \cdot OX$ and $OD = 2 \cdot OY$, so there exist real numbers x and y with $1 \le x, y \le 2$ and $OX' = x \cdot OX$ and $OY' = y \cdot OY$. Corresponding sides of R_3 and R_2 are parallel which yields

$$|R_3| = xy \cdot |R_2|. \tag{2}$$

The side of R_3 containing X' contains at least one point X^* of P; due to the convexity of P we have $AX^*B \subset P$. Since this side of the parallelogram R_3 is parallel to AB we have $|AX^*B| = |AX'B|$, so $|OAX'B|$ does not exceed the area of P confined to the sector defined by the rays OB and OA. In a similar way we conclude that $|OB'Y'A'|$ does not exceed the area of P confined to the sector defined by the rays OB and OA'. Putting things together we have $|OAX'B| = x \cdot |OAB|$, $|OBDA'| = y \cdot |OBA'|$. Since $|OAB| = |OBA'|$, we conclude that $|P| \ge 2 \cdot |AX'BY'A'| = 2 \cdot (x \cdot |OAB| + y \cdot |OBA'|) = 4 \cdot \frac{x+y}{2} \cdot |OAB| = \frac{x+y}{2} \cdot R_2$; this is in short

$$\frac{x+y}{2} \cdot |R_2| \le |P|. \tag{3}$$

Since all numbers concerned are positive, we can combine (1)–(3). Using the arithmetic-geometric-mean inequality we obtain

$$|R_1| \cdot |R_3| = 2 \cdot |R_2| \cdot xy \cdot |R_2| \le 2 \cdot |R_2|^2 \left(\frac{x+y}{2}\right)^2 \le 2 \cdot |P|^2.$$

This implies immediately the desired result $|R_1| \le \sqrt{2} \cdot |P|$ or $|R_3| \le \sqrt{2} \cdot |P|$.

Solution 2. We construct the parallelograms R_1, R_2 and R_3 in the same way as in Solution 1 and will show that $\frac{|R_1|}{|P|} \le \sqrt{2}$ or $\frac{|R_3|}{|P|} \le \sqrt{2}$.

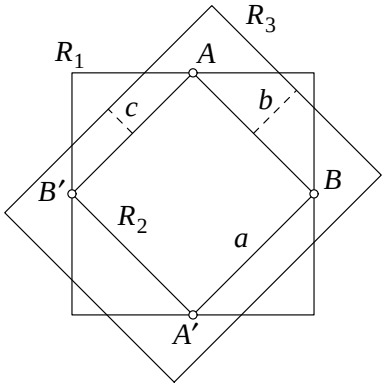


Figure 2

Recall that affine one-to-one maps of the plane preserve the ratio of areas of subsets of the plane. On the other hand, every parallelogram can be transformed with an affine map onto a square. It follows that without loss of generality we may assume that R_1 is a square (see Figure 2).

Then R_2, whose vertices are the midpoints of the sides of R_1, is a square too, and R_3, whose sides are parallel to the diagonals of R_1, is a rectangle.

Let $a > 0$, $b \ge 0$ and $c \ge 0$ be the distances introduced in Figure 2. Then $|R_1| = 2a^2$ and

$|R_3| = (a+2b)(a+2c)$.

Points A, A', B and B' are in the convex polygon P. Hence the square $ABA'B'$ is a subset of P. Moreover, each of the sides of the rectangle R_3 contains a point of P, otherwise R_3 would not be minimal. It follows that

$$|P| \geq a^2 + 2 \cdot \frac{ab}{2} + 2 \cdot \frac{ac}{2} = a(a+b+c).$$

Now assume that both $\frac{|R_1|}{|P|} > \sqrt{2}$ and $\frac{|R_3|}{|P|} > \sqrt{2}$, then

$$2a^2 = |R_1| > \sqrt{2} \cdot |P| \geq \sqrt{2} \cdot a(a+b+c)$$

and

$$(a+2b)(a+2c) = |R_3| > \sqrt{2} \cdot |P| \geq \sqrt{2} \cdot a(a+b+c).$$

All numbers concerned are positive, so after multiplying these inequalities we get

$$2a^2(a+2b)(a+2c) > 2a^2(a+b+c)^2.$$

But the arithmetic-geometric-mean inequality implies the contradictory result

$$2a^2(a+2b)(a+2c) \leq 2a^2 \left(\frac{(a+2b)+(a+2c)}{2} \right)^2 = 2a^2(a+b+c)^2.$$

Hence $\frac{|R_1|}{|P|} \leq \sqrt{2}$ or $\frac{|R_3|}{|P|} \leq \sqrt{2}$, as desired.

G6 UKR (Ukraine)

Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P. Points O_1 and O_2 are the circumcenters and points H_1 and H_2 are the orthocenters of triangles ABP and DCP, respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2, respectively. Prove that the perpendicular from E_1 on CD, the perpendicular from E_2 on AB and the line H_1H_2 are concurrent.

Solution 1. We keep triangle ABP fixed and move the line CD parallel to itself uniformly, i.e. linearly dependent on a single parameter λ (see Figure 1). Then the points C and D also move uniformly. Hence, the points O_2, H_2 and E_2 move uniformly, too. Therefore also the perpendicular from E_2 on AB moves uniformly. Obviously, the points O_1, H_1, E_1 and the perpendicular from E_1 on CD do not move at all. Hence, the intersection point S of these two perpendiculars moves uniformly. Since H_1 does not move, while H_2 and S move uniformly along parallel lines (both are perpendicular to CD), it is sufficient to prove their collinearity for two different positions of CD.

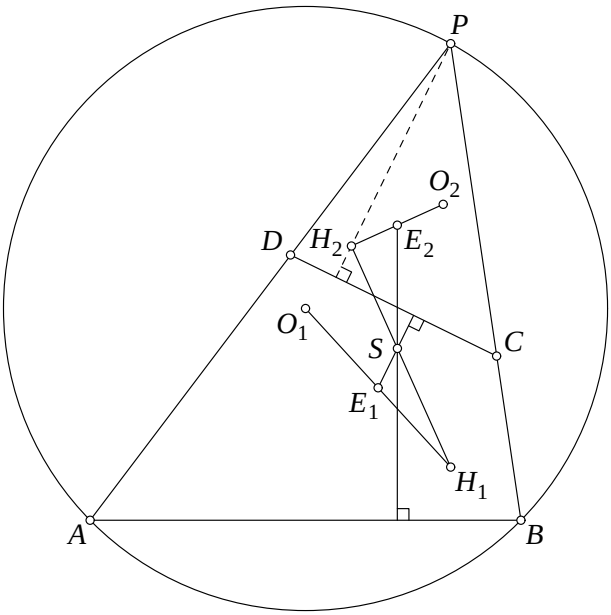


Figure 1

Let CD pass through either point A or point B. Note that by hypothesis these two cases are different. We will consider the case $A \in CD$, i.e. $A = D$. So we have to show that the perpendiculars from E_1 on AC and from E_2 on AB intersect on the altitude AH of triangle ABC (see Figure 2).

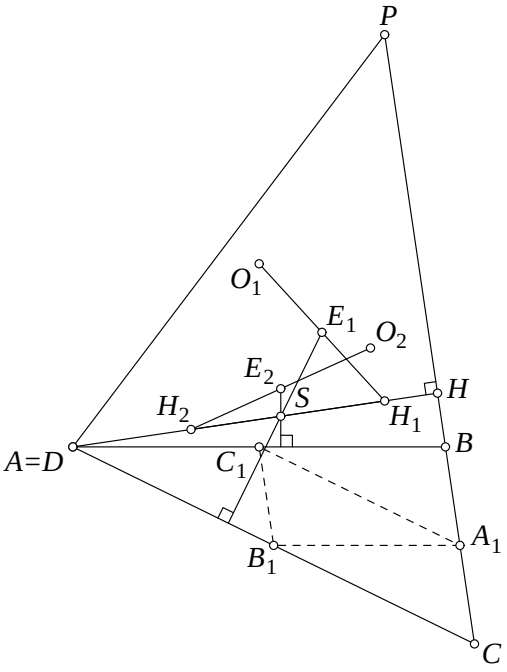


Figure 2

To this end, we consider the midpoints A_1, B_1, C_1 of BC, CA, AB, respectively. As E_1 is the center of Feuerbach's circle (nine-point circle) of $\triangle ABP$, we have $E_1C_1 = E_1H$. Similarly, $E_2B_1 = E_2H$. Note further that a point X lies on the perpendicular from E_1 on A_1C_1 if and only if

$$XC_1^2 - XA_1^2 = E_1C_1^2 - E_1A_1^2.$$

Similarly, the perpendicular from E_2 on A_1B_1 is characterized by

$$XA_1^2 - XB_1^2 = E_2A_1^2 - E_2B_1^2.$$

The line H_1H_2, which is perpendicular to B_1C_1 and contains A, is given by

$$XB_1^2 - XC_1^2 = AB_1^2 - AC_1^2.$$

The three lines are concurrent if and only if

$$\begin{aligned} 0 &= XC_1^2 - XA_1^2 + XA_1^2 - XB_1^2 + XB_1^2 - XC_1^2 \\ &= E_1C_1^2 - E_1A_1^2 + E_2A_1^2 - E_2B_1^2 + AB_1^2 - AC_1^2 \\ &= -E_1A_1^2 + E_2A_1^2 + E_1H^2 - E_2H^2 + AB_1^2 - AC_1^2, \end{aligned}$$

i.e. it suffices to show that

$$E_1A_1^2 - E_2A_1^2 - E_1H^2 + E_2H^2 = \frac{AC^2 - AB^2}{4}.$$

We have

$$\frac{AC^2 - AB^2}{4} = \frac{HC^2 - HB^2}{4} = \frac{(HC + HB)(HC - HB)}{4} = \frac{HA_1 \cdot BC}{2}.$$

Let F_1, F_2 be the projections of E_1, E_2 on BC. Obviously, these are the midpoints of HP_1,

HP_2, where P_1, P_2 are the midpoints of PB and PC respectively. Then

$$\begin{aligned}
&E_1A_1^2 - E_2A_1^2 - E_1H^2 + E_2H^2 \\
&= F_1A_1^2 - F_1H^2 - F_2A_1^2 + F_2H^2 \\
&= (F_1A_1 - F_1H)(F_1A_1 + F_1H) - (F_2A_1 - F_2H)(F_2A_1 + F_2H) \\
&= A_1H \cdot (A_1P_1 - A_1P_2) \\
&= \frac{A_1H \cdot BC}{2} \\
&= \frac{AC^2 - AB^2}{4},
\end{aligned}$$

which proves the claim.

Solution 2. Let the perpendicular from E_1 on CD meet PH_1 at X, and the perpendicular from E_2 on AB meet PH_2 at Y (see Figure 3). Let φ be the intersection angle of AB and CD. Denote by M, N the midpoints of PH_1, PH_2 respectively.

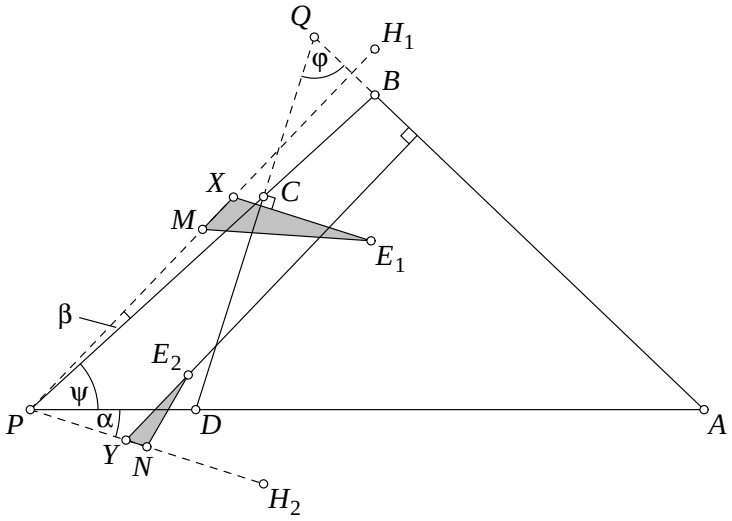


Figure 3

We will prove now that triangles E_1XM and E_2YN have equal angles at E_1, E_2, and supplementary angles at X, Y.

In the following, angles are understood as oriented, and equalities of angles modulo 180°.

Let $\alpha = \angle H_2PD$, $\psi = \angle DPC$, $\beta = \angle CPH_1$. Then $\alpha + \psi + \beta = \varphi$, $\angle E_1XH_1 = \angle H_2YE_2 = \varphi$, thus $\angle MXE_1 + \angle NYE_2 = 180°$.

By considering the Feuerbach circle of $\triangle ABP$ whose center is E_1 and which goes through M, we have $\angle E_1MH_1 = \psi + 2\beta$. Analogous considerations with the Feuerbach circle of $\triangle DCP$ yield $\angle H_2NE_2 = \psi + 2\alpha$. Hence indeed $\angle XE_1M = \varphi - (\psi + 2\beta) = (\psi + 2\alpha) - \varphi = \angle YE_2N$.

It follows now that

$$\frac{XM}{ME_1} = \frac{YN}{NE_2}\ .$$

Furthermore, ME_1 is half the circumradius of $\triangle ABP$, while PH_1 is the distance of P to the orthocenter of that triangle, which is twice the circumradius times the cosine of ψ. Together

with analogous reasoning for $\triangle DCP$ we have

$$\frac{ME_1}{PH_1} = \frac{1}{4\cos\psi} = \frac{NE_2}{PH_2}\,.$$

By multiplication,

$$\frac{XM}{PH_1} = \frac{YN}{PH_2},$$

and therefore

$$\frac{PX}{XH_1} = \frac{H_2Y}{YP}.$$

Let E_1X, E_2Y meet H_1H_2 in R, S respectively.

Applying the intercept theorem to the parallels E_1X, PH_2 and center H_1 gives

$$\frac{H_2R}{RH_1} = \frac{PX}{XH_1},$$

while with parallels E_2Y, PH_1 and center H_2 we obtain

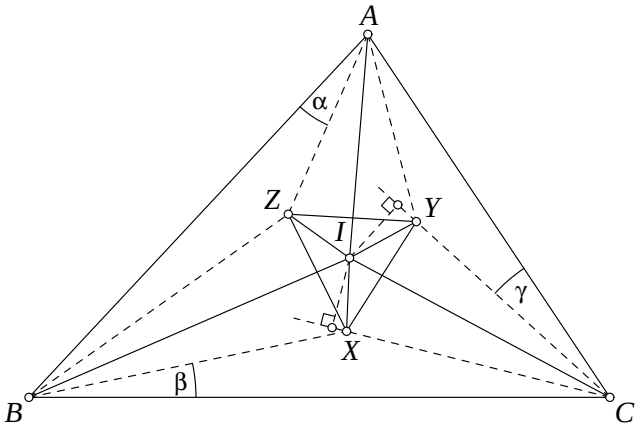
$$\frac{H_2S}{SH_1} = \frac{H_2Y}{YP}.$$

Combination of the last three equalities yields that R and S coincide.

G7 IRN (Islamic Republic of Iran)

Let ABC be a triangle with incenter I and let X, Y and Z be the incenters of the triangles BIC, CIA and AIB, respectively. Let the triangle XYZ be equilateral. Prove that ABC is equilateral too.

Solution. AZ, AI and AY divide $\angle BAC$ into four equal angles; denote them by α. In the same way we have four equal angles β at B and four equal angles γ at C. Obviously $\alpha+\beta+\gamma=\frac{180^\circ}{4}=45^\circ$; and $0^\circ<\alpha,\beta,\gamma<45^\circ$.



Easy calculations in various triangles yield $\angle BIC=180^\circ-2\beta-2\gamma=180^\circ-(90^\circ-2\alpha)=90^\circ+2\alpha$, hence (for X is the incenter of triangle BCI, so IX bisects $\angle BIC$) we have $\angle XIC=\angle BIX=\frac{1}{2}\angle BIC=45^\circ+\alpha$ and with similar aguments $\angle CIY=\angle YIA=45^\circ+\beta$ and $\angle AIZ=\angle ZIB=45^\circ+\gamma$. Furthermore, we have $\angle XIY=\angle XIC+\angle CIY=(45^\circ+\alpha)+(45^\circ+\beta)=135^\circ-\gamma$, $\angle YIZ=135^\circ-\alpha$, and $\angle ZIX=135^\circ-\beta$.

Now we calculate the lengths of IX, IY and IZ in terms of α, β and γ. The perpendicular from I on CX has length $IX\cdot\sin\angle CXI=IX\cdot\sin(90^\circ+\beta)=IX\cdot\cos\beta$. But CI bisects $\angle YCX$, so the perpendicular from I on CY has the same length, and we conclude

$$IX\cdot\cos\beta=IY\cdot\cos\alpha.$$

To make calculations easier we choose a length unit that makes $IX=\cos\alpha$. Then $IY=\cos\beta$ and with similar arguments $IZ=\cos\gamma$.

Since XYZ is equilateral we have $ZX=ZY$. The law of Cosines in triangles XYI, YZI yields

$$\begin{aligned}
& ZX^2=ZY^2\\
\Longrightarrow\quad & IZ^2+IX^2-2\cdot IZ\cdot IX\cdot\cos\angle ZIX=IZ^2+IY^2-2\cdot IZ\cdot IY\cdot\cos\angle YIZ\\
\Longrightarrow\quad & IX^2-IY^2=2\cdot IZ\cdot(IX\cdot\cos\angle ZIX-IY\cdot\cos\angle YIZ)\\
\Longrightarrow\quad & \underbrace{\cos^2\alpha-\cos^2\beta}_{\text{L.H.S.}}=\underbrace{2\cdot\cos\gamma\cdot(\cos\alpha\cdot\cos(135^\circ-\beta)-\cos\beta\cdot\cos(135^\circ-\alpha))}_{\text{R.H.S.}}.
\end{aligned}$$

A transformation of the left-hand side (L.H.S.) yields

$$\begin{aligned}
\text{L.H.S.}&=\cos^2\alpha\cdot\left(\sin^2\beta+\cos^2\beta\right)-\cos^2\beta\cdot\left(\sin^2\alpha+\cos^2\alpha\right)\\
&=\cos^2\alpha\cdot\sin^2\beta-\cos^2\beta\cdot\sin^2\alpha
\end{aligned}$$

$$= (\cos\alpha \cdot \sin\beta + \cos\beta \cdot \sin\alpha) \cdot (\cos\alpha \cdot \sin\beta - \cos\beta \cdot \sin\alpha)$$
$$= \sin(\beta + \alpha) \cdot \sin(\beta - \alpha) = \sin(45^\circ - \gamma) \cdot \sin(\beta - \alpha)$$

whereas a transformation of the right-hand side (R.H.S.) leads to

$$\begin{aligned}\text{R.H.S.} &= 2 \cdot \cos\gamma \cdot (\cos\alpha \cdot (-\cos(45^\circ + \beta)) - \cos\beta \cdot (-\cos(45^\circ + \alpha)))\\ &= 2 \cdot \frac{\sqrt{2}}{2} \cdot \cos\gamma \cdot (\cos\alpha \cdot (\sin\beta - \cos\beta) + \cos\beta \cdot (\cos\alpha - \sin\alpha))\\ &= \sqrt{2} \cdot \cos\gamma \cdot (\cos\alpha \cdot \sin\beta - \cos\beta \cdot \sin\alpha)\\ &= \sqrt{2} \cdot \cos\gamma \cdot \sin(\beta - \alpha).\end{aligned}$$

Equating L.H.S. and R.H.S. we obtain

$$\begin{aligned}& \sin(45^\circ - \gamma) \cdot \sin(\beta - \alpha) = \sqrt{2} \cdot \cos\gamma \cdot \sin(\beta - \alpha)\\ \Longrightarrow\quad & \sin(\beta - \alpha) \cdot \Big(\sqrt{2} \cdot \cos\gamma - \sin(45^\circ - \gamma)\Big) = 0\\ \Longrightarrow\quad & \alpha = \beta \quad \text{or} \quad \sqrt{2} \cdot \cos\gamma = \sin(45^\circ - \gamma).\end{aligned}$$

But $\gamma < 45^\circ$; so $\sqrt{2} \cdot \cos\gamma > \cos\gamma > \cos 45^\circ = \sin 45^\circ > \sin(45^\circ - \gamma)$. This leaves $\alpha = \beta$. With similar reasoning we have $\alpha = \gamma$, which means triangle ABC must be equilateral.

G8 BGR (Bulgaria)

Let $ABCD$ be a circumscribed quadrilateral. Let g be a line through A which meets the segment BC in M and the line CD in N. Denote by I_1, I_2, and I_3 the incenters of $\triangle ABM$, $\triangle MNC$, and $\triangle NDA$, respectively. Show that the orthocenter of $\triangle I_1I_2I_3$ lies on g.

Solution 1. Let k_1, k_2 and k_3 be the incircles of triangles ABM, MNC, and NDA, respectively (see Figure 1). We shall show that the tangent h from C to k_1 which is different from CB is also tangent to k_3.

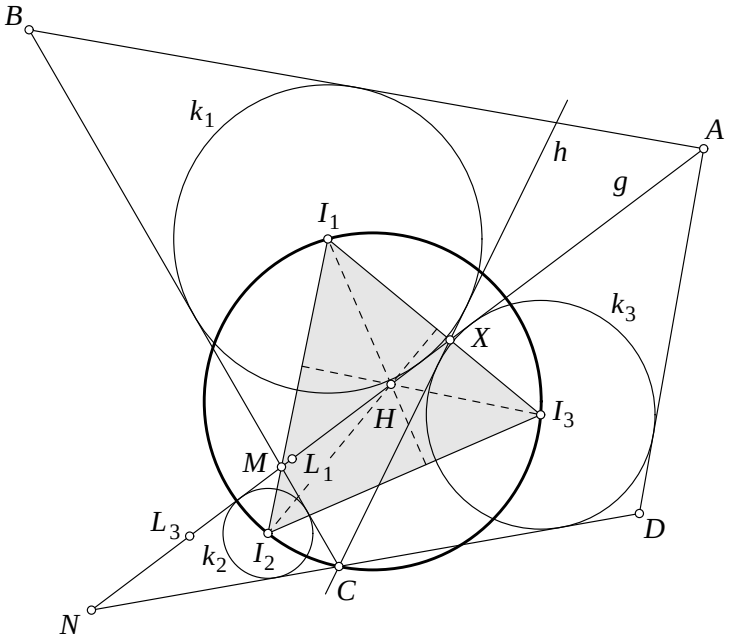


Figure 1

To this end, let X denote the point of intersection of g and h. Then $ABCX$ and $ABCD$ are circumscribed quadrilaterals, whence

$$CD - CX = (AB + CD) - (AB + CX) = (BC + AD) - (BC + AX) = AD - AX,$$

i.e.

$$AX + CD = CX + AD$$

which in turn reveals that the quadrilateral $AXCD$ is also circumscribed. Thus h touches indeed the circle k_3.

Moreover, we find that $\angle I_3CI_1 = \angle I_3CX + \angle XCI_1 = \frac{1}{2}(\angle DCX + \angle XCB) = \frac{1}{2}\angle DCB = \frac{1}{2}(180^\circ - \angle MCN) = 180^\circ - \angle MI_2N = \angle I_3I_2I_1$, from which we conclude that C, I_1, I_2, I_3 are concyclic.

Let now L_1 and L_3 be the reflection points of C with respect to the lines I_2I_3 and I_1I_2 respectively. Since I_1I_2 is the angle bisector of $\angle NMC$, it follows that L_3 lies on g. By analogous reasoning, L_1 lies on g.

Let H be the orthocenter of $\triangle I_1I_2I_3$. We have $\angle I_2L_3I_1 = \angle I_1CI_2 = \angle I_1I_3I_2 = 180^\circ - \angle I_1HI_2$, which entails that the quadrilateral $I_2HI_1L_3$ is cyclic. Analogously, $I_3HL_1I_2$ is cyclic.

Then, working with oriented angles modulo 180°, we have

$$\angle L_3HI_2 = \angle L_3I_1I_2 = \angle I_2I_1C = \angle I_2I_3C = \angle L_1I_3I_2 = \angle L_1HI_2,$$

whence L_1, L_3, and H are collinear. By $L_1 \neq L_3$, the claim follows.

Comment. The last part of the argument essentially reproves the following fact: The Simson line of a point P lying on the circumcircle of a triangle ABC with respect to that triangle bisects the line segment connecting P with the orthocenter of ABC.

Solution 2. We start by proving that C, I_1, I_2, and I_3 are concyclic.

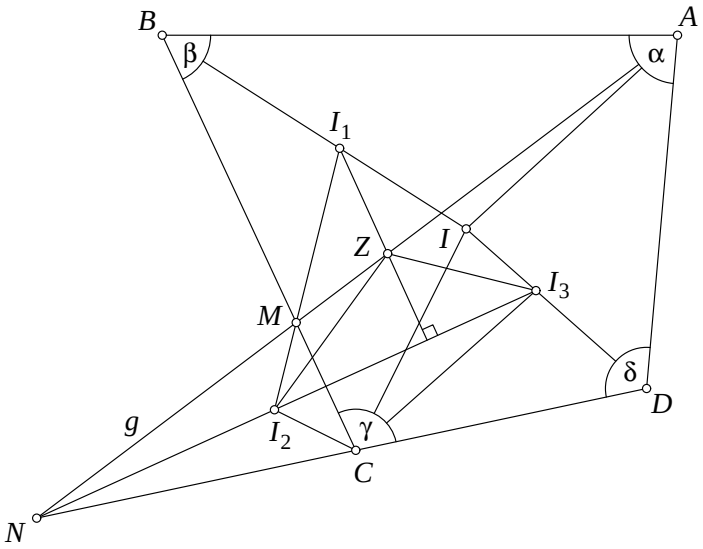


Figure 2

To this end, notice first that I_2, M, I_1 are collinear, as are N, I_2, I_3 (see Figure 2). Denote by α, β, γ, δ the internal angles of $ABCD$. By considerations in triangle CMN, it follows that $\angle I_3I_2I_1 = \frac{\gamma}{2}$. We will show that $\angle I_3CI_1 = \frac{\gamma}{2}$, too. Denote by I the incenter of $ABCD$. Clearly, $I_1 \in BI$, $I_3 \in DI$, $\angle I_1AI_3 = \frac{\alpha}{2}$.

Using the abbreviation $[X, YZ]$ for the distance from point X to the line YZ, we have because of $\angle BAI_1 = \angle IAI_3$ and $\angle I_1AI = \angle I_3AD$ that

$$\frac{[I_1, AB]}{[I_1, AI]} = \frac{[I_3, AI]}{[I_3, AD]}\,.$$

Furthermore, consideration of the angle sums in AIB, BIC, CID and DIA implies $\angle AIB + \angle CID = \angle BIC + \angle DIA = 180°$, from which we see

$$\frac{[I_1, AI]}{[I_3, CI]} = \frac{I_1I}{I_3I} = \frac{[I_1, CI]}{[I_3, AI]}\,.$$

Because of $[I_1, AB] = [I_1, BC]$, $[I_3, AD] = [I_3, CD]$, multiplication yields

$$\frac{[I_1, BC]}{[I_3, CI]} = \frac{[I_1, CI]}{[I_3, CD]}\,.$$

By $\angle DCI = \angle ICB = \gamma/2$ it follows that $\angle I_1CB = \angle I_3CI$ which concludes the proof of the

above statement.

Let the perpendicular from I_1 on I_2I_3 intersect g at Z. Then $\angle MI_1Z = 90^\circ - \angle I_3I_2I_1 = 90^\circ - \gamma/2 = \angle MCI_2$. Since we have also $\angle ZMI_1 = \angle I_2MC$, triangles MZI_1 and MI_2C are similar. From this one easily proves that also MI_2Z and MCI_1 are similar. Because C, I_1, I_2, and I_3 are concyclic, $\angle MZI_2 = \angle MI_1C = \angle NI_3C$, thus NI_2Z and NCI_3 are similar, hence NCI_2 and NI_3Z are similar. We conclude $\angle ZI_3I_2 = \angle I_2CN = 90^\circ - \gamma/2$, hence $I_1I_2 \perp ZI_3$. This completes the proof.

Number Theory

N1 AUS (Australia)

A social club has n members. They have the membership numbers $1, 2, \ldots, n$, respectively. From time to time members send presents to other members, including items they have already received as presents from other members. In order to avoid the embarrassing situation that a member might receive a present that he or she has sent to other members, the club adds the following rule to its statutes at one of its annual general meetings:

"A member with membership number a is permitted to send a present to a member with membership number b if and only if $a(b-1)$ is a multiple of n."

Prove that, if each member follows this rule, none will receive a present from another member that he or she has already sent to other members.

Alternative formulation: Let G be a directed graph with n vertices $v_1, v_2, \ldots, v_n$, such that there is an edge going from v_a to v_b if and only if a and b are distinct and $a(b-1)$ is a multiple of n. Prove that this graph does not contain a directed cycle.

Solution 1. Suppose there is an edge from v_i to v_j. Then $i(j-1) = ij - i = kn$ for some integer k, which implies $i = ij - kn$. If $\gcd(i,n) = d$ and $\gcd(j,n) = e$, then e divides $ij - kn = i$ and thus e also divides d. Hence, if there is an edge from v_i to v_j, then $\gcd(j,n) | \gcd(i,n)$.

If there is a cycle in G, say $v_{i_1} \to v_{i_2} \to \cdots \to v_{i_r} \to v_{i_1}$, then we have

$$\gcd(i_1,n) | \gcd(i_r,n) | \gcd(i_{r-1},n) | \ldots | \gcd(i_2,n) | \gcd(i_1,n),$$

which implies that all these greatest common divisors must be equal, say be equal to t.

Now we pick any of the i_k, without loss of generality let it be i_1. Then $i_r(i_1 - 1)$ is a multiple of n and hence also (by dividing by t), $i_1 - 1$ is a multiple of $\frac{n}{t}$. Since i_1 and $i_1 - 1$ are relatively prime, also t and $\frac{n}{t}$ are relatively prime. So, by the Chinese remainder theorem, the value of i_1 is uniquely determined modulo $n = t \cdot \frac{n}{t}$ by the value of t. But, as i_1 was chosen arbitrarily among the i_k, this implies that all the i_k have to be equal, a contradiction.

Solution 2. If a, b, c are integers such that $ab - a$ and $bc - b$ are multiples of n, then also $ac - a = a(bc - b) + (ab - a) - (ab - a)c$ is a multiple of n. This implies that if there is an edge from v_a to v_b and an edge from v_b to v_c, then there also must be an edge from v_a to v_c. Therefore, if there are any cycles at all, the smallest cycle must have length 2. But suppose the vertices v_a and v_b form such a cycle, i. e., $ab - a$ and $ab - b$ are both multiples of n. Then $a - b$ is also a multiple of n, which can only happen if $a = b$, which is impossible.

Solution 3. Suppose there was a cycle $v_{i_1} \to v_{i_2} \to \cdots \to v_{i_r} \to v_{i_1}$. Then $i_1(i_2 - 1)$ is a multiple of n, i. e., $i_1 \equiv i_1 i_2 \bmod n$. Continuing in this manner, we get $i_1 \equiv i_1 i_2 \equiv i_1 i_2 i_3 \equiv i_1 i_2 i_3 \ldots i_r \bmod n$. But the same holds for all i_k, i. e., $i_k \equiv i_1 i_2 i_3 \ldots i_r \bmod n$. Hence $i_1 \equiv i_2 \equiv \cdots \equiv i_r \bmod n$, which means $i_1 = i_2 = \cdots = i_r$, a contradiction.

Solution 4. Let $n = k$ be the smallest value of n for which the corresponding graph has a cycle. We show that k is a prime power.
If k is not a prime power, it can be written as a product $k = de$ of relatively prime integers greater than 1. Reducing all the numbers modulo d yields a single vertex or a cycle in the corresponding graph on d vertices, because if $a(b-1) \equiv 0 \bmod k$ then this equation also holds modulo d. But since the graph on d vertices has no cycles, by the minimality of k, we must have that all the indices of the cycle are congruent modulo d. The same holds modulo e and hence also modulo $k = de$. But then all the indices are equal, which is a contradiction.
Thus k must be a prime power $k = p^m$. There are no edges ending at v_k, so v_k is not contained in any cycle. All edges not starting at v_k end at a vertex belonging to a non-multiple of p, and all edges starting at a non-multiple of p must end at v_1. But there is no edge starting at v_1. Hence there is no cycle.

Solution 5. Suppose there was a cycle $v_{i_1} \to v_{i_2} \to \cdots \to v_{i_r} \to v_{i_1}$. Let $q = p^m$ be a prime power dividing n. We claim that either $i_1 \equiv i_2 \equiv \cdots \equiv i_r \equiv 0 \bmod q$ or $i_1 \equiv i_2 \equiv \cdots \equiv i_r \equiv 1 \bmod q$.

Suppose that there is an i_s not divisible by q. Then, as $i_s(i_{s+1} - 1)$ is a multiple of q, $i_{s+1} \equiv 1 \bmod p$. Similarly, we conclude $i_{s+2} \equiv 1 \bmod p$ and so on. So none of the labels is divisible by p, but since $i_s(i_{s+1} - 1)$ is a multiple of $q = p^m$ for all s, all i_{s+1} are congruent to 1 modulo q. This proves the claim.

Now, as all the labels are congruent modulo all the prime powers dividing n, they must all be equal by the Chinese remainder theorem. This is a contradiction.

N2 PER (Peru)

A positive integer N is called *balanced*, if $N = 1$ or if N can be written as a product of an even number of not necessarily distinct primes. Given positive integers a and b, consider the polynomial P defined by $P(x) = (x + a)(x + b)$.

(a) Prove that there exist distinct positive integers a and b such that all the numbers $P(1), P(2), \ldots, P(50)$ are balanced.

(b) Prove that if $P(n)$ is balanced for all positive integers n, then $a = b$.

Solution. Define a function f on the set of positive integers by $f(n) = 0$ if n is balanced and $f(n) = 1$ otherwise. Clearly, $f(nm) \equiv f(n) + f(m) \mod 2$ for all positive integers n, m.

(a) Now for each positive integer n consider the binary sequence $(f(n+1), f(n+2), \ldots, f(n+50))$. As there are only 2^{50} different such sequences there are two different positive integers a and b such that

$$(f(a+1), f(a+2), \ldots, f(a+50)) = (f(b+1), f(b+2), \ldots, f(b+50)).$$

But this implies that for the polynomial $P(x) = (x+a)(x+b)$ all the numbers $P(1)$, $P(2)$, $\ldots$, $P(50)$ are balanced, since for all $1 \le k \le 50$ we have $f(P(k)) \equiv f(a+k)+f(b+k) \equiv 2f(a+k) \equiv 0 \mod 2$.

(b) Now suppose $P(n)$ is balanced for all positive integers n and $a < b$. Set $n = k(b-a)-a$ for sufficiently large k, such that n is positive. Then $P(n) = k(k+1)(b-a)^2$, and this number can only be balanced, if $f(k) = f(k+1)$ holds. Thus, the sequence $f(k)$ must become constant for sufficiently large k. But this is not possible, as for every prime p we have $f(p) = 1$ and for every square t^2 we have $f(t^2) = 0$.

Hence $a = b$.

Comment. Given a positive integer k, a computer search for the pairs of positive integers (a, b), for which $P(1), P(2), \ldots, P(k)$ are all balanced yields the following results with minimal sum $a + b$ and $a < b$:

k	3	4	5	10	20
(a, b)	$(2, 4)$	$(6, 11)$	$(8, 14)$	$(20, 34)$	$(1751, 3121)$

Therefore, trying to find a and b in part (a) of the problem cannot be done by elementary calculations.

N3 EST (Estonia)

Let f be a non-constant function from the set of positive integers into the set of positive integers, such that $a-b$ divides $f(a)-f(b)$ for all distinct positive integers a, b. Prove that there exist infinitely many primes p such that p divides $f(c)$ for some positive integer c.

Solution 1. Denote by $v_p(a)$ the exponent of the prime p in the prime decomposition of a.

Assume that there are only finitely many primes $p_1, p_2, \ldots, p_m$ that divide some function value produced of f.

There are infinitely many positive integers a such that $v_{p_i}(a) > v_{p_i}(f(1))$ for all $i = 1, 2, \ldots, m$, e.g. $a = (p_1 p_2 \ldots p_m)^\alpha$ with α sufficiently large. Pick any such a. The condition of the problem then yields $a \mid (f(a+1) - f(1))$. Assume $f(a+1) \neq f(1)$. Then we must have $v_{p_i}(f(a+1)) \neq v_{p_i}(f(1))$ for at least one i. This yields $v_{p_i}(f(a+1) - f(1)) = \min\{v_{p_i}(f(a+1)), v_{p_i}(f(1))\} \leq v_{p_1}(f(1)) < v_{p_i}(a)$. But this contradicts the fact that $a \mid (f(a+1) - f(1))$.

Hence we must have $f(a+1) = f(1)$ for all such a.

Now, for any positive integer b and all such a, we have $(a+1-b) | (f(a+1) - f(b))$, i.e., $(a+1-b) | (f(1) - f(b))$. Since this is true for infinitely many positive integers a we must have $f(b) = f(1)$. Hence f is a constant function, a contradiction. Therefore, our initial assumption was false and there are indeed infinitely many primes p dividing $f(c)$ for some positive integer c.

Solution 2. Assume that there are only finitely many primes $p_1, p_2, \ldots, p_m$ that divide some function value of f. Since f is not identically 1, we must have $m \geq 1$.

Then there exist non-negative integers $\alpha_1, \ldots, \alpha_m$ such that

$$f(1) = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_m^{\alpha_m}.$$

We can pick a positive integer r such that $f(r) \neq f(1)$. Let

$$M = 1 + p_1^{\alpha_1+1} p_2^{\alpha_2+1} \ldots p_m^{\alpha_m+1} \cdot (f(r) + r).$$

Then for all $i \in \{1, \ldots, m\}$ we have that $p_i^{\alpha_i+1}$ divides $M-1$ and hence by the condition of the problem also $f(M) - f(1)$. This implies that $f(M)$ is divisible by $p_i^{\alpha_i}$ but not by $p_i^{\alpha_i+1}$ for all i and therefore $f(M) = f(1)$.

Hence

$$\begin{aligned} M - r &> p_1^{\alpha_1+1} p_2^{\alpha_2+1} \ldots p_m^{\alpha_m+1} \cdot (f(r) + r) - r \\ &\geq p_1^{\alpha_1+1} p_2^{\alpha_2+1} \ldots p_m^{\alpha_m+1} + (f(r) + r) - r \\ &> p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_m^{\alpha_m} + f(r) \\ &\geq |f(M) - f(r)|. \end{aligned}$$

But since $M - r$ divides $f(M) - f(r)$ this can only be true if $f(r) = f(M) = f(1)$, which contradicts the choice of r.

Comment. In the case that f is a polynomial with integer coefficients the result is well-known, see e.g. W. Schwarz, Einführung in die Methoden der Primzahltheorie, 1969.

N4 PRK (Democratic People's Republic of Korea)

Find all positive integers n such that there exists a sequence of positive integers $a_1, a_2, \ldots, a_n$ satisfying

$$a_{k+1} = \frac{a_k^2 + 1}{a_{k-1} + 1} - 1$$

for every k with $2 \leq k \leq n-1$.

Solution 1. Such a sequence exists for $n = 1, 2, 3, 4$ and no other n. Since the existence of such a sequence for some n implies the existence of such a sequence for all smaller n, it suffices to prove that $n = 5$ is not possible and $n = 4$ is possible.

Assume first that for $n = 5$ there exists a sequence of positive integers $a_1, a_2, \ldots, a_5$ satisfying the conditions

$$a_2^2 + 1 = (a_1 + 1)(a_3 + 1),$$
$$a_3^2 + 1 = (a_2 + 1)(a_4 + 1),$$
$$a_4^2 + 1 = (a_3 + 1)(a_5 + 1).$$

Assume a_1 is odd, then a_2 has to be odd as well and as then $a_2^2 + 1 \equiv 2 \mod 4$, a_3 has to be even. But this is a contradiction, since then the even number $a_2 + 1$ cannot divide the odd number $a_3^2 + 1$.

Hence a_1 is even.

If a_2 is odd, $a_3^2 + 1$ is even (as a multiple of $a_2 + 1$) and hence a_3 is odd, too. Similarly we must have a_4 odd as well. But then $a_3^2 + 1$ is a product of two even numbers $(a_2 + 1)(a_4 + 1)$ and thus is divisible by 4, which is a contradiction as for odd a_3 we have $a_3^2 + 1 \equiv 2 \mod 4$.

Hence a_2 is even. Furthermore $a_3 + 1$ divides the odd number $a_2^2 + 1$ and so a_3 is even. Similarly, a_4 and a_5 are even as well.

Now set $x = a_2$ and $y = a_3$. From the given condition we get $(x+1)|(y^2+1)$ and $(y+1)|(x^2+1)$. We will prove that there is no pair of positive even numbers (x, y) satisfying these two conditions, thus yielding a contradiction to the assumption.

Assume there exists a pair (x_0, y_0) of positive even numbers satisfying the two conditions $(x_0 + 1)|(y_0^2 + 1)$ and $(y_0 + 1)|(x_0^2 + 1)$.

Then one has $(x_0 + 1)|(y_0^2 + 1 + x_0^2 - 1)$, i.e., $(x_0 + 1)|(x_0^2 + y_0^2)$, and similarly $(y_0 + 1)|(x_0^2 + y_0^2)$. Any common divisor d of $x_0 + 1$ and $y_0 + 1$ must hence also divide the number $(x_0^2 + 1) + (y_0^2 + 1) - (x_0^2 + y_0^2) = 2$. But as $x_0 + 1$ and $y_0 + 1$ are both odd, we must have $d = 1$. Thus $x_0 + 1$ and $y_0 + 1$ are relatively prime and therefore there exists a positive integer k such that

$$k(x + 1)(y + 1) = x^2 + y^2$$

has the solution (x_0, y_0). We will show that the latter equation has no solution (x, y) in positive even numbers.

Assume there is a solution. Pick the solution (x_1, y_1) with the smallest sum $x_1 + y_1$ and assume $x_1 \geq y_1$. Then x_1 is a solution to the quadratic equation

$$x^2 - k(y_1 + 1)x + y_1^2 - k(y_1 + 1) = 0.$$

Let x_2 be the second solution, which by VIETA's theorem fulfills $x_1 + x_2 = k(y_1 + 1)$ and $x_1x_2 = y_1^2 - k(y_1 + 1)$. If $x_2 = 0$, the second equation implies $y_1^2 = k(y_1 + 1)$, which is impossible, as $y_1 + 1 > 1$ cannot divide the relatively prime number y_1^2. Therefore $x_2 \neq 0$.

Also we get $(x_1 + 1)(x_2 + 1) = x_1x_2 + x_1 + x_2 + 1 = y_1^2 + 1$ which is odd, and hence x_2 must be even and positive. Also we have $x_2 + 1 = \frac{y_1^2+1}{x_1+1} \leq \frac{y_1^2+1}{y_1+1} \leq y_1 \leq x_1$. But this means that the pair (x', y') with $x' = y_1$ and $y' = x_2$ is another solution of $k(x + 1)(y + 1) = x^2 + y^2$ in even positive numbers with $x' + y' < x_1 + y_1$, a contradiction.

Therefore we must have $n \leq 4$.

When $n = 4$, a possible example of a sequence is $a_1 = 4$, $a_2 = 33$, $a_3 = 217$ and $a_4 = 1384$.

Solution 2. It is easy to check that for $n = 4$ the sequence $a_1 = 4$, $a_2 = 33$, $a_3 = 217$ and $a_4 = 1384$ is possible.

Now assume there is a sequence with $n \geq 5$. Then we have in particular

$$\begin{aligned} a_2^2 + 1 &= (a_1 + 1)(a_3 + 1), \\ a_3^2 + 1 &= (a_2 + 1)(a_4 + 1), \\ a_4^2 + 1 &= (a_3 + 1)(a_5 + 1). \end{aligned}$$

Also assume without loss of generality that among all such quintuples $(a_1, a_2, a_3, a_4, a_5)$ we have chosen one with minimal a_1.

One shows quickly the following fact:

> If three positive integers x, y, z fulfill $y^2 + 1 = (x + 1)(z + 1)$ and if y is even, then x and z are even as well and either $x < y < z$ or $z < y < x$ holds. (1)

Indeed, the first part is obvious and from $x < y$ we conclude

$$z + 1 = \frac{y^2 + 1}{x + 1} \geq \frac{y^2 + 1}{y} > y,$$

and similarly in the other case.

Now, if a_3 was odd, then $(a_2 + 1)(a_4 + 1) = a_3^2 + 1 \equiv 2 \mod 4$ would imply that one of a_2 or a_4 is even, this contradicts (1). Thus a_3 and hence also a_1, a_2, a_4 and a_5 are even. According to (1), one has $a_1 < a_2 < a_3 < a_4 < a_5$ or $a_1 > a_2 > a_3 > a_4 > a_5$ but due to the minimality of a_1 the first series of inequalities must hold.

Consider the identity

$$(a_3+1)(a_1+a_3) = a_3^2-1+(a_1+1)(a_3+1) = a_2^2+a_3^2 = a_2^2-1+(a_2+1)(a_4+1) = (a_2+1)(a_2+a_4).$$

Any common divisor of the two odd numbers $a_2 + 1$ and $a_3 + 1$ must also divide $(a_2 + 1)(a_4 + 1) - (a_3 + 1)(a_3 - 1) = 2$, so these numbers are relatively prime. Hence the last identity shows that $a_1 + a_3$ must be a multiple of $a_2 + 1$, i.e. there is an integer k such that

$$a_1 + a_3 = k(a_2 + 1). \tag{2}$$

Now set $a_0 = k(a_1 + 1) - a_2$. This is an integer and we have

$$\begin{aligned} (a_0 + 1)(a_2 + 1) &= k(a_1 + 1)(a_2 + 1) - (a_2 - 1)(a_2 + 1) \\ &= (a_1 + 1)(a_1 + a_3) - (a_1 + 1)(a_3 + 1) + 2 \\ &= (a_1 + 1)(a_1 - 1) + 2 = a_1^2 + 1. \end{aligned}$$

Thus $a_0 \geq 0$. If $a_0 > 0$, then by (1) we would have $a_0 < a_1 < a_2$ and then the quintuple $(a_0, a_1, a_2, a_3, a_4)$ would contradict the minimality of a_1.

Hence $a_0 = 0$, implying $a_2 = a_1^2$. But also $a_2 = k(a_1 + 1)$, which finally contradicts the fact that $a_1 + 1 > 1$ is relatively prime to a_1^2 and thus cannot be a divisior of this number.

Hence $n \geq 5$ is not possible.

Comment 1. Finding the example for $n = 4$ is not trivial and requires a tedious calculation, but it can be reduced to checking a few cases. The equations $(a_1 + 1)(a_3 + 1) = a_2^2 + 1$ and $(a_2 + 1)(a_4 + 1) = a_3^2 + 1$ imply, as seen in the proof, that a_1 is even and a_2, a_3, a_4 are odd. The case $a_1 = 2$ yields $a_2^2 \equiv -1 \mod 3$ which is impossible. Hence $a_1 = 4$ is the smallest possibility. In this case $a_2^2 \equiv -1 \mod 5$ and a_2 is odd, which implies $a_2 \equiv 3$ or $a_2 \equiv 7 \mod 10$. Hence we have to start checking $a_2 = 7, 13, 17, 23, 27, 33$ and in the last case we succeed.

Comment 2. The choice of $a_0 = k(a_1 + 1) - a_2$ in the second solution appears more natural if one considers that by the previous calculations one has $a_1 = k(a_2+1)-a_3$ and $a_2 = k(a_3+1)-a_4$.

Alternatively, one can solve the equation (2) for a_3 and use $a_2^2 + 1 = (a_1 + 1)(a_3 + 1)$ to get $a_2^2 - k(a_1 + 1)a_2 + a_1^2 - k(a_1 + 1) = 0$. Now a_0 is the second solution to this quadratic equation in a_2 (Vieta jumping).

N5 HUN (Hungary)

Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function T from the set of integers into the set of integers such that the number of integers x with $T^n(x) = x$ is equal to $P(n)$ for every $n \geq 1$, where T^n denotes the n-fold application of T.

Solution 1. Assume there is a polynomial P of degree at least 1 with the desired property for a given function T. Let $A(n)$ denote the set of all $x \in \mathbb{Z}$ such that $T^n(x) = x$ and let $B(n)$ denote the set of all $x \in \mathbb{Z}$ for which $T^n(x) = x$ and $T^k(x) \neq x$ for all $1 \leq k < n$. Both sets are finite under the assumption made. For each $x \in A(n)$ there is a smallest $k \geq 1$ such that $T^k(x) = x$, i.e., $x \in B(k)$. Let $d = \gcd(k, n)$. There are positive integers r, s such that $rk - sn = d$ and hence $x = T^{rk}(x) = T^{sn+d}(x) = T^d(T^{sn}(x)) = T^d(x)$. The minimality of k implies $d = k$, i.e., $k|n$. On the other hand one clearly has $B(k) \subset A(n)$ if $k|n$ and thus we have $A(n) = \bigcup_{d|n} B(d)$ as a disjoint union and hence

$$|A(n)| = \sum_{d|n} |B(d)|.$$

Furthermore, for every $x \in B(n)$ the elements $x, T^1(x), T^2(x), \ldots, T^{n-1}(x)$ are n distinct elements of $B(n)$. The fact that they are in $A(n)$ is obvious. If for some $k < n$ and some $0 \leq i < n$ we had $T^k(T^i(x)) = T^i(x)$, i.e. $T^{k+i}(x) = T^i(x)$, that would imply $x = T^n(x) = T^{n-i}(T^i(x)) = T^{n-i}(T^{k+i}(x)) = T^k(T^n(x)) = T^k(x)$ contradicting the minimality of n. Thus $T^i(x) \in B(n)$ and $T^i(x) \neq T^j(x)$ for $0 \leq i < j \leq n-1$.

So indeed, T permutes the elements of $B(n)$ in (disjoint) cycles of length n and in particular one has $n \big| |B(n)|$.

Now let $P(x) = \sum_{i=0}^{k} a_i x^i$, $a_i \in \mathbb{Z}$, $k \geq 1$, $a_k \neq 0$ and suppose that $|A(n)| = P(n)$ for all $n \geq 1$. Let p be any prime. Then

$$p^2 \big| |B(p^2)| = |A(p^2)| - |A(p)| = a_1(p^2 - p) + a_2(p^4 - p^2) + \ldots$$

Hence $p|a_1$ and since this is true for all primes we must have $a_1 = 0$.

Now consider any two different primes p and q. Since $a_1 = 0$ we have that

$$|A(p^2q)| - |A(pq)| = a_2(p^4q^2 - p^2q^2) + a_3(p^6q^3 - p^3q^3) + \ldots$$

is a multiple of p^2q. But we also have

$$p^2q \big| |B(p^2q)| = |A(p^2q)| - |A(pq)| - |B(p^2)|.$$

This implies

$$p^2q \big| |B(p^2)| = |A(p^2)| - |A(p)| = a_2(p^4 - p^2) + a_3(p^6 - p^3) + \cdots + a_k(p^{2k} - p^k).$$

Since this is true for every prime q we must have $a_2(p^4 - p^2) + a_3(p^6 - p^3) + \cdots + a_k(p^{2k} - p^k) = 0$ for every prime p. Since this expression is a polynomial in p of degree $2k$ (because $a_k \neq 0$) this is a contradiction, as such a polynomial can have at most $2k$ zeros.

Comment. The last contradiction can also be reached via

$$a_k = \lim_{p\to\infty} \frac{1}{p^{2k}} \left(a_2(p^4 - p^2) + a_3(p^6 - p^3) + \cdots + a_k(p^{2k} - p^k)\right) = 0.$$

Solution 2. As in the first solution define $A(n)$ and $B(n)$ and assume that a polynomial P with the required property exists. This again implies that $|A(n)|$ and $|B(n)|$ is finite for all positive integers n and that

$$P(n) = |A(n)| = \sum_{d|n} |B(d)| \qquad \text{and} \qquad n \Big| |B(n)|.$$

Now, for any two distinct primes p and q, we have

$$P(0) \equiv P(pq) \equiv |B(1)| + |B(p)| + |B(q)| + |B(pq)| \equiv |B(1)| + |B(p)| \mod q.$$

Thus, for any fixed p, the expression $P(0) - |B(1)| - |B(p)|$ is divisible by arbitrarily large primes q which means that $P(0) = |B(1)| + |B(p)| = P(p)$ for any prime p. This implies that the polynomial P is constant, a contradiction.

N6 TUR (Turkey)

Let k be a positive integer. Show that if there exists a sequence $a_0, a_1, \ldots$ of integers satisfying the condition

$$a_n = \frac{a_{n-1} + n^k}{n} \quad \text{for all } n \geq 1,$$

then $k - 2$ is divisible by 3.

Solution 1. *Part A.* For each positive integer k, there exists a polynomial P_k of degree $k-1$ with integer coefficients, i. e., $P_k \in \mathbb{Z}[x]$, and an integer q_k such that the polynomial identity

$$xP_k(x) = x^k + P_k(x-1) + q_k \tag{I_k}$$

is satisfied. To prove this, for fixed k we write

$$P_k(x) = b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$$

and determine the coefficients $b_{k-1}, b_{k-2}, \ldots, b_0$ and the number q_k successively. Obviously, we have $b_{k-1} = 1$. For $m = k-1, k-2, \ldots, 1$, comparing the coefficients of x^m in the identity (I_k) results in an expression of b_{m-1} as an integer linear combination of $b_{k-1}, \ldots, b_m$, and finally $q_k = -P_k(-1)$.

Part B. Let k be a positive integer, and let $a_0, a_1, \ldots$ be a sequence of real numbers satisfying the recursion given in the problem. This recursion can be written as

$$a_n - P_k(n) = \frac{a_{n-1} - P_k(n-1)}{n} - \frac{q_k}{n} \quad \text{for all } n \geq 1,$$

which by induction gives

$$a_n - P_k(n) = \frac{a_0 - P_k(0)}{n!} - q_k \sum_{i=0}^{n-1} \frac{i!}{n!} \quad \text{for all } n \geq 1.$$

Therefore, the numbers a_n are integers for all $n \geq 1$ only if

$$a_0 = P_k(0) \quad \text{and} \quad q_k = 0.$$

Part C. Multiplying the identity (I_k) by $x^2 + x$ and subtracting the identities (I_{k+1}), (I_{k+2}) and $q_k x^2 = q_k x^2$ therefrom, we obtain

$$xT_k(x) = T_k(x-1) + 2x\big(P_k(x-1) + q_k\big) - (q_{k+2} + q_{k+1} + q_k),$$

where the polynomials $T_k \in \mathbb{Z}[x]$ are defined by $T_k(x) = (x^2+x)P_k(x) - P_{k+1}(x) - P_{k+2}(x) - q_k x$. Thus

$$xT_k(x) \equiv T_k(x-1) + q_{k+2} + q_{k+1} + q_k \bmod 2, \quad k = 1, 2, \ldots$$

Comparing the degrees, we easily see that this is only possible if T_k is the zero polynomial modulo 2, and

$$q_{k+2} \equiv q_{k+1} + q_k \bmod 2 \quad \text{for } k = 1, 2, \ldots$$

Since $q_1 = -1$ and $q_2 = 0$, these congruences finish the proof.

Solution 2. *Part A and B.* Let k be a positive integer, and suppose there is a sequence $a_0, a_1, \ldots$ as required. We prove: There exists a polynomial $P \in \mathbb{Z}[x]$, i. e., with integer coefficients, such that $a_n = P(n)$, $n = 0, 1, \ldots$, and $xP(x) = x^k + P(x-1)$.
To prove this, we write $P(x) = b_{k-1}x^{k-1} + \cdots + b_1 x + b_0$ and determine the coefficients $b_{k-1}, b_{k-2}, \ldots, b_0$ successively such that

$$xP(x) - x^k - P(x-1) = q,$$

where $q = q_k$ is an integer. Comparing the coefficients of x^m results in an expression of b_{m-1} as an integer linear combination of $b_{k-1}, \ldots, b_m$.
Defining $c_n = a_n - P(n)$, we get

$$\begin{aligned} P(n) + c_n &= \frac{P(n-1) + c_{n-1} + n^k}{n}, \quad \text{i. e.,} \\ q + nc_n &= c_{n-1}, \end{aligned}$$

hence

$$c_n = \frac{c_0}{n!} - q \cdot \frac{0! + 1! + \cdots + (n-1)!}{n!}.$$

We conclude $\lim_{n\to\infty} c_n = 0$, which, using $c_n \in \mathbb{Z}$, implies $c_n = 0$ for sufficiently large n. Therefore, we get $q = 0$ and $c_n = 0$, $n = 0, 1, \ldots$.

Part C. Suppose that $q = q_k = 0$, i. e. $xP(x) = x^k + P(x-1)$. To consider this identity for arguments $x \in \mathbb{F}_4$, we write $\mathbb{F}_4 = \{0, 1, \alpha, \alpha+1\}$. Then we get

$$\begin{aligned} \alpha P_k(\alpha) &= \alpha^k + P_k(\alpha+1) \quad \text{and} \\ (\alpha+1)P_k(\alpha+1) &= (\alpha+1)^k + P_k(\alpha), \end{aligned}$$

hence

$$\begin{aligned} P_k(\alpha) &= 1 \cdot P_k(\alpha) = (\alpha+1)\alpha P_k(\alpha) \\ &= (\alpha+1)P_k(\alpha+1) + (\alpha+1)\alpha^k \\ &= P_k(\alpha) + (\alpha+1)^k + (\alpha+1)\alpha^k. \end{aligned}$$

Now, $(\alpha+1)^{k-1} = \alpha^k$ implies $k \equiv 2 \bmod 3$.

Comment 1. For $k = 2$, the sequence given by $a_n = n+1$, $n = 0, 1, \ldots$, satisfies the conditions of the problem.

Comment 2. The first few polynomials P_k and integers q_k are

$$\begin{aligned}
&P_1(x) = 1, \quad q_1 = -1,\\
&P_2(x) = x + 1, \quad q_2 = 0,\\
&P_3(x) = x^2 + x - 1, \quad q_3 = 1,\\
&P_4(x) = x^3 + x^2 - 2x - 1, \quad q_4 = -1,\\
&P_5(x) = x^4 + x^3 - 3x^2 + 5, \quad q_5 = -2,\\
&P_6(x) = x^5 + x^4 - 4x^3 + 2x^2 + 10x - 5, \quad q_6 = 9,\\
&q_7 = -9, \quad q_8 = -50, \quad q_9 = 267, \quad q_{10} = -413, \quad q_{11} = -2180.
\end{aligned}$$

A lookup in the On-Line Encyclopedia of Integer Sequences (A000587) reveals that the sequence $q_1, -q_2, q_3, -q_4, q_5, \ldots$ is known as Uppuluri-Carpenter numbers. The result that $q_k = 0$ implies $k \equiv 2 \bmod 3$ is contained in
Murty, Summer: On the p-adic series $\sum_{n=0}^{\infty} n^k \cdot n!$. CRM Proc. and Lecture Notes 36, 2004.
As shown by Alexander (Non-Vanishing of Uppuluri-Carpenter Numbers, Preprint 2006), Uppuluri-Carpenter numbers are zero at most twice.

Comment 3. The numbers q_k can be written in terms of the Stirling numbers of the second kind. To show this, we fix the notation such that

$$\begin{aligned}
x^k = &S_{k-1,k-1}x(x-1)\cdots(x-k+1)\\
&+ S_{k-1,k-2}x(x-1)\cdots(x-k+2)\\
&+ \cdots + S_{k-1,0}x,
\end{aligned} \tag{$*$}$$

e. g., $S_{2,2} = 1$, $S_{2,1} = 3$, $S_{2,0} = 1$, and we define

$$\Omega_k = S_{k-1,k-1} - S_{k-1,k-2} + - \cdots .$$

Replacing x by $-x$ in $(*)$ results in

$$\begin{aligned}
x^k = &S_{k-1,k-1}x(x+1)\cdots(x+k-1)\\
&- S_{k-1,k-2}x(x+1)\cdots(x+k-2)\\
&+ - \cdots \pm S_{k-1,0}x.
\end{aligned}$$

Defining

$$\begin{aligned}
P(x) = &S_{k-1,k-1}(x+1)\cdots(x+k-1)\\
&+ (S_{k-1,k-1} - S_{k-1,k-2})(x+1)\cdots(x+k-2)\\
&+ (S_{k-1,k-1} - S_{k-1,k-2} + S_{k-1,k-3})(x+1)\cdots(x+k-3)\\
&+ \cdots + \Omega_k,
\end{aligned}$$

we obtain

$$\begin{aligned}
xP(x) - P(x-1) &= S_{k-1,k-1}x(x+1)\cdots(x+k-1)\\
&\quad - S_{k-1,k-2}x(x+1)\cdots(x+k-2)\\
&\quad + - \cdots \pm S_{k-1,0}x - \Omega_k\\
&= x^k - \Omega_k,
\end{aligned}$$

hence $q_k = -\Omega_k$.

N7 MNG (Mongolia)

Let a and b be distinct integers greater than 1. Prove that there exists a positive integer n such that $(a^n-1)(b^n-1)$ is not a perfect square.

Solution 1. At first we notice that

$$\begin{aligned}(1-\alpha)^{\frac{1}{2}}(1-\beta)^{\frac{1}{2}} &= \left(1-\tfrac{1}{2}\cdot\alpha-\tfrac{1}{8}\cdot\alpha^2-\cdots\right)\left(1-\tfrac{1}{2}\cdot\beta-\tfrac{1}{8}\cdot\beta^2-\cdots\right)\\ &= \sum_{k,\ell\geq 0} c_{k,\ell}\cdot\alpha^k\beta^\ell \quad \text{for all } \alpha,\beta\in(0,1),\end{aligned} \tag{1}$$

where $c_{0,0}=1$ and $c_{k,\ell}$ are certain coefficients.

For an indirect proof, we suppose that $x_n=\sqrt{(a^n-1)(b^n-1)}\in\mathbb{Z}$ for all positive integers n. Replacing a by a^2 and b by b^2 if necessary, we may assume that a and b are perfect squares, hence $\sqrt{ab}$ is an integer.

At first we shall assume that $a^\mu\neq b^\nu$ for all positive integers μ,ν. We have

$$x_n=(\sqrt{ab})^n\left(1-\frac{1}{a^n}\right)^{\frac{1}{2}}\left(1-\frac{1}{b^n}\right)^{\frac{1}{2}}=\sum_{k,\ell\geq 0}c_{k,\ell}\left(\frac{\sqrt{ab}}{a^kb^\ell}\right)^n. \tag{2}$$

Choosing k_0 and ℓ_0 such that $a^{k_0}>\sqrt{ab}$, $b^{\ell_0}>\sqrt{ab}$, we define the polynomial

$$P(x)=\prod_{k=0,\ell=0}^{k_0-1,\ell_0-1}(a^kb^\ell x-\sqrt{ab})=:\sum_{i=0}^{k_0\cdot\ell_0}d_ix^i$$

with integer coefficients d_i. By our assumption, the zeros

$$\frac{\sqrt{ab}}{a^kb^\ell},\quad k=0,\ldots,k_0-1,\quad \ell=0,\ldots,\ell_0-1,$$

of P are pairwise distinct.

Furthermore, we consider the integer sequence

$$y_n=\sum_{i=0}^{k_0\cdot\ell_0}d_ix_{n+i},\quad n=1,2,\ldots \tag{3}$$

By the theory of linear recursions, we obtain

$$y_n=\sum_{\substack{k,\ell\geq 0\\ k\geq k_0 \text{ or } \ell\geq\ell_0}}e_{k,\ell}\left(\frac{\sqrt{ab}}{a^kb^\ell}\right)^n,\quad n=1,2,\ldots, \tag{4}$$

with real numbers $e_{k,\ell}$. We have

$$|y_n|\leq\sum_{\substack{k,\ell\geq 0\\ k\geq k_0 \text{ or } \ell\geq\ell_0}}|e_{k,\ell}|\left(\frac{\sqrt{ab}}{a^kb^\ell}\right)^n=:M_n.$$

Because the series in (4) is obtained by a finite linear combination of the absolutely convergent series (1), we conclude that in particular $M_1 < \infty$. Since

$$\frac{\sqrt{ab}}{a^k b^\ell} \le \lambda := \max\left\{\frac{\sqrt{ab}}{a^{k_0}}, \frac{\sqrt{ab}}{b^{\ell_0}}\right\} \quad \text{for all } k, \ell \ge 0 \text{ such that } k \ge k_0 \text{ or } \ell \ge \ell_0,$$

we get the estimates $M_{n+1} \le \lambda M_n$, $n = 1, 2, \ldots$ Our choice of k_0 and ℓ_0 ensures $\lambda < 1$, which implies $M_n \to 0$ and consequently $y_n \to 0$ as $n \to \infty$. It follows that $y_n = 0$ for all sufficiently large n.

So, equation (3) reduces to $\sum_{i=0}^{k_0 \cdot \ell_0} d_i x_{n+i} = 0$.

Using the theory of linear recursions again, for sufficiently large n we have

$$x_n = \sum_{k=0,\ell=0}^{k_0-1,\ell_0-1} f_{k,\ell}\left(\frac{\sqrt{ab}}{a^k b^\ell}\right)^n$$

for certain real numbers $f_{k,\ell}$.

Comparing with (2), we see that $f_{k,\ell} = c_{k,\ell}$ for all $k, \ell \ge 0$ with $k < k_0$, $\ell < \ell_0$, and $c_{k,\ell} = 0$ if $k \ge k_0$ or $\ell \ge \ell_0$, since we assumed that $a^\mu \ne b^\nu$ for all positive integers μ, ν.

In view of (1), this means

$$(1-\alpha)^{\frac{1}{2}}(1-\beta)^{\frac{1}{2}} = \sum_{k=0,\ell=0}^{k_0-1,\ell_0-1} c_{k,\ell} \cdot \alpha^k \beta^\ell \tag{5}$$

for all real numbers $\alpha, \beta \in (0, 1)$. We choose $k^* < k_0$ maximal such that there is some i with $c_{k^*,i} \ne 0$. Squaring (5) and comparing coefficients of $\alpha^{2k^*}\beta^{2i^*}$, where i^* is maximal with $c_{k^*,i^*} \ne 0$, we see that $k^* = 0$. This means that the right hand side of (5) is independent of α, which is clearly impossible.

We are left with the case that $a^\mu = b^\nu$ for some positive integers μ and ν. We may assume that μ and ν are relatively prime. Then there is some positive integer c such that $a = c^\nu$ and $b = c^\mu$. Now starting with the expansion (2), i. e.,

$$x_n = \sum_{j \ge 0} g_j \left(\frac{\sqrt{c^{\mu+\nu}}}{c^j}\right)^n$$

for certain coefficients g_j, and repeating the arguments above, we see that $g_j = 0$ for sufficiently large j, say $j > j_0$. But this means that

$$(1-x^\mu)^{\frac{1}{2}}(1-x^\nu)^{\frac{1}{2}} = \sum_{j=0}^{j_0} g_j x^j$$

for all real numbers $x \in (0, 1)$. Squaring, we see that

$$(1-x^\mu)(1-x^\nu)$$

is the square of a polynomial in x. In particular, all its zeros are of order at least 2, which implies $\mu = \nu$ by looking at roots of unity. So we obtain $\mu = \nu = 1$, i. e., $a = b$, a contradiction.

Solution 2. We set $a^2 = A$, $b^2 = B$, and $z_n = \sqrt{(A^n-1)(B^n-1)}$. Let us assume that z_n is an integer for $n = 1, 2, \ldots$ Without loss of generality, we may suppose that $b < a$. We determine an integer $k \geq 2$ such that $b^{k-1} \leq a < b^k$, and define a sequence $\gamma_1, \gamma_2, \ldots$ of rational numbers such that

$$2\gamma_1 = 1 \qquad \text{and} \qquad 2\gamma_{n+1} = \sum_{i=1}^{n} \gamma_i \gamma_{n-i} \quad \text{for} \quad n = 1, 2, \ldots$$

It could easily be shown that $\gamma_n = \frac{1\cdot 1\cdot 3\ldots(2n-3)}{2\cdot 4\cdot 6\ldots 2n}$, for instance by reading VANDERMONDEs convolution as an equation between polynomials, but we shall have no use for this fact.

Using LANDAUs O–Notation in the usual way, we have

$$\begin{aligned}
&\left\{(ab)^n - \gamma_1\left(\frac{a}{b}\right)^n - \gamma_2\left(\frac{a}{b^3}\right)^n - \cdots - \gamma_k\left(\frac{a}{b^{2k-1}}\right)^n + O\left(\frac{b}{a}\right)^n\right\}^2 \\
&= A^nB^n - 2\gamma_1 A^n - \sum_{i=2}^{k}\left(2\gamma_i - \sum_{j=1}^{i-1}\gamma_j\gamma_{i-j}\right)\left(\frac{A}{B^{i-1}}\right)^n + O\left(\frac{A}{B^k}\right)^n + O\left(B^n\right) \\
&= A^nB^n - A^n + O\left(B^n\right),
\end{aligned}$$

whence

$$z_n = (ab)^n - \gamma_1\left(\frac{a}{b}\right)^n - \gamma_2\left(\frac{a}{b^3}\right)^n - \cdots - \gamma_k\left(\frac{a}{b^{2k-1}}\right)^n + O\left(\frac{b}{a}\right)^n.$$

Now choose rational numbers $r_1, r_2, \ldots, r_{k+1}$ such that

$$(x - ab)\cdot(x - \tfrac{a}{b})\ldots(x - \tfrac{a}{b^{2k-1}}) = x^{k+1} - r_1x^k + - \cdots \pm r_{k+1},$$

and then a natural number M for which $Mr_1, Mr_2, \ldots Mr_{k+1}$ are integers. For known reasons,

$$M(z_{n+k+1} - r_1z_{n+k} + - \cdots \pm r_{k+1}z_n) = O\left(\frac{b}{a}\right)^n$$

for all $n \in \mathbb{N}$ and thus there is a natural number N which is so large, that

$$z_{n+k+1} = r_1z_{n+k} - r_2z_{n+k-1} + - \cdots \mp r_{k+1}z_n$$

holds for all $n \geqslant N$. Now the theory of linear recursions reveals that there are some rational numbers $\delta_0, \delta_1, \delta_2, \ldots, \delta_k$ such that

$$z_n = \delta_0(ab)^n - \delta_1\left(\frac{a}{b}\right)^n - \delta_2\left(\frac{a}{b^3}\right)^n - \cdots - \delta_k\left(\frac{a}{b^{2k-1}}\right)^n$$

for sufficiently large n, where $\delta_0 > 0$ as $z_n > 0$. As before, one obtains

$$\begin{aligned}
&A^nB^n - A^n - B^n + 1 = z_n^2 \\
&= \left\{\delta_0(ab)^n - \delta_1\left(\frac{a}{b}\right)^n - \delta_2\left(\frac{a}{b^3}\right)^n - \cdots - \delta_k\left(\frac{a}{b^{2k-1}}\right)^n\right\}^2 \\
&= \delta_0^2A^nB^n - 2\delta_0\delta_1A^n - \sum_{i=2}^{i=k}\left(2\delta_0\delta_i - \sum_{j=1}^{j=i-1}\delta_j\delta_{i-j}\right)\left(\frac{A}{B^{i-1}}\right)^n + O\left(\frac{A}{B^k}\right)^n.
\end{aligned}$$

Easy asymptotic calculations yield $\delta_0 = 1$, $\delta_1 = \frac{1}{2}$, $\delta_i = \frac{1}{2}\sum_{j=1}^{j=i-1}\delta_j\delta_{i-j}$ for $i = 2, 3, \ldots, k-2$, and then $a = b^{k-1}$. It follows that $k > 2$ and there is some $P \in \mathbb{Q}[X]$ for which $(X-1)(X^{k-1}-1) = P(X)^2$. But this cannot occur, for instance as $X^{k-1} - 1$ has no double zeros. Thus our

assumption that z_n was an integer for $n = 1, 2, \ldots$ turned out to be wrong, which solves the problem.

Original formulation of the problem. a, b are positive integers such that $a \cdot b$ is not a square of an integer. Prove that there exists a (infinitely many) positive integer n such that $(a^n-1)(b^n-1)$ is not a square of an integer.

Solution. *Lemma.* Let c be a positive integer, which is not a perfect square. Then there exists an odd prime p such that c is not a quadratic residue modulo p.
Proof. Denoting the square-free part of c by c', we have the equality $\left(\frac{c'}{p}\right) = \left(\frac{c}{p}\right)$ of the corresponding LEGENDRE symbols. Suppose that $c' = q_1 \cdots q_m$, where $q_1 < \cdots < q_m$ are primes. Then we have

$$\left(\frac{c'}{p}\right) = \left(\frac{q_1}{p}\right) \cdots \left(\frac{q_m}{p}\right).$$

Case 1. Let q_1 be odd. We choose a quadratic nonresidue r_1 modulo q_1 and quadratic residues r_i modulo q_i for $i = 2, \ldots, m$. By the Chinese remainder theorem and the DIRICHLET theorem, there exists a (infinitely many) prime p such that

$$\begin{aligned} p &\equiv r_1 \bmod q_1, \\ p &\equiv r_2 \bmod q_2, \\ &\vdots \quad \vdots \\ p &\equiv r_m \bmod q_m, \\ p &\equiv 1 \bmod 4. \end{aligned}$$

By our choice of the residues, we obtain

$$\left(\frac{p}{q_i}\right) = \left(\frac{r_i}{q_i}\right) = \begin{cases} -1, & i = 1, \\ 1, & i = 2, \ldots, m. \end{cases}$$

The congruence $p \equiv 1 \bmod 4$ implies that $\left(\frac{q_i}{p}\right) = \left(\frac{p}{q_i}\right)$, $i = 1, \ldots, m$, by the law of quadratic reciprocity. Thus

$$\left(\frac{c'}{p}\right) = \left(\frac{q_1}{p}\right) \cdots \left(\frac{q_m}{p}\right) = -1.$$

Case 2. Suppose $q_1 = 2$. We choose quadratic residues r_i modulo q_i for $i = 2, \ldots, m$. Again, by the Chinese remainder theorem and the DIRICHLET theorem, there exists a prime p such that

$$\begin{aligned} p &\equiv r_2 \bmod q_2, \\ &\vdots \quad \vdots \\ p &\equiv r_m \bmod q_m, \\ p &\equiv 5 \bmod 8. \end{aligned}$$

By the choice of the residues, we obtain $\left(\frac{p}{q_i}\right) = \left(\frac{r_i}{q_i}\right) = 1$ for $i = 2, \ldots, m$. Since $p \equiv 1 \bmod 4$ we have $\left(\frac{q_i}{p}\right) = \left(\frac{p}{q_i}\right)$, $i = 2, \ldots, m$, by the law of quadratic reciprocity. The congruence $p \equiv 5 \bmod 8$

implies that $\left(\frac{2}{p}\right) = -1$. Thus

$$\left(\frac{c'}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{q_2}{p}\right)\cdots\left(\frac{q_m}{p}\right) = -1,$$

and the lemma is proved.

Applying the lemma for $c = a \cdot b$, we find an odd prime p such that

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\cdot\left(\frac{b}{p}\right) = -1.$$

This implies either

$$a^{\frac{p-1}{2}} \equiv 1 \bmod p, \quad b^{\frac{p-1}{2}} \equiv -1 \bmod p, \qquad \text{or} \qquad a^{\frac{p-1}{2}} \equiv -1 \bmod p, \quad b^{\frac{p-1}{2}} \equiv 1 \bmod p.$$

Without loss of generality, suppose that $a^{\frac{p-1}{2}} \equiv 1 \bmod p$ and $b^{\frac{p-1}{2}} \equiv -1 \bmod p$. The second congruence implies that $b^{\frac{p-1}{2}} - 1$ is not divisible by p. Hence, if the exponent $\nu_p(a^{\frac{p-1}{2}} - 1)$ of p in the prime decomposition of $(a^{\frac{p-1}{2}} - 1)$ is odd, then $(a^{\frac{p-1}{2}} - 1)(b^{\frac{p-1}{2}} - 1)$ is not a perfect square. If $\nu_p(a^{\frac{p-1}{2}} - 1)$ is even, then $\nu_p(a^{\frac{p-1}{2}p} - 1)$ is odd by the well-known power lifting property

$$\nu_p\left(a^{\frac{p-1}{2}p} - 1\right) = \nu_p\left(a^{\frac{p-1}{2}} - 1\right) + 1.$$

In this case, $(a^{\frac{p-1}{2}p} - 1)(b^{\frac{p-1}{2}p} - 1)$ is not a perfect square.

Comment 1. In 1998, the following problem appeared in Crux Mathematicorum:
Problem 2344. Find all positive integers N that are quadratic residues modulo all primes greater than N.
The published solution (Crux Mathematicorum, 25(1999)4) is the same as the proof of the lemma given above, see also `http://www.mathlinks.ro/viewtopic.php?t=150495`.

Comment 2. There is also an elementary proof of the lemma. We cite Theorem 3 of Chapter 5 and its proof from the book
Ireland, Rosen: A Classical Introduction to Modern Number Theory, Springer 1982.

Theorem. Let a be a nonsquare integer. Then there are infinitely many primes p for which a is a quadratic nonresidue.

Proof. It is easily seen that we may assume that a is square-free. Let $a = 2^e q_1 q_2 \cdots q_n$, where q_i are distinct odd primes and $e = 0$ or 1. The case $a = 2$ has to be dealt with separately. We shall assume to begin with that $n \geq 1$, i. e., that a is divisible by an odd prime.

Let $\ell_1, \ell_2, \ldots, \ell_k$ be a finite set of odd primes not including any q_i. Let s be any quadratic nonresidue modq_n, and find a simultaneous solution to the congruences

$$\begin{aligned} x &\equiv 1 \bmod \ell_i, \quad i = 1, \ldots, k, \\ x &\equiv 1 \bmod 8, \\ x &\equiv 1 \bmod q_i, \quad i = 1, \ldots, n-1, \\ x &\equiv s \bmod q_n. \end{aligned}$$

Call the solution b. b is odd. Suppose that $b = p_1 p_2 \cdots p_m$ is its prime decomposition. Since

$b \equiv 1 \bmod 8$ we have $\left(\frac{2}{b}\right) = 1$ and $\left(\frac{q_i}{b}\right) = \left(\frac{b}{q_i}\right)$ by a result on JACOBI symbols. Thus

$$\left(\frac{a}{b}\right) = \left(\frac{2}{b}\right)^e \left(\frac{q_1}{b}\right) \cdots \left(\frac{q_{n-1}}{b}\right)\left(\frac{q_n}{b}\right) = \left(\frac{b}{q_1}\right) \cdots \left(\frac{b}{q_{n-1}}\right)\left(\frac{b}{q_n}\right) = \left(\frac{1}{q_1}\right) \cdots \left(\frac{1}{q_{n-1}}\right)\left(\frac{s}{q_n}\right) = -1.$$

On the other hand, by the definition of $\left(\frac{a}{b}\right)$, we have $\left(\frac{a}{b}\right) = \left(\frac{a}{p_1}\right)\left(\frac{a}{p_2}\right) \cdots \left(\frac{a}{p_m}\right)$. It follows that $\left(\frac{a}{p_i}\right) = -1$ for some i.

Notice that ℓ_j does not divide b. Thus $p_i \notin \{\ell_1, \ell_2, \ldots, \ell_k\}$.

To summarize, if a is a nonsquare, divisible by an odd prime, we have found a prime p, outside of a given finite set of primes $\{2, \ell_1, \ell_2, \ldots, \ell_k\}$, such that $\left(\frac{a}{p}\right) = -1$. This proves the theorem in this case.

It remains to consider the case $a = 2$. Let $\ell_1, \ell_2, \ldots, \ell_k$ be a finite set of primes, excluding 3, for which $\left(\frac{2}{\ell_i}\right) = -1$. Let $b = 8\ell_1\ell_2 \cdots \ell_k + 3$. b is not divisible by 3 or any ℓ_i. Since $b \equiv 3 \bmod 8$ we have $\left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}} = -1$. Suppose that $b = p_1p_2 \cdots p_m$ is the prime decomposition of b. Then, as before, we see that $\left(\frac{2}{p_i}\right) = -1$ for some i. $p_i \notin \{3, \ell_1, \ell_2, \ldots, \ell_k\}$. This proves the theorem for $a = 2$.

This proof has also been posted to `mathlinks`, see `http://www.mathlinks.ro/viewtopic.php?t=150495` again.

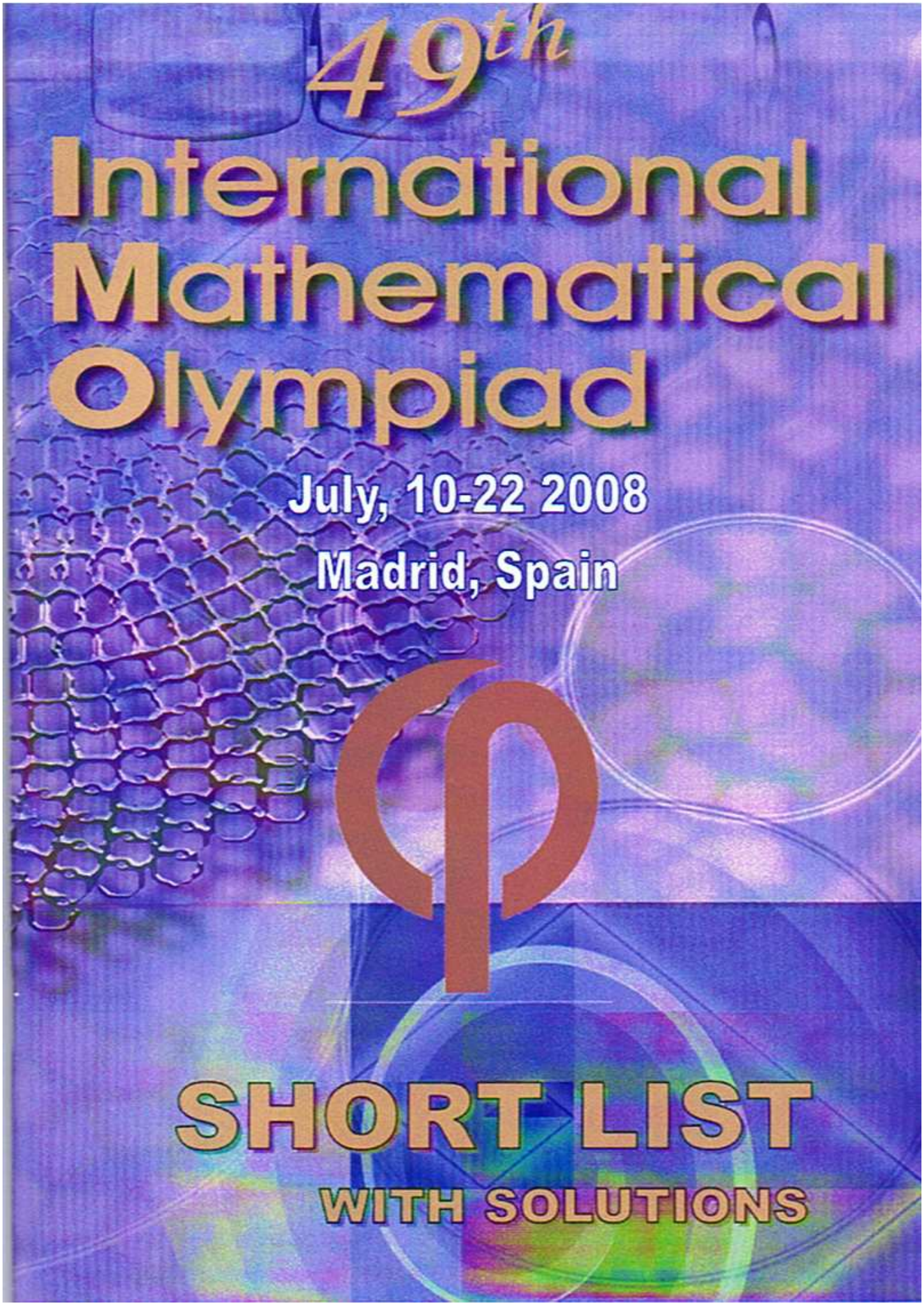
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Contents

Contributing Countries & Problem Selection Committee **5**

Algebra **7**
Problem A1 . . . 7
Problem A2 . . . 9
Problem A3 . . . 11
Problem A4 . . . 12
Problem A5 . . . 14
Problem A6 . . . 15
Problem A7 . . . 17

Combinatorics **21**
Problem C1 . . . 21
Problem C2 . . . 23
Problem C3 . . . 24
Problem C4 . . . 25
Problem C5 . . . 26
Problem C6 . . . 27

Geometry **29**
Problem G1 . . . 29
Problem G2 . . . 31
Problem G3 . . . 32
Problem G4 . . . 34
Problem G5 . . . 36
Problem G6 . . . 37
Problem G7 . . . 40

Number Theory **43**
Problem N1 . . . 43
Problem N2 . . . 45
Problem N3 . . . 46
Problem N4 . . . 47
Problem N5 . . . 49
Problem N6 . . . 50

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Algebra

A1. Find all functions $f\colon (0,\infty)\to(0,\infty)$ such that

$$\frac{f(p)^2+f(q)^2}{f(r^2)+f(s^2)}=\frac{p^2+q^2}{r^2+s^2}$$

for all $p,q,r,s>0$ with $pq=rs$.

Solution. Let f satisfy the given condition. Setting $p=q=r=s=1$ yields $f(1)^2=f(1)$ and hence $f(1)=1$. Now take any $x>0$ and set $p=x$, $q=1$, $r=s=\sqrt{x}$ to obtain

$$\frac{f(x)^2+1}{2f(x)}=\frac{x^2+1}{2x}\,.$$

This recasts into

$$\begin{gathered}xf(x)^2+x=x^2f(x)+f(x),\\ \big(xf(x)-1\big)\big(f(x)-x\big)=0.\end{gathered}$$

And thus,

$$\text{for every } \ x>0, \quad \text{either } \ f(x)=x \ \text{ or } \ f(x)=\frac{1}{x}\,. \tag{1}$$

Obviously, if

$$f(x)=x \quad \text{for all } x>0 \qquad \text{or} \qquad f(x)=\frac{1}{x} \quad \text{for all } x>0 \tag{2}$$

then the condition of the problem is satisfied. We show that actually these two functions are the only solutions.

So let us assume that there exists a function f satisfying the requirement, other than those in (2). Then $f(a)\neq a$ and $f(b)\neq 1/b$ for some $a,b>0$. By (1), these values must be $f(a)=1/a$, $f(b)=b$. Applying now the equation with $p=a$, $q=b$, $r=s=\sqrt{ab}$ we obtain $(a^{-2}+b^2)/2f(ab)=(a^2+b^2)/2ab$; equivalently,

$$f(ab)=\frac{ab(a^{-2}+b^2)}{a^2+b^2}\,. \tag{3}$$

We know however (see (1)) that $f(ab)$ must be either ab or $1/ab$. If $f(ab)=ab$ then by (3) $a^{-2}+b^2=a^2+b^2$, so that $a=1$. But, as $f(1)=1$, this contradicts the relation $f(a)\neq a$. Likewise, if $f(ab)=1/ab$ then (3) gives $a^2b^2(a^{-2}+b^2)=a^2+b^2$, whence $b=1$, in contradiction to $f(b)\neq 1/b$. Thus indeed the functions listed in (2) are the only two solutions.

8

Comment. The equation has as many as four variables with only one constraint $pq = rs$, leaving three degrees of freedom and providing a lot of information. Various substitutions force various useful properties of the function searched. We sketch one more method to reach conclusion (1); certainly there are many others.

Noticing that $f(1) = 1$ and setting, first, $p = q = 1$, $r = \sqrt{x}$, $s = 1/\sqrt{x}$, and then $p = x$, $q = 1/x$, $r = s = 1$, we obtain two relations, holding for every $x > 0$,

$$f(x) + f\left(\frac{1}{x}\right) = x + \frac{1}{x} \qquad \text{and} \qquad f(x)^2 + f\left(\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2}\,. \tag{4}$$

Squaring the first and subtracting the second gives $2f(x)f(1/x) = 2$. Subtracting this from the second relation of (4) leads to

$$\left(f(x) - f\left(\frac{1}{x}\right)\right)^2 = \left(x - \frac{1}{x}\right)^2 \qquad \text{or} \qquad f(x) - f\left(\frac{1}{x}\right) = \pm\left(x - \frac{1}{x}\right).$$

The last two alternatives combined with the first equation of (4) imply the two alternatives of (1).

A2. (a) Prove the inequality

$$\frac{x^2}{(x-1)^2}+\frac{y^2}{(y-1)^2}+\frac{z^2}{(z-1)^2}\geq 1$$

for real numbers $x, y, z \neq 1$ satisfying the condition $xyz = 1$.

(b) Show that there are infinitely many triples of rational numbers x, y, z for which this inequality turns into equality.

Solution 1. (a) We start with the substitution

$$\frac{x}{x-1}=a,\quad \frac{y}{y-1}=b,\quad \frac{z}{z-1}=c,\quad \text{i.e.,}\quad x=\frac{a}{a-1},\quad y=\frac{b}{b-1},\quad z=\frac{c}{c-1}.$$

The inequality to be proved reads $a^2+b^2+c^2\geq 1$. The new variables are subject to the constraints $a, b, c \neq 1$ and the following one coming from the condition $xyz = 1$,

$$(a-1)(b-1)(c-1)=abc.$$

This is successively equivalent to

$$\begin{aligned} a+b+c-1 &= ab+bc+ca,\\ 2(a+b+c-1) &= (a+b+c)^2-(a^2+b^2+c^2),\\ a^2+b^2+c^2-2 &= (a+b+c)^2-2(a+b+c),\\ a^2+b^2+c^2-1 &= (a+b+c-1)^2. \end{aligned}$$

Thus indeed $a^2+b^2+c^2\geq 1$, as desired.

(b) From the equation $a^2+b^2+c^2-1=(a+b+c-1)^2$ we see that the proposed inequality becomes an equality if and only if both sums $a^2+b^2+c^2$ and $a+b+c$ have value 1. The first of them is equal to $(a+b+c)^2-2(ab+bc+ca)$. So the instances of equality are described by the system of two equations

$$a+b+c=1,\qquad ab+bc+ca=0$$

plus the constraint $a, b, c \neq 1$. Elimination of c leads to $a^2+ab+b^2=a+b$, which we regard as a quadratic equation in b,

$$b^2+(a-1)b+a(a-1)=0,$$

with discriminant

$$\Delta=(a-1)^2-4a(a-1)=(1-a)(1+3a).$$

We are looking for rational triples (a, b, c); it will suffice to have a rational such that $1-a$ and $1+3a$ are both squares of rational numbers (then Δ will be so too). Set $a = k/m$. We want $m-k$ and $m+3k$ to be squares of integers. This is achieved for instance by taking $m=k^2-k+1$ (clearly nonzero); then $m-k=(k-1)^2$, $m+3k=(k+1)^2$. Note that distinct integers k yield distinct values of $a = k/m$.

And thus, if k is any integer and $m=k^2-k+1$, $a=k/m$ then $\Delta=(k^2-1)^2/m^2$ and the quadratic equation has rational roots $b=(m-k\pm k^2\mp 1)/(2m)$. Choose e.g. the larger root,

$$b=\frac{m-k+k^2-1}{2m}=\frac{m+(m-2)}{2m}=\frac{m-1}{m}.$$

Computing c from $a+b+c=1$ then gives $c=(1-k)/m$. The condition $a,b,c\neq 1$ eliminates only $k=0$ and $k=1$. Thus, as k varies over integers greater than 1, we obtain an infinite family of rational triples (a,b,c)—and coming back to the original variables ($x=a/(a-1)$ etc.)—an infinite family of rational triples (x,y,z) with the needed property. (A short calculation shows that the resulting triples are $x=-k/(k-1)^2$, $y=k-k^2$, $z=(k-1)/k^2$; but the proof was complete without listing them.)

Comment 1. There are many possible variations in handling the equation system $a^2+b^2+c^2=1$, $a+b+c=1$ $(a,b,c\neq 1)$ which of course describes a circle in the (a,b,c)-space (with three points excluded), and finding infinitely many rational points on it.

Also the initial substitution $x=a/(a-1)$ (etc.) can be successfully replaced by other similar substitutions, e.g. $x=1-1/\alpha$ (etc.); or $x=x'-1$ (etc.); or $1-yz=u$ (etc.)—eventually reducing the inequality to $(\cdots)^2\geq 0$, the expression in the parentheses depending on the actual substitution.

Depending on the method chosen, one arrives at various sequences of rational triples (x,y,z) as needed; let us produce just one more such example: $x=(2r-2)/(r+1)^2$, $y=(2r+2)/(r-1)^2$, $z=(r^2-1)/4$ where r can be any rational number different from 1 or -1.

Solution 2 (an outline)**.** (a) Without changing variables, just setting $z=1/xy$ and clearing fractions, the proposed inequality takes the form

$$(xy-1)^2\big(x^2(y-1)^2+y^2(x-1)^2\big)+(x-1)^2(y-1)^2\geq (x-1)^2(y-1)^2(xy-1)^2.$$

With the notation $p=x+y$, $q=xy$ this becomes, after lengthy routine manipulation and a lot of cancellation

$$q^4-6q^3+2pq^2+9q^2-6pq+p^2\geq 0.$$

It is not hard to notice that the expression on the left is just $(q^2-3q+p)^2$, hence nonnegative.

(Without introducing p and q, one is of course led with some more work to the same expression, just written in terms of x and y; but then it is not that easy to see that it is a square.)

(b) To have equality, one needs $q^2-3q+p=0$. Note that x and y are the roots of the quadratic trinomial (in a formal variable t): t^2-pt+q. When $q^2-3q+p=0$, the discriminant equals

$$\delta=p^2-4q=(3q-q^2)^2-4q=q(q-1)^2(q-4).$$

Now it suffices to have both q and $q-4$ squares of rational numbers (then $p=3q-q^2$ and $\sqrt{\delta}$ are also rational, and so are the roots of the trinomial). On setting $q=(n/m)^2=4+(l/m)^2$ the requirement becomes $4m^2+l^2=n^2$ (with l,m,n being integers). This is just the Pythagorean equation, known to have infinitely many integer solutions.

Comment 2. Part (a) alone might also be considered as a possible contest problem (in the category of easy problems).

A3. Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a *Spanish Couple* on S, if they satisfy the following conditions:

(i) Both functions are strictly increasing, i.e. $f(x) < f(y)$ and $g(x) < g(y)$ for all $x, y \in S$ with $x < y$;

(ii) The inequality $f(g(g(x))) < g(f(x))$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple

(a) on the set $S = \mathbb{N}$ of positive integers;

(b) on the set $S = \{a - 1/b : \ a, b \in \mathbb{N}\}$.

Solution. We show that the answer is NO for part (a), and YES for part (b).

(a) Throughout the solution, we will use the notation $g_k(x) = \overbrace{g(g(\ldots g}^{k}(x)\ldots))$, including $g_0(x) = x$ as well.

Suppose that there exists a Spanish Couple (f, g) on the set $\mathbb{N}$. From property (i) we have $f(x) \geq x$ and $g(x) \geq x$ for all $x \in \mathbb{N}$.

We claim that $g_k(x) \leq f(x)$ for all $k \geq 0$ and all positive integers x. The proof is done by induction on k. We already have the base case $k = 0$ since $x \leq f(x)$. For the induction step from k to $k + 1$, apply the induction hypothesis on $g_2(x)$ instead of x, then apply (ii):

$$g(g_{k+1}(x)) = g_k\big(g_2(x)\big) \leq f\big(g_2(x)\big) < g(f(x)).$$

Since g is increasing, it follows that $g_{k+1}(x) < f(x)$. The claim is proven.

If $g(x) = x$ for all $x \in \mathbb{N}$ then $f(g(g(x))) = f(x) = g(f(x))$, and we have a contradiction with (ii). Therefore one can choose an $x_0 \in S$ for which $x_0 < g(x_0)$. Now consider the sequence $x_0, x_1, \ldots$ where $x_k = g_k(x_0)$. The sequence is increasing. Indeed, we have $x_0 < g(x_0) = x_1$, and $x_k < x_{k+1}$ implies $x_{k+1} = g(x_k) < g(x_{k+1}) = x_{k+2}$.

Hence, we obtain a strictly increasing sequence $x_0 < x_1 < \ldots$ of positive integers which on the other hand has an upper bound, namely $f(x_0)$. This cannot happen in the set $\mathbb{N}$ of positive integers, thus no Spanish Couple exists on $\mathbb{N}$.

(b) We present a Spanish Couple on the set $S = \{a - 1/b : \ a, b \in \mathbb{N}\}$.

Let

$$\begin{aligned} f(a - 1/b) &= a + 1 - 1/b, \\ g(a - 1/b) &= a - 1/(b + 3^a). \end{aligned}$$

These functions are clearly increasing. Condition (ii) holds, since

$$f(g(g(a - 1/b))) = (a + 1) - 1/(b + 2 \cdot 3^a) < (a + 1) - 1/(b + 3^{a+1}) = g(f(a - 1/b)).$$

Comment. Another example of a Spanish couple is $f(a - 1/b) = 3a - 1/b$, $g(a - 1/b) = a - 1/(a+b)$. More generally, postulating $f(a - 1/b) = h(a) - 1/b$, $g(a - 1/b) = a - 1/G(a, b)$ with h increasing and G increasing in both variables, we get that $f \circ g \circ g < g \circ f$ holds if $G\big(a, G(a, b)\big) < G\big(h(a), b\big)$. A search just among linear functions $h(a) = Ca$, $G(a, b) = Aa + Bb$ results in finding that any integers $A > 0$, $C > 2$ and $B = 1$ produce a Spanish couple (in the example above, $A = 1$, $C = 3$). The proposer's example results from taking $h(a) = a + 1$, $G(a, b) = 3^a + b$.

A4. For an integer m, denote by $t(m)$ the unique number in $\{1,2,3\}$ such that $m+t(m)$ is a multiple of 3. A function $f\colon \mathbb{Z}\to\mathbb{Z}$ satisfies $f(-1)=0$, $f(0)=1$, $f(1)=-1$ and

$$f(2^n+m)=f(2^n-t(m))-f(m) \quad \text{for all integers } m,n\ge 0 \text{ with } 2^n>m.$$

Prove that $f(3p)\ge 0$ holds for all integers $p\ge 0$.

Solution. The given conditions determine f uniquely on the positive integers. The signs of $f(1),f(2),\dots$ seem to change quite erratically. However values of the form $f(2^n-t(m))$ are sufficient to compute directly any functional value. Indeed, let $n>0$ have base 2 representation $n=2^{a_0}+2^{a_1}+\cdots+2^{a_k}$, $a_0>a_1>\cdots>a_k\ge 0$, and let $n_j=2^{a_j}+2^{a_{j-1}}+\cdots+2^{a_k}$, $j=0,\dots,k$. Repeated applications of the recurrence show that $f(n)$ is an alternating sum of the quantities $f(2^{a_j}-t(n_{j+1}))$ plus $(-1)^{k+1}$. (The exact formula is not needed for our proof.)

So we focus attention on the values $f(2^n-1)$, $f(2^n-2)$ and $f(2^n-3)$. Six cases arise; more specifically,

$$t(2^{2k}-3)=2,\ t(2^{2k}-2)=1,\ t(2^{2k}-1)=3,\ t(2^{2k+1}-3)=1,\ t(2^{2k+1}-2)=3,\ t(2^{2k+1}-1)=2.$$

Claim. For all integers $k\ge 0$ the following equalities hold:

$$\begin{aligned} &f(2^{2k+1}-3)=0, && f(2^{2k+1}-2)=3^k, && f(2^{2k+1}-1)=-3^k,\\ &f(2^{2k+2}-3)=-3^k, && f(2^{2k+2}-2)=-3^k, && f(2^{2k+2}-1)=2\cdot 3^k. \end{aligned}$$

Proof. By induction on k. The base $k=0$ comes down to checking that $f(2)=-1$ and $f(3)=2$; the given values $f(-1)=0$, $f(0)=1$, $f(1)=-1$ are also needed. Suppose the claim holds for $k-1$. For $f(2^{2k+1}-t(m))$, the recurrence formula and the induction hypothesis yield

$$\begin{aligned} f(2^{2k+1}-3)&=f(2^{2k}+(2^{2k}-3))=f(2^{2k}-2)-f(2^{2k}-3)=-3^{k-1}+3^{k-1}=0,\\ f(2^{2k+1}-2)&=f(2^{2k}+(2^{2k}-2))=f(2^{2k}-1)-f(2^{2k}-2)=2\cdot 3^{k-1}+3^{k-1}=3^k,\\ f(2^{2k+1}-1)&=f(2^{2k}+(2^{2k}-1))=f(2^{2k}-3)-f(2^{2k}-1)=-3^{k-1}-2\cdot 3^{k-1}=-3^k. \end{aligned}$$

For $f(2^{2k+2}-t(m))$ we use the three equalities just established:

$$\begin{aligned} f(2^{2k+2}-3)&=f(2^{2k+1}+(2^{2k+1}-3))=f(2^{2k+1}-1)-f(2^{2k+1}-3)=-3^k-0=-3^k,\\ f(2^{2k+2}-2)&=f(2^{2k+1}+(2^{2k+1}-2))=f(2^{2k+1}-3)-f(2^{2k}-2)=0-3^k=-3^k,\\ f(2^{2k+2}-1)&=f(2^{2k+1}+(2^{2k+1}-1))=f(2^{2k+1}-2)-f(2^{2k+1}-1)=3^k+3^k=2\cdot 3^k. \end{aligned}$$

The claim follows.

A closer look at the six cases shows that $f(2^n-t(m))\ge 3^{(n-1)/2}$ if $2^n-t(m)$ is divisible by 3, and $f(2^n-t(m))\le 0$ otherwise. On the other hand, note that $2^n-t(m)$ is divisible by 3 if and only if 2^n+m is. Therefore, for all nonnegative integers m and n,

(i) $f(2^n-t(m))\ge 3^{(n-1)/2}$ if 2^n+m is divisible by 3;

(ii) $f(2^n-t(m))\le 0$ if 2^n+m is not divisible by 3.

One more (direct) consequence of the claim is that $|f(2^n-t(m))|\le \frac{2}{3}\cdot 3^{n/2}$ for all $m,n\ge 0$.

The last inequality enables us to find an upper bound for $|f(m)|$ for m less than a given power of 2. We prove by induction on n that $|f(m)|\le 3^{n/2}$ holds true for all integers $m,n\ge 0$ with $2^n>m$.

The base $n = 0$ is clear as $f(0) = 1$. For the inductive step from n to $n + 1$, let m and n satisfy $2^{n+1} > m$. If $m < 2^n$, we are done by the inductive hypothesis. If $m \geq 2^n$ then $m = 2^n + k$ where $2^n > k \geq 0$. Now, by $|f(2^n - t(k))| \leq \frac{2}{3} \cdot 3^{n/2}$ and the inductive assumption,

$$|f(m)| = |f(2^n - t(k)) - f(k)| \leq |f(2^n - t(k))| + |f(k)| \leq \frac{2}{3} \cdot 3^{n/2} + 3^{n/2} < 3^{(n+1)/2}.$$

The induction is complete.

We proceed to prove that $f(3p) \geq 0$ for all integers $p \geq 0$. Since $3p$ is not a power of 2, its binary expansion contains at least two summands. Hence one can write $3p = 2^a + 2^b + c$ where $a > b$ and $2^b > c \geq 0$. Applying the recurrence formula twice yields

$$f(3p) = f(2^a + 2^b + c) = f(2^a - t(2^b + c)) - f(2^b - t(c)) + f(c).$$

Since $2^a + 2^b + c$ is divisible by 3, we have $f(2^a - t(2^b + c)) \geq 3^{(a-1)/2}$ by (i). Since $2^b + c$ is not divisible by 3, we have $f(2^b - t(c)) \leq 0$ by (ii). Finally $|f(c)| \leq 3^{b/2}$ as $2^b > c \geq 0$, so that $f(c) \geq -3^{b/2}$. Therefore $f(3p) \geq 3^{(a-1)/2} - 3^{b/2}$ which is nonnegative because $a > b$.

14

A5. Let a, b, c, d be positive real numbers such that

$$abcd = 1 \quad \text{and} \quad a + b + c + d > \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

Prove that

$$a + b + c + d < \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}.$$

Solution. We show that if $abcd = 1$, the sum $a + b + c + d$ cannot exceed a certain weighted mean of the expressions $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$ and $\frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}$.

By applying the AM-GM inequality to the numbers $\frac{a}{b}$, $\frac{a}{b}$, $\frac{b}{c}$ and $\frac{a}{d}$, we obtain

$$a = \sqrt[4]{\frac{a^4}{abcd}} = \sqrt[4]{\frac{a}{b} \cdot \frac{a}{b} \cdot \frac{b}{c} \cdot \frac{a}{d}} \le \frac{1}{4}\left(\frac{a}{b} + \frac{a}{b} + \frac{b}{c} + \frac{a}{d}\right).$$

Analogously,

$$b \le \frac{1}{4}\left(\frac{b}{c} + \frac{b}{c} + \frac{c}{d} + \frac{b}{a}\right), \quad c \le \frac{1}{4}\left(\frac{c}{d} + \frac{c}{d} + \frac{d}{a} + \frac{c}{b}\right) \quad \text{and} \quad d \le \frac{1}{4}\left(\frac{d}{a} + \frac{d}{a} + \frac{a}{b} + \frac{d}{c}\right).$$

Summing up these estimates yields

$$a + b + c + d \le \frac{3}{4}\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}\right) + \frac{1}{4}\left(\frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}\right).$$

In particular, if $a + b + c + d > \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$ then $a + b + c + d < \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}$.

Comment. The estimate in the above solution was obtained by applying the AM-GM inequality to each column of the 4×4 array

$$\begin{array}{cccc} a/b & b/c & c/d & d/a \\ a/b & b/c & c/d & d/a \\ b/c & c/d & d/a & a/b \\ a/d & b/a & c/b & d/c \end{array}$$

and adding up the resulting inequalities. The same table yields a stronger bound: If $a, b, c, d > 0$ and $abcd = 1$ then

$$\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}\right)^3 \left(\frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}\right) \ge (a + b + c + d)^4.$$

It suffices to apply Hölder's inequality to the sequences in the four rows, with weights $1/4$:

$$\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}\right)^{1/4} \left(\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}\right)^{1/4} \left(\frac{b}{c} + \frac{c}{d} + \frac{d}{a} + \frac{a}{b}\right)^{1/4} \left(\frac{a}{d} + \frac{b}{a} + \frac{c}{b} + \frac{d}{c}\right)^{1/4}$$
$$\ge \left(\frac{aaba}{bbcd}\right)^{1/4} + \left(\frac{bbcb}{ccda}\right)^{1/4} + \left(\frac{ccdc}{ddab}\right)^{1/4} + \left(\frac{ddad}{aabc}\right)^{1/4} = a + b + c + d.$$

A6. Let $f\colon \mathbb{R} \to \mathbb{N}$ be a function which satisfies

$$f\left(x + \frac{1}{f(y)}\right) = f\left(y + \frac{1}{f(x)}\right) \qquad \text{for all } x, y \in \mathbb{R}. \tag{1}$$

Prove that there is a positive integer which is not a value of f.

Solution. Suppose that the statement is false and $f(\mathbb{R}) = \mathbb{N}$. We prove several properties of the function f in order to reach a contradiction.

To start with, observe that one can assume $f(0) = 1$. Indeed, let $a \in \mathbb{R}$ be such that $f(a) = 1$, and consider the function $g(x) = f(x + a)$. By substituting $x + a$ and $y + a$ for x and y in (1), we have

$$g\left(x + \frac{1}{g(y)}\right) = f\left(x + a + \frac{1}{f(y + a)}\right) = f\left(y + a + \frac{1}{f(x + a)}\right) = g\left(y + \frac{1}{g(x)}\right).$$

So g satisfies the functional equation (1), with the additional property $g(0) = 1$. Also, g and f have the same set of values: $g(\mathbb{R}) = f(\mathbb{R}) = \mathbb{N}$. Henceforth we assume $f(0) = 1$.

Claim 1. For an arbitrary fixed $c \in \mathbb{R}$ we have $\left\{f\left(c + \frac{1}{n}\right) : n \in \mathbb{N}\right\} = \mathbb{N}$.

Proof. Equation (1) and $f(\mathbb{R}) = \mathbb{N}$ imply

$$f(\mathbb{R}) = \left\{f\left(x + \frac{1}{f(c)}\right) : x \in \mathbb{R}\right\} = \left\{f\left(c + \frac{1}{f(x)}\right) : x \in \mathbb{R}\right\} \subset \left\{f\left(c + \frac{1}{n}\right) : n \in \mathbb{N}\right\} \subset f(\mathbb{R}).$$

The claim follows.

We will use Claim 1 in the special cases $c = 0$ and $c = 1/3$:

$$\left\{f\left(\frac{1}{n}\right) : n \in \mathbb{N}\right\} = \left\{f\left(\frac{1}{3} + \frac{1}{n}\right) : n \in \mathbb{N}\right\} = \mathbb{N}. \tag{2}$$

Claim 2. If $f(u) = f(v)$ for some $u, v \in \mathbb{R}$ then $f(u+q) = f(v+q)$ for all nonnegative rational q. Furthermore, if $f(q) = 1$ for some nonnegative rational q then $f(kq) = 1$ for all $k \in \mathbb{N}$.

Proof. For all $x \in \mathbb{R}$ we have by (1)

$$f\left(u + \frac{1}{f(x)}\right) = f\left(x + \frac{1}{f(u)}\right) = f\left(x + \frac{1}{f(v)}\right) = f\left(v + \frac{1}{f(x)}\right).$$

Since $f(x)$ attains all positive integer values, this yields $f(u + 1/n) = f(v + 1/n)$ for all $n \in \mathbb{N}$. Let $q = k/n$ be a positive rational number. Then k repetitions of the last step yield

$$f(u + q) = f\left(u + \frac{k}{n}\right) = f\left(v + \frac{k}{n}\right) = f(v + q).$$

Now let $f(q) = 1$ for some nonnegative rational q, and let $k \in \mathbb{N}$. As $f(0) = 1$, the previous conclusion yields successively $f(q) = f(2q), f(2q) = f(3q), \ldots, f\left((k-1)q\right) = f(kq)$, as needed.

Claim 3. The equality $f(q) = f(q + 1)$ holds for all nonnegative rational q.

Proof. Let m be a positive integer such that $f(1/m) = 1$. Such an m exists by (2). Applying the second statement of Claim 2 with $q = 1/m$ and $k = m$ yields $f(1) = 1$.

Given that $f(0) = f(1) = 1$, the first statement of Claim 2 implies $f(q) = f(q + 1)$ for all nonnegative rational q.

Claim 4. The equality $f\left(\frac{1}{n}\right) = n$ holds for every $n \in \mathbb{N}$.

Proof. For a nonnegative rational q we set $x = q$, $y = 0$ in (1) and use Claim 3 to obtain

$$f\left(\frac{1}{f(q)}\right) = f\left(q + \frac{1}{f(0)}\right) = f(q+1) = f(q).$$

By (2), for each $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f(1/k) = n$. Applying the last equation with $q = 1/k$, we have

$$n = f\left(\frac{1}{k}\right) = f\left(\frac{1}{f(1/k)}\right) = f\left(\frac{1}{n}\right).$$

Now we are ready to obtain a contradiction. Let $n \in \mathbb{N}$ be such that $f(1/3 + 1/n) = 1$. Such an n exists by (2). Let $1/3 + 1/n = s/t$, where $s, t \in \mathbb{N}$ are coprime. Observe that $t > 1$ as $1/3 + 1/n$ is not an integer. Choose $k, l \in \mathbb{N}$ so that that $ks - lt = 1$.

Because $f(0) = f(s/t) = 1$, Claim 2 implies $f(ks/t) = 1$. Now $f(ks/t) = f(1/t + l)$; on the other hand $f(1/t + l) = f(1/t)$ by l successive applications of Claim 3. Finally, $f(1/t) = t$ by Claim 4, leading to the impossible $t = 1$. The solution is complete.

A7. Prove that for any four positive real numbers a, b, c, d the inequality

$$\frac{(a-b)(a-c)}{a+b+c}+\frac{(b-c)(b-d)}{b+c+d}+\frac{(c-d)(c-a)}{c+d+a}+\frac{(d-a)(d-b)}{d+a+b}\geq 0$$

holds. Determine all cases of equality.

Solution 1. Denote the four terms by

$$A=\frac{(a-b)(a-c)}{a+b+c},\quad B=\frac{(b-c)(b-d)}{b+c+d},\quad C=\frac{(c-d)(c-a)}{c+d+a},\quad D=\frac{(d-a)(d-b)}{d+a+b}.$$

The expression $2A$ splits into two summands as follows,

$$2A=A'+A''\qquad\text{where}\qquad A'=\frac{(a-c)^2}{a+b+c},\qquad A''=\frac{(a-c)(a-2b+c)}{a+b+c};$$

this is easily verified. We analogously represent $2B=B'+B''$, $2C=C'+C''$, $2B=D'+D''$ and examine each of the sums $A'+B'+C'+D'$ and $A''+B''+C''+D''$ separately.

Write $s=a+b+c+d$; the denominators become $s-d$, $s-a$, $s-b$, $s-c$. By the Cauchy-Schwarz inequality,

$$\begin{aligned}&\left(\frac{|a-c|}{\sqrt{s-d}}\cdot\sqrt{s-d}+\frac{|b-d|}{\sqrt{s-a}}\cdot\sqrt{s-a}+\frac{|c-a|}{\sqrt{s-b}}\cdot\sqrt{s-b}+\frac{|d-b|}{\sqrt{s-c}}\cdot\sqrt{s-c}\right)^2\\&\qquad\leq\left(\frac{(a-c)^2}{s-d}+\frac{(b-d)^2}{s-a}+\frac{(c-a)^2}{s-b}+\frac{(d-b)^2}{s-c}\right)\big(4s-s\big)=3s\big(A'+B'+C'+D'\big).\end{aligned}$$

Hence

$$A'+B'+C'+D'\geq\frac{\big(2|a-c|+2|b-d|\big)^2}{3s}\geq\frac{16\cdot|a-c|\cdot|b-d|}{3s}.\tag{1}$$

Next we estimate the absolute value of the other sum. We couple A'' with C'' to obtain

$$\begin{aligned}A''+C''&=\frac{(a-c)(a+c-2b)}{s-d}+\frac{(c-a)(c+a-2d)}{s-b}\\&=\frac{(a-c)(a+c-2b)(s-b)+(c-a)(c+a-2d)(s-d)}{(s-d)(s-b)}\\&=\frac{(a-c)\big(-2b(s-b)-b(a+c)+2d(s-d)+d(a+c)\big)}{s(a+c)+bd}\\&=\frac{3(a-c)(d-b)(a+c)}{M},\quad\text{with}\quad M=s(a+c)+bd.\end{aligned}$$

Hence by cyclic shift

$$B''+D''=\frac{3(b-d)(a-c)(b+d)}{N},\quad\text{with}\quad N=s(b+d)+ca.$$

Thus

$$A''+B''+C''+D''=3(a-c)(b-d)\left(\frac{b+d}{N}-\frac{a+c}{M}\right)=\frac{3(a-c)(b-d)W}{MN}\tag{2}$$

where

$$W=(b+d)M-(a+c)N=bd(b+d)-ac(a+c).\tag{3}$$

Note that

$$MN > \big(ac(a+c) + bd(b+d)\big)s \geq |W| \cdot s. \tag{4}$$

Now (2) and (4) yield

$$|A'' + B'' + C'' + D''| \leq \frac{3 \cdot |a-c| \cdot |b-d|}{s}. \tag{5}$$

Combined with (1) this results in

$$\begin{aligned} 2(A+B+C+D) &= (A'+B'+C'+D') + (A''+B''+C''+D'') \\ &\geq \frac{16 \cdot |a-c| \cdot |b-d|}{3s} - \frac{3 \cdot |a-c| \cdot |b-d|}{s} = \frac{7 \cdot |a-c| \cdot |b-d|}{3(a+b+c+d)} \geq 0. \end{aligned}$$

This is the required inequality. From the last line we see that equality can be achieved only if either $a = c$ or $b = d$. Since we also need equality in (1), this implies that actually $a = c$ and $b = d$ must hold simultaneously, which is obviously also a sufficient condition.

Solution 2. We keep the notations A, B, C, D, s, and also M, N, W from the preceding solution; the definitions of M, N, W and relations (3), (4) in that solution did not depend on the foregoing considerations. Starting from

$$2A = \frac{(a-c)^2 + 3(a+c)(a-c)}{a+b+c} - 2a + 2c,$$

we get

$$\begin{aligned} 2(A+C) &= (a-c)^2 \left(\frac{1}{s-d} + \frac{1}{s-b}\right) + 3(a+c)(a-c)\left(\frac{1}{s-d} - \frac{1}{s-b}\right) \\ &= (a-c)^2 \, \frac{2s-b-d}{M} + 3(a+c)(a-c) \cdot \frac{d-b}{M} = \frac{p(a-c)^2 - 3(a+c)(a-c)(b-d)}{M} \end{aligned}$$

where $p = 2s - b - d = s + a + c$. Similarly, writing $q = s + b + d$ we have

$$2(B+D) = \frac{q(b-d)^2 - 3(b+d)(b-d)(c-a)}{N};$$

specific grouping of terms in the numerators has its aim. Note that $pq > 2s^2$. By adding the fractions expressing $2(A+C)$ and $2(B+D)$,

$$2(A+B+C+D) = \frac{p(a-c)^2}{M} + \frac{3(a-c)(b-d)W}{MN} + \frac{q(b-d)^2}{N}$$

with W defined by (3).

Substitution $x = (a-c)/M$, $y = (b-d)/N$ brings the required inequality to the form

$$2(A+B+C+D) = Mpx^2 + 3Wxy + Nqy^2 \geq 0. \tag{6}$$

It will be enough to verify that the discriminant $\Delta = 9W^2 - 4MNpq$ of the quadratic trinomial $Mpt^2 + 3Wt + Nq$ is negative; on setting $t = x/y$ one then gets (6). The first inequality in (4) together with $pq > 2s^2$ imply $4MNpq > 8s^3\big(ac(a+c) + bd(b+d)\big)$. Since

$$(a+c)s^3 > (a+c)^4 \geq 4ac(a+c)^2 \quad \text{and likewise} \quad (b+d)s^3 > 4bd(b+d)^2,$$

the estimate continues as follows,

$$4MNpq > 8\big(4(ac)^2(a+c)^2 + 4(bd)^2(b+d)^2\big) > 32\big(bd(b+d) - ac(a+c)\big)^2 = 32W^2 \geq 9W^2.$$

Thus indeed $\Delta < 0$. The desired inequality (6) hence results. It becomes an equality if and only if $x = y = 0$; equivalently, if and only if $a = c$ and simultaneously $b = d$.

Comment. The two solutions presented above do not differ significantly; large portions overlap. The properties of the number W turn out to be crucial in both approaches. The Cauchy-Schwarz inequality, applied in the first solution, is avoided in the second, which requires no knowledge beyond quadratic trinomials.

The estimates in the proof of $\Delta < 0$ in the second solution seem to be very wasteful. However, they come close to sharp when the terms in one of the pairs (a, c), (b, d) are equal and much bigger than those in the other pair.

In attempts to prove the inequality by just considering the six cases of arrangement of the numbers a, b, c, d on the real line, one soon discovers that the cases which create real trouble are precisely those in which a and c are both greater or both smaller than b and d.

Solution 3.

$$(a-b)(a-c)(a+b+d)(a+c+d)(b+c+d) =$$
$$= \Big((a-b)(a+b+d)\Big)\Big((a-c)(a+c+d)\Big)(b+c+d) =$$
$$= (a^2+ad-b^2-bd)(a^2+ad-c^2-cd)(b+c+d) =$$
$$= \big(a^4+2a^3d-a^2b^2-a^2bd-a^2c^2-a^2cd+a^2d^2-ab^2d-abd^2-ac^2d-acd^2+b^2c^2+b^2cd+bc^2d+bcd^2\big)(b+c+d) =$$
$$= a^4b+a^4c+a^4d+(b^3c^2+a^2d^3)-a^2c^3+(2a^3d^2-b^3a^2+c^3b^2)+$$
$$+(b^3cd-c^3da-d^3ab)+(2a^3bd+c^3db-d^3ac)+(2a^3cd-b^3da+d^3bc)$$
$$+(-a^2b^2c+3b^2c^2d-2ac^2d^2)+(-2a^2b^2d+2bc^2d^2)+(-a^2bc^2-2a^2c^2d-2ab^2d^2+2b^2cd^2)+$$
$$+(-2a^2bcd-ab^2cd-abc^2d-2abcd^2)$$

Introducing the notation $S_{xyzw} = \sum\limits_{cyc} a^x b^y c^z d^w$, one can write

$$\sum_{cyc}(a-b)(a-c)(a+b+d)(a+c+d)(b+c+d) =$$
$$= S_{4100}+S_{4010}+S_{4001}+2S_{3200}-S_{3020}+2S_{3002}-S_{3110}+2S_{3101}+2S_{3011}-3S_{2120}-6S_{2111} =$$
$$+\left(S_{4100}+S_{4001}+\frac{1}{2}S_{3110}+\frac{1}{2}S_{3011}-3S_{2120}\right)+$$
$$+\left(S_{4010}-S_{3020}-\frac{3}{2}S_{3110}+\frac{3}{2}S_{3011}+\frac{9}{16}S_{2210}+\frac{9}{16}S_{2201}-\frac{9}{8}S_{2111}\right)+$$
$$+\frac{9}{16}\Big(S_{3200}-S_{2210}-S_{2201}+S_{3002}\Big)+\frac{23}{16}\Big(S_{3200}-2S_{3101}+S_{3002}\Big)+\frac{39}{8}\Big(S_{3101}-S_{2111}\Big),$$

where the expressions

$$S_{4100}+S_{4001}+\frac{1}{2}S_{3110}+\frac{1}{2}S_{3011}-3S_{2120} = \sum_{cyc}\left(a^4b+bc^4+\frac{1}{2}a^3bc+\frac{1}{2}abc^3-3a^2bc^2\right),$$

$$S_{4010}-S_{3020}-\frac{3}{2}S_{3110}+\frac{3}{2}S_{3011}+\frac{9}{16}S_{2210}+\frac{9}{16}S_{2201}-\frac{9}{8}S_{2111} = \sum_{cyc}a^2c\left(a-c-\frac{3}{4}b+\frac{3}{4}d\right)^2,$$

$$S_{3200}-S_{2210}-S_{2201}+S_{3002} = \sum_{cyc}b^2(a^3-a^2c-ac^2+c^3) = \sum_{cyc}b^2(a+c)(a-c)^2,$$

$$S_{3200}-2S_{3101}+S_{3002} = \sum_{cyc}a^3(b-d)^2 \qquad \text{and} \qquad S_{3101}-S_{2111} = \frac{1}{3}\sum_{cyc}bd(2a^3+c^3-3a^2c)$$

are all nonnegative.

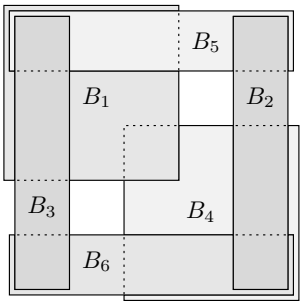
20

Combinatorics

C1. In the plane we consider rectangles whose sides are parallel to the coordinate axes and have positive length. Such a rectangle will be called a *box*. Two boxes *intersect* if they have a common point in their interior or on their boundary.

Find the largest n for which there exist n boxes $B_1, \ldots, B_n$ such that B_i and B_j intersect if and only if $i \not\equiv j \pm 1 \pmod{n}$.

Solution. The maximum number of such boxes is 6. One example is shown in the figure.



Now we show that 6 is the maximum. Suppose that boxes $B_1, \ldots, B_n$ satisfy the condition. Let the closed intervals I_k and J_k be the projections of B_k onto the x- and y-axis, for $1 \le k \le n$.

If B_i and B_j intersect, with a common point (x, y), then $x \in I_i \cap I_j$ and $y \in J_i \cap J_j$. So the intersections $I_i \cap I_j$ and $J_i \cap J_j$ are nonempty. Conversely, if $x \in I_i \cap I_j$ and $y \in J_i \cap J_j$ for some real numbers x, y, then (x, y) is a common point of B_i and B_j. Putting it around, B_i and B_j are disjoint if and only if their projections on at least one coordinate axis are disjoint.

For brevity we call two boxes or intervals adjacent if their indices differ by 1 modulo n, and nonadjacent otherwise.

The adjacent boxes B_k and B_{k+1} do not intersect for each $k = 1, \ldots, n$. Hence (I_k, I_{k+1}) or (J_k, J_{k+1}) is a pair of disjoint intervals, $1 \le k \le n$. So there are at least n pairs of disjoint intervals among $(I_1, I_2), \ldots, (I_{n-1}, I_n), (I_n, I_1); (J_1, J_2), \ldots, (J_{n-1}, J_n), (J_n, J_1)$.

Next, every two nonadjacent boxes intersect, hence their projections on both axes intersect, too. Then the claim below shows that at most 3 pairs among $(I_1, I_2), \ldots, (I_{n-1}, I_n), (I_n, I_1)$ are disjoint, and the same holds for $(J_1, J_2), \ldots, (J_{n-1}, J_n), (J_n, J_1)$. Consequently $n \le 3 + 3 = 6$, as stated. Thus we are left with the claim and its justification.

Claim. Let $\Delta_1, \Delta_2, \ldots, \Delta_n$ be intervals on a straight line such that every two nonadjacent intervals intersect. Then Δ_k and Δ_{k+1} are disjoint for at most three values of $k = 1, \ldots, n$.

Proof. Denote $\Delta_k = [a_k, b_k]$, $1 \le k \le n$. Let $\alpha = \max(a_1, \ldots, a_n)$ be the rightmost among the left endpoints of $\Delta_1, \ldots, \Delta_n$, and let $\beta = \min(b_1, \ldots, b_n)$ be the leftmost among their right endpoints. Assume that $\alpha = a_2$ without loss of generality.

If $\alpha \le \beta$ then $a_i \le \alpha \le \beta \le b_i$ for all i. Every Δ_i contains α, and thus no disjoint pair (Δ_i, Δ_{i+1}) exists.

If $\beta < \alpha$ then $\beta = b_i$ for some i such that $a_i < b_i = \beta < \alpha = a_2 < b_2$, hence Δ_2 and Δ_i are disjoint. Now Δ_2 intersects all remaining intervals except possibly Δ_1 and Δ_3, so Δ_2 and Δ_i can be disjoint only if $i = 1$ or $i = 3$. Suppose by symmetry that $i = 3$; then $\beta = b_3$. Since each of the intervals $\Delta_4, \ldots, \Delta_n$ intersects Δ_2, we have $a_i \le \alpha \le b_i$ for $i = 4, \ldots, n$. Therefore $\alpha \in \Delta_4 \cap \ldots \cap \Delta_n$, in particular $\Delta_4 \cap \ldots \cap \Delta_n \neq \emptyset$. Similarly, $\Delta_5, \ldots, \Delta_n, \Delta_1$ all intersect Δ_3, so that $\Delta_5 \cap \ldots \cap \Delta_n \cap \Delta_1 \neq \emptyset$ as $\beta \in \Delta_5 \cap \ldots \cap \Delta_n \cap \Delta_1$. This leaves (Δ_1, Δ_2), (Δ_2, Δ_3) and (Δ_3, Δ_4) as the only candidates for disjoint interval pairs, as desired.

Comment. The problem is a two-dimensional version of the original proposal which is included below. The extreme shortage of easy and appropriate submissions forced the Problem Selection Committee to shortlist a simplified variant. The same one-dimensional Claim is used in both versions.

Original proposal. We consider parallelepipeds in three-dimensional space, with edges parallel to the coordinate axes and of positive length. Such a parallelepiped will be called a *box*. Two boxes *intersect* if they have a common point in their interior or on their boundary.

Find the largest n for which there exist n boxes $B_1, \ldots, B_n$ such that B_i and B_j intersect if and only if $i \not\equiv j \pm 1 \pmod{n}$.

The maximum number of such boxes is 9. Suppose that boxes $B_1, \ldots, B_n$ satisfy the condition. Let the closed intervals I_k, J_k and K_k be the projections of box B_k onto the x-, y- and z-axis, respectively, for $1 \le k \le n$. As before, B_i and B_j are disjoint if and only if their projections on at least one coordinate axis are disjoint.

We call again two boxes or intervals adjacent if their indices differ by 1 modulo n, and nonadjacent otherwise.

The adjacent boxes B_i and B_{i+1} do not intersect for each $i = 1, \ldots, n$. Hence at least one of the pairs (I_i, I_{i+1}), (J_i, J_{i+1}) and (K_i, K_{i+1}) is a pair of disjoint intervals. So there are at least n pairs of disjoint intervals among $(I_i, I_{i+1}), (J_i, J_{i+1}), (K_i, K_{i+1})$, $1 \le i \le n$.

Next, every two nonadjacent boxes intersect, hence their projections on the three axes intersect, too. Referring to the Claim in the solution of the two-dimensional version, we cocnclude that at most 3 pairs among $(I_1, I_2), \ldots, (I_{n-1}, I_n), (I_n, I_1)$ are disjoint; the same holds for $(J_1, J_2), \ldots, (J_{n-1}, J_n), (J_n, J_1)$ and $(K_1, K_2), \ldots, (K_{n-1}, K_n), (K_n, K_1)$. Consequently $n \le 3 + 3 + 3 = 9$, as stated.

For $n = 9$, the desired system of boxes exists. Consider the intervals in the following table:

i	I_i	J_i	K_i
1	$[1,4]$	$[1,6]$	$[3,6]$
2	$[5,6]$	$[1,6]$	$[1,6]$
3	$[1,2]$	$[1,6]$	$[1,6]$
4	$[3,6]$	$[1,4]$	$[1,6]$
5	$[1,6]$	$[5,6]$	$[1,6]$
6	$[1,6]$	$[1,2]$	$[1,6]$
7	$[1,6]$	$[3,6]$	$[1,4]$
8	$[1,6]$	$[1,6]$	$[5,6]$
9	$[1,6]$	$[1,6]$	$[1,2]$

We have $I_1 \cap I_2 = I_2 \cap I_3 = I_3 \cap I_4 = \emptyset$, $J_4 \cap J_5 = J_5 \cap J_6 = J_6 \cap J_7 = \emptyset$, and finally $K_7 \cap K_8 = K_8 \cap K_9 = K_9 \cap K_1 = \emptyset$. The intervals in each column intersect in all other cases. It follows that the boxes $B_i = I_i \times J_i \times K_i$, $i = 1, \ldots, 9$, have the stated property.

C2. For every positive integer n determine the number of permutations $(a_1, a_2, \ldots, a_n)$ of the set $\{1, 2, \ldots, n\}$ with the following property:

$$2(a_1 + a_2 + \cdots + a_k) \quad \text{is divisible by } k \quad \text{for } k = 1, 2, \ldots, n.$$

Solution. For each n let F_n be the number of permutations of $\{1, 2, \ldots, n\}$ with the required property; call them *nice*. For $n = 1, 2, 3$ every permutation is nice, so $F_1 = 1$, $F_2 = 2$, $F_3 = 6$.

Take an $n > 3$ and consider any nice permutation $(a_1, a_2, \ldots, a_n)$ of $\{1, 2, \ldots, n\}$. Then $n - 1$ must be a divisor of the number

$$\begin{aligned} 2(a_1 + a_2 + \cdots + a_{n-1}) &= 2\big((1 + 2 + \cdots + n) - a_n\big) \\ &= n(n+1) - 2a_n = (n+2)(n-1) + (2 - 2a_n). \end{aligned}$$

So $2a_n - 2$ must be divisible by $n - 1$, hence equal to 0 or $n - 1$ or $2n - 2$. This means that

$$a_n = 1 \qquad \text{or} \qquad a_n = \frac{n+1}{2} \qquad \text{or} \qquad a_n = n.$$

Suppose that $a_n = (n + 1)/2$. Since the permutation is nice, taking $k = n - 2$ we get that $n - 2$ has to be a divisor of

$$\begin{aligned} 2(a_1 + a_2 + \cdots + a_{n-2}) &= 2\big((1 + 2 + \cdots + n) - a_n - a_{n-1}\big) \\ &= n(n+1) - (n+1) - 2a_{n-1} = (n+2)(n-2) + (3 - 2a_{n-1}). \end{aligned}$$

So $2a_{n-1} - 3$ should be divisible by $n - 2$, hence equal to 0 or $n - 2$ or $2n - 4$. Obviously 0 and $2n - 4$ are excluded because $2a_{n-1} - 3$ is odd. The remaining possibility ($2a_{n-1} - 3 = n - 2$) leads to $a_{n-1} = (n + 1)/2 = a_n$, which also cannot hold. This eliminates $(n + 1)/2$ as a possible value of a_n. Consequently $a_n = 1$ or $a_n = n$.

If $a_n = n$ then $(a_1, a_2, \ldots, a_{n-1})$ is a nice permutation of $\{1, 2, \ldots, n-1\}$. There are F_{n-1} such permutations. Attaching n to any one of them at the end creates a nice permutation of $\{1, 2, \ldots, n\}$.

If $a_n = 1$ then $(a_1-1, a_2-1, \ldots, a_{n-1}-1)$ is a permutation of $\{1, 2, \ldots, n-1\}$. It is also nice because the number

$$2\big((a_1-1) + \cdots + (a_k-1)\big) = 2(a_1 + \cdots + a_k) - 2k$$

is divisible by k, for any $k \le n - 1$. And again, any one of the F_{n-1} nice permutations $(b_1, b_2, \ldots, b_{n-1})$ of $\{1, 2, \ldots, n-1\}$ gives rise to a nice permutation of $\{1, 2, \ldots, n\}$ whose last term is 1, namely $(b_1+1, b_2+1, \ldots, b_{n-1}+1, 1)$.

The bijective correspondences established in both cases show that there are F_{n-1} nice permutations of $\{1, 2, \ldots, n\}$ with the last term 1 and also F_{n-1} nice permutations of $\{1, 2, \ldots, n\}$ with the last term n. Hence follows the recurrence $F_n = 2F_{n-1}$. With the base value $F_3 = 6$ this gives the outcome formula $F_n = 3 \cdot 2^{n-2}$ for $n \ge 3$.

24

C3. In the coordinate plane consider the set S of all points with integer coordinates. For a positive integer k, two distinct points $A, B \in S$ will be called *k-friends* if there is a point $C \in S$ such that the area of the triangle ABC is equal to k. A set $T \subset S$ will be called a *k-clique* if every two points in T are k-friends. Find the least positive integer k for which there exists a k-clique with more than 200 elements.

Solution. To begin, let us describe those points $B \in S$ which are k-friends of the point $(0,0)$. By definition, $B = (u, v)$ satisfies this condition if and only if there is a point $C = (x, y) \in S$ such that $\frac{1}{2}|uy - vx| = k$. (This is a well-known formula expressing the area of triangle ABC when A is the origin.)

To say that there exist integers x, y for which $|uy - vx| = 2k$, is equivalent to saying that the greatest common divisor of u and v is also a divisor of $2k$. Summing up, a point $B = (u, v) \in S$ is a k-friend of $(0,0)$ if and only if $\gcd(u, v)$ divides $2k$.

Translation by a vector with integer coordinates does not affect k-friendship; if two points are k-friends, so are their translates. It follows that two points $A, B \in S$, $A = (s, t)$, $B = (u, v)$, are k-friends if and only if the point $(u - s, v - t)$ is a k-friend of $(0,0)$; i.e., if $\gcd(u - s, v - t)|2k$.

Let n be a positive integer which does not divide $2k$. We claim that a k-clique cannot have more than n^2 elements.

Indeed, all points $(x, y) \in S$ can be divided into n^2 classes determined by the remainders that x and y leave in division by n. If a set T has more than n^2 elements, some two points $A, B \in T$, $A = (s, t)$, $B = (u, v)$, necessarily fall into the same class. This means that $n|u - s$ and $n|v - t$. Hence $n|d$ where $d = \gcd(u - s, v - t)$. And since n does not divide $2k$, also d does not divide $2k$. Thus A and B are not k-friends and the set T is not a k-clique.

Now let $M(k)$ be the *least* positive integer which does not divide $2k$. Write $M(k) = m$ for the moment and consider the set T of all points (x, y) with $0 \le x, y < m$. There are m^2 of them. If $A = (s, t)$, $B = (u, v)$ are two distinct points in T then both differences $|u - s|$, $|v - t|$ are integers less than m and at least one of them is positive. By the definition of m, every positive integer less than m divides $2k$. Therefore $u - s$ (if nonzero) divides $2k$, and the same is true of $v - t$. So $2k$ is divisible by $\gcd(u - s, v - t)$, meaning that A, B are k-friends. Thus T is a k-clique.

It follows that the maximum size of a k-clique is $M(k)^2$, with $M(k)$ defined as above. We are looking for the minimum k such that $M(k)^2 > 200$.

By the definition of $M(k)$, $2k$ is divisible by the numbers $1, 2, \ldots, M(k)-1$, but not by $M(k)$ itself. If $M(k)^2 > 200$ then $M(k) \ge 15$. Trying to hit $M(k) = 15$ we get a contradiction immediately ($2k$ would have to be divisible by 3 and 5, but not by 15).

So let us try $M(k) = 16$. Then $2k$ is divisible by the numbers $1, 2, \ldots, 15$, hence also by their least common multiple L, but not by 16. And since L is not a multiple of 16, we infer that $k = L/2$ is the least k with $M(k) = 16$.

Finally, observe that if $M(k) \ge 17$ then $2k$ must be divisible by the least common multiple of $1, 2, \ldots, 16$, which is equal to $2L$. Then $2k \ge 2L$, yielding $k > L/2$.

In conclusion, the least k with the required property is equal to $L/2 = 180180$.

C4. Let n and k be fixed positive integers of the same parity, $k \geq n$. We are given $2n$ lamps numbered 1 through $2n$; each of them can be *on* or *off*. At the beginning all lamps are *off*. We consider sequences of k steps. At each step one of the lamps is switched (from *off* to *on* or from *on* to *off*).

Let N be the number of k-step sequences ending in the state: lamps $1, \ldots, n$ *on*, lamps $n{+}1, \ldots, 2n$ *off*.

Let M be the number of k-step sequences leading to the same state and not touching lamps $n{+}1, \ldots, 2n$ at all.

Find the ratio N/M.

Solution. A sequence of k switches ending in the state as described in the problem statement (lamps $1, \ldots, n$ *on*, lamps $n{+}1, \ldots, 2n$ *off*) will be called an *admissible process*. If, moreover, the process does not touch the lamps $n{+}1, \ldots, 2n$, it will be called *restricted*. So there are N admissible processes, among which M are restricted.

In every admissible process, restricted or not, each one of the lamps $1, \ldots, n$ goes from *off* to *on*, so it is switched an odd number of times; and each one of the lamps $n{+}1, \ldots, 2n$ goes from *off* to *off*, so it is switched an even number of times.

Notice that $M > 0$; i.e., restricted admissible processes do exist (it suffices to switch each one of the lamps $1, \ldots, n$ just once and then choose one of them and switch it $k - n$ times, which by hypothesis is an even number).

Consider any restricted admissible process $\mathbf{p}$. Take any lamp ℓ, $1 \leq \ell \leq n$, and suppose that it was switched k_ℓ times. As noticed, k_ℓ must be odd. Select arbitrarily an even number of these k_ℓ switches and replace each of them by the switch of lamp $n{+}\ell$. This can be done in $2^{k_\ell - 1}$ ways (because a k_ℓ-element set has $2^{k_\ell - 1}$ subsets of even cardinality). Notice that $k_1 + \cdots + k_n = k$.

These actions are independent, in the sense that the action involving lamp ℓ does not affect the action involving any other lamp. So there are $2^{k_1 - 1} \cdot 2^{k_2 - 1} \cdots 2^{k_n - 1} = 2^{k-n}$ ways of combining these actions. In any of these combinations, each one of the lamps $n{+}1, \ldots, 2n$ gets switched an even number of times and each one of the lamps $1, \ldots, n$ remains switched an odd number of times, so the final state is the same as that resulting from the original process $\mathbf{p}$.

This shows that every restricted admissible process $\mathbf{p}$ can be modified in 2^{k-n} ways, giving rise to 2^{k-n} distinct admissible processes (with all lamps allowed).

Now we show that every admissible process $\mathbf{q}$ can be achieved in that way. Indeed, it is enough to replace every switch of a lamp with a label $\ell > n$ that occurs in $\mathbf{q}$ by the switch of the corresponding lamp $\ell{-}n$; in the resulting process $\mathbf{p}$ the lamps $n{+}1, \ldots, 2n$ are not involved.

Switches of each lamp with a label $\ell > n$ had occurred in $\mathbf{q}$ an even number of times. So the performed replacements have affected each lamp with a label $\ell \leq n$ also an even number of times; hence in the overall effect the final state of each lamp has remained the same. This means that the resulting process $\mathbf{p}$ is admissible—and clearly restricted, as the lamps $n{+}1, \ldots, 2n$ are not involved in it any more.

If we now take process $\mathbf{p}$ and reverse all these replacements, then we obtain process $\mathbf{q}$. These reversed replacements are nothing else than the modifications described in the foregoing paragraphs.

Thus there is a one–to–(2^{k-n}) correspondence between the M restricted admissible processes and the total of N admissible processes. Therefore $N/M = 2^{k-n}$.

C5. Let $S = \{x_1, x_2, \ldots, x_{k+\ell}\}$ be a $(k+\ell)$-element set of real numbers contained in the interval $[0,1]$; k and ℓ are positive integers. A k-element subset $A \subset S$ is called *nice* if

$$\left| \frac{1}{k} \sum_{x_i \in A} x_i - \frac{1}{\ell} \sum_{x_j \in S \setminus A} x_j \right| \leq \frac{k+\ell}{2k\ell}.$$

Prove that the number of nice subsets is at least $\dfrac{2}{k+\ell}\binom{k+\ell}{k}$.

Solution. For a k-element subset $A \subset S$, let $f(A) = \dfrac{1}{k} \sum_{x_i \in A} x_i - \dfrac{1}{\ell} \sum_{x_j \in S \setminus A} x_j$. Denote $\dfrac{k+\ell}{2k\ell} = d$. By definition a subset A is nice if $|f(A)| \leq d$.

To each permutation $(y_1, y_2, \ldots, y_{k+\ell})$ of the set $S = \{x_1, x_2, \ldots, x_{k+\ell}\}$ we assign $k+\ell$ subsets of S with k elements each, namely $A_i = \{y_i, y_{i+1}, \ldots, y_{i+k-1}\}$, $i = 1, 2, \ldots, k+\ell$. Indices are taken modulo $k+\ell$ here and henceforth. In other words, if $y_1, y_2, \ldots, y_{k+\ell}$ are arranged around a circle in this order, the sets in question are all possible blocks of k consecutive elements.

Claim. At least two nice sets are assigned to every permutation of S.

Proof. Adjacent sets A_i and A_{i+1} differ only by the elements y_i and y_{i+k}, $i = 1, \ldots, k+\ell$. By the definition of f, and because $y_i, y_{i+k} \in [0,1]$,

$$|f(A_{i+1}) - f(A_i)| = \left| \left(\frac{1}{k} + \frac{1}{\ell} \right) (y_{i+k} - y_i) \right| \leq \frac{1}{k} + \frac{1}{\ell} = 2d.$$

Each element $y_i \in S$ belongs to exactly k of the sets $A_1, \ldots, A_{k+\ell}$. Hence in k of the expressions $f(A_1), \ldots, f(A_{k+\ell})$ the coefficient of y_i is $1/k$; in the remaining ℓ expressions, its coefficient is $-1/\ell$. So the contribution of y_i to the sum of all $f(A_i)$ equals $k \cdot 1/k - \ell \cdot 1/\ell = 0$. Since this holds for all i, it follows that $f(A_1) + \cdots + f(A_{k+\ell}) = 0$.

If $f(A_p) = \min f(A_i)$, $f(A_q) = \max f(A_i)$, we obtain in particular $f(A_p) \leq 0$, $f(A_q) \geq 0$. Let $p < q$ (the case $p > q$ is analogous; and the claim is true for $p = q$ as $f(A_i) = 0$ for all i).

We are ready to prove that at least two of the sets $A_1, \ldots, A_{k+\ell}$ are nice. The interval $[-d, d]$ has length $2d$, and we saw that adjacent numbers in the circular arrangement $f(A_1), \ldots, f(A_{k+\ell})$ differ by at most $2d$. Suppose that $f(A_p) < -d$ and $f(A_q) > d$. Then one of the numbers $f(A_{p+1}), \ldots, f(A_{q-1})$ lies in $[-d, d]$, and also one of the numbers $f(A_{q+1}), \ldots, f(A_{p-1})$ lies there. Consequently, one of the sets $A_{p+1}, \ldots, A_{q-1}$ is nice, as well as one of the sets $A_{q+1}, \ldots, A_{p-1}$. If $-d \leq f(A_p)$ and $f(A_q) \leq d$ then A_p and A_q are nice.

Let now $f(A_p) < -d$ and $f(A_q) \leq d$. Then $f(A_p) + f(A_q) < 0$, and since $\sum f(A_i) = 0$, there is an $r \neq q$ such that $f(A_r) > 0$. We have $0 < f(A_r) \leq f(A_q) \leq d$, so the sets $f(A_r)$ and $f(A_q)$ are nice. The only case remaining, $-d \leq f(A_p)$ and $d < f(A_q)$, is analogous.

Apply the claim to each of the $(k+\ell)!$ permutations of $S = \{x_1, x_2, \ldots, x_{k+\ell}\}$. This gives at least $2(k+\ell)!$ nice sets, counted with repetitions: each nice set is counted as many times as there are permutations to which it is assigned.

On the other hand, each k-element set $A \subset S$ is assigned to exactly $(k+\ell)\, k!\, \ell!$ permutations. Indeed, such a permutation $(y_1, y_2, \ldots, y_{k+\ell})$ is determined by three independent choices: an index $i \in \{1, 2, \ldots, k+\ell\}$ such that $A = \{y_i, y_{i+1}, \ldots, y_{i+k-1}\}$, a permutation $(y_i, y_{i+1}, \ldots, y_{i+k-1})$ of the set A, and a permutation $(y_{i+k}, y_{i+k+1}, \ldots, y_{i-1})$ of the set $S \setminus A$.

In summary, there are at least $\dfrac{2(k+\ell)!}{(k+\ell)\, k!\, \ell!} = \dfrac{2}{k+\ell}\binom{k+\ell}{k}$ nice sets.

C6. For $n \geq 2$, let $S_1, S_2, \ldots, S_{2^n}$ be 2^n subsets of $A = \{1, 2, 3, \ldots, 2^{n+1}\}$ that satisfy the following property: There do not exist indices a and b with $a < b$ and elements $x, y, z \in A$ with $x < y < z$ such that $y, z \in S_a$ and $x, z \in S_b$. Prove that at least one of the sets $S_1, S_2, \ldots, S_{2^n}$ contains no more than $4n$ elements.

Solution 1. We prove that there exists a set S_a with at most $3n + 1$ elements.

Given a $k \in \{1, \ldots, n\}$, we say that an element $z \in A$ is *k-good* to a set S_a if $z \in S_a$ and S_a contains two other elements x and y with $x < y < z$ such that $z - y < 2^k$ and $z - x \geq 2^k$. Also, $z \in A$ will be called *good* to S_a if z is k-good to S_a for some $k = 1, \ldots, n$.

We claim that each $z \in A$ can be k-good to at most one set S_a. Indeed, suppose on the contrary that z is k-good simultaneously to S_a and S_b, with $a < b$. Then there exist $y_a \in S_a$, $y_a < z$, and $x_b \in S_b$, $x_b < z$, such that $z - y_a < 2^k$ and $z - x_b \geq 2^k$. On the other hand, since $z \in S_a \cap S_b$, by the condition of the problem there is no element of S_a strictly between x_b and z. Hence $y_a \leq x_b$, implying $z - y_a \geq z - x_b$. However this contradicts $z - y_a < 2^k$ and $z - x_b \geq 2^k$. The claim follows.

As a consequence, a fixed $z \in A$ can be good to at most n of the given sets (no more than one of them for each $k = 1, \ldots, n$).

Furthermore, let $u_1 < u_2 < \cdots < u_m < \cdots < u_p$ be all elements of a fixed set S_a that are not good to S_a. We prove that $u_m - u_1 > 2(u_{m-1} - u_1)$ for all $m \geq 3$.

Indeed, assume that $u_m - u_1 \leq 2(u_{m-1} - u_1)$ holds for some $m \geq 3$. This inequality can be written as $2(u_m - u_{m-1}) \leq u_m - u_1$. Take the unique k such that $2^k \leq u_m - u_1 < 2^{k+1}$. Then $2(u_m - u_{m-1}) \leq u_m - u_1 < 2^{k+1}$ yields $u_m - u_{m-1} < 2^k$. However the elements $z = u_m$, $x = u_1$, $y = u_{m-1}$ of S_a then satisfy $z - y < 2^k$ and $z - x \geq 2^k$, so that $z = u_m$ is k-good to S_a.

Thus each term of the sequence $u_2 - u_1, u_3 - u_1, \ldots, u_p - u_1$ is more than twice the previous one. Hence $u_p - u_1 > 2^{p-1}(u_2 - u_1) \geq 2^{p-1}$. But $u_p \in \{1, 2, 3, \ldots, 2^{n+1}\}$, so that $u_p \leq 2^{n+1}$. This yields $p - 1 \leq n$, i. e. $p \leq n + 1$.

In other words, each set S_a contains at most $n + 1$ elements that are not good to it.

To summarize the conclusions, mark with red all elements in the sets S_a that are good to the respective set, and with blue the ones that are not good. Then the total number of red elements, counting multiplicities, is at most $n \cdot 2^{n+1}$ (each $z \in A$ can be marked red in at most n sets). The total number of blue elements is at most $(n + 1)2^n$ (each set S_a contains at most $n + 1$ blue elements). Therefore the sum of cardinalities of $S_1, S_2, \ldots, S_{2^n}$ does not exceed $(3n + 1)2^n$. By averaging, the smallest set has at most $3n + 1$ elements.

Solution 2. We show that one of the sets S_a has at most $2n + 1$ elements. In the sequel $|\cdot|$ denotes the cardinality of a (finite) set.

Claim. For $n \geq 2$, suppose that k subsets $S_1, \ldots, S_k$ of $\{1, 2, \ldots, 2^n\}$ (not necessarily different) satisfy the condition of the problem. Then

$$\sum_{i=1}^{k} (|S_i| - n) \leq (2n - 1)2^{n-2}.$$

Proof. Observe that if the sets S_i $(1 \leq i \leq k)$ satisfy the condition then so do their arbitrary subsets T_i $(1 \leq i \leq k)$. The condition also holds for the sets $t + S_i = \{t + x \mid x \in S_i\}$ where t is arbitrary.

Note also that a set may occur more than once among $S_1, \ldots, S_k$ only if its cardinality is less than 3, in which case its contribution to the sum $\sum_{i=1}^{k} (|S_i| - n)$ is nonpositive (as $n \geq 2$).

The proof is by induction on n. In the base case $n = 2$ we have subsets S_i of $\{1, 2, 3, 4\}$. Only the ones of cardinality 3 and 4 need to be considered by the remark above; each one of

them occurs at most once among $S_1, \ldots, S_k$. If $S_i = \{1, 2, 3, 4\}$ for some i then no S_j is a 3-element subset in view of the condition, hence $\sum_{i=1}^{k}(|S_i| - 2) \le 2$. By the condition again, it is impossible that $S_i = \{1, 3, 4\}$ and $S_j = \{2, 3, 4\}$ for some i, j. So if $|S_i| \le 3$ for all i then at most 3 summands $|S_i| - 2$ are positive, corresponding to 3-element subsets. This implies $\sum_{i=1}^{k}(|S_i| - 2) \le 3$, therefore the conclusion is true for $n = 2$.

Suppose that the claim holds for some $n \ge 2$, and let the sets $S_1, \ldots, S_k \subseteq \{1, 2, \ldots, 2^{n+1}\}$ satisfy the given property. Denote $U_i = S_i \cap \{1, 2, \ldots, 2^n\}$, $V_i = S_i \cap \{2^n + 1, \ldots, 2^{n+1}\}$. Let

$$I = \{i \mid 1 \le i \le k,\ |U_i| \ne 0\}, \qquad J = \{1, \ldots, k\} \setminus I.$$

The sets S_j with $j \in J$ are all contained in $\{2^n + 1, \ldots, 2^{n+1}\}$, so the induction hypothesis applies to their translates $-2^n + S_j$ which have the same cardinalities. Consequently, this gives $\sum_{j\in J}(|S_j| - n) \le (2n - 1)2^{n-2}$, so that

$$\sum_{j\in J}(|S_j| - (n + 1)) \le \sum_{j\in J}(|S_j| - n) \le (2n - 1)2^{n-2}. \tag{1}$$

For $i \in I$, denote by v_i the least element of V_i. Observe that if V_a and V_b intersect, with $a < b$, $a, b \in I$, then v_a is their unique common element. Indeed, let $z \in V_a \cap V_b \subseteq S_a \cap S_b$ and let m be the least element of S_b. Since $b \in I$, we have $m \le 2^n$. By the condition, there is no element of S_a strictly between $m \le 2^n$ and $z > 2^n$, which implies $z = v_a$.

It follows that if the element v_i is removed from each V_i, a family of pairwise disjoint sets $W_i = V_i \setminus \{v_i\}$ is obtained, $i \in I$ (we assume $W_i = \emptyset$ if $V_i = \emptyset$). As $W_i \subseteq \{2^n + 1, \ldots, 2^{n+1}\}$ for all i, we infer that $\sum_{i\in I} |W_i| \le 2^n$. Therefore $\sum_{i\in I}(|V_i| - 1) \le \sum_{i\in I} |W_i| \le 2^n$.

On the other hand, the induction hypothesis applies directly to the sets U_i, $i \in I$, so that $\sum_{i\in \mathcal{I}}(|U_i| - n) \le (2n - 1)2^{n-2}$. In summary,

$$\sum_{i\in I}(|S_i| - (n + 1)) = \sum_{i\in I}(|U_i| - n) + \sum_{i\in I}(|V_i| - 1) \le (2n - 1)2^{n-2} + 2^n. \tag{2}$$

The estimates (1) and (2) are sufficient to complete the inductive step:

$$\begin{aligned}\sum_{i=1}^{k}(|S_i| - (n + 1)) &= \sum_{i\in I}(|S_i| - (n + 1)) + \sum_{j\in J}(|S_j| - (n + 1)) \\ &\le (2n - 1)2^{n-2} + 2^n + (2n - 1)2^{n-2} = (2n + 1)2^{n-1}.\end{aligned}$$

Returning to the problem, consider $k = 2^n$ subsets $S_1, S_2, \ldots, S_{2^n}$ of $\{1, 2, 3, \ldots, 2^{n+1}\}$. If they satisfy the given condition, the claim implies $\sum_{i=1}^{2^n}(|S_i| - (n + 1)) \le (2n + 1)2^{n-1}$. By averaging again, we see that the smallest set has at most $2n + 1$ elements.

Comment. It can happen that each set S_i has cardinality at least $n + 1$. Here is an example by the proposer.

For $i = 1, \ldots, 2^n$, let $S_i = \{i + 2^k \mid 0 \le k \le n\}$. Then $|S_i| = n + 1$ for all i. Suppose that there exist $a < b$ and $x < y < z$ such that $y, z \in S_a$ and $x, z \in S_b$. Hence $z = a + 2^k = b + 2^l$ for some $k > l$. Since $y \in S_a$ and $y < z$, we have $y \le a + 2^{k-1}$. So the element $x \in S_b$ satisfies

$$x < y \le a + 2^{k-1} = z - 2^{k-1} \le z - 2^l = b.$$

However the least element of S_b is $b + 1$, a contradiction.

Geometry

G1. In an acute-angled triangle ABC, point H is the orthocentre and A_0, B_0, C_0 are the midpoints of the sides BC, CA, AB, respectively. Consider three circles passing through H: ω_a around A_0, ω_b around B_0 and ω_c around C_0. The circle ω_a intersects the line BC at A_1 and A_2; ω_b intersects CA at B_1 and B_2; ω_c intersects AB at C_1 and C_2. Show that the points A_1, A_2, B_1, B_2, C_1, C_2 lie on a circle.

Solution 1. The perpendicular bisectors of the segments A_1A_2, B_1B_2, C_1C_2 are also the perpendicular bisectors of BC, CA, AB. So they meet at O, the circumcentre of ABC. Thus O is the only point that can possibly be the centre of the desired circle.

From the right triangle OA_0A_1 we get

$$OA_1^2 = OA_0^2 + A_0A_1^2 = OA_0^2 + A_0H^2. \tag{1}$$

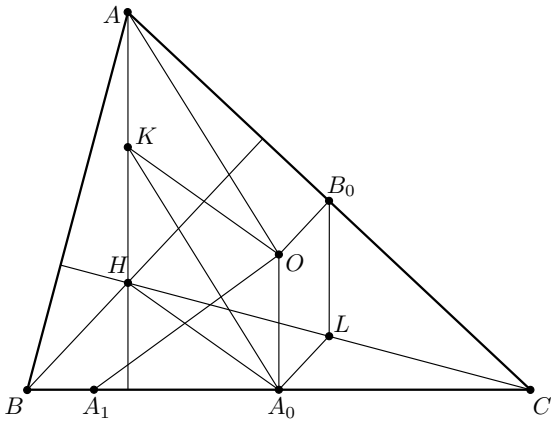
Let K be the midpoint of AH and let L be the midpoint of CH. Since A_0 and B_0 are the midpoints of BC and CA, we see that $A_0L \| BH$ and $B_0L \| AH$. Thus the segments A_0L and B_0L are perpendicular to AC and BC, hence parallel to OB_0 and OA_0, respectively. Consequently OA_0LB_0 is a parallelogram, so that OA_0 and B_0L are equal and parallel. Also, the midline B_0L of triangle AHC is equal and parallel to AK and KH.

It follows that AKA_0O and HA_0OK are parallelograms. The first one gives $A_0K = OA = R$, where R is the circumradius of ABC. From the second one we obtain

$$2(OA_0^2 + A_0H^2) = OH^2 + A_0K^2 = OH^2 + R^2. \tag{2}$$

(In a parallelogram, the sum of squares of the diagonals equals the sum of squares of the sides).

From (1) and (2) we get $OA_1^2 = (OH^2 + R^2)/2$. By symmetry, the same holds for the distances OA_2, OB_1, OB_2, OC_1 and OC_2. Thus A_1, A_2, B_1, B_2, C_1, C_2 all lie on a circle with centre at O and radius $(OH^2 + R^2)/2$.

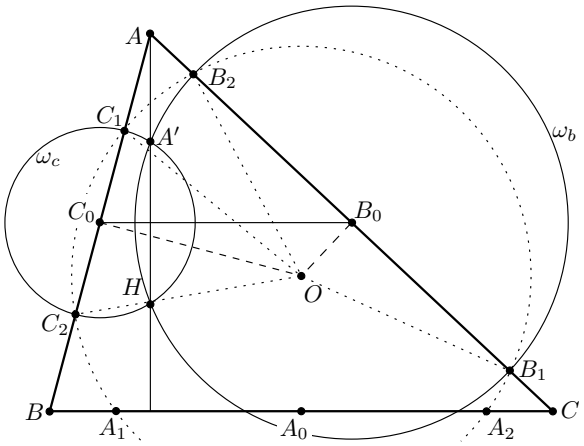


Solution 2. We are going to show again that the circumcentre O is equidistant from the six points in question.

Let A' be the second intersection point of ω_b and ω_c. The line B_0C_0, which is the line of centers of circles ω_b and ω_c, is a midline in triangle ABC, parallel to BC and perpendicular to the altitude AH. The points A' and H are symmetric with respect to the line of centers. Therefore A' lies on the line AH.

From the two circles ω_b and ω_c we obtain $AC_1 \cdot AC_2 = AA' \cdot AH = AB_1 \cdot AB_2$. So the quadrilateral $B_1B_2C_1C_2$ is cyclic. The perpendicular bisectors of the sides B_1B_2 and C_1C_2 meet at O. Hence O is the circumcentre of $B_1B_2C_1C_2$ and so $OB_1 = OB_2 = OC_1 = OC_2$.

Analogous arguments yield $OA_1 = OA_2 = OB_1 = OB_2$ and $OA_1 = OA_2 = OC_1 = OC_2$. Thus A_1, A_2, B_1, B_2, C_1, C_2 lie on a circle centred at O.



Comment. The problem can be solved without much difficulty in many ways by calculation, using trigonometry, coordinate geometry or complex numbers. As an example we present a short proof using vectors.

Solution 3. Let again O and R be the circumcentre and circumradius. Consider the vectors

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}, \quad \overrightarrow{OC} = \mathbf{c}, \quad \text{where} \quad \mathbf{a}^2 = \mathbf{b}^2 = \mathbf{c}^2 = R^2.$$

It is well known that $\overrightarrow{OH} = \mathbf{a} + \mathbf{b} + \mathbf{c}$. Accordingly,

$$\overrightarrow{A_0H} = \overrightarrow{OH} - \overrightarrow{OA_0} = (\mathbf{a} + \mathbf{b} + \mathbf{c}) - \frac{\mathbf{b} + \mathbf{c}}{2} = \frac{2\mathbf{a} + \mathbf{b} + \mathbf{c}}{2},$$

and

$$OA_1^2 = OA_0^2 + A_0A_1^2 = OA_0^2 + A_0H^2 = \left(\frac{\mathbf{b} + \mathbf{c}}{2}\right)^2 + \left(\frac{2\mathbf{a} + \mathbf{b} + \mathbf{c}}{2}\right)^2$$

$$= \frac{1}{4}(\mathbf{b}^2 + 2\mathbf{bc} + \mathbf{c}^2) + \frac{1}{4}(4\mathbf{a}^2 + 4\mathbf{ab} + 4\mathbf{ac} + \mathbf{b}^2 + 2\mathbf{bc} + \mathbf{c}^2) = 2R^2 + (\mathbf{ab} + \mathbf{ac} + \mathbf{bc});$$

here $\mathbf{ab}$, $\mathbf{bc}$, etc. denote dot products of vectors. We get the same for the distances OA_2, OB_1, OB_2, OC_1 and OC_2.

G2. Given trapezoid $ABCD$ with parallel sides AB and CD, assume that there exist points E on line BC outside segment BC, and F inside segment AD, such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF, and by J the point of intersection of AB and EF. Let K be the midpoint of segment EF; assume it does not lie on line AB.

Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ.

Solution. Assume that the disposition of points is as in the diagram.

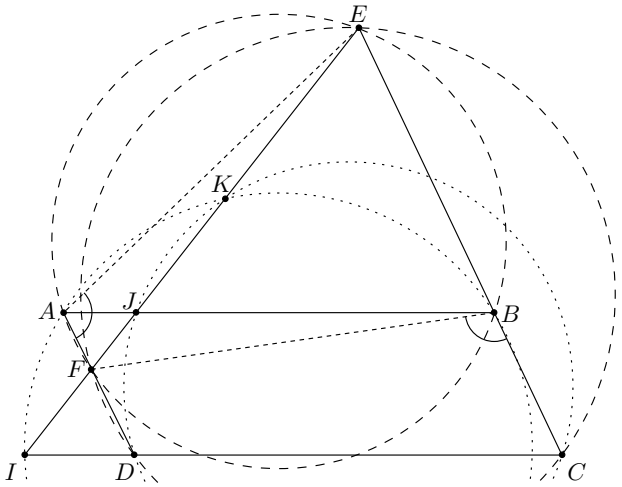
Since $\angle EBF = 180^\circ - \angle CBF = 180^\circ - \angle EAF$ by hypothesis, the quadrilateral $AEBF$ is cyclic. Hence $AJ \cdot JB = FJ \cdot JE$. In view of this equality, I belongs to the circumcircle of ABK if and only if $IJ \cdot JK = FJ \cdot JE$. Expressing $IJ = IF + FJ$, $JE = FE - FJ$, and $JK = \frac{1}{2}FE - FJ$, we find that I belongs to the circumcircle of ABK if and only if

$$FJ = \frac{IF \cdot FE}{2IF + FE}\,.$$

Since $AEBF$ is cyclic and AB, CD are parallel, $\angle FEC = \angle FAB = 180^\circ - \angle CDF$. Then $CDFE$ is also cyclic, yielding $ID \cdot IC = IF \cdot IE$. It follows that K belongs to the circumcircle of CDJ if and only if $IJ \cdot IK = IF \cdot IE$. Expressing $IJ = IF + FJ$, $IK = IF + \frac{1}{2}FE$, and $IE = IF + FE$, we find that K is on the circumcircle of CDJ if and only if

$$FJ = \frac{IF \cdot FE}{2IF + FE}\,.$$

The conclusion follows.



Comment. While the figure shows B inside segment CE, it is possible that C is inside segment BE. Consequently, I would be inside segment EF and J outside segment EF. The position of point K on line EF with respect to points I, J may also vary.

Some case may require that an angle φ be replaced by $180^\circ - \varphi$, and in computing distances, a sum may need to become a difference. All these cases can be covered by the proposed solution if it is clearly stated that *signed* distances and angles are used.

32

G3. Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

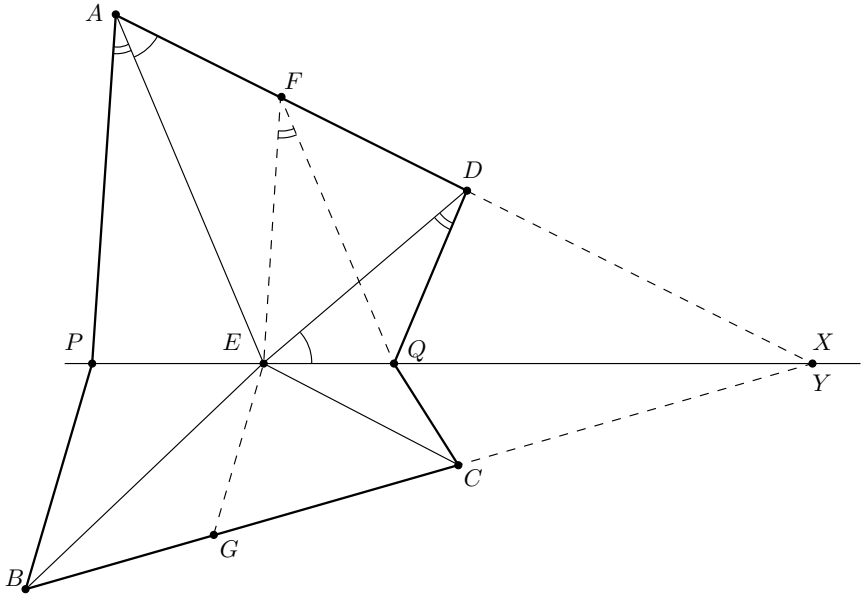
Solution 1. Let F be the point on the line AD such that $EF \| PA$. By hypothesis, the quadrilateral $PQDA$ is cyclic. So if F lies between A and D then $\angle EFD = \angle PAD = 180^\circ - \angle EQD$; the points F and Q are on distinct sides of the line DE and we infer that $EFDQ$ is a cyclic quadrilateral. And if D lies between A and F then a similar argument shows that $\angle EFD = \angle EQD$; but now the points F and Q lie on the same side of DE, so that $EDFQ$ is a cyclic quadrilateral.

In either case we obtain the equality $\angle EFQ = \angle EDQ = \angle PAE$ which implies that $FQ \| AE$. So the triangles EFQ and PAE are either homothetic or parallel-congruent. More specifically, triangle EFQ is the image of PAE under the mapping f which carries the points P, E respectively to E, Q and is either a homothety or translation by a vector. Note that f is uniquely determined by these conditions and the position of the points P, E, Q alone.

Let now G be the point on the line BC such that $EG \| PB$. The same reasoning as above applies to points B, C in place of A, D, implying that the triangle EGQ is the image of PBE under the same mapping f. So f sends the four points A, P, B, E respectively to F, E, G, Q.

If $PE \neq QE$, so that f is a homothety with a centre X, then the lines AF, PE, BG—i.e. the lines AD, PQ, BC—are concurrent at X. And since $PQDA$ and $QPBC$ are cyclic quadrilaterals, the equalities $XA \cdot XD = XP \cdot XQ = XB \cdot XC$ hold, showing that the quadrilateral $ABCD$ is cyclic.

Finally, if $PE = QE$, so that f is a translation, then $AD \| PQ \| BC$. Thus $PQDA$ and $QPBC$ are isosceles trapezoids. Then also $ABCD$ is an isosceles trapezoid, hence a cyclic quadrilateral.



Solution 2. Here is another way to reach the conclusion that the lines AD, BC and PQ are either concurrent or parallel. From the cyclic quadrilateral $PQDA$ we get

$$\angle PAD = 180^\circ - \angle PQD = \angle QDE + \angle QED = \angle PAE + \angle QED.$$

Hence $\angle QED = \angle PAD - \angle PAE = \angle EAD$. This in view of the tangent-chord theorem means that the circumcircle of triangle EAD is tangent to the line PQ at E. Analogously, the circumcircle of triangle EBC is tangent to PQ at E.

Suppose that the line AD intersects PQ at X. Since XE is tangent to the circle (EAD), $XE^2 = XA \cdot XD$. Also, $XA \cdot XD = XP \cdot XQ$ because P, Q, D, A lie on a circle. Therefore $XE^2 = XP \cdot XQ$.

It is not hard to see that this equation determines the position of the point X on the line PQ uniquely. Thus, if BC also cuts PQ, say at Y, then the analogous equation for Y yields $X = Y$, meaning that the three lines indeed concur. In this case, as well as in the case where $AD \| PQ \| BC$, the concluding argument is the same as in the first solution.

It remains to eliminate the possibility that e.g. AD meets PQ at X while $BC \| PQ$. Indeed, $QPBC$ would then be an isosceles trapezoid and the angle equality $\angle PBE = \angle QCE$ would force that E is the midpoint of PQ. So the length of XE, which is the geometric mean of the lengths of XP and XQ, should also be their arithmetic mean—impossible, as $XP \neq XQ$. The proof is now complete.

Comment. After reaching the conclusion that the circles (EDA) and (EBC) are tangent to PQ one may continue as follows. Denote the circles $(PQDA)$, (EDA), (EBC), $(QPBC)$ by ω_1, ω_2, ω_3, ω_4 respectively. Let ℓ_{ij} be the radical axis of the pair (ω_i, ω_j) for $i < j$. As is well-known, the lines ℓ_{12}, ℓ_{13}, ℓ_{23} concur, possibly at infinity (let this be the meaning of the word *concur* in this comment). So do the lines ℓ_{12}, ℓ_{14}, ℓ_{24}. Note however that ℓ_{23} and ℓ_{14} both coincide with the line PQ. Hence the pair ℓ_{12}, PQ is in both triples; thus the four lines ℓ_{12}, ℓ_{13}, ℓ_{24} and PQ are concurrent.

Similarly, ℓ_{13}, ℓ_{14}, ℓ_{34} concur, ℓ_{23}, ℓ_{24}, ℓ_{34} concur, and since $\ell_{14} = \ell_{23} = PQ$, the four lines ℓ_{13}, ℓ_{24}, ℓ_{34} and PQ are concurrent. The lines ℓ_{13} and ℓ_{24} are present in both quadruples, therefore all the lines ℓ_{ij} are concurrent. Hence the result.

34

G4. In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the points A and F are tangent to the line BC at the points P and Q so that B lies between C and Q. Prove that the lines PE and QF intersect on the circumcircle of triangle AEF.

Solution 1. To approach the desired result we need some information about the slopes of the lines PE and QF; this information is provided by formulas (1) and (2) which we derive below.

The tangents BP and BQ to the two circles passing through A and F are equal, as $BP^2 = BA \cdot BF = BQ^2$. Consider the altitude AD of triangle ABC and its orthocentre H. From the cyclic quadrilaterals $CDFA$ and $CDHE$ we get $BA \cdot BF = BC \cdot BD = BE \cdot BH$. Thus $BP^2 = BE \cdot BH$, or $BP/BH = BE/BP$, implying that the triangles BPH and BEP are similar. Hence

$$\angle BPE = \angle BHP. \tag{1}$$

The point P lies between D and C; this follows from the equality $BP^2 = BC \cdot BD$. In view of this equality, and because $BP = BQ$,

$$DP \cdot DQ = (BP - BD) \cdot (BP + BD) = BP^2 - BD^2 = BD \cdot (BC - BD) = BD \cdot DC.$$

Also $AD \cdot DH = BD \cdot DC$, as is seen from the similar triangles BDH and ADC. Combining these equalities we obtain $AD \cdot DH = DP \cdot DQ$. Therefore $DH/DP = DQ/DA$, showing that the triangles HDP and QDA are similar. Hence $\angle HPD = \angle QAD$, which can be rewritten as $\angle BPH = \angle BAD + \angle BAQ$. And since BQ is tangent to the circumcircle of triangle FAQ,

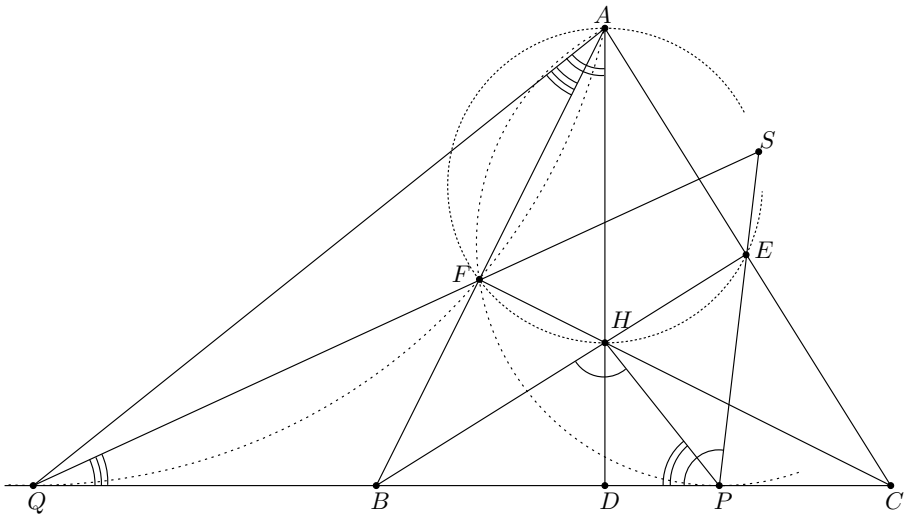
$$\angle BQF = \angle BAQ = \angle BPH - \angle BAD. \tag{2}$$

From (1) and (2) we deduce

$$\begin{aligned}\angle BPE + \angle BQF &= (\angle BHP + \angle BPH) - \angle BAD = (180^\circ - \angle PBH) - \angle BAD \\ &= (90^\circ + \angle BCA) - (90^\circ - \angle ABC) = \angle BCA + \angle ABC = 180^\circ - \angle CAB.\end{aligned}$$

Thus $\angle BPE + \angle BQF < 180^\circ$, which means that the rays PE and QF meet. Let S be the point of intersection. Then $\angle PSQ = 180^\circ - (\angle BPE + \angle BQF) = \angle CAB = \angle EAF$.

If S lies between P and E then $\angle PSQ = 180^\circ - \angle ESF$; and if E lies between P and S then $\angle PSQ = \angle ESF$. In either case the equality $\angle PSQ = \angle EAF$ which we have obtained means that S lies on the circumcircle of triangle AEF.



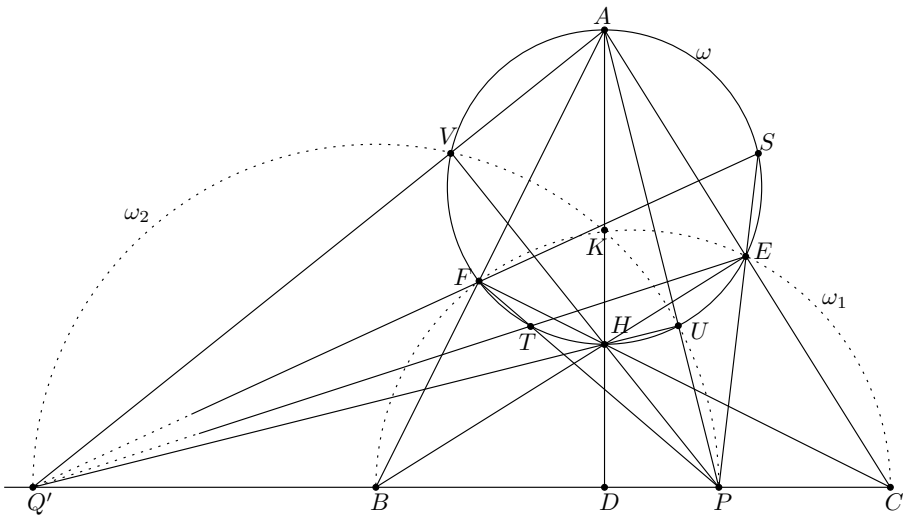
Solution 2. Let H be the orthocentre of triangle ABC and let ω be the circle with diameter AH, passing through E and F. Introduce the points of intersection of ω with the following lines emanating from P: $PA \cap \omega = \{A, U\}$, $PH \cap \omega = \{H, V\}$, $PE \cap \omega = \{E, S\}$. The altitudes of triangle AHP are contained in the lines AV, HU, BC, meeting at its orthocentre Q'.

By Pascal's theorem applied to the (tied) hexagon $AESFHV$, the points $AE \cap FH = C$, $ES \cap HV = P$ and $SF \cap VA$ are collinear, so FS passes through Q'.

Denote by ω_1 and ω_2 the circles with diameters BC and PQ', respectively. Let D be the foot of the altitude from A in triangle ABC. Suppose that AD meets the circles ω_1 and ω_2 at the respective points K and L.

Since H is the orthocentre of ABC, the triangles BDH and ADC are similar, and so $DA \cdot DH = DB \cdot DC = DK^2$; the last equality holds because BKC is a right triangle. Since H is the orthocentre also in triangle $AQ'P$, we analogously have $DL^2 = DA \cdot DH$. Therefore $DK = DL$ and $K = L$.

Also, $BD \cdot BC = BA \cdot BF$, from the similar triangles ABD, CBF. In the right triangle BKC we have $BK^2 = BD \cdot BC$. Hence, and because $BA \cdot BF = BP^2 = BQ^2$ (by the definition of P and Q in the problem statement), we obtain $BK = BP = BQ$. It follows that B is the centre of ω_2 and hence $Q' = Q$. So the lines PE and QF meet at the point S lying on the circumcircle of triangle AEF.



Comment 1. If T is the point defined by $PF \cap \omega = \{F, T\}$, Pascal's theorem for the hexagon $AFTEHV$ will analogously lead to the conclusion that the line ET goes through Q'. In other words, the lines PF and QE also concur on ω.

Comment 2. As is known from algebraic geometry, the points of the circle ω form a commutative groups with the operation defined as follows. Choose any point $0 \in \omega$ (to be the neutral element of the group) and a line ℓ exterior to the circle. For $X, Y \in \omega$, draw the line from the point $XY \cap \ell$ through 0 to its second intersection with ω and define this point to be $X + Y$.

In our solution we have chosen H to be the neutral element in this group and line BC to be ℓ. The fact that the lines AV, HU, ET, FS are concurrent can be deduced from the identities $A + A = 0$, $F = E + A$, $V = U + A = S + E = T + F$.

Comment 3. The problem was submitted in the following equivalent formulation:

Let BE and CF be altitudes of an acute triangle ABC. We choose P on the side BC and Q on the extension of CB beyond B such that $BQ^2 = BP^2 = BF \cdot AB$. If QF and PE intersect at S, prove that $ESAF$ is cyclic.

G5. Let k and n be integers with $0 \le k \le n-2$. Consider a set L of n lines in the plane such that no two of them are parallel and no three have a common point. Denote by I the set of intersection points of lines in L. Let O be a point in the plane not lying on any line of L.

A point $X \in I$ is colored red if the open line segment OX intersects at most k lines in L. Prove that I contains at least $\frac{1}{2}(k+1)(k+2)$ red points.

Solution. There are at least $\frac{1}{2}(k+1)(k+2)$ points in the intersection set I in view of the condition $n \ge k+2$.

For each point $P \in I$, define its *order* as the number of lines that intersect the open line segment OP. By definition, P is red if its order is at most k. Note that there is always at least one point $X \in I$ of order 0. Indeed, the lines in L divide the plane into regions, bounded or not, and O belongs to one of them. Clearly any corner of this region is a point of I with order 0.

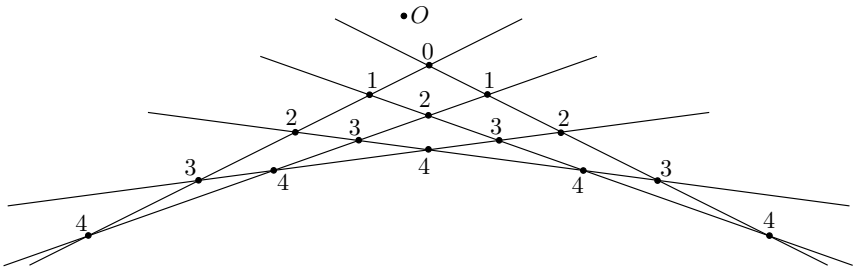
Claim. Suppose that two points $P, Q \in I$ lie on the same line of L, and no other line of L intersects the open line segment PQ. Then the orders of P and Q differ by at most 1.

Proof. Let P and Q have orders p and q, respectively, with $p \ge q$. Consider triangle OPQ. Now p equals the number of lines in L that intersect the interior of side OP. None of these lines intersects the interior of side PQ, and at most one can pass through Q. All remaining lines must intersect the interior of side OQ, implying that $q \ge p-1$. The conclusion follows.

We prove the main result by induction on k. The base $k=0$ is clear since there is a point of order 0 which is red. Assuming the statement true for $k-1$, we pass on to the inductive step. Select a point $P \in I$ of order 0, and consider one of the lines $\ell \in L$ that pass through P. There are $n-1$ intersection points on ℓ, one of which is P. Out of the remaining $n-2$ points, the k closest to P have orders not exceeding k by the Claim. It follows that there are at least $k+1$ red points on ℓ.

Let us now consider the situation with ℓ removed (together with all intersection points it contains). By hypothesis of induction, there are at least $\frac{1}{2}k(k+1)$ points of order not exceeding $k-1$ in the resulting configuration. Restoring ℓ back produces at most one new intersection point on each line segment joining any of these points to O, so their order is at most k in the original configuration. The total number of points with order not exceeding k is therefore at least $(k+1)+\frac{1}{2}k(k+1)=\frac{1}{2}(k+1)(k+2)$. This completes the proof.

Comment. The steps of the proof can be performed in reverse order to obtain a configuration of n lines such that equality holds simultaneously for all $0 \le k \le n-2$. Such a set of lines is illustrated in the Figure.



G6. There is given a convex quadrilateral $ABCD$. Prove that there exists a point P inside the quadrilateral such that

$$\angle PAB + \angle PDC = \angle PBC + \angle PAD = \angle PCD + \angle PBA = \angle PDA + \angle PCB = 90^\circ \qquad (1)$$

if and only if the diagonals AC and BD are perpendicular.

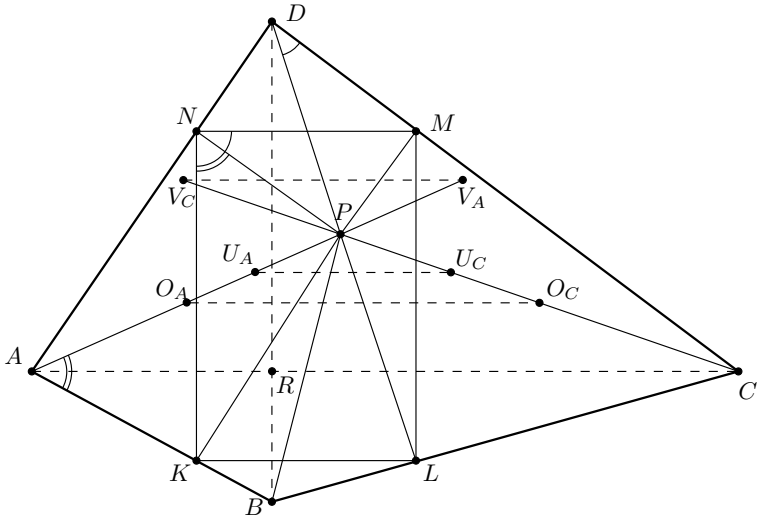
Solution 1. For a point P in $ABCD$ which satisfies (1), let K, L, M, N be the feet of perpendiculars from P to lines AB, BC, CD, DA, respectively. Note that K, L, M, N are interior to the sides as all angles in (1) are acute. The cyclic quadrilaterals $AKPN$ and $DNPM$ give

$$\angle PAB + \angle PDC = \angle PNK + \angle PNM = \angle KNM.$$

Analogously, $\angle PBC + \angle PAD = \angle LKN$ and $\angle PCD + \angle PBA = \angle MLK$. Hence the equalities (1) imply $\angle KNM = \angle LKN = \angle MLK = 90^\circ$, so that $KLMN$ is a rectangle. The converse also holds true, provided that K, L, M, N are interior to sides AB, BC, CD, DA.

(i) Suppose that there exists a point P in $ABCD$ such that $KLMN$ is a rectangle. We show that AC and BD are parallel to the respective sides of $KLMN$.

Let O_A and O_C be the circumcentres of the cyclic quadrilaterals $AKPN$ and $CMPL$. Line O_AO_C is the common perpendicular bisector of LM and KN, therefore O_AO_C is parallel to KL and MN. On the other hand, O_AO_C is the midline in the triangle ACP that is parallel to AC. Therefore the diagonal AC is parallel to the sides KL and MN of the rectangle. Likewise, BD is parallel to KN and LM. Hence AC and BD are perpendicular.



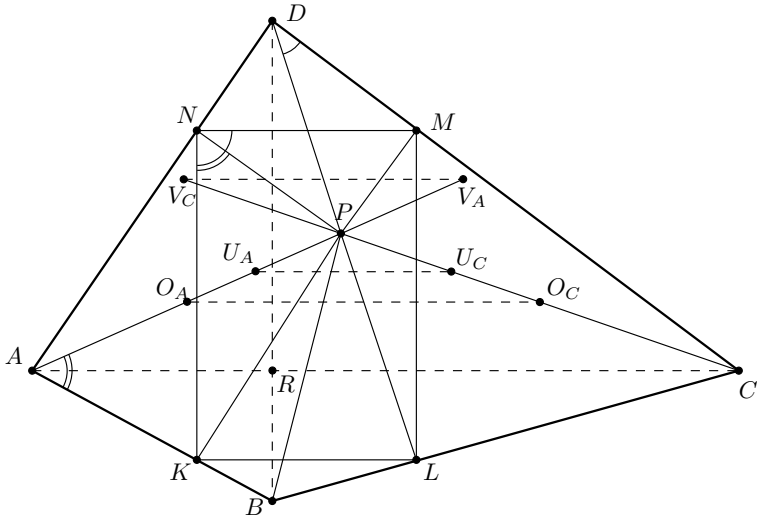
(ii) Suppose that AC and BD are perpendicular and meet at R. If $ABCD$ is a rhombus, P can be chosen to be its centre. So assume that $ABCD$ is not a rhombus, and let $BR < DR$ without loss of generality.

Denote by U_A and U_C the circumcentres of the triangles ABD and CDB, respectively. Let AV_A and CV_C be the diameters through A and C of the two circumcircles. Since AR is an altitude in triangle ADB, lines AC and AV_A are isogonal conjugates, i. e. $\angle DAV_A = \angle BAC$. Now $BR < DR$ implies that ray AU_A lies in $\angle DAC$. Similarly, ray CU_C lies in $\angle DCA$. Both diameters AV_A and CV_C intersect BD as the angles at B and D of both triangles are acute. Also U_AU_C is parallel to AC as it is the perpendicular bisector of BD. Hence V_AV_C is parallel to AC, too. We infer that AV_A and CV_C intersect at a point P inside triangle ACD, hence inside $ABCD$.

Construct points K, L, M, N, O_A and O_C in the same way as in the introduction. It follows from the previous paragraph that K, L, M, N are interior to the respective sides. Now O_AO_C is a midline in triangle ACP again. Therefore lines AC, O_AO_C and U_AU_C are parallel.

The cyclic quadrilateral $AKPN$ yields $\angle NKP = \angle NAP$. Since $\angle NAP = \angle DAU_A = \angle BAC$, as specified above, we obtain $\angle NKP = \angle BAC$. Because PK is perpendicular to AB, it follows that NK is perpendicular to AC, hence parallel to BD. Likewise, LM is parallel to BD.

Consider the two homotheties with centres A and C which transform triangles ABD and CDB into triangles AKN and CML, respectively. The images of points U_A and U_C are O_A and O_C, respectively. Since U_AU_C and O_AO_C are parallel to AC, the two ratios of homothety are the same, equal to $\lambda = AN/AD = AK/AB = AO_A/AU_A = CO_C/CU_C = CM/CD = CL/CB$. It is now straightforward that $DN/DA = DM/DC = BK/BA = BL/BC = 1 - \lambda$. Hence KL and MN are parallel to AC, implying that $KLMN$ is a rectangle and completing the proof.



Solution 2. For a point P distinct from A, B, C, D, let circles (APD) and (BPC) intersect again at Q ($Q = P$ if the circles are tangent). Next, let circles (AQB) and (CQD) intersect again at R. We show that if P lies in $ABCD$ and satisfies (1) then AC and BD intersect at R and are perpendicular; the converse is also true. It is convenient to use directed angles. Let $\measuredangle(UV, XY)$ denote the angle of counterclockwise rotation that makes line UV parallel to line XY. Recall that four noncollinear points U, V, X, Y are concyclic if and only if $\measuredangle(UX, VX) = \measuredangle(UY, VY)$.

The definitions of points P, Q and R imply

$$\begin{aligned}
\measuredangle(AR, BR) &= \measuredangle(AQ, BQ) = \measuredangle(AQ, PQ) + \measuredangle(PQ, BQ) = \measuredangle(AD, PD) + \measuredangle(PC, BC),\\
\measuredangle(CR, DR) &= \measuredangle(CQ, DQ) = \measuredangle(CQ, PQ) + \measuredangle(PQ, DQ) = \measuredangle(CB, PB) + \measuredangle(PA, DA),\\
\measuredangle(BR, CR) &= \measuredangle(BR, RQ) + \measuredangle(RQ, CR) = \measuredangle(BA, AQ) + \measuredangle(DQ, CD)\\
&= \measuredangle(BA, AP) + \measuredangle(AP, AQ) + \measuredangle(DQ, DP) + \measuredangle(DP, CD)\\
&= \measuredangle(BA, AP) + \measuredangle(DP, CD).
\end{aligned}$$

Observe that the whole construction is reversible. One may start with point R, define Q as the second intersection of circles (ARB) and (CRD), and then define P as the second intersection of circles (AQD) and (BQC). The equalities above will still hold true.

Assume in addition that P is interior to $ABCD$. Then

$$\measuredangle(AD, PD) = \angle PDA,\ \measuredangle(PC, BC) = \angle PCB,\ \measuredangle(CB, PB) = \angle PBC,\ \measuredangle(PA, DA) = \angle PAD,$$
$$\measuredangle(BA, AP) = \angle PAB,\ \measuredangle(DP, CD) = \angle PDC.$$

(i) Suppose that P lies in $ABCD$ and satisfies (1). Then $\measuredangle(AR, BR) = \angle PDA + \angle PCB = 90^\circ$ and similarly $\measuredangle(BR, CR) = \measuredangle(CR, DR) = 90^\circ$. It follows that R is the common point of lines AC and BD, and that these lines are perpendicular.
(ii) Suppose that AC and BD are perpendicular and intersect at R. We show that the point P defined by the reverse construction (starting with R and ending with P) lies in $ABCD$. This is enough to finish the solution, because then the angle equalities above will imply (1).

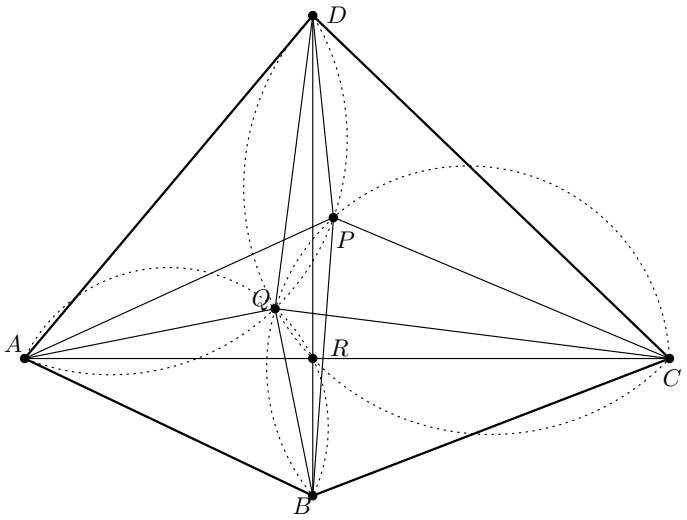
One can assume that Q, the second common point of circles (ABR) and (CDR), lies in $\angle ARD$. Then in fact Q lies in triangle ADR as angles AQR and DQR are obtuse. Hence $\angle AQD$ is obtuse, too, so that B and C are outside circle (ADQ) ($\angle ABD$ and $\angle ACD$ are acute).

Now $\angle CAB + \angle CDB = \angle BQR + \angle CQR = \angle CQB$ implies $\angle CAB < \angle CQB$ and $\angle CDB < \angle CQB$. Hence A and D are outside circle (BCQ). In conclusion, the second common point P of circles (ADQ) and (BCQ) lies on their arcs ADQ and BCQ.

We can assume that P lies in $\angle CQD$. Since

$$\angle QPC + \angle QPD = (180^\circ - \angle QBC) + (180^\circ - \angle QAD) =$$
$$= 360^\circ - (\angle RBC + \angle QBR) - (\angle RAD - \angle QAR) = 360^\circ - \angle RBC - \angle RAD > 180^\circ,$$

point P lies in triangle CDQ, and hence in $ABCD$. The proof is complete.



G7. Let $ABCD$ be a convex quadrilateral with $AB \neq BC$. Denote by ω_1 and ω_2 the incircles of triangles ABC and ADC. Suppose that there exists a circle ω inscribed in angle ABC, tangent to the extensions of line segments AD and CD. Prove that the common external tangents of ω_1 and ω_2 intersect on ω.

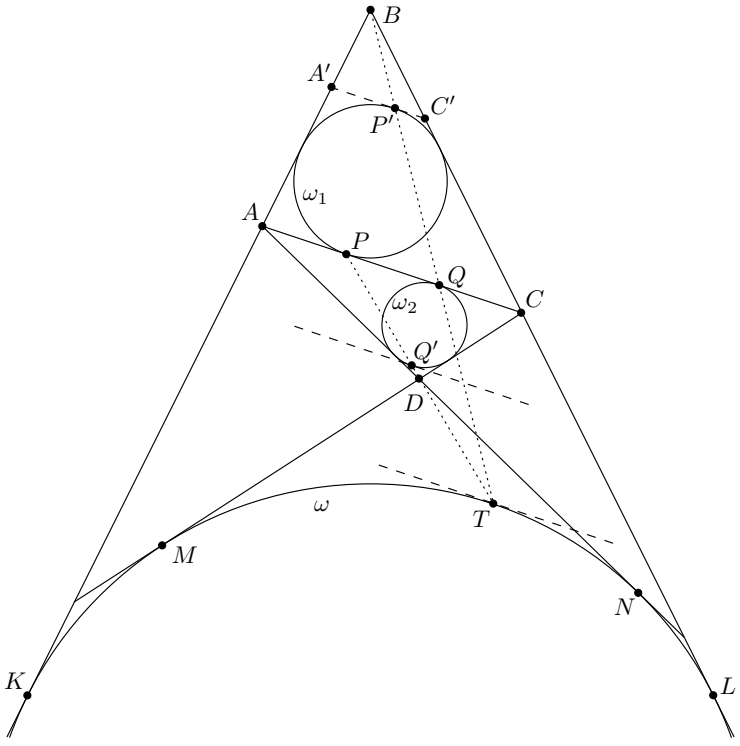
Solution. The proof below is based on two known facts.

Lemma 1. Given a convex quadrilateral $ABCD$, suppose that there exists a circle which is inscribed in angle ABC and tangent to the extensions of line segments AD and CD. Then $AB + AD = CB + CD$.

Proof. The circle in question is tangent to each of the lines AB, BC, CD, DA, and the respective points of tangency K, L, M, N are located as with circle ω in the figure. Then

$$AB + AD = (BK - AK) + (AN - DN), \quad CB + CD = (BL - CL) + (CM - DM).$$

Also $BK = BL$, $DN = DM$, $AK = AN$, $CL = CM$ by equalities of tangents. It follows that $AB + AD = CB + CD$.



For brevity, in the sequel we write "excircle AC" for the excircle of a triangle with side AC which is tangent to line segment AC and the extensions of the other two sides.

Lemma 2. The incircle of triangle ABC is tangent to its side AC at P. Let PP' be the diameter of the incircle through P, and let line BP' intersect AC at Q. Then Q is the point of tangency of side AC and excircle AC.

Proof. Let the tangent at P' to the incircle ω_1 meet BA and BC at A' and C'. Now ω_1 is the excircle $A'C'$ of triangle $A'BC'$, and it touches side $A'C'$ at P'. Since $A'C' \parallel AC$, the homothety with centre B and ratio BQ/BP' takes ω_1 to the excircle AC of triangle ABC. Because this homothety takes P' to Q, the lemma follows.

Recall also that if the incircle of a triangle touches its side AC at P, then the tangency point Q of the same side and excircle AC is the unique point on line segment AC such that $AP = CQ$.

We pass on to the main proof. Let ω_1 and ω_2 touch AC at P and Q, respectively; then $AP = (AC + AB - BC)/2$, $CQ = (CA + CD - AD)/2$. Since $AB - BC = CD - AD$ by Lemma 1, we obtain $AP = CQ$. It follows that in triangle ABC side AC and excircle AC are tangent at Q. Likewise, in triangle ADC side AC and excircle AC are tangent at P. Note that $P \neq Q$ as $AB \neq BC$.

Let PP' and QQ' be the diameters perpendicular to AC of ω_1 and ω_2, respectively. Then Lemma 2 shows that points B, P' and Q are collinear, and so are points D, Q' and P.

Consider the diameter of ω perpendicular to AC and denote by T its endpoint that is closer to AC. The homothety with centre B and ratio BT/BP' takes ω_1 to ω. Hence B, P' and T are collinear. Similarly, D, Q' and T are collinear since the homothety with centre D and ratio $-DT/DQ'$ takes ω_2 to ω.

We infer that points T, P' and Q are collinear, as well as T, Q' and P. Since $PP' \parallel QQ'$, line segments PP' and QQ' are then homothetic with centre T. The same holds true for circles ω_1 and ω_2 because they have PP' and QQ' as diameters. Moreover, it is immediate that T lies on the same side of line PP' as Q and Q', hence the ratio of homothety is positive. In particular ω_1 and ω_2 are not congruent.

In summary, T is the centre of a homothety with positive ratio that takes circle ω_1 to circle ω_2. This completes the solution, since the only point with the mentioned property is the intersection of the the common external tangents of ω_1 and ω_2.

42

Number Theory

N1. Let n be a positive integer and let p be a prime number. Prove that if a, b, c are integers (not necessarily positive) satisfying the equations

$$a^n + pb = b^n + pc = c^n + pa,$$

then $a = b = c$.

Solution 1. If two of a, b, c are equal, it is immediate that all the three are equal. So we may assume that $a \neq b \neq c \neq a$. Subtracting the equations we get $a^n - b^n = -p(b-c)$ and two cyclic copies of this equation, which upon multiplication yield

$$\frac{a^n - b^n}{a-b} \cdot \frac{b^n - c^n}{b-c} \cdot \frac{c^n - a^n}{c-a} = -p^3. \tag{1}$$

If n is odd then the differences $a^n - b^n$ and $a - b$ have the same sign and the product on the left is positive, while $-p^3$ is negative. So n must be even.

Let d be the greatest common divisor of the three differences $a-b$, $b-c$, $c-a$, so that $a - b = du$, $b - c = dv$, $c - a = dw$; $\gcd(u, v, w) = 1$, $u + v + w = 0$.

From $a^n - b^n = -p(b-c)$ we see that $(a-b)|p(b-c)$, i.e., $u|pv$; and cyclically $v|pw$, $w|pu$. As $\gcd(u, v, w) = 1$ and $u + v + w = 0$, at most one of u, v, w can be divisible by p. Supposing that the prime p does not divide any one of them, we get $u|v$, $v|w$, $w|u$, whence $|u| = |v| = |w| = 1$; but this quarrels with $u + v + w = 0$.

Thus p must divide exactly one of these numbers. Let e.g. $p|u$ and write $u = pu_1$. Now we obtain, similarly as before, $u_1|v$, $v|w$, $w|u_1$ so that $|u_1| = |v| = |w| = 1$. The equation $pu_1 + v + w = 0$ forces that the prime p must be even; i.e. $p = 2$. Hence $v + w = -2u_1 = \pm 2$, implying $v = w\,(= \pm 1)$ and $u = -2v$. Consequently $a - b = -2(b-c)$.

Knowing that n is even, say $n = 2k$, we rewrite the equation $a^n - b^n = -p(b-c)$ with $p = 2$ in the form

$$(a^k + b^k)(a^k - b^k) = -2(b-c) = a - b.$$

The second factor on the left is divisible by $a - b$, so the first factor $(a^k + b^k)$ must be ± 1. Then exactly one of a and b must be odd; yet $a - b = -2(b-c)$ is even. Contradiction ends the proof.

Solution 2. The beginning is as in the first solution. Assuming that a, b, c are not all equal, hence are all distinct, we derive equation (1) with the conclusion that n is even. Write $n = 2k$.

Suppose that p is odd. Then the integer

$$\frac{a^n - b^n}{a-b} = a^{n-1} + a^{n-2}b + \cdots + b^{n-1},$$

which is a factor in (1), must be odd as well. This sum of $n = 2k$ summands is odd only if a and b have different parities. The same conclusion holding for b, c and for c, a, we get that a, b, c, a alternate in their parities, which is clearly impossible.

Thus $p = 2$. The original system shows that a, b, c must be of the same parity. So we may divide (1) by p^3, i.e. 2^3, to obtain the following product of six integer factors:

$$\frac{a^k + b^k}{2} \cdot \frac{a^k - b^k}{a - b} \cdot \frac{b^k + c^k}{2} \cdot \frac{b^k - c^k}{b - c} \cdot \frac{c^k + a^k}{2} \cdot \frac{c^k - a^k}{c - a} = -1. \tag{2}$$

Each one of the factors must be equal to ± 1. In particular, $a^k + b^k = \pm 2$. If k is even, this becomes $a^k + b^k = 2$ and yields $|a| = |b| = 1$, whence $a^k - b^k = 0$, contradicting (2).

Let now k be odd. Then the sum $a^k + b^k$, with value ± 2, has $a + b$ as a factor. Since a and b are of the same parity, this means that $a + b = \pm 2$; and cyclically, $b + c = \pm 2$, $c + a = \pm 2$. In some two of these equations the signs must coincide, hence some two of a, b, c are equal. This is the desired contradiction.

Comment. Having arrived at the equation (1) one is tempted to write down all possible decompositions of $-p^3$ (cube of a prime) into a product of three integers. This leads to cumbersome examination of many cases, some of which are unpleasant to handle. One may do that just for $p = 2$, having earlier in some way eliminated odd primes from consideration.

However, the second solution shows that the condition of p being a prime is far too strong. What is actually being used in that solution, is that p is either a positive odd integer or $p = 2$.

N2. Let $a_1, a_2, \ldots, a_n$ be distinct positive integers, $n \geq 3$. Prove that there exist distinct indices i and j such that $a_i + a_j$ does not divide any of the numbers $3a_1, 3a_2, \ldots, 3a_n$.

Solution. Without loss of generality, let $0 < a_1 < a_2 < \cdots < a_n$. One can also assume that $a_1, a_2, \ldots, a_n$ are coprime. Otherwise division by their greatest common divisor reduces the question to the new sequence whose terms are coprime integers.

Suppose that the claim is false. Then for each $i < n$ there exists a j such that $a_n + a_i$ divides $3a_j$. If $a_n + a_i$ is not divisible by 3 then $a_n + a_i$ divides a_j which is impossible as $0 < a_j \leq a_n < a_n + a_i$. Thus $a_n + a_i$ is a multiple of 3 for $i = 1, \ldots, n-1$, so that $a_1, a_2, \ldots, a_{n-1}$ are all congruent (to $-a_n$) modulo 3.

Now a_n is not divisible by 3 or else so would be all remaining a_i's, meaning that $a_1, a_2, \ldots, a_n$ are not coprime. Hence $a_n \equiv r \pmod 3$ where $r \in \{1, 2\}$, and $a_i \equiv 3 - r \pmod 3$ for all $i = 1, \ldots, n-1$.

Consider a sum $a_{n-1} + a_i$ where $1 \leq i \leq n-2$. There is at least one such sum as $n \geq 3$. Let j be an index such that $a_{n-1} + a_i$ divides $3a_j$. Observe that $a_{n-1} + a_i$ is not divisible by 3 since $a_{n-1} + a_i \equiv 2a_i \not\equiv 0 \pmod 3$. It follows that $a_{n-1} + a_i$ divides a_j, in particular $a_{n-1} + a_i \leq a_j$. Hence $a_{n-1} < a_j \leq a_n$, implying $j = n$. So a_n is divisible by all sums $a_{n-1} + a_i$, $1 \leq i \leq n-2$. In particular $a_{n-1} + a_i \leq a_n$ for $i = 1, \ldots, n-2$.

Let j be such that $a_n + a_{n-1}$ divides $3a_j$. If $j \leq n-2$ then $a_n + a_{n-1} \leq 3a_j < a_j + 2a_{n-1}$. This yields $a_n < a_{n-1} + a_j$; however $a_{n-1} + a_j \leq a_n$ for $j \leq n-2$. Therefore $j = n-1$ or $j = n$.

For $j = n-1$ we obtain $3a_{n-1} = k(a_n + a_{n-1})$ with k an integer, and it is straightforward that $k = 1$ ($k \leq 0$ and $k \geq 3$ contradict $0 < a_{n-1} < a_n$; $k = 2$ leads to $a_{n-1} = 2a_n > a_{n-1}$). Thus $3a_{n-1} = a_n + a_{n-1}$, i. e. $a_n = 2a_{n-1}$.

Similarly, if $j = n$ then $3a_n = k(a_n + a_{n-1})$ for some integer k, and only $k = 2$ is possible. Hence $a_n = 2a_{n-1}$ holds true in both cases remaining, $j = n-1$ and $j = n$.

Now $a_n = 2a_{n-1}$ implies that the sum $a_{n-1} + a_1$ is strictly between $a_n/2$ and a_n. But a_{n-1} and a_1 are distinct as $n \geq 3$, so it follows from the above that $a_{n-1} + a_1$ divides a_n. This provides the desired contradiction.

N3. Let $a_0, a_1, a_2, \ldots$ be a sequence of positive integers such that the greatest common divisor of any two consecutive terms is greater than the preceding term; in symbols, $\gcd(a_i, a_{i+1}) > a_{i-1}$. Prove that $a_n \geq 2^n$ for all $n \geq 0$.

Solution. Since $a_i \geq \gcd(a_i, a_{i+1}) > a_{i-1}$, the sequence is strictly increasing. In particular $a_0 \geq 1$, $a_1 \geq 2$. For each $i \geq 1$ we also have $a_{i+1} - a_i \geq \gcd(a_i, a_{i+1}) > a_{i-1}$, and consequently $a_{i+1} \geq a_i + a_{i-1} + 1$. Hence $a_2 \geq 4$ and $a_3 \geq 7$. The equality $a_3 = 7$ would force equalities in the previous estimates, leading to $\gcd(a_2, a_3) = \gcd(4, 7) > a_1 = 2$, which is false. Thus $a_3 \geq 8$; the result is valid for $n = 0, 1, 2, 3$. These are the base cases for a proof by induction.

Take an $n \geq 3$ and assume that $a_i \geq 2^i$ for $i = 0, 1, \ldots, n$. We must show that $a_{n+1} \geq 2^{n+1}$. Let $\gcd(a_n, a_{n+1}) = d$. We know that $d > a_{n-1}$. The induction claim is reached immediately in the following cases:

$$\begin{aligned}
&\text{if } a_{n+1} \geq 4d \quad \text{then } a_{n+1} > 4a_{n-1} \geq 4 \cdot 2^{n-1} = 2^{n+1};\\
&\text{if } a_n \geq 3d \quad \text{then } a_{n+1} \geq a_n + d \geq 4d > 4a_{n-1} \geq 4 \cdot 2^{n-1} = 2^{n+1};\\
&\text{if } a_n = d \quad \text{then } a_{n+1} \geq a_n + d = 2a_n \geq 2 \cdot 2^n = 2^{n+1}.
\end{aligned}$$

The only remaining possibility is that $a_n = 2d$ and $a_{n+1} = 3d$, which we assume for the sequel. So $a_{n+1} = \frac{3}{2}a_n$.

Let now $\gcd(a_{n-1}, a_n) = d'$; then $d' > a_{n-2}$. Write $a_n = md'$ (m an integer). Keeping in mind that $d' \leq a_{n-1} < d$ and $a_n = 2d$, we get that $m \geq 3$. Also $a_{n-1} < d = \frac{1}{2}md'$, $a_{n+1} = \frac{3}{2}md'$. Again we single out the cases which imply the induction claim immediately:

$$\begin{aligned}
&\text{if } m \geq 6 \quad \text{then } a_{n+1} = \tfrac{3}{2}md' \geq 9d' > 9a_{n-2} \geq 9 \cdot 2^{n-2} > 2^{n+1};\\
&\text{if } 3 \leq m \leq 4 \text{ then } a_{n-1} < \tfrac{1}{2} \cdot 4d', \text{ and hence } a_{n-1} = d',\\
&\qquad\qquad a_{n+1} = \tfrac{3}{2}ma_{n-1} \geq \tfrac{3}{2} \cdot 3a_{n-1} \geq \tfrac{9}{2} \cdot 2^{n-1} > 2^{n+1}.
\end{aligned}$$

So we are left with the case $m = 5$, which means that $a_n = 5d'$, $a_{n+1} = \frac{15}{2}d'$, $a_{n-1} < d = \frac{5}{2}d'$. The last relation implies that a_{n-1} is either d' or $2d'$. Anyway, $a_{n-1} | 2d'$.

The same pattern repeats once more. We denote $\gcd(a_{n-2}, a_{n-1}) = d''$; then $d'' > a_{n-3}$. Because d'' is a divisor of a_{n-1}, hence also of $2d'$, we may write $2d' = m'd''$ (m' an integer). Since $d'' \leq a_{n-2} < d'$, we get $m' \geq 3$. Also, $a_{n-2} < d' = \frac{1}{2}m'd''$, $a_{n+1} = \frac{15}{2}d' = \frac{15}{4}m'd''$. As before, we consider the cases:

$$\begin{aligned}
&\text{if } m' \geq 5 \quad \text{then } a_{n+1} = \tfrac{15}{4}m'd'' \geq \tfrac{75}{4}d'' > \tfrac{75}{4}a_{n-3} \geq \tfrac{75}{4} \cdot 2^{n-3} > 2^{n+1};\\
&\text{if } 3 \leq m' \leq 4 \text{ then } a_{n-2} < \tfrac{1}{2} \cdot 4d'', \text{ and hence } a_{n-2} = d'',\\
&\qquad\qquad a_{n+1} = \tfrac{15}{4}m'a_{n-2} \geq \tfrac{15}{4} \cdot 3a_{n-2} \geq \tfrac{45}{4} \cdot 2^{n-2} > 2^{n+1}.
\end{aligned}$$

Both of them have produced the induction claim. But now there are no cases left. Induction is complete; the inequality $a_n \geq 2^n$ holds for all n.

N4. Let n be a positive integer. Show that the numbers

$$\binom{2^n-1}{0}, \quad \binom{2^n-1}{1}, \quad \binom{2^n-1}{2}, \quad \dots, \quad \binom{2^n-1}{2^{n-1}-1}$$

are congruent modulo 2^n to $1, 3, 5, \dots, 2^n-1$ in some order.

Solution 1. It is well-known that all these numbers are odd. So the assertion that their remainders (mod 2^n) make up a permutation of $\{1, 3, \dots, 2^n-1\}$ is equivalent just to saying that these remainders are all distinct. We begin by showing that

$$\binom{2^n-1}{2k} + \binom{2^n-1}{2k+1} \equiv 0 \pmod{2^n} \quad \text{and} \quad \binom{2^n-1}{2k} \equiv (-1)^k \binom{2^{n-1}-1}{k} \pmod{2^n}. \tag{1}$$

The first relation is immediate, as the sum on the left is equal to $\binom{2^n}{2k+1} = \frac{2^n}{2k+1}\binom{2^n-1}{2k}$, hence is divisible by 2^n. The second relation:

$$\binom{2^n-1}{2k} = \prod_{j=1}^{2k} \frac{2^n-j}{j} = \prod_{i=1}^{k} \frac{2^n-(2i-1)}{2i-1} \cdot \prod_{i=1}^{k} \frac{2^{n-1}-i}{i} \equiv (-1)^k \binom{2^{n-1}-1}{k} \pmod{2^n}.$$

This prepares ground for a proof of the required result by induction on n. The base case $n = 1$ is obvious. Assume the assertion is true for $n-1$ and pass to n, denoting $a_k = \binom{2^{n-1}-1}{k}$, $b_m = \binom{2^n-1}{m}$. The induction hypothesis is that all the numbers a_k $(0 \le k < 2^{n-2})$ are distinct (mod 2^{n-1}); the claim is that all the numbers b_m $(0 \le m < 2^{n-1})$ are distinct (mod 2^n).

The congruence relations (1) are restated as

$$b_{2k} \equiv (-1)^k a_k \equiv -b_{2k+1} \pmod{2^n}. \tag{2}$$

Shifting the exponent in the first relation of (1) from n to $n-1$ we also have the congruence $a_{2i+1} \equiv -a_{2i} \pmod{2^{n-1}}$. We hence conclude:

If, for some $j, k < 2^{n-2}$, $a_k \equiv -a_j \pmod{2^{n-1}}$, then $\{j, k\} = \{2i, 2i+1\}$ for some i. (3)

This is so because in the sequence $(a_k : k < 2^{n-2})$ each term a_j is complemented to 0 (mod 2^{n-1}) by only one other term a_k, according to the induction hypothesis.

From (2) we see that $b_{4i} \equiv a_{2i}$ and $b_{4i+3} \equiv a_{2i+1} \pmod{2^n}$. Let

$$M = \{m : 0 \le m < 2^{n-1},\ m \equiv 0 \text{ or } 3 \pmod 4\}, \quad L = \{l : 0 \le l < 2^{n-1},\ l \equiv 1 \text{ or } 2 \pmod 4\}.$$

The last two congruences take on the unified form

$$b_m \equiv a_{\lfloor m/2 \rfloor} \pmod{2^n} \quad \text{for all} \quad m \in M. \tag{4}$$

Thus all the numbers b_m for $m \in M$ are distinct (mod 2^n) because so are the numbers a_k (they are distinct (mod 2^{n-1}), hence also (mod 2^n)).

Every $l \in L$ is paired with a unique $m \in M$ into a pair of the form $\{2k, 2k+1\}$. So (2) implies that also all the b_l for $l \in L$ are distinct (mod 2^n). It remains to eliminate the possibility that $b_m \equiv b_l \pmod{2^n}$ for some $m \in M$, $l \in L$.

Suppose that such a situation occurs. Let $m' \in M$ be such that $\{m', l\}$ is a pair of the form $\{2k, 2k+1\}$, so that (see (2)) $b_{m'} \equiv -b_l \pmod{2^n}$. Hence $b_{m'} \equiv -b_m \pmod{2^n}$. Since both m' and m are in M, we have by (4) $b_{m'} \equiv a_j$, $b_m \equiv a_k \pmod{2^n}$ for $j = \lfloor m'/2 \rfloor$, $k = \lfloor m/2 \rfloor$.

Then $a_j \equiv -a_k \pmod{2^n}$. Thus, according to (3), $j = 2i$, $k = 2i+1$ for some i (or vice versa). The equality $a_{2i+1} \equiv -a_{2i} \pmod{2^n}$ now means that $\binom{2^{n-1}-1}{2i} + \binom{2^{n-1}-1}{2i+1} \equiv 0 \pmod{2^n}$. However, the sum on the left is equal to $\binom{2^{n-1}}{2i+1}$. A number of this form cannot be divisible by 2^n. This is a contradiction which concludes the induction step and proves the result.

Solution 2. We again proceed by induction, writing for brevity $N = 2^{n-1}$ and keeping notation $a_k = \binom{N-1}{k}$, $b_m = \binom{2N-1}{m}$. Assume that the result holds for the sequence $(a_0, a_1, a_2, \ldots, a_{N/2-1})$. In view of the symmetry $a_{N-1-k} = a_k$ this sequence is a permutation of $(a_0, a_2, a_4, \ldots, a_{N-2})$. So the induction hypothesis says that this latter sequence, taken (mod N), is a permutation of $(1, 3, 5, \ldots, N-1)$. Similarly, the induction claim is that $(b_0, b_2, b_4, \ldots, b_{2N-2})$, taken (mod $2N$), is a permutation of $(1, 3, 5, \ldots, 2N-1)$.

In place of the congruence relations (2) we now use the following ones,

$$b_{4i} \equiv a_{2i} \pmod{N} \qquad \text{and} \qquad b_{4i+2} \equiv b_{4i} + N \pmod{2N}. \tag{5}$$

Given this, the conclusion is immediate: the first formula of (5) together with the induction hypothesis tells us that $(b_0, b_4, b_8, \ldots, b_{2N-4})$ (mod N) is a permutation of $(1, 3, 5, \ldots, N-1)$. Then the second formula of (5) shows that $(b_2, b_6, b_{10}, \ldots, b_{2N-2})$ (mod N) is exactly the same permutation; moreover, this formula distinguishes (mod $2N$) each b_{4i} from b_{4i+2}.

Consequently, these two sequences combined represent (mod $2N$) a permutation of the sequence $(1, 3, 5, \ldots, N-1,\ N+1, N+3, N+5, \ldots, N+N-1)$, and this is precisely the induction claim.

Now we prove formulas (5); we begin with the second one. Since $b_{m+1} = b_m \cdot \frac{2N-m-1}{m+1}$,

$$b_{4i+2} = b_{4i} \cdot \frac{2N-4i-1}{4i+1} \cdot \frac{2N-4i-2}{4i+2} = b_{4i} \cdot \frac{2N-4i-1}{4i+1} \cdot \frac{N-2i-1}{2i+1}.$$

The desired congruence $b_{4i+2} \equiv b_{4i} + N$ may be multiplied by the odd number $(4i+1)(2i+1)$, giving rise to a chain of successively equivalent congruences:

$$\begin{aligned} b_{4i}(2N-4i-1)(N-2i-1) &\equiv (b_{4i}+N)(4i+1)(2i+1) && \pmod{2N}, \\ b_{4i}(2i+1-N) &\equiv (b_{4i}+N)(2i+1) && \pmod{2N}, \\ (b_{4i}+2i+1)N &\equiv 0 && \pmod{2N}; \end{aligned}$$

and the last one is satisfied, as b_{4i} is odd. This settles the second relation in (5).

The first one is proved by induction on i. It holds for $i = 0$. Assume $b_{4i} \equiv a_{2i}$ (mod $2N$) and consider $i+1$:

$$b_{4i+4} = b_{4i+2} \cdot \frac{2N-4i-3}{4i+3} \cdot \frac{2N-4i-4}{4i+4}; \qquad a_{2i+2} = a_{2i} \cdot \frac{N-2i-1}{2i+1} \cdot \frac{N-2i-2}{2i+2}.$$

Both expressions have the fraction $\frac{N-2i-2}{2i+2}$ as the last factor. Since $2i+2 < N = 2^{n-1}$, this fraction reduces to ℓ/m with ℓ and m odd. In showing that $b_{4i+4} \equiv a_{2i+2}$ (mod $2N$), we may ignore this common factor ℓ/m. Clearing other odd denominators reduces the claim to

$$b_{4i+2}(2N-4i-3)(2i+1) \equiv a_{2i}(N-2i-1)(4i+3) \pmod{2N}.$$

By the inductive assumption (saying that $b_{4i} \equiv a_{2i}$ (mod $2N$)) and by the second relation of (5), this is equivalent to

$$(b_{4i}+N)(2i+1) \equiv b_{4i}(2i+1-N) \pmod{2N},$$

a congruence which we have already met in the preceding proof a few lines above. This completes induction (on i) and the proof of (5), hence also the whole solution.

Comment. One can avoid the words *congruent modulo* in the problem statement by rephrasing the assertion into: *Show that these numbers leave distinct remainders in division by* 2^n.

N5. For every $n \in \mathbb{N}$ let $d(n)$ denote the number of (positive) divisors of n. Find all functions $f\colon \mathbb{N} \to \mathbb{N}$ with the following properties:

(i) $d(f(x)) = x$ for all $x \in \mathbb{N}$;

(ii) $f(xy)$ divides $(x-1)y^{xy-1}f(x)$ for all $x, y \in \mathbb{N}$.

Solution. There is a unique solution: the function $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(1) = 1$ and

$$f(n) = p_1^{p_1^{a_1}-1} p_2^{p_2^{a_2}-1} \cdots p_k^{p_k^{a_k}-1} \text{ where } n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \text{ is the prime factorization of } n > 1. \quad (1)$$

Direct verification shows that this function meets the requirements.

Conversely, let $f\colon \mathbb{N} \to \mathbb{N}$ satisfy (i) and (ii). Applying (i) for $x = 1$ gives $d(f(1)) = 1$, so $f(1) = 1$. In the sequel we prove that (1) holds for all $n > 1$. Notice that $f(m) = f(n)$ implies $m = n$ in view of (i). The formula $d\left(p_1^{b_1} \cdots p_k^{b_k}\right) = (b_1+1)\cdots(b_k+1)$ will be used throughout.

Let p be a prime. Since $d(f(p)) = p$, the formula just mentioned yields $f(p) = q^{p-1}$ for some prime q; in particular $f(2) = q^{2-1} = q$ is a prime. We prove that $f(p) = p^{p-1}$ for all primes p.

Suppose that p is odd and $f(p) = q^{p-1}$ for a prime q. Applying (ii) first with $x = 2$, $y = p$ and then with $x = p$, $y = 2$ shows that $f(2p)$ divides both $(2-1)p^{2p-1}f(2) = p^{2p-1}f(2)$ and $(p-1)2^{2p-1}f(p) = (p-1)2^{2p-1}q^{p-1}$. If $q \neq p$ then the odd prime p does not divide $(p-1)2^{2p-1}q^{p-1}$, hence the greatest common divisor of $p^{2p-1}f(2)$ and $(p-1)2^{2p-1}q^{p-1}$ is a divisor of $f(2)$. Thus $f(2p)$ divides $f(2)$ which is a prime. As $f(2p) > 1$, we obtain $f(2p) = f(2)$ which is impossible. So $q = p$, i. e. $f(p) = p^{p-1}$.

For $p = 2$ the same argument with $x = 2$, $y = 3$ and $x = 3$, $y = 2$ shows that $f(6)$ divides both $3^5 f(2)$ and $2^6 f(3) = 2^6 3^2$. If the prime $f(2)$ is odd then $f(6)$ divides $3^2 = 9$, so $f(6) \in \{1, 3, 9\}$. However then $6 = d(f(6)) \in \{d(1), d(3), d(9)\} = \{1, 2, 3\}$ which is false. In conclusion $f(2) = 2$.

Next, for each $n > 1$ the prime divisors of $f(n)$ are among the ones of n. Indeed, let p be the least prime divisor of n. Apply (ii) with $x = p$ and $y = n/p$ to obtain that $f(n)$ divides $(p-1)y^{n-1}f(p) = (p-1)y^{n-1}p^{p-1}$. Write $f(n) = \ell P$ where ℓ is coprime to n and P is a product of primes dividing n. Since ℓ divides $(p-1)y^{n-1}p^{p-1}$ and is coprime to $y^{n-1}p^{p-1}$, it divides $p-1$; hence $d(\ell) \leq \ell < p$. But (i) gives $n = d(f(n)) = d(\ell P)$, and $d(\ell P) = d(\ell)d(P)$ as ℓ and P are coprime. Therefore $d(\ell)$ is a divisor of n less than p, meaning that $\ell = 1$ and proving the claim.

Now (1) is immediate for prime powers. If p is a prime and $a \geq 1$, by the above the only prime factor of $f(p^a)$ is p (a prime factor does exist as $f(p^a) > 1$). So $f(p^a) = p^b$ for some $b \geq 1$, and (i) yields $p^a = d(f(p^a)) = d\left(p^b\right) = b + 1$. Hence $f(p^a) = p^{p^a-1}$, as needed.

Let us finally show that (1) is true for a general $n > 1$ with prime factorization $n = p_1^{a_1} \cdots p_k^{a_k}$. We saw that the prime factorization of $f(n)$ has the form $f(n) = p_1^{b_1} \cdots p_k^{b_k}$. For $i = 1, \ldots, k$, set $x = p_i^{a_i}$ and $y = n/x$ in (ii) to infer that $f(n)$ divides $(p_i^{a_i} - 1)\, y^{n-1} f(p_i^{a_i})$. Hence $p_i^{b_i}$ divides $(p_i^{a_i} - 1)\, y^{n-1} f(p_i^{a_i})$, and because $p_i^{b_i}$ is coprime to $(p_i^{a_i} - 1)\, y^{n-1}$, it follows that $p_i^{b_i}$ divides $f(p_i^{a_i}) = p_i^{p_i^{a_i}-1}$. So $b_i \leq p_i^{a_i} - 1$ for all $i = 1, \ldots, k$. Combined with (i), these conclusions imply

$$p_1^{a_1} \cdots p_k^{a_k} = n = d(f(n)) = d\left(p_1^{b_1} \cdots p_k^{b_k}\right) = (b_1+1)\cdots(b_k+1) \leq p_1^{a_1} \cdots p_k^{a_k}.$$

Hence all inequalities $b_i \leq p_i^{a_i} - 1$ must be equalities, $i = 1, \ldots, k$, implying that (1) holds true. The proof is complete.

N6. Prove that there exist infinitely many positive integers n such that n^2+1 has a prime divisor greater than $2n+\sqrt{2n}$.

Solution. Let $p \equiv 1 \pmod 8$ be a prime. The congruence $x^2 \equiv -1 \pmod p$ has two solutions in $[1, p-1]$ whose sum is p. If n is the smaller one of them then p divides n^2+1 and $n \le (p-1)/2$. We show that $p > 2n+\sqrt{10n}$.

Let $n = (p-1)/2 - \ell$ where $\ell \ge 0$. Then $n^2 \equiv -1 \pmod p$ gives

$$\left(\frac{p-1}{2} - \ell\right)^2 \equiv -1 \pmod p \qquad \text{or} \qquad (2\ell+1)^2 + 4 \equiv 0 \pmod p.$$

Thus $(2\ell+1)^2+4 = rp$ for some $r \ge 0$. As $(2\ell+1)^2 \equiv 1 \equiv p \pmod 8$, we have $r \equiv 5 \pmod 8$, so that $r \ge 5$. Hence $(2\ell+1)^2 + 4 \ge 5p$, implying $\ell \ge \big(\sqrt{5p-4} - 1\big)/2$. Set $\sqrt{5p-4} = u$ for clarity; then $\ell \ge (u-1)/2$. Therefore

$$n = \frac{p-1}{2} - \ell \le \frac{1}{2}\big(p-u\big).$$

Combined with $p = (u^2+4)/5$, this leads to $u^2 - 5u - 10n + 4 \ge 0$. Solving this quadratic inequality with respect to $u \ge 0$ gives $u \ge \big(5 + \sqrt{40n+9}\big)/2$. So the estimate $n \le \big(p-u\big)/2$ leads to

$$p \ge 2n + u \ge 2n + \frac{1}{2}\big(5 + \sqrt{40n+9}\big) > 2n + \sqrt{10n}.$$

Since there are infinitely many primes of the form $8k+1$, it follows easily that there are also infinitely many n with the stated property.

Comment. By considering the prime factorization of the product $\prod_{n=1}^{N}(n^2+1)$, it can be obtained that its greatest prime divisor is at least $cN\log N$. This could improve the statement as $p > n\log n$.

However, the proof applies some advanced information about the distribution of the primes of the form $4k+1$, which is inappropriate for high schools contests.

51

52